DICE Embeddings

Release 0.1.3.2

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Contents:

1 Dicee Manual **2**

2 Installation **3**
2.1 Installation from Source 3

3 Download Knowledge Graphs **3**

4 Knowledge Graph Embedding Models **3**

5 How to Train **3**

6 Creating an Embedding Vector Database **5**
6.1 Learning Embeddings 5
6.2 Loading Embeddings into Qdrant Vector Database 6
6.3 Launching Webservice 6

7 Answering Complex Queries **6**

8 Predicting Missing Links **8**

9 Downloading Pretrained Models **8**

10 How to Deploy **8**

11 Docker **8**

12 Coverage Report **8**

13 How to cite **10**

14 dicee **12**
14.1 Submodules 12
14.2 Attributes 281
14.3 Classes 281
14.4 Package Contents 282

Python Module Index **296**

DICE Embeddings[1]: Hardware-agnostic Framework for Large-scale Knowledge Graph Embeddings:

1 Dicee Manual

Version: dicee 0.3.2

GitHub repository: https://github.com/dice-group/dice-embeddings

Publisher and maintainer: Caglar Demir[2]

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Dicee is a hardware-agnostic framework for large-scale knowledge graph embeddings.

Knowledge graph embedding research has mainly focused on learning continuous representations of knowledge graphs towards the link prediction problem. Recently developed frameworks can be effectively applied in a wide range of research-related applications. Yet, using these frameworks in real-world applications becomes more challenging as the size of the knowledge graph grows

We developed the DICE Embeddings framework (dicee) to compute embeddings for large-scale knowledge graphs in a hardware-agnostic manner. To achieve this goal, we rely on

1. **Pandas**[3] **& Co.** to use parallelism at preprocessing a large knowledge graph,
2. **PyTorch**[4] **& Co.** to learn knowledge graph embeddings via multi-CPUs, GPUs, TPUs or computing cluster, and
3. **Huggingface**[5] to ease the deployment of pre-trained models.

Why Pandas[6] **& Co. ?** A large knowledge graph can be read and preprocessed (e.g. removing literals) by pandas, modin, or polars in parallel. Through polars, a knowledge graph having more than 1 billion triples can be read in parallel fashion. Importantly, using these frameworks allow us to perform all necessary computations on a single CPU as well as a cluster of computers.

Why PyTorch[7] **& Co. ?** PyTorch is one of the most popular machine learning frameworks available at the time of writing. PytorchLightning facilitates scaling the training procedure of PyTorch without boilerplate. In our framework, we combine PyTorch[8] & PytorchLightning[9]. Users can choose the trainer class (e.g., DDP by Pytorch) to train large knowledge graph embedding models with billions of parameters. PytorchLightning allows us to use state-of-the-art model parallelism techniques (e.g. Fully Sharded Training, FairScale, or DeepSpeed) without extra effort. With our framework, practitioners can directly use PytorchLightning for model parallelism to train gigantic embedding models.

Why Huggingface[10]**?** Seamlessly deploy and share pre-trained embedding models through the Huggingface ecosystem.

[1] https://github.com/dice-group/dice-embeddings
[2] https://github.com/Demirrr
[3] https://pandas.pydata.org/
[4] https://pytorch.org/
[5] https://huggingface.co/
[6] https://pandas.pydata.org/
[7] https://pytorch.org/
[8] https://pytorch.org/
[9] https://www.pytorchlightning.ai/
[10] https://huggingface.co/

2 Installation

2.1 Installation from Source

```
git clone https://github.com/dice-group/dice-embeddings.git
conda create -n dice python=3.10.13 --no-default-packages && conda activate dice &&
↪cd dice-embeddings &&
pip3 install -e .
```

or

```
pip install dicee
```

3 Download Knowledge Graphs

```
wget https://files.dice-research.org/datasets/dice-embeddings/KGs.zip --no-check-
↪certificate && unzip KGs.zip
```

To test the Installation

```
python -m pytest -p no:warnings -x # Runs >114 tests leading to > 15 mins
python -m pytest -p no:warnings --lf # run only the last failed test
python -m pytest -p no:warnings --ff # to run the failures first and then the rest of
↪the tests.
```

4 Knowledge Graph Embedding Models

1. TransE, DistMult, ComplEx, ConEx, QMult, OMult, ConvO, ConvQ, Keci
2. All 44 models available in https://github.com/pykeen/pykeen#models

   For more, please refer to `examples`.

5 How to Train

To Train a KGE model (KECI) and evaluate it on the train, validation, and test sets of the UMLS benchmark dataset.

```
from dicee.executer import Execute
from dicee.config import Namespace
args = Namespace()
args.model = 'Keci'
args.scoring_technique = "KvsAll"  # 1vsAll, or AllvsAll, or NegSample
args.dataset_dir = "KGs/UMLS"
args.path_to_store_single_run = "Keci_UMLS"
args.num_epochs = 100
args.embedding_dim = 32
args.batch_size = 1024
reports = Execute(args).start()
print(reports["Train"]["MRR"]) # => 0.9912
print(reports["Test"]["MRR"]) # => 0.8155
# See the Keci_UMLS folder embeddings and all other files
```

where the data is in the following form

```
$ head -3 KGs/UMLS/train.txt
acquired_abnormality    location_of     experimental_model_of_disease
anatomical_abnormality  manifestation_of        physiologic_function
alga    isa     entity
```

A KGE model can also be trained from the command line

```
dicee --dataset_dir "KGs/UMLS" --model Keci --eval_model "train_val_test"
```

dicee automaticaly detects available GPUs and trains a model with distributed data parallels technique. Under the hood, dicee uses lighning as a default trainer.

```
# Train a model by only using the GPU-0
CUDA_VISIBLE_DEVICES=0 dicee --dataset_dir "KGs/UMLS" --model Keci --eval_model
↪"train_val_test"
# Train a model by only using GPU-1
CUDA_VISIBLE_DEVICES=1 dicee --dataset_dir "KGs/UMLS" --model Keci --eval_model
↪"train_val_test"
NCCL_P2P_DISABLE=1 CUDA_VISIBLE_DEVICES=0,1 python dicee/scripts/run.py --trainer PL -
↪-dataset_dir "KGs/UMLS" --model Keci --eval_model "train_val_test"
```

Under the hood, dicee executes run.py script and uses lighning as a default trainer

```
# Two equivalent executions
# (1)
dicee --dataset_dir "KGs/UMLS" --model Keci --eval_model "train_val_test"
# Evaluate Keci on Train set: Evaluate Keci on Train set
# {'H@1': 0.9518788343558282, 'H@3': 0.9988496932515337, 'H@10': 1.0, 'MRR': 0.
↪9753123402351737}
# Evaluate Keci on Validation set: Evaluate Keci on Validation set
# {'H@1': 0.6932515337423313, 'H@3': 0.9041411042944786, 'H@10': 0.9754601226993865,
↪'MRR': 0.8072362996241839}
# Evaluate Keci on Test set: Evaluate Keci on Test set
# {'H@1': 0.6951588502269289, 'H@3': 0.9039334341906202, 'H@10': 0.9750378214826021,
↪'MRR': 0.8064032293278861}

# (2)
CUDA_VISIBLE_DEVICES=0,1 python dicee/scripts/run.py --trainer PL --dataset_dir "KGs/
↪UMLS" --model Keci --eval_model "train_val_test"
# Evaluate Keci on Train set: Evaluate Keci on Train set
# {'H@1': 0.9518788343558282, 'H@3': 0.9988496932515337, 'H@10': 1.0, 'MRR': 0.
↪9753123402351737}
# Evaluate Keci on Train set: Evaluate Keci on Train set
# Evaluate Keci on Validation set: Evaluate Keci on Validation set
# {'H@1': 0.6932515337423313, 'H@3': 0.9041411042944786, 'H@10': 0.9754601226993865,
↪'MRR': 0.8072362996241839}
# Evaluate Keci on Test set: Evaluate Keci on Test set
# {'H@1': 0.6951588502269289, 'H@3': 0.9039334341906202, 'H@10': 0.9750378214826021,
↪'MRR': 0.8064032293278861}
```

Similarly, models can be easily trained with torchrun

```
torchrun --standalone --nnodes=1 --nproc_per_node=gpu dicee/scripts/run.py --trainer␣
↪torchDDP --dataset_dir "KGs/UMLS" --model Keci --eval_model "train_val_test"
# Evaluate Keci on Train set: Evaluate Keci on Train set: Evaluate Keci on Train set
# {'H@1': 0.9518788343558282, 'H@3': 0.9988496932515337, 'H@10': 1.0, 'MRR': 0.
↪9753123402351737}
# Evaluate Keci on Validation set: Evaluate Keci on Validation set
# {'H@1': 0.6932515337423313, 'H@3': 0.9041411042944786, 'H@10': 0.9754601226993865,
↪'MRR': 0.8072499937521418}
# Evaluate Keci on Test set: Evaluate Keci on Test set
{'H@1': 0.6951588502269289, 'H@3': 0.9039334341906202, 'H@10': 0.9750378214826021,
↪'MRR': 0.8064032293278861}
```

You can also train a model in multi-node multi-gpu setting.

```
torchrun --nnodes 2 --nproc_per_node=gpu  --node_rank 0 --rdzv_id 455 --rdzv_backend␣
↪c10d --rdzv_endpoint=nebula  dicee/scripts/run.py --trainer torchDDP --dataset_dir␣
↪KGs/UMLS
torchrun --nnodes 2 --nproc_per_node=gpu  --node_rank 1 --rdzv_id 455 --rdzv_backend␣
↪c10d --rdzv_endpoint=nebula dicee/scripts/run.py --trainer torchDDP --dataset_dir␣
↪KGs/UMLS
```

Train a KGE model by providing the path of a single file and store all parameters under newly created directory called `KeciFamilyRun`.

```
dicee --path_single_kg "KGs/Family/family-benchmark_rich_background.owl" --model Keci␣
↪--path_to_store_single_run KeciFamilyRun --backend rdflib
```

where the data is in the following form

```
$ head -3 KGs/Family/train.txt
_:1 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://www.w3.org/2002/07/owl
↪#Ontology> .
<http://www.benchmark.org/family#hasChild> <http://www.w3.org/1999/02/22-rdf-syntax-ns
↪#type> <http://www.w3.org/2002/07/owl#ObjectProperty> .
<http://www.benchmark.org/family#hasParent> <http://www.w3.org/1999/02/22-rdf-syntax-
↪ns#type> <http://www.w3.org/2002/07/owl#ObjectProperty> .
```

Apart from n-triples or standard link prediction dataset formats, we support [“owl”, “nt”, “turtle”, “rdf/xml”, “n3”]*. Moreover, a KGE model can be also trained by providing **an endpoint of a triple store**.

```
dicee --sparql_endpoint "http://localhost:3030/mutagenesis/" --model Keci
```

For more, please refer to `examples`.

6 Creating an Embedding Vector Database

6.1 Learning Embeddings

```
# Train an embedding model
dicee --dataset_dir KGs/Countries-S1 --path_to_store_single_run CountryEmbeddings --
↪model Keci --p 0 --q 1 --embedding_dim 32 --adaptive_swa
```

6.2 Loading Embeddings into Qdrant Vector Database

```
# Ensure that Qdrant available
# docker pull qdrant/qdrant && docker run -p 6333:6333 -p 6334:6334      -v $(pwd)/
↪qdrant_storage:/qdrant/storage:z      qdrant/qdrant
diceeindex --path_model "CountryEmbeddings" --collection_name "dummy" --location
↪"localhost"
```

6.3 Launching Webservice

```
diceeserve --path_model "CountryEmbeddings" --collection_name "dummy" --collection_
↪location "localhost"
```

Retrieve and Search

Get embedding of germany

```
curl -X 'GET' 'http://0.0.0.0:8000/api/get?q=germany' -H 'accept: application/json'
```

Get most similar things to europe

```
curl -X 'GET' 'http://0.0.0.0:8000/api/search?q=europe' -H 'accept: application/json'
{"result":[{"hit":"europe","score":1.0},
{"hit":"northern_europe","score":0.67126536},
{"hit":"western_europe","score":0.6010134},
{"hit":"puerto_rico","score":0.5051694},
{"hit":"southern_europe","score":0.4829831}]}
```

7 Answering Complex Queries

```
# pip install dicee
# wget https://files.dice-research.org/datasets/dice-embeddings/KGs.zip --no-check-
↪certificate & unzip KGs.zip
from dicee.executer import Execute
from dicee.config import Namespace
from dicee.knowledge_graph_embeddings import KGE
# (1) Train a KGE model
args = Namespace()
args.model = 'Keci'
args.p=0
args.q=1
args.optim = 'Adam'
args.scoring_technique = "AllvsAll"
args.path_single_kg = "KGs/Family/family-benchmark_rich_background.owl"
args.backend = "rdflib"
args.num_epochs = 200
args.batch_size = 1024
args.lr = 0.1
args.embedding_dim = 512
result = Execute(args).start()
# (2) Load the pre-trained model
```

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```
pre_trained_kge = KGE(path=result['path_experiment_folder'])
# (3) Single-hop query answering
# Query: ?E : \exist E.hasSibling(E, F9M167)
# Question: Who are the siblings of F9M167?
# Answer: [F9M157, F9F141], as (F9M167, hasSibling, F9M157) and (F9M167, hasSibling,␣
↪F9F141)
predictions = pre_trained_kge.answer_multi_hop_query(query_type="1p",
                                                     query=('http://www.benchmark.org/
↪family#F9M167',
                                                            ('http://www.benchmark.
↪org/family#hasSibling',)),
                                                     tnorm="min", k=3)
top_entities = [topk_entity for topk_entity, query_score in predictions]
assert "http://www.benchmark.org/family#F9F141" in top_entities
assert "http://www.benchmark.org/family#F9M157" in top_entities
# (2) Two-hop query answering
# Query: ?D : \exist E.Married(D, E) \land hasSibling(E, F9M167)
# Question: To whom a sibling of F9M167 is married to?
# Answer: [F9F158, F9M142] as (F9M157 #married F9F158) and (F9F141 #married F9M142)
predictions = pre_trained_kge.answer_multi_hop_query(query_type="2p",
                                                     query=("http://www.benchmark.org/
↪family#F9M167",
                                                            ("http://www.benchmark.
↪org/family#hasSibling",
                                                             "http://www.benchmark.
↪org/family#married")),
                                                     tnorm="min", k=3)
top_entities = [topk_entity for topk_entity, query_score in predictions]
assert "http://www.benchmark.org/family#F9M142" in top_entities
assert "http://www.benchmark.org/family#F9F158" in top_entities
# (3) Three-hop query answering
# Query: ?T : \exist D.type(D,T) \land Married(D,E) \land hasSibling(E, F9M167)
# Question: What are the type of people who are married to a sibling of F9M167?
# (3) Answer: [Person, Male, Father] since  F9M157 is [Brother Father Grandfather␣
↪Male] and F9M142 is [Male Grandfather Father]

predictions = pre_trained_kge.answer_multi_hop_query(query_type="3p", query=("http://
↪www.benchmark.org/family#F9M167",
                                                                             ("http://
↪www.benchmark.org/family#hasSibling",
                                                                              "http://
↪www.benchmark.org/family#married",
                                                                              "http://
↪www.w3.org/1999/02/22-rdf-syntax-ns#type")),
                                                     tnorm="min", k=5)
top_entities = [topk_entity for topk_entity, query_score in predictions]
print(top_entities)
assert "http://www.benchmark.org/family#Person" in top_entities
assert "http://www.benchmark.org/family#Father" in top_entities
assert "http://www.benchmark.org/family#Male" in top_entities
```

For more, please refer to `examples/multi_hop_query_answering`.

8 Predicting Missing Links

```
from dicee import KGE
# (1) Train a knowledge graph embedding model..
# (2) Load a pretrained model
pre_trained_kge = KGE(path='..')
# (3) Predict missing links through head entity rankings
pre_trained_kge.predict_topk(h=[".."],r=[".."],topk=10)
# (4) Predict missing links through relation rankings
pre_trained_kge.predict_topk(h=[".."],t=[".."],topk=10)
# (5) Predict missing links through tail entity rankings
pre_trained_kge.predict_topk(r=[".."],t=[".."],topk=10)
```

9 Downloading Pretrained Models

```
from dicee import KGE
# (1) Load a pretrained ConEx on DBpedia
model = KGE(url="https://files.dice-research.org/projects/DiceEmbeddings/KINSHIP-Keci-
↪dim128-epoch256-KvsAll")
```

- For more please look at dice-research.org/projects/DiceEmbeddings/[11]

10 How to Deploy

```
from dicee import KGE
KGE(path='...').deploy(share=True,top_k=10)
```

11 Docker

To build the Docker image:

```
docker build -t dice-embeddings .
```

To test the Docker image:

```
docker run --rm -v ~/.local/share/dicee/KGs:/dicee/KGs dice-embeddings ./main.py --
↪model AConEx --embedding_dim 16
```

12 Coverage Report

The coverage report is generated using coverage.py[12]:

```
Name                                                    Stmts   Miss  Cover   Missing
-------------------------------------------------------------------------------------
dicee/__init__.py                                           7      0   100%
dicee/abstracts.py                                        338    115    66%   112-113,␣
```

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[11] https://files.dice-research.org/projects/DiceEmbeddings/
[12] https://coverage.readthedocs.io/en/7.6.0/

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```
↪131, 154-155, 160, 173, 197, 240-254, 290, 303-306, 309-313, 353-364, 379-387, 402,␣
↪413-417, 427-428, 434-436, 442-445, 448-453, 576-596, 602-606, 610-612, 631, 658-696
dicee/callbacks.py                                    248    103    58%   50-55,␣
↪67-73, 76, 88-93, 98-103, 106-109, 116-133, 138-142, 146-147, 247, 281-285, 291-292,
↪ 310-316, 319, 324-325, 337-343, 349-358, 363-365, 410, 421-434, 438-473, 485-491
dicee/config.py                                        97      2    98%   146-147
dicee/dataset_classes.py                              430    146    66%   16, 44,␣
↪57, 89-98, 104, 111-118, 121, 124, 127-151, 207-213, 216, 219-221, 324, 335-338,␣
↪354, 420-421, 439, 562-581, 583, 587-599, 606-615, 618, 622-636, 780-787, 790-794,␣
↪845, 866-878, 902-915, 937, 941-954, 964-967, 973, 985, 987, 989, 1012-1022
dicee/eval_static_funcs.py                            256    100    61%   104, 109,
↪ 114, 261-356, 363-414, 442, 465-468
dicee/evaluator.py                                    267     48    82%   48, 53,␣
↪58, 77, 82-83, 86, 102, 119, 130, 134, 139, 173-184, 191-202, 310, 340-358, 452,␣
↪462, 480-485
dicee/executer.py                                     134     16    88%   53-57,␣
↪166-176, 235-236, 283
dicee/knowledge_graph.py                               82     10    88%   84, 94-
↪95, 124, 128, 132-134, 137-138, 140
dicee/knowledge_graph_embeddings.py                   654    415    37%   25, 28-
↪29, 37-50, 55-88, 91-125, 129-137, 171, 173-229, 261, 265, 276-277, 301-303, 311,␣
↪339-362, 493, 497-519, 523-547, 580, 656, 665, 710-716, 748, 806-1171, 1202-1263,␣
↪1267-1295, 1326, 1332
dicee/models/__init__.py                               9      0   100%
dicee/models/adopt.py                                 187    172     8%   50-86,␣
↪99-110, 129-185, 195-242, 266-322, 346-448, 484-517
dicee/models/base_model.py                            240     35    85%   30-35,␣
↪64, 66, 92, 99-116, 171, 204, 244, 250, 259, 262, 266, 273, 277, 279, 294, 307-308,␣
↪362, 365, 438, 450
dicee/models/clifford.py                              470    278    41%   10, 12,␣
↪16, 24-25, 52-56, 79-87, 101-103, 108-109, 140-160, 184, 191, 195-256, 273-277, 289,
↪ 292, 297, 302, 346-361, 377-444, 464-470, 483, 486, 491, 496, 525-531, 544, 547,␣
↪552, 557, 567-576, 592-593, 613-685, 696-699, 724-749, 773-806, 842-846, 859, 869,␣
↪872, 877, 882, 887, 891, 895, 904-905, 935, 942, 947, 975-979, 1007-1016, 1026-1034,
↪ 1052-1054, 1072-1074, 1090-1092
dicee/models/complex.py                               162     25    85%   86-109,␣
↪273-287
dicee/models/dualE.py                                  59     10    83%   93-102,␣
↪142-156
dicee/models/ensemble.py                               89     67    25%   7-29, 31,
↪ 34, 37, 40, 43, 46, 49, 52-54, 56-58, 64-68, 71-90, 93-94, 97-112, 131
dicee/models/function_space.py                        262    221    16%   10-23,␣
↪27-36, 39-48, 52-69, 76-87, 90-99, 102-111, 115-127, 135-157, 160-166, 169-186, 189-
↪195, 198-206, 209, 214-235, 244-247, 251-255, 259-268, 272-293, 302-308, 312-329,␣
↪333-336, 345-353, 356, 367-373, 393-407, 425-439, 444-454, 462-466, 475-479
dicee/models/literal.py                                33      1    97%   82
dicee/models/octonion.py                              227     83    63%   21-44,␣
↪320-329, 334-345, 348-370, 374-416, 426-474
dicee/models/pykeen_models.py                          55      5    91%   77-80,␣
↪135
dicee/models/quaternion.py                            192     69    64%   7-21, 30-
↪55, 68-72, 107, 185, 328-342, 345-364, 368-389, 399-426
```

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```
dicee/models/real.py                                    61     12    80%   37-42,␣
↪70-73, 91, 107-110
dicee/models/static_funcs.py                            10      0   100%
dicee/models/transformers.py                           234    189    19%   20-39,␣
↪42, 56-71, 80-98, 101-112, 119-121, 124, 130-147, 151-176, 182-186, 189-193, 199-
↪203, 206-208, 225-252, 261-264, 267-272, 275-300, 306-311, 315-368, 372-394, 400-410
dicee/query_generator.py                               374    346     7%   17-51,␣
↪55, 61-64, 68-69, 77-91, 99-146, 154-187, 191-205, 211-268, 273-302, 306-442, 452-
↪471, 479-502, 509-513, 518, 523-529
dicee/read_preprocess_save_load_kg/__init__.py           3      0   100%
dicee/read_preprocess_save_load_kg/preprocess.py       243     40    84%   33, 39,␣
↪76, 100-125, 131, 136-149, 175, 205, 380-381
dicee/read_preprocess_save_load_kg/read_from_disk.py    36     11    69%   34, 38-
↪40, 47, 55, 58-72
dicee/read_preprocess_save_load_kg/save_load_disk.py    53     21    60%   29-30,␣
↪38, 47-68
dicee/read_preprocess_save_load_kg/util.py             236    125    47%   159, 173-
↪175, 179-180, 198-204, 207-209, 214-216, 230, 244-247, 252-260, 265-271, 276-281,␣
↪286-291, 303-324, 330-386, 390-394, 398-399, 403, 407-408, 436, 441, 448-449
dicee/sanity_checkers.py                                47     19    60%   8-12, 21-
↪31, 46, 51, 58, 69-79
dicee/static_funcs.py                                  483    194    60%   42, 52,␣
↪58-63, 85, 92-96, 109-119, 129-131, 136, 143, 167, 172, 184, 190, 198, 202, 229-233,
↪ 295, 303-309, 320-330, 341-361, 389, 413-414, 419-420, 437-438, 440-441, 443-444,␣
↪452, 470-474, 491-494, 498-503, 507-511, 515-516, 522-524, 539-553, 558-561, 566-
↪569, 578-629, 634-646, 663-680, 683-691, 695-713, 724
dicee/static_funcs_training.py                         155     66    57%   7-10,␣
↪222-319, 327-328
dicee/static_preprocess_funcs.py                        98     43    56%   17-25,␣
↪50, 57, 59, 70, 83-107, 112-115, 120-123, 128-131
dicee/trainer/__init__.py                                1      0   100%
dicee/trainer/dice_trainer.py                          151     18    88%   22, 30-
↪31, 33-35, 97, 104, 109-114, 152, 237, 280-283
dicee/trainer/model_parallelism.py                      99     87    12%   10-25,␣
↪30-116, 121-132, 136, 141-197
dicee/trainer/torch_trainer.py                          77      6    92%   31, 102,␣
↪168, 179-181
dicee/trainer/torch_trainer_ddp.py                      89     71    20%   11-14,␣
↪43, 47-67, 78-94, 113-122, 126-136, 151-158, 168-191
-------------------------------------------------------------------------------
TOTAL                                                 6948   3169    54%
```

13 How to cite

Currently, we are working on our manuscript describing our framework. If you really like our work and want to cite it now, feel free to chose one :)

```
# Keci
@inproceedings{demir2023clifford,
  title={Clifford Embeddings--A Generalized Approach for Embedding in Normed Algebras}
↪,
```

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(continued from previous page)

```
  author={Demir, Caglar and Ngonga Ngomo, Axel-Cyrille},
  booktitle={Joint European Conference on Machine Learning and Knowledge Discovery in␣
↪Databases},
  pages={567--582},
  year={2023},
  organization={Springer}
}
# LitCQD
@inproceedings{demir2023litcqd,
  title={LitCQD: Multi-Hop Reasoning in Incomplete Knowledge Graphs with Numeric␣
↪Literals},
  author={Demir, Caglar and Wiebesiek, Michel and Lu, Renzhong and Ngonga Ngomo, Axel-
↪Cyrille and Heindorf, Stefan},
  booktitle={Joint European Conference on Machine Learning and Knowledge Discovery in␣
↪Databases},
  pages={617--633},
  year={2023},
  organization={Springer}
}
# DICE Embedding Framework
@article{demir2022hardware,
  title={Hardware-agnostic computation for large-scale knowledge graph embeddings},
  author={Demir, Caglar and Ngomo, Axel-Cyrille Ngonga},
  journal={Software Impacts},
  year={2022},
  publisher={Elsevier}
}
# KronE
@inproceedings{demir2022kronecker,
  title={Kronecker decomposition for knowledge graph embeddings},
  author={Demir, Caglar and Lienen, Julian and Ngonga Ngomo, Axel-Cyrille},
  booktitle={Proceedings of the 33rd ACM Conference on Hypertext and Social Media},
  pages={1--10},
  year={2022}
}
# QMult, OMult, ConvQ, ConvO
@InProceedings{pmlr-v157-demir21a,
  title =          {Convolutional Hypercomplex Embeddings for Link Prediction},
  author =       {Demir, Caglar and Moussallem, Diego and Heindorf, Stefan and Ngonga␣
↪Ngomo, Axel-Cyrille},
  booktitle =          {Proceedings of The 13th Asian Conference on Machine Learning},
  pages =          {656--671},
  year =          {2021},
  editor =          {Balasubramanian, Vineeth N. and Tsang, Ivor},
  volume =          {157},
  series =          {Proceedings of Machine Learning Research},
  month =          {17--19 Nov},
  publisher =    {PMLR},
  pdf =          {https://proceedings.mlr.press/v157/demir21a/demir21a.pdf},
  url =          {https://proceedings.mlr.press/v157/demir21a.html},
}
# ConEx
```

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```
@inproceedings{demir2021convolutional,
title={Convolutional Complex Knowledge Graph Embeddings},
author={Caglar Demir and Axel-Cyrille Ngonga Ngomo},
booktitle={Eighteenth Extended Semantic Web Conference - Research Track},
year={2021},
url={https://openreview.net/forum?id=6T45-4TFqaX}}
# Shallom
@inproceedings{demir2021shallow,
  title={A shallow neural model for relation prediction},
  author={Demir, Caglar and Moussallem, Diego and Ngomo, Axel-Cyrille Ngonga},
  booktitle={2021 IEEE 15th International Conference on Semantic Computing (ICSC)},
  pages={179--182},
  year={2021},
  organization={IEEE}
```

For any questions or wishes, please contact: caglar.demir@upb.de

14 dicee

DICE Embeddings - Knowledge Graph Embedding Library.

A library for training and using knowledge graph embedding models with support for various scoring techniques and training strategies.

Submodules:

evaluation: Model evaluation functions and Evaluator class models: KGE model implementations trainer: Training orchestration scripts: Utility scripts

14.1 Submodules

dicee.__main__

dicee.abstracts

Attributes

logger

Classes

AbstractTrainer	Abstract base class for KGE model trainers.
BaseInteractiveKGE	Base class for interactive, post-training use of KGE models.
InteractiveQueryDecomposition	Mixin that provides fuzzy-logic operators for multi-hop EPFO query answering.
AbstractCallback	Abstract base class for KGE training lifecycle callbacks.
AbstractPPECallback	Abstract base class for Post-training Parameter Ensembling (PPE) callbacks.
BaseInteractiveTrainKGE	Abstract/base class for training knowledge graph embedding models interactively.

Module Contents

dicee.abstracts.**logger**

class dicee.abstracts.**AbstractTrainer**(*args*, *callbacks*)

Abstract base class for KGE model trainers.

Provides the callback dispatch mechanism shared by all concrete trainer implementations (TorchTrainer, TorchDDPTrainer, etc.). Sub-classes call the `on_*` hooks at the appropriate points in the training loop so that any registered *AbstractCallback* can react.

Parameters

- **args** (*argparse.Namespace or similar*) – Processed configuration object. Must expose at least `random_seed` (int).
- **callbacks** (*list of AbstractCallback*) – Ordered list of callback instances to invoke at each lifecycle hook.

attributes

callbacks

is_global_zero = True

global_rank = 0

local_rank = 0

strategy = None

on_fit_start(**args*, ***kwargs*)

Dispatch `on_fit_start` to all registered callbacks.

Called once before the first training epoch begins.

on_fit_end(**args*, ***kwargs*)

Dispatch `on_fit_end` to all registered callbacks.

Called once after the last training epoch completes.

on_train_epoch_start(**args*, ***kwargs*)

Dispatch `on_train_epoch_start` to all registered callbacks.

Called at the beginning of every epoch.

on_train_epoch_end(**args*, ***kwargs*)

Dispatch `on_train_epoch_end` to all registered callbacks.

Called at the end of every epoch after the loss has been accumulated.

on_train_batch_end(**args*, ***kwargs*)

Dispatch `on_train_batch_end` to all registered callbacks.

Called after each mini-batch gradient update.

static save_checkpoint(*full_path: str*, *model*) → None

Persist model weights to disk.

Parameters

- **full_path** (*str*) – Absolute or relative file path (including filename) where the `state_dict` will be written, e.g. `'Experiments/run1/model.pt'`.

- **model** (*torch.nn.Module*) – The model whose `state_dict` is to be saved.

class dicee.abstracts.**BaseInteractiveKGE**(*path: str | None = None*, *url: str | None = None*, *construct_ensemble: bool = False*, *model_name: str | None = None*, *apply_semantic_constraint: bool = False*)

Base class for interactive, post-training use of KGE models.

Loads a pre-trained model from disk (or a remote URL) together with its entity/relation index mappings and exposes the prediction API used by *KGE*.

Parameters

- **path** (*str, optional*) – Path to the experiment directory produced by `Execute`. Must contain `model.pt`, `configuration.json`, `entity_to_idx.csv` and `relation_to_idx.csv`.
- **url** (*str, optional*) – Remote URL of a pre-trained model to download. Mutually exclusive with *path*.
- **construct_ensemble** (*bool, optional*) – When `True`, load all checkpoint files in *path* and average their weights to form an ensemble model. Defaults to `False`.
- **model_name** (*str, optional*) – Filename (without extension) of the checkpoint to load when multiple `.pt` files exist in *path*.
- **apply_semantic_constraint** (*bool, optional*) – Reserved for future use. Defaults to `False`.

construct_ensemble = False

apply_semantic_constraint = False

configs

get_eval_report() → dict

get_bpe_token_representation(*str_entity_or_relation: List[str] | str*) → List[List[int]] | List[int]

Parameters

str_entity_or_relation (*corresponds to a str or a list of strings to be tokenized via BPE and shaped.*)

Return type

A list integer(s) or a list of lists containing integer(s)

get_padded_bpe_triple_representation(*triples: List[List[str]]*) → Tuple[List, List, List]

Parameters

triples

set_model_train_mode() → None

Switch the underlying model to training mode.

Calls `model.train()` and re-enables gradient computation for all parameters so that subsequent calls to optimisation steps work correctly after a period of inference.

set_model_eval_mode() → None

Switch the underlying model to evaluation mode.

Calls `model.eval()` and freezes all parameters (`requires_grad = False`) so that dropout and batch-norm layers behave deterministically during inference.

property name

sample_entity(*n: int*) → List[str]

Return *n* random entity strings without replacement.

Parameters
n (*int*) – Number of entities to sample. Must be non-negative and at most `num_entities`.

Returns
Randomly selected entity string labels.

Return type
List[str]

sample_relation(*n: int*) → List[str]

Return *n* random relation strings without replacement.

Parameters
n (*int*) – Number of relations to sample. Must be non-negative and at most `num_relations`.

Returns
Randomly selected relation string labels.

Return type
List[str]

is_seen(*entity: str | None = None*, *relation: str | None = None*) → bool

Check whether an entity or relation was present in the training set.

Exactly one of *entity* or *relation* should be provided.

Parameters

- **entity** (*str, optional*) – Entity string label to look up.
- **relation** (*str, optional*) – Relation string label to look up.

Returns
`True` if the given string is in the respective index mapping, `False` otherwise.

Return type
bool

save() → None

Persist the current model weights to the experiment directory.

The checkpoint filename encodes the current timestamp so successive calls do not overwrite each other. Ensemble models are saved with an `_ensemble_` infix in the filename.

get_entity_index(*x: str*) → int

Return the integer index for a given entity string.

Parameters
x (*str*) – Entity string label (must have been seen during training).

Returns
Corresponding row index in the entity embedding matrix.

Return type
int

get_relation_index(*x: str*) → int

Return the integer index for a given relation string.

Parameters
x (*str*) – Relation string label (must have been seen during training).

Returns
Corresponding row index in the relation embedding matrix.

Return type
int

index_triple(*head_entity: List[str]*, *relation: List[str]*, *tail_entity: List[str]*) → Tuple[torch.LongTensor, torch.LongTensor, torch.LongTensor]

Convert string triple lists to integer index tensors.

Parameters

- **head_entity** (*List[str]*) – Head entity string labels.
- **relation** (*List[str]*) – Relation string labels.
- **tail_entity** (*List[str]*) – Tail entity string labels.

Returns
idx_head_entity, idx_relation, idx_tail_entity – Each has shape `(n, 1)` containing the integer indices for the corresponding strings.

Return type
torch.LongTensor

add_new_entity_embeddings(*entity_name: str | None = None*, *embeddings: torch.FloatTensor | None = None*) → None

Extend the entity embedding table with a new entity at inference time.

The new entity is appended to both `entity_to_idx` / `idx_to_entity` mappings and the `entity_embeddings` weight tensor so that subsequent calls to prediction methods can reference it by name.

Parameters

- **entity_name** (*str*) – String label for the new entity. If the entity already exists in the index no modification is made.
- **embeddings** (*torch.FloatTensor*) – 1-D float tensor of length `embedding_dim` containing the pre-computed embedding for the new entity.

get_entity_embeddings(*items: List[str]*) → torch.FloatTensor

Return the embedding vectors for the given entity strings.

For standard (non-BPE) models the method looks up each string in `entity_to_idx` and returns the corresponding rows of the entity embedding matrix. For BPE models subword token embeddings are fetched and flattened into a single vector per entity.

Parameters
items (*List[str]*) – Entity string labels to retrieve.

Returns
Shape `(len(items), embedding_dim)`.

Return type
torch.FloatTensor

get_relation_embeddings(*items: List[str]*) → torch.FloatTensor

Return the embedding vectors for the given relation strings.

Parameters

items (*List[str]*) – Relation string labels to retrieve.

Returns

Shape `(len(items), embedding_dim)`.

Return type

torch.FloatTensor

construct_input_and_output(*head_entity: List[str]*, *relation: List[str]*, *tail_entity: List[str]*, *labels*) → Tuple[torch.LongTensor, torch.FloatTensor]

Build an indexed triple tensor and a label tensor from string inputs.

Parameters

- **head_entity** (*List[str]*) – Head entity string labels.
- **relation** (*List[str]*) – Relation string labels.
- **tail_entity** (*List[str]*) – Tail entity string labels.
- **labels** (*list or array-like*) – Binary or soft labels (one per triple) used as training targets.

Returns

- **x** (*torch.LongTensor*) – Shape `(n, 3)` integer-indexed triples.
- **labels** (*torch.FloatTensor*) – Shape `(n,)` float label tensor.

parameters()

class dicee.abstracts.**InteractiveQueryDecomposition**

Mixin that provides fuzzy-logic operators for multi-hop EPFO query answering.

The three families of operators — T-norm, T-conorm, and negation norm — are applied element-wise over entity score tensors to compose complex queries from atomic link-prediction results (e.g. 2p, 3p, 2i, ip, up).

t_norm(*tens_1: torch.Tensor*, *tens_2: torch.Tensor*, *tnorm: str = 'min'*) → torch.Tensor

Apply a T-norm to combine two entity score distributions.

Parameters

- **tens_1** (*torch.Tensor*) – Score tensors of identical shape, values in `[0, 1]`.
- **tens_2** (*torch.Tensor*) – Score tensors of identical shape, values in `[0, 1]`.
- **tnorm** (*str*) – Operator to use. `'min'` applies the Gödel (min) T-norm; `'prod'` applies the product T-norm.

Returns

Element-wise combined scores of the same shape as the inputs.

Return type

torch.Tensor

tensor_t_norm(*subquery_scores: torch.FloatTensor*, *tnorm: str = 'min'*) → torch.FloatTensor

Compute T-norm over [0,1] ^{n imes d} where n denotes the number of hops and d denotes number of entities

t_conorm(*tens_1: torch.Tensor*, *tens_2: torch.Tensor*, *tconorm: str = 'min'*) → torch.Tensor

Apply a T-conorm (S-norm) to combine two score distributions (union).

Parameters

- **tens_1** (*torch.Tensor*) – Score tensors of identical shape, values in `[0, 1]`.
- **tens_2** (*torch.Tensor*) – Score tensors of identical shape, values in `[0, 1]`.
- **tconorm** (*str*) – Operator to use. `'min'` applies the Gödel (max) T-conorm; `'prod'` applies the probabilistic sum T-conorm.

Returns

Element-wise combined scores of the same shape as the inputs.

Return type

torch.Tensor

negnorm(*tens_1: torch.Tensor*, *lambda_: float*, *neg_norm: str = 'standard'*) → torch.Tensor

Apply a negation norm (complement) to an entity score distribution.

Parameters

- **tens_1** (*torch.Tensor*) – Input score tensor, values in `[0, 1]`.
- **lambda** (*float*) – Shape parameter used by the Sugeno and Yager negation norms. Ignored for the standard complement.
- **neg_norm** (*str*) – Which negation to apply: `'standard'` (`1 - x`), `'sugeno'`, or `'yager'`.

Returns

Complemented score tensor of the same shape as *tens_1*.

Return type

torch.Tensor

class dicee.abstracts.**AbstractCallback**

Bases: `abc.ABC`, `lightning.pytorch.callbacks.Callback`

Abstract base class for KGE training lifecycle callbacks.

Concrete sub-classes override one or more hook methods to perform custom actions at specific points during training (e.g. weight averaging, periodic evaluation, model checkpointing). All hooks have empty default implementations so sub-classes only need to override the hooks they care about.

Callbacks are registered by passing them to the trainer's *callbacks* list. They are also compatible with PyTorch Lightning trainers because this class extends `lightning.pytorch.callbacks.Callback`.

on_init_start(**args*, ***kwargs*)

Called when the trainer is about to be constructed.

Override to perform setup that must happen before any trainer state is initialised.

on_init_end(**args*, ***kwargs*)

Called immediately after the trainer has been constructed.

Override to perform setup that requires a fully initialised trainer.

on_fit_start(*trainer*, *model*)

Called once before the first training epoch.

Parameters

- **trainer** (AbstractTrainer *or pl.Trainer*) – The active trainer instance.

- **model** (BaseKGE) – The model about to be trained.

on_train_epoch_end(*trainer*, *model*)

Called at the end of each training epoch.

Parameters

- **trainer** (AbstractTrainer *or* *pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model being trained. `model.loss_history` contains the per-epoch average losses accumulated so far.

on_train_batch_end(**args*, ***kwargs*)

Called after each mini-batch gradient update.

Override to inspect or modify the model at a finer granularity than epoch-level hooks.

on_fit_end(**args*, ***kwargs*)

Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

class dicee.abstracts.**AbstractPPECallback**(*num_epochs*, *path*, *epoch_to_start*, *last_percent_to_consider*)

Bases: *AbstractCallback*

Abstract base class for Post-training Parameter Ensembling (PPE) callbacks.

Sub-classes implement weight-averaging strategies (SWA, EMA, SWAG, …) by overriding `on_train_epoch_end()` and *on_fit_end()*. Common book-keeping (epoch counter, sample counter, alpha weights) is managed here.

Parameters

- **num_epochs** (*int*) – Total number of training epochs.
- **path** (*str*) – Experiment directory where averaged checkpoints will be written.
- **epoch_to_start** (*int*) – First epoch at which the averaging procedure should begin.
- **last_percent_to_consider** (*float*) – Fraction of the total training epochs (counted from the end) whose checkpoints are included in the ensemble.

num_epochs

path

sample_counter = 0

epoch_count = 0

alphas = None

on_fit_start(*trainer*, *model*)

Called once before the first training epoch.

Parameters

- **trainer** (AbstractTrainer *or* *pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model about to be trained.

on_fit_end(*trainer*, *model*)

Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

store_ensemble(*param_ensemble*) → None

class dicee.abstracts.**BaseInteractiveTrainKGE**

Abstract/base class for training knowledge graph embedding models interactively. This class provides methods for re-training KGE models and also Literal Embedding model.

train_triples(*h: List[str]*, *r: List[str]*, *t: List[str]*, *labels: List[float]*, *iteration=2*, *optimizer=None*)

train_k_vs_all(*h*, *r*, *iteration=1*, *lr=0.001*)

Train k vs all :param head_entity: :param relation: :param iteration: :param lr: :return:

train(*kg*, *lr=0.1*, *epoch=10*, *batch_size=32*, *neg_sample_ratio=10*, *num_workers=1*) → None

Retrained a pretrain model on an input KG via negative sampling.

train_literals(*train_file_path: str | None = None*, *num_epochs: int = 100*, *lit_lr: float = 0.001*, *lit_normalization_type: str = 'z-norm'*, *batch_size: int = 1024*, *sampling_ratio: float | None = None*, *random_seed=1*, *loader_backend: str = 'pandas'*, *freeze_entity_embeddings: bool = True*, *gate_residual: bool = True*, *device: str | None = None*, *suffle_data: bool = True*)

Trains the Literal Embeddings model using literal data.

Parameters

- **train_file_path** (*str*) – Path to the training data file.
- **num_epochs** (*int*) – Number of training epochs.
- **lit_lr** (*float*) – Learning rate for the literal model.
- **norm_type** (*str*) – Normalization type to use (‘z-norm’, ‘min-max’, or None).
- **batch_size** (*int*) – Batch size for training.
- **sampling_ratio** (*float*) – Ratio of training triples to use.
- **loader_backend** (*str*) – Backend for loading the dataset (‘pandas’ or ‘rdflib’).
- **freeze_entity_embeddings** (*bool*) – If True, freeze the entity embeddings during training.
- **gate_residual** (*bool*) – If True, use gate residual connections in the model.
- **device** (*str*) – Device to use for training (‘cuda’ or ‘cpu’). If None, will use available GPU or CPU.
- **suffle_data** (*bool*) – If True, shuffle the dataset before training.

dicee.analyse_experiments

This script should be moved to dicee/scripts Example: python dicee/analyse_experiments.py –dir Experiments –features “model” “trainMRR” “testMRR”

Attributes

logger

Classes

Experiment

Functions

get_default_arguments()
analyse(args)

Module Contents

dicee.analyse_experiments.**logger**

dicee.analyse_experiments.**get_default_arguments**()

class dicee.analyse_experiments.**Experiment**

model_name = []

callbacks = []

embedding_dim = []

num_params = []

num_epochs = []

batch_size = []

lr = []

byte_pair_encoding = []

aswa = []

path_dataset_folder = []

full_storage_path = []

pq = []

train_mrr = []

train_h1 = []

train_h3 = []

train_h10 = []

val_mrr = []

val_h1 = []

val_h3 = []

val_h10 = []

test_mrr = []

test_h1 = []

test_h3 = []

test_h10 = []

runtime = []

normalization = []

scoring_technique = []

save_experiment(*x*)

to_df()

dicee.analyse_experiments.**analyse**(*args*)

dicee.callbacks

Callbacks for training lifecycle events.

Provides callback classes for various training events including epoch end, model saving, weight averaging, and evaluation.

Attributes

logger

Classes

AccumulateEpochLossCallback	Callback to store epoch losses to a CSV file.
PrintCallback	Callback that prints training start/end times and total runtime.
KGESaveCallback	Callback that periodically saves model checkpoints during training.
PseudoLabellingCallback	Callback that augments the training set with pseudo-labelled triples.
Eval	Abstract base class for KGE training lifecycle callbacks.
KronE	Abstract base class for KGE training lifecycle callbacks.
Perturb	A callback for a three-Level Perturbation
PeriodicEvalCallback	Callback to periodically evaluate the model and optionally save checkpoints during training.
LRScheduler	Callback for managing learning rate scheduling and model snapshots.

Functions

estimate_q(eps)	estimate rate of convergence q from sequence esp
compute_convergence(seq, i)	

Module Contents

dicee.callbacks.**logger**

class dicee.callbacks.**AccumulateEpochLossCallback**(*path: str*)

Bases: *dicee.abstracts.AbstractCallback*

Callback to store epoch losses to a CSV file.

Parameters
path – Directory path where the loss file will be saved.

path

on_fit_end(*trainer*, *model*) → None

Store epoch loss history to CSV file.

Parameters

- **trainer** – The trainer instance.
- **model** – The model being trained.

class dicee.callbacks.**PrintCallback**

Bases: *dicee.abstracts.AbstractCallback*

Callback that prints training start/end times and total runtime.

start_time

on_fit_start(*trainer*, *pl_module*)

Called once before the first training epoch.

Parameters

- **trainer** (*AbstractTrainer or pl.Trainer*) – The active trainer instance.
- **model** (*BaseKGE*) – The model about to be trained.

on_fit_end(*trainer*, *pl_module*)

Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

on_train_batch_end(**args*, ***kwargs*)

Called after each mini-batch gradient update.

Override to inspect or modify the model at a finer granularity than epoch-level hooks.

on_train_epoch_end(**args*, ***kwargs*)

Called at the end of each training epoch.

Parameters

- **trainer** (*AbstractTrainer or pl.Trainer*) – The active trainer instance.

- **model** (BaseKGE) – The model being trained. `model.loss_history` contains the per-epoch average losses accumulated so far.

class dicee.callbacks.**KGESaveCallback**(*every_x_epoch: int*, *max_epochs: int*, *path: str*)

Bases: *dicee.abstracts.AbstractCallback*

Callback that periodically saves model checkpoints during training.

Parameters

- **every_x_epoch** (*int or None*) – Save a checkpoint every *every_x_epoch* epochs. When `None`, the interval defaults to `max(max_epochs // 2, 1)`.
- **max_epochs** (*int*) – Total number of training epochs (used to compute the default interval when *every_x_epoch* is `None`).
- **path** (*str*) – Directory where checkpoint files will be written.

every_x_epoch

max_epochs

epoch_counter = 0

path

on_train_batch_end(**args*, ***kwargs*)

Called after each mini-batch gradient update.

Override to inspect or modify the model at a finer granularity than epoch-level hooks.

on_fit_start(*trainer*, *pl_module*)

Called once before the first training epoch.

Parameters

- **trainer** (AbstractTrainer *or pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model about to be trained.

on_train_epoch_end(**args*, ***kwargs*)

Called at the end of each training epoch.

Parameters

- **trainer** (AbstractTrainer *or pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model being trained. `model.loss_history` contains the per-epoch average losses accumulated so far.

on_fit_end(**args*, ***kwargs*)

Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

on_epoch_end(*model*, *trainer*, ***kwargs*)

class dicee.callbacks.**PseudoLabellingCallback**(*data_module*, *kg*, *batch_size*)

Bases: *dicee.abstracts.AbstractCallback*

Callback that augments the training set with pseudo-labelled triples.

At the end of each epoch the current model scores a batch of unlabelled triples and those with a predicted probability >= 0.90 are appended to the training dataset (semi-supervised self-training).

Parameters

- **data_module** (*object*) – Dataset module exposing a `train_set_idx` attribute and a `train_dataloader()` method.
- **kg** (KG) – The knowledge graph, providing `num_entities`, `num_relations`, and `unlabelled_set`.
- **batch_size** (*int*) – Number of unlabelled triples to sample and score each epoch.

data_module

kg

num_of_epochs = 0

unlabelled_size

batch_size

create_random_data()

on_epoch_end(*trainer*, *model*)

dicee.callbacks.**estimate_q**(*eps*)

estimate rate of convergence q from sequence esp

dicee.callbacks.**compute_convergence**(*seq*, *i*)

class dicee.callbacks.**Eval**(*path*, *epoch_ratio: int | None = None*)

Bases: *dicee.abstracts.AbstractCallback*

Abstract base class for KGE training lifecycle callbacks.

Concrete sub-classes override one or more hook methods to perform custom actions at specific points during training (e.g. weight averaging, periodic evaluation, model checkpointing). All hooks have empty default implementations so sub-classes only need to override the hooks they care about.

Callbacks are registered by passing them to the trainer's *callbacks* list. They are also compatible with PyTorch Lightning trainers because this class extends `lightning.pytorch.callbacks.Callback`.

path

reports = []

epoch_ratio

epoch_counter = 0

on_fit_start(*trainer*, *model*)

Called once before the first training epoch.

Parameters

- **trainer** (AbstractTrainer *or pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model about to be trained.

on_fit_end(*trainer*, *model*)

Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

on_train_epoch_end(*trainer*, *model*)

Called at the end of each training epoch.

Parameters

- **trainer** (*AbstractTrainer or pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model being trained. `model.loss_history` contains the per-epoch average losses accumulated so far.

on_train_batch_end(**args*, ***kwargs*)

Called after each mini-batch gradient update.

Override to inspect or modify the model at a finer granularity than epoch-level hooks.

class dicee.callbacks.**KronE**

Bases: *dicee.abstracts.AbstractCallback*

Abstract base class for KGE training lifecycle callbacks.

Concrete sub-classes override one or more hook methods to perform custom actions at specific points during training (e.g. weight averaging, periodic evaluation, model checkpointing). All hooks have empty default implementations so sub-classes only need to override the hooks they care about.

Callbacks are registered by passing them to the trainer's *callbacks* list. They are also compatible with PyTorch Lightning trainers because this class extends `lightning.pytorch.callbacks.Callback`.

f = None

static batch_kronecker_product(*a*, *b*)

Kronecker product of matrices a and b with leading batch dimensions. Batch dimensions are broadcast. The number of them mush :type a: torch.Tensor :type b: torch.Tensor :rtype: torch.Tensor

get_kronecker_triple_representation(*indexed_triple: torch.LongTensor*)

Get kronecker embeddings

on_fit_start(*trainer*, *model*)

Called once before the first training epoch.

Parameters

- **trainer** (*AbstractTrainer or pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model about to be trained.

class dicee.callbacks.**Perturb**(*level: str = 'input'*, *ratio: float = 0.0*, *method: str | None = None*, *scaler: float | None = None*, *frequency=None*)

Bases: *dicee.abstracts.AbstractCallback*

A callback for a three-Level Perturbation

Input Perturbation: During training an input x is perturbed by randomly replacing its element. In the context of knowledge graph embedding models, x can denote a triple, a tuple of an entity and a relation, or a tuple of two entities. A perturbation means that a component of x is randomly replaced by an entity or a relation.

Parameter Perturbation:

Output Perturbation:

level = 'input'

ratio = 0.0

method = None

scaler = None

frequency = None

on_train_batch_start(*trainer*, *model*, *batch*, *batch_idx*)

Called when the train batch begins.

class dicee.callbacks.**PeriodicEvalCallback**(*experiment_path: str*, *max_epochs: int*, *eval_every_n_epoch: int = 0*, *eval_at_epochs: list | None = None*, *save_model_every_n_epoch: bool = True*, *n_epochs_eval_model: str = 'val_test'*)

Bases: *dicee.abstracts.AbstractCallback*

Callback to periodically evaluate the model and optionally save checkpoints during training.

Evaluates at regular intervals (every N epochs) or at explicitly specified epochs. Stores evaluation reports and model checkpoints.

experiment_dir

max_epochs

epoch_counter = 0

save_model_every_n_epoch = True

reports

n_epochs_eval_model = 'val_test'

default_eval_model = None

eval_epochs

on_fit_end(*trainer*, *model*)

Called at the end of training. Saves final evaluation report.

on_train_epoch_end(*trainer*, *model*)

Called at the end of each training epoch. Performs evaluation and checkpointing if scheduled.

class dicee.callbacks.**LRScheduler**(*adaptive_lr_config: dict*, *total_epochs: int*, *experiment_dir: str*, *eta_max: float = 0.1*, *snapshot_dir: str = 'snapshots'*)

Bases: *dicee.abstracts.AbstractCallback*

Callback for managing learning rate scheduling and model snapshots.

Supports cosine annealing (“cca”), MMCCLR (“mmcclr”), and their deferred (warmup) variants: - “deferred_cca” - “deferred_mmcclr”

At the end of each learning rate cycle, the model can optionally be saved as a snapshot.

total_epochs

experiment_dir

snapshot_dir

batches_per_epoch = None

total_steps = None

cycle_length = None

warmup_steps = None

lr_lambda = None

scheduler = None

step_count = 0

snapshot_loss

on_train_start(*trainer*, *model*)

Initialize training parameters and LR scheduler at start of training.

on_train_batch_end(*trainer*, *model*, *outputs*, *batch*, *batch_idx*)

Step the LR scheduler and save model snapshot if needed after each batch.

on_fit_end(*trainer*, *model*)

Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

dicee.config

Configuration module for DICE embeddings.

Provides the Namespace class with default configuration values for training knowledge graph embedding models.

Classes

Namespace	Extended Namespace with default KGE training configuration.

Module Contents

class dicee.config.**Namespace**(***kwargs*)

Bases: argparse.Namespace

Extended Namespace with default KGE training configuration.

Provides sensible defaults for all training parameters while allowing easy customization through command-line arguments or direct assignment.

dataset_dir: str | None = None

The path of a folder containing train.txt, and/or valid.txt and/or test.txt

save_embeddings_as_csv: bool = False

Embeddings of entities and relations are stored into CSV files to facilitate easy usage.

storage_path: str = 'Experiments'

A directory named with time of execution under –storage_path that contains related data about embeddings.

path_to_store_single_run: str | None = None

A single directory created that contains related data about embeddings.

path_single_kg = None

Path of a file corresponding to the input knowledge graph

sparql_endpoint = None

An endpoint of a triple store.

ultra_checkpoint: str | None = None

ultra_dim: int = 64

ultra_num_layers: int = 6

ultra_query_batch_size: int = 8

trix_checkpoint: str | None = None

trix_dim: int = 32

trix_query_batch_size: int = 8

flock_checkpoint: str | None = None

flock_dim: int = 64

flock_walk_num: int = 128

flock_walk_len: int = 128

flock_refinements: int = 6

flock_num_layers: int = 1

flock_attention_heads: int = 4

flock_test_samples: int = 1

flock_query_batch_size: int = 1

flock_seed: int | None = None

flock_prefetch_walks: bool = True

flock_compact_state: bool = True

flock_compile_sampler: bool = True

flock_pack_walks: bool = True

graph_inference_backend: str = 'auto'

graph_relation_cache_mb: int = 64

graph_projection_cache_mb: int = 64

graph_inference_compile: bool = False

grouped_negative_sampling: bool = False

Sample a fixed-size group of negatives per positive (positive in column 0) via GroupedNegativeSampling-Dataset instead of the flat NegSample layout. Requires scoring_technique='NegSample', entity prediction, no byte_pair_encoding.

strict_negative_sampling: bool = False

With grouped/adversarial sampling, exclude every known-true triple from the negative candidate pool instead of sampling entities uniformly at random.

adversarial_temperature: float | None = None

Softmax temperature for self-adversarial weighting of negatives by their own score (dicee.models.sampled_loss.grouped_adversarial_bce); implies grouped sampling. 0 weights negatives uniformly; higher values weight high-scoring (harder) negatives more. None disables adversarial weighting.

model: str = 'Keci'

KGE model

optim: str = 'Adam'

Optimizer

embedding_dim: int = 64

Size of continuous vector representation of an entity/relation

num_epochs: int = 150

Number of pass over the training data

batch_size: int = 1024

Mini-batch size if it is None, an automatic batch finder technique applied

lr: float = 0.1

Learning rate

add_noise_rate: float | None = None

The ratio of added random triples into training dataset

gpus = None

Number GPUs to be used during training

callbacks

10}}

Type

Callbacks, e.g., {“PPE”

Type

{ “last_percent_to_consider”

backend: str = 'pandas'

Backend to read, process, and index input knowledge graph. pandas, polars and rdflib available

separator: str = '\\s+'

separator for extracting head, relation and tail from a triple

trainer: str = 'torchCPUTrainer'

‘torchCPUTrainer’ (CPU/single GPU), ‘PL’ (PyTorch Lightning multi-GPU), ‘torchDDP’ (native DDP), ‘TP’ (Tensor Parallelism - implements ‘Multiple Run Ensemble Learning with Low-Dimensional Knowledge Graph Embeddings’)

Type

Trainer for knowledge graph embedding model. Options

scoring_technique: str = 'KvsAll'

Scoring technique for knowledge graph embedding models

neg_ratio: int = 0

Negative ratio for a true triple in NegSample training_technique

weight_decay: float = 0.0

Weight decay for all trainable params

normalization: str = 'None'

LayerNorm, BatchNorm1d, or None

init_param: str | None = None

xavier_normal or None

gradient_accumulation_steps: int = 0

Not tested e

num_folds_for_cv: int = 0

Number of folds for CV

eval_model: str = 'train_val_test'

["None", "train", "train_val", "train_val_test", "test"]

Type

Evaluate trained model choices

eval_tie_policy: str = 'sort'

sort (legacy), optimistic, random, or pessimistic.

Type

Prediction ties

eval_tie_seed: int | None = None

Independent random tie seed; None uses random_seed.

save_model_at_every_epoch: int | None = None

Not tested

label_smoothing_rate: float = 0.0

Smooths dataset targets toward a uniform distribution at construction time (dicee/dataset_classes/_label_based.py, _negative_sampling.py), independent of loss_fn. Also passed as smoothness_ratio to loss_fn="LS"/"CombinedLSandLR"/ "AggregatedLSandLR" (see dicee/losses/custom_losses.py).

num_core: int = 0

Number of CPUs to be used in the mini-batch loading process

random_seed: int = 0

Random Seed

log_level: str = 'INFO'

DEBUG, INFO, WARNING, ERROR, or CRITICAL. Dataset info, timing, and checkpoint messages are logged at INFO; set to WARNING or higher to silence them.

Type

Logging verbosity

sample_triples_ratio: float | None = None

Read some triples that are uniformly at random sampled. Ratio being between 0 and 1

read_only_few: int | None = None

Read only first few triples

pykeen_model_kwargs

Additional keyword arguments for pykeen models

pl_trainer_kwargs

Additional keyword arguments for the PyTorch Lightning Trainer

kernel_size: int = 3

Size of a square kernel in a convolution operation

num_of_output_channels: int = 32

Number of slices in the generated feature map by convolution.

margin: float = 4.0

Margin used by margin-based scoring functions (e.g. TransE, TransH, RotatE), and by torch.nn.MarginRankingLoss when scoring_technique='NegSampleMargin'

p: int = 0

P parameter of Clifford Embeddings

q: int = 1

Q parameter of Clifford Embeddings

input_dropout_rate: float = 0.0

Dropout rate on embeddings of input triples

hidden_dropout_rate: float = 0.0

Dropout rate on hidden representations of input triples

feature_map_dropout_rate: float = 0.0

Dropout rate on a feature map generated by a convolution operation

byte_pair_encoding: bool = False

Byte pair encoding

Type

WIP

adaptive_swa: bool = False

Adaptive stochastic weight averaging

swa: bool = False

Stochastic weight averaging

swag: bool = False

Stochastic weight averaging - Gaussian

ema: bool = False

Exponential Moving Average

twa: bool = False

Trainable weight averaging

block_size: int | None = None

block size of LLM

continual_learning: str | None = None

Path of a pretrained model size of LLM

auto_batch_finding: bool = False

A flag for using auto batch finding

eval_every_n_epochs: int = 0

Evaluate model every n epochs. If 0, no evaluation is applied.

save_every_n_epochs: bool = False

Save model every n epochs. If True, save model at every epoch.

eval_at_epochs: list | None = None

List of epoch numbers at which to evaluate the model (e.g., 1 5 10).

n_epochs_eval_model: str = 'val_test'

Evaluating link prediction performance on data splits while performing periodic evaluation.

adaptive_lr

"cca"}'

Type

Adaptive learning rate parameters, e.g., '{"scheduler_name"

swa_start_epoch: int | None = None

Epoch at which to start applying stochastic weight averaging.

swa_c_epochs: int = 1

Number of epochs to average over for SWA, SWAG, EMA, TWA.

loss_fn: str | None = None

Overrides the default entity/relation-prediction loss with one of the classes in dicee/losses/custom_losses.py: "LS", "LRLoss", "BCELoss", "WeightedBCELoss", "CombinedLSandLR", "AdaptiveLabelSmoothingLoss", "AdaptiveLabelRelaxationLoss", "ConfidenceBasedAdaptiveLabelRelaxationLoss", "CombinedAdaptiveLSandAdaptiveLR", "AggregatedLSandLR", "ACLS". None (default) keeps the framework's own BCEWithLogitsLoss/CrossEntropyLoss/MarginRankingLoss selection. See the docstring of each class for what it actually does and known caveats.

label_relaxation_alpha: float = 0.1

alpha for loss_fn="LRLoss"/"CombinedLSandLR"/"AggregatedLSandLR" (dicee.losses.custom_losses.LabelRelaxationLoss).

amwa: bool = False

Adaptive Momentum Weight Averaging

amwa_start_epoch: int = 0

Epoch at which to start applying AMWA.

amwa_c_epochs: int = 1

Number of epochs between AMWA updates.

amwa_monitor: str = 'MRR'

Validation metric used by AMWA to compare BaseNet and StableNet.

amwa_maximize: bool = True

Whether larger values of amwa_monitor are better.

amwa_beta: float | None = None

Fixed beta for AMWA. If None, use adaptive beta_n.

amwa_beta_window: int = 10

Number of recent delta values used to estimate beta_n.

amwa_beta_init: float = 1.0

Initial beta used before enough delta history exists.

amwa_beta_floor: float = 1e-08

Numerical floor for beta_n.

__iter__()

dicee.dataset_classes

Dataset classes for knowledge graph embedding training.

This package groups the various `torch.utils.data.Dataset` implementations used throughout DICE Embeddings into thematic sub-modules:

- `_bpe` – Byte-pair-encoding related datasets.
- `_negative_sampling` – Negative sampling based datasets.
- `_label_based` – Multi-label / multi-class scoring datasets.
- `_literal` – Literal (numeric) embedding dataset.
- `_factory` – `construct_dataset` / `reload_dataset` helpers.

All public names are re-exported here so that existing `from dicee.dataset_classes import …` statements continue to work unchanged.

Classes

BPE_NegativeSamplingDataset	Dataset for negative sampling with byte-pair encoded triples.
MultiClassClassificationDataset	Dataset for autoregressive multi-class classification on sub-word units.
MultiLabelDataset	Multi-label dataset for BPE-based KvsAll / AllvsAll training.
AllvsAll	Dataset for AllvsAll training (multi-label, exhaustive).
FSDP1vsSampleDataset	Positive-triple dataset for FSDP 1vsSample training with true-negative sampling.
KvsAll	Dataset for KvsAll training (multi-label).
KvsSampleDataset	Dataset for KvsSample training (dynamic multi-label).
OnevsAllDataset	Dataset for the 1-vs-All training strategy (multi-class).
LiteralDataset	Dataset for loading and processing literal data for Literal Embedding models.
FixedNegSampleDataset	Pre-computed (fixed) negative sampling dataset.
OnevsSample	Dataset for 1-vs-Sample training (dynamic multi-class with negatives).
TriplePredictionDataset	Dataset for triple prediction with on-the-fly negative sampling.

Functions

construct_dataset(→ torch.utils.data.Dataset)	Build the appropriate dataset for the given training configuration.
reload_dataset(path, form_of_labelling, ...)	Reload training data from disk and construct a Pytorch dataset.

Package Contents

class dicee.dataset_classes.**BPE_NegativeSamplingDataset**(*train_set: torch.LongTensor*, *ordered_shaped_bpe_entities: torch.LongTensor*, *neg_ratio: int*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Dataset for negative sampling with byte-pair encoded triples.

Each sample is a BPE-encoded triple. The custom collate_fn constructs negatives by corrupting head or tail entities with random BPE entities.

Parameters

- **train_set** (*torch.LongTensor*) – Integer-encoded triples of shape (N, 3).
- **ordered_shaped_bpe_entities** (*torch.LongTensor*) – All BPE entity representations, ordered by entity index.
- **neg_ratio** (*int*) – Number of negative samples per positive triple.

train_set

ordered_bpe_entities

num_bpe_entities

neg_ratio

num_datapoints

__len__()

__getitem__(*idx*)

collate_fn(*batch_shaped_bpe_triples: List[Tuple[torch.Tensor, torch.Tensor]]*)

class dicee.dataset_classes.**MultiClassClassificationDataset**(*subword_units: numpy.ndarray*, *block_size: int = 8*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Dataset for autoregressive multi-class classification on sub-word units.

Splits a flat sequence of sub-word token ids into overlapping windows of size block_size for next-token prediction.

Parameters

- **subword_units** (*numpy.ndarray*) – 1-D array of sub-word token ids.
- **block_size** (*int, optional*) – Context window length (default 8).

train_data

block_size = 8

num_of_data_points

collate_fn = None

__len__()

__getitem__(*idx*)

class dicee.dataset_classes.**MultiLabelDataset**(*train_set: torch.LongTensor*, *train_indices_target*, *target_dim: int*, *torch_ordered_shaped_bpe_entities: torch.LongTensor*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Multi-label dataset for BPE-based KvsAll / AllvsAll training.

Each sample is a BPE-encoded `(head, relation)` pair together with a binary multi-label target vector over all entities.

Parameters

- **train_set** (*torch.LongTensor*) – BPE-encoded input pairs of shape `(N, 2, token_length)`.
- **train_indices_target** (*sequence of integer sequences or RaggedIndices*) – Per-sample lists of positive target entity indices.
- **target_dim** (*int*) – Dimensionality of the target vector (number of entities).
- **torch_ordered_shaped_bpe_entities** (*torch.LongTensor*) – Ordered BPE entity representations.

train_set

train_indices_target

target_dim

num_datapoints

torch_ordered_shaped_bpe_entities

collate_fn = None

__len__()

__getitem__(*idx*)

dicee.dataset_classes.**construct_dataset**(** (Keyword-only parameters separator (PEP 3102))*, *train_set: numpy.ndarray | list*, *valid_set=None*, *test_set=None*, *ordered_bpe_entities=None*, *train_target_indices=None*, *target_dim: int | None = None*, *entity_to_idx: dict*, *relation_to_idx: dict*, *form_of_labelling: str*, *scoring_technique: str*, *neg_ratio: int*, *label_smoothing_rate: float*, *byte_pair_encoding=None*, *block_size: int | None = None*, *seed: int | None = None*, *sort_train_set: bool = True*, *grouped_negative_sampling: bool = False*, *strict_negative_sampling: bool = False*, *adversarial_temperature: float | None = None*) → torch.utils.data.Dataset

Build the appropriate dataset for the given training configuration.

Parameters

- **train_set** (*numpy.ndarray or list*) – Raw integer-indexed triples.
- **entity_to_idx** (*dict*) – Name → index mappings.

- **relation_to_idx** (*dict*) – Name → index mappings.
- **form_of_labelling** (*str*) – `'EntityPrediction'` or `'RelationPrediction'`.
- **scoring_technique** (*str*) – One of `'NegSample'`, `'NegSampleMargin'`, `'Fixed-NegSample'`, `'1vsAll'`, `'1vsSample'`, `'KvsAll'`, `'AllvsAll'`, `'KvsSample'`.
- **neg_ratio** (*int*) – Negative sample ratio.
- **label_smoothing_rate** (*float*) – Label smoothing coefficient.

Return type
torch.utils.data.Dataset

dicee.dataset_classes.**reload_dataset**(*path: str*, *form_of_labelling*, *scoring_technique*, *neg_ratio*, *label_smoothing_rate*)

Reload training data from disk and construct a Pytorch dataset.

class dicee.dataset_classes.**AllvsAll**(*train_set_idx: numpy.ndarray*, *entity_idxs*, *relation_idxs*, *label_smoothing_rate=0.0*)

Bases: `dicee.dataset_classes._storage.WorkerDataset`

Dataset for AllvsAll training (multi-label, exhaustive).

Extends the `KvsAll` idea: every *possible* `(entity, relation)` combination is included — not just those observed in the KG. Pairs without any known tail entities receive an all-zeros label vector.

Parameters

- **train_set_idx** (*numpy.ndarray*) – `(N, 3)` integer-indexed triples.
- **entity_idxs** (*dict*) – Entity-name → index mapping.
- **relation_idxs** (*dict*) – Relation-name → index mapping.
- **label_smoothing_rate** (*float, optional*) – Label smoothing coefficient (default `0.0`).

train_data: torch.Tensor

train_target: dicee.dataset_classes._storage.RaggedIndices

label_smoothing_rate

collate_fn = None

target_dim

__len__()

__getitem__(*idx*)

class dicee.dataset_classes.**FSDP1vsSampleDataset**(*train_set_idx: numpy.ndarray*, *entity_idxs*, *relation_idxs*, *form*, *neg_ratio=None*, *label_smoothing_rate: float = 0.0*)

Bases: `dicee.dataset_classes._storage.WorkerDataset`

Positive-triple dataset for FSDP 1vsSample training with true-negative sampling.

Each dataset item is a single positive triple (h, r, t). The collate_fn builds the full (source, target_idx, labels) batch in a DataLoader worker, sampling true negatives via index remapping so they never coincide with any known positive tail for that (h, r) pair.

Fixed batch width follows KvsSample:
max_num_of_classes = max_positives_per_pair + neg_ratio

Each sample contributes 1 positive and (max_num_of_classes - 1) negatives.

train_data

num_entities

num_relations

neg_ratio = None

label_smoothing_rate = 0.0

max_num_of_classes

num_negatives

collate_fn

__len__()

__getitem__(*idx*)

class dicee.dataset_classes.**KvsAll**(*train_set_idx: numpy.ndarray*, *entity_idxs*, *relation_idxs*, *form*, *store=None*, *label_smoothing_rate: float = 0.0*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Dataset for KvsAll training (multi-label).

D := {(x, y)_i}_{i=1}^{N} where

- x = (h, r) is a unique (entity, relation) pair observed in the KG,
- y ∈ [0, 1]^{|E|} is a multi-label vector with y_j = 1 iff (h, r, e_j) ∈ KG.

Parameters

- **train_set_idx** (*numpy.ndarray*) – `(N, 3)` integer-indexed triples.
- **entity_idxs** (*dict*) – Entity-name → index mapping.
- **relation_idxs** (*dict*) – Relation-name → index mapping.
- **form** (*str*) – `'EntityPrediction'` or `'RelationPrediction'`.
- **label_smoothing_rate** (*float, optional*) – Label smoothing coefficient (default `0.0`).

train_data: torch.Tensor

train_target: dicee.dataset_classes._storage.RaggedIndices

label_smoothing_rate

collate_fn = None

__len__()

__getitem__(*idx*)

class dicee.dataset_classes.**KvsSampleDataset**(*train_set_idx: numpy.ndarray*, *entity_idxs*, *relation_idxs*, *form*, *store=None*, *neg_ratio=None*, *label_smoothing_rate: float = 0.0*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Dataset for KvsSample training (dynamic multi-label).

Like KvsAll but sub-samples the target vector at each access to keep mini-batch sizes manageable when the entity set is large.

Parameters

- **train_set_idx** (*numpy.ndarray*) – (N, 3) integer-indexed triples.
- **entity_idxs** (*dict*) – Entity-name → index mapping.
- **relation_idxs** (*dict*) – Relation-name → index mapping.
- **form** (*str*) – 'EntityPrediction'.
- **neg_ratio** (*int*) – Number of negative samples per positive target.
- **label_smoothing_rate** (*float, optional*) – Label smoothing coefficient (default 0.0).

train_data: torch.Tensor

train_target: dicee.dataset_classes._storage.RaggedIndices

neg_ratio = None

num_entities

label_smoothing_rate

collate_fn = None

max_num_of_classes

__len__()

__getitem__(*idx*)

class dicee.dataset_classes.**OnevsAllDataset**(*train_set_idx: numpy.ndarray*, *entity_idxs*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Dataset for the 1-vs-All training strategy (multi-class).

Each sample is a (head, relation) pair with a one-hot target vector whose single active position corresponds to the true tail entity.

Parameters

- **train_set_idx** (*numpy.ndarray*) – (N, 3) integer-indexed triples.
- **entity_idxs** (*dict*) – Entity-name → index mapping (used to determine the target dimension).

train_data

target_dim

collate_fn = None

__len__()

__getitem__(*idx*)

class dicee.dataset_classes.**LiteralDataset**(*file_path: str*, *ent_idx: dict | None = None*, *normalization_type: str = 'z-norm'*, *sampling_ratio: float | None = None*, *loader_backend: str = 'pandas'*)

Bases: `torch.utils.data.Dataset`

Dataset for loading and processing literal data for Literal Embedding models.

Handles loading, normalization, and preparation of `(entity, attribute, value)` triples. Supports z-score and min-max normalization as well as optional sub-sampling for ablation studies.

Parameters

- **file_path** (*str*) – Path to the training data file (CSV / TSV / RDF).
- **ent_idx** (*dict*) – Entity-name → index mapping.
- **normalization_type** (*str, optional*) – `'z-norm'`, `'min-max'`, or `None` (default `'z-norm'`).
- **sampling_ratio** (*float or None, optional*) – Fraction of the training set to keep (default `None` = use all).
- **loader_backend** (*str, optional*) – `'pandas'` or `'rdflib'` (default `'pandas'`).

train_file_path

loader_backend = 'pandas'

normalization_type = 'z-norm'

normalization_params

sampling_ratio = None

entity_to_idx = None

num_entities

__getitem__(*index*)

__len__()

static load_and_validate_literal_data(*file_path: str | None = None*, *loader_backend: str = 'pandas'*) → pandas.DataFrame

Load and validate a literal data file.

Parameters

- **file_path** (*str*) – Path to the data file.
- **loader_backend** (*str*) – `'pandas'` or `'rdflib'`.

Returns

Three-column DataFrame with columns `head`, `attribute`, `value`.

Return type

pandas.DataFrame

static denormalize(*preds_norm*, *attributes*, *normalization_params*) → numpy.ndarray

Reverse the normalization applied during training.

Parameters

- **preds_norm** (*numpy.ndarray*) – Normalized predictions.
- **attributes** (*list*) – Attribute names corresponding to each prediction.
- **normalization_params** (*dict*) – Parameters stored during training.

Returns
Denormalized predictions.

Return type
numpy.ndarray

class dicee.dataset_classes.**FixedNegSampleDataset**(*train_set: numpy.ndarray*, *num_entities: int*, *num_relations: int*, *neg_sample_ratio: int = 1*, *label_smoothing_rate: float = 0.0*, *seed: int | None = None*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Pre-computed (fixed) negative sampling dataset.

At construction time every positive triple is paired with one random negative (head- or tail-corrupted) using vectorized operations for efficiency. The pairs are stored so that __getitem__ is a simple lookup.

This is useful when you want deterministic negatives across epochs (e.g., for reproducibility or debugging).

Parameters

- **train_set** (*numpy.ndarray*) – (N, 3) integer-indexed triples.
- **num_entities** (*int*) – Total number of entities.
- **num_relations** (*int*) – Total number of relations.
- **neg_sample_ratio** (*int, optional*) – Number of negative samples per positive triple (default 1).
- **label_smoothing_rate** (*float, optional*) – Label smoothing coefficient (default 0.0).

neg_sample_ratio = 1

num_entities

num_relations

label_smoothing_rate = 0.0

collate_fn = None

seed = None

train_triples

length

__len__() → int

__getitem__(*idx: int*) → Tuple[torch.Tensor, torch.Tensor]

class dicee.dataset_classes.**OnevsSample**(*train_set: numpy.ndarray*, *num_entities: int*, *num_relations: int*, *neg_sample_ratio: int | None = None*, *label_smoothing_rate: float = 0.0*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Dataset for 1-vs-Sample training (dynamic multi-class with negatives).

For every positive triple `(h, r, t)` the dataset draws `neg_sample_ratio` random entities as negatives and returns a label vector that marks the true tail and the negatives.

Parameters

- **train_set** (*numpy.ndarray*) – `(N, 3)` integer-indexed triples.
- **num_entities** (*int*) – Total number of entities.
- **num_relations** (*int*) – Total number of relations.
- **neg_sample_ratio** (*int*) – Number of negative samples per positive.
- **label_smoothing_rate** (*float, optional*) – Label smoothing coefficient (default `0.0`).

train_data

num_entities

num_relations

neg_sample_ratio = None

label_smoothing_rate

collate_fn = None

__len__()

__getitem__(*idx*)

class dicee.dataset_classes.**TriplePredictionDataset**(*train_set: numpy.ndarray*, *num_entities: int*, *num_relations: int*, *neg_sample_ratio: int = 1*, *label_smoothing_rate: float = 0.0*, *seed: int | None = None*, *sort_train_set: bool = True*)

Bases: dicee.dataset_classes._storage.WorkerDataset

Dataset for triple prediction with on-the-fly negative sampling.

Each item is a single positive triple; the custom `collate_fn` generates a batch of mixed positive and negative triples.

Parameters

- **train_set** (*numpy.ndarray*) – `(N, 3)` integer-indexed triples.
- **num_entities** (*int*) – Total number of entities.
- **num_relations** (*int*) – Total number of relations.
- **neg_sample_ratio** (*int, optional*) – Number of negative samples per positive triple (default `1`).
- **label_smoothing_rate** (*float, optional*) – Label smoothing coefficient (default `0.0`).

label_smoothing_rate

neg_sample_ratio

seed = None

length

num_entities

num_relations

__len__()

__getitem__(*idx*)

collate_fn(*batch: List[torch.Tensor]*)

dicee.eval_static_funcs

Static evaluation functions for KGE models.

This module provides backward compatibility by re-exporting from the new dicee.evaluation module.

Deprecated since version Use: `dicee.evaluation` submodules instead. This module will be removed in a future version.

Functions

evaluate_ensemble_link_prediction_performa	Evaluate link prediction performance of an ensemble of KGE models.
evaluate_link_prediction_performance(→ Dict[str, float])	Evaluate link prediction performance with head and tail prediction.
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with BPE encoding (head and tail).
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with BPE encoding and reciprocals.
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with reciprocal relations.
evaluate_lp_bpe_k_vs_all(→ Dict[str, float])	Evaluate BPE link prediction with KvsAll scoring.
evaluate_literal_prediction(→ Optional[pandas.DataFrame])	Evaluate trained literal prediction model on a test file.

Module Contents

dicee.eval_static_funcs.**evaluate_ensemble_link_prediction_performance**(*models: List*, *triples*, *er_vocab: Dict[Tuple, List]*, *weights: List[float] | None = None*, *batch_size: int = 512*, *weighted_averaging: bool = True*, *normalize_scores: bool = True*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction performance of an ensemble of KGE models.

Combines predictions from multiple models using weighted or simple averaging, with optional score normalization.

Parameters

- **models** – List of KGE models (e.g., snapshots from training).
- **triples** – Test triples as numpy array or list, shape (N, 3), with integer indices (head, relation, tail).

- **er_vocab** – Mapping (head_idx, rel_idx) -> list of tail indices for filtered evaluation.
- **weights** – Weights for model averaging. Required if weighted_averaging is True. Must sum to 1 for proper averaging.
- **batch_size** – Batch size for processing triples.
- **weighted_averaging** – If True, use weighted averaging of predictions. If False, use simple mean.
- **normalize_scores** – If True, normalize scores to [0, 1] range per sample before averaging.
- **tie_policy** – Exact prediction ties: sort (legacy), optimistic, random, or pessimistic. Applied after combining model scores.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

Raises

AssertionError – If weighted_averaging is True but weights are not provided or have wrong length.

Example

```
>>> from dicee.evaluation import evaluate_ensemble_link_prediction_performance
>>> models = [model1, model2, model3]
>>> weights = [0.5, 0.3, 0.2]
>>> results = evaluate_ensemble_link_prediction_performance(
...     models, test_triples, er_vocab,
...     weights=weights, weighted_averaging=True
... )
>>> print(f"MRR: {results['MRR']:.4f}")
```

dicee.eval_static_funcs.**evaluate_link_prediction_performance**(*model*, *triples*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction performance with head and tail prediction.

Performs filtered evaluation where known correct answers are filtered out before computing ranks.

Parameters

- **model** – KGE model wrapper with entity/relation mappings.
- **triples** – Test triples as list of (head, relation, tail) strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **re_vocab** – Mapping (relation, entity) -> list of valid head entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.eval_static_funcs.**evaluate_link_prediction_performance_with_bpe**(*model*, *within_entities: List[str]*, *triples: List[Tuple[str]]*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE encoding (head and tail).

Parameters

- **model** – KGE model wrapper with BPE support.
- **within_entities** – List of entities to evaluate within.
- **triples** – Test triples as list of (head, relation, tail) tuples.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **re_vocab** – Mapping (relation, entity) -> list of valid head entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.eval_static_funcs.**evaluate_link_prediction_performance_with_bpe_reciprocals**(*model*, *within_entities: List[str]*, *triples: List[List[str]]*, *er_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE encoding and reciprocals.

Parameters

- **model** – KGE model wrapper with BPE support.
- **within_entities** – List of entities to evaluate within.
- **triples** – Test triples as list of [head, relation, tail] strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.eval_static_funcs.**evaluate_link_prediction_performance_with_reciprocals**(*model*, *triples*, *er_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with reciprocal relations.

Optimized for models trained with reciprocal triples where only tail prediction is needed.

Parameters

- **model** – KGE model wrapper.
- **triples** – Test triples as list of (head, relation, tail) strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.eval_static_funcs.**evaluate_lp_bpe_k_vs_all**(*model*, *triples: List[List[str]]*, *er_vocab: Dict | None = None*, *batch_size: int | None = None*, *func_triple_to_bpe_representation: Callable | None = None*, *str_to_bpe_entity_to_idx: Dict | None = None*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate BPE link prediction with KvsAll scoring.

Parameters

- **model** – The KGE model wrapper.
- **triples** – List of string triples.
- **er_vocab** – Entity-relation vocabulary for filtering.
- **batch_size** – Batch size for processing.
- **func_triple_to_bpe_representation** – Function to convert triples to BPE.
- **str_to_bpe_entity_to_idx** – Mapping from string entities to BPE indices.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

Raises
ValueError – If batch_size is not provided.

dicee.eval_static_funcs.**evaluate_literal_prediction**(*kge_model*, *eval_file_path: str | None = None*, *store_lit_preds: bool = True*, *eval_literals: bool = True*, *loader_backend: str = 'pandas'*, *return_attr_error_metrics: bool = False*) → pandas.DataFrame | None

Evaluate trained literal prediction model on a test file.

Evaluates the literal prediction capabilities of a KGE model by computing MAE and RMSE metrics for each attribute.

Parameters

- **kge_model** – Trained KGE model with literal prediction capability.
- **eval_file_path** – Path to the evaluation file containing test literals.
- **store_lit_preds** – If True, stores predictions to CSV file.
- **eval_literals** – If True, evaluates and prints error metrics.
- **loader_backend** – Backend for loading dataset ('pandas' or 'rdflib').
- **return_attr_error_metrics** – If True, returns the metrics DataFrame.

Returns
DataFrame with per-attribute MAE and RMSE if return_attr_error_metrics is True, otherwise None.

Raises

- **RuntimeError** – If the KGE model doesn't have a trained literal model.

- **AssertionError** – If model is invalid or test set has no valid data.

Example

```
>>> from dicee import KGE
>>> from dicee.evaluation import evaluate_literal_prediction
>>> model = KGE(path="pretrained_model")
>>> metrics = evaluate_literal_prediction(
...     model,
...     eval_file_path="test_literals.csv",
...     return_attr_error_metrics=True
... )
>>> print(metrics)
```

dicee.evaluation

Evaluation module for knowledge graph embedding models.

This module provides comprehensive evaluation capabilities for KGE models, including link prediction, literal prediction, and ensemble evaluation.

Modules:

link_prediction: Functions for evaluating link prediction performance literal_prediction: Functions for evaluating literal/attribute prediction ensemble: Functions for ensemble model evaluation evaluator: Main Evaluator class for integrated evaluation utils: Shared utility functions for evaluation _filtering: Internal helpers for filtered ranking (not exported)

Example

```
>>> from dicee.evaluation import Evaluator
>>> from dicee.evaluation.link_prediction import evaluate_link_prediction_performance
>>> from dicee.evaluation.ensemble import evaluate_ensemble_link_prediction_
↪performance
```

Submodules

dicee.evaluation.ensemble

Ensemble evaluation functions.

This module provides functions for evaluating ensemble models, including weighted averaging and score normalization.

Functions

evaluate_ensemble_link_prediction_performa	Evaluate link prediction performance of an ensemble of KGE models.

Module Contents

dicee.evaluation.ensemble.**evaluate_ensemble_link_prediction_performance**(*models: List*, *triples*, *er_vocab: Dict[Tuple, List]*, *weights: List[float] | None = None*, *batch_size: int = 512*, *weighted_averaging: bool = True*, *normalize_scores: bool = True*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction performance of an ensemble of KGE models.

Combines predictions from multiple models using weighted or simple averaging, with optional score normalization.

Parameters

- **models** – List of KGE models (e.g., snapshots from training).
- **triples** – Test triples as numpy array or list, shape (N, 3), with integer indices (head, relation, tail).
- **er_vocab** – Mapping (head_idx, rel_idx) -> list of tail indices for filtered evaluation.
- **weights** – Weights for model averaging. Required if weighted_averaging is True. Must sum to 1 for proper averaging.
- **batch_size** – Batch size for processing triples.
- **weighted_averaging** – If True, use weighted averaging of predictions. If False, use simple mean.
- **normalize_scores** – If True, normalize scores to [0, 1] range per sample before averaging.
- **tie_policy** – Exact prediction ties: sort (legacy), optimistic, random, or pessimistic. Applied after combining model scores.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

Raises

AssertionError – If weighted_averaging is True but weights are not provided or have wrong length.

Example

```
>>> from dicee.evaluation import evaluate_ensemble_link_prediction_performance
>>> models = [model1, model2, model3]
>>> weights = [0.5, 0.3, 0.2]
>>> results = evaluate_ensemble_link_prediction_performance(
...     models, test_triples, er_vocab,
...     weights=weights, weighted_averaging=True
... )
>>> print(f"MRR: {results['MRR']:.4f}")
```

dicee.evaluation.evaluator

Main Evaluator class for KGE model evaluation.

This module provides the Evaluator class which orchestrates evaluation of knowledge graph embedding models across different datasets and scoring techniques.

Attributes

logger
VALID_SCORING_TECHNIQUES

Classes

Evaluator	Evaluator class for KGE models in various downstream tasks.

Module Contents

dicee.evaluation.evaluator.**logger**

dicee.evaluation.evaluator.**VALID_SCORING_TECHNIQUES**

class dicee.evaluation.evaluator.**Evaluator**(*args*, *is_continual_training: bool = False*)

Evaluator class for KGE models in various downstream tasks.

Orchestrates link prediction evaluation with different scoring techniques including standard evaluation and byte-pair encoding based evaluation.

er_vocab

Entity-relation to tail vocabulary for filtered ranking.

re_vocab

Relation-entity (tail) to head vocabulary.

ee_vocab

Entity-entity to relation vocabulary.

num_entities

Total number of entities in the knowledge graph.

num_relations

Total number of relations in the knowledge graph.

args

Configuration arguments.

report

Dictionary storing evaluation results.

during_training

Whether evaluation is happening during training.

Example

```
>>> from dicee.evaluation import Evaluator
>>> evaluator = Evaluator(args)
>>> results = evaluator.eval(dataset, model, 'EntityPrediction')
>>> print(f"Test MRR: {results['Test']['MRR']:.4f}")
```

re_vocab: Dict | None = None

er_vocab: Dict | None = None

ee_vocab: Dict | None = None

func_triple_to_bpe_representation = None

is_continual_training = False

num_entities: int | None = None

num_relations: int | None = None

domain_constraints_per_rel = None

range_constraints_per_rel = None

args

report: Dict

during_training = False

vocab_preparation(*dataset*) → None

Prepare vocabularies from the dataset for evaluation.

Resolves any future objects and saves vocabularies to disk.

Parameters

dataset – Knowledge graph dataset with vocabulary attributes.

eval(*dataset*, *trained_model*, *form_of_labelling: str*, *during_training: bool = False*) → Dict | None

Evaluate the trained model on the dataset.

Parameters

- **dataset** – Knowledge graph dataset (KG instance).
- **trained_model** – The trained KGE model.
- **form_of_labelling** – Type of labelling ('EntityPrediction' or 'RelationPrediction').
- **during_training** – Whether evaluation is during training.

Returns

Dictionary of evaluation metrics, or None if evaluation is skipped.

eval_rank_of_head_and_tail_entity(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*) → None

Evaluate with negative sampling scoring.

eval_rank_of_head_and_tail_byte_pair_encoded_entity(*, *train_set=None*, *valid_set=None*, *test_set=None*, *ordered_bpe_entities*, *trained_model*) → None

Evaluate with BPE-encoded entities and negative sampling.

eval_with_byte(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate BytE model with generation.

eval_with_bpe_vs_all(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with BPE and KvsAll scoring.

eval_with_vs_all(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with KvsAll or 1vsAll scoring.

evaluate_lp_k_vs_all(*model*, *triple_idx*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Filtered link prediction evaluation with KvsAll scoring.

Parameters

- **model** – The trained model to evaluate.
- **triple_idx** – Integer-indexed test triples.
- **info** – Description to print.
- **form_of_labelling** – 'EntityPrediction' or 'RelationPrediction'.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp_with_byte(*model*, *triples: List[List[str]]*, *info: str | None = None*) → Dict[str, float]

Evaluate BytE model with text generation.

Parameters

- **model** – BytE model.
- **triples** – String triples.
- **info** – Description to print.

Returns

Dictionary with placeholder metrics (-1 values).

evaluate_lp_bpe_k_vs_all(*model*, *triples: List[List[str]]*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Evaluate BPE model with KvsAll scoring.

Parameters

- **model** – BPE-enabled model.
- **triples** – String triples.
- **info** – Description to print.
- **form_of_labelling** – Type of labelling.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp(*model*, *triple_idx*, *info: str*) → Dict[str, float]

Evaluate link prediction with negative sampling.

Parameters

- **model** – The model to evaluate.
- **triple_idx** – Integer-indexed triples.
- **info** – Description to print.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dummy_eval(*trained_model*, *form_of_labelling: str*) → None

Run evaluation from saved data (for continual training).

Parameters

- **trained_model** – The trained model.
- **form_of_labelling** – Type of labelling.

eval_with_data(*dataset*, *trained_model*, *triple_idx: numpy.ndarray*, *form_of_labelling: str*) → Dict[str, float]

Evaluate a trained model on a given dataset.

Parameters

- **dataset** – Knowledge graph dataset.
- **trained_model** – The trained model.
- **triple_idx** – Integer-indexed triples to evaluate.
- **form_of_labelling** – Type of labelling.

Returns

Dictionary with evaluation metrics.

Raises

ValueError – If scoring technique is invalid.

dicee.evaluation.link_prediction

Link prediction evaluation functions.

This module provides various functions for evaluating link prediction performance of knowledge graph embedding models.

Attributes

logger

Functions

evaluate_link_prediction_performance(→ Dict[str, float])	Evaluate link prediction performance with head and tail prediction.
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with reciprocal relations.
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with BPE encoding and reciprocals.
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with BPE encoding (head and tail).
evaluate_lp(→ Dict[str, float])	Evaluate link prediction with batched processing.
evaluate_bpe_lp(→ Dict[str, float])	Evaluate link prediction with BPE-encoded entities.
evaluate_lp_bpe_k_vs_all(→ Dict[str, float])	Evaluate BPE link prediction with KvsAll scoring.

Module Contents

dicee.evaluation.link_prediction.**logger**

dicee.evaluation.link_prediction.**evaluate_link_prediction_performance**(*model*, *triples*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction performance with head and tail prediction.

Performs filtered evaluation where known correct answers are filtered out before computing ranks.

Parameters

- **model** – KGE model wrapper with entity/relation mappings.
- **triples** – Test triples as list of (head, relation, tail) strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **re_vocab** – Mapping (relation, entity) -> list of valid head entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.link_prediction.**evaluate_link_prediction_performance_with_reciprocals**(*model*, *triples*, *er_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with reciprocal relations.

Optimized for models trained with reciprocal triples where only tail prediction is needed.

Parameters

- **model** – KGE model wrapper.
- **triples** – Test triples as list of (head, relation, tail) strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.link_prediction.**evaluate_link_prediction_performance_with_bpe_reciprocals**(*model*, *within_entities: List[str]*, *triples: List[List[str]]*, *er_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE encoding and reciprocals.

Parameters

- **model** – KGE model wrapper with BPE support.
- **within_entities** – List of entities to evaluate within.
- **triples** – Test triples as list of [head, relation, tail] strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.link_prediction.**evaluate_link_prediction_performance_with_bpe**(*model*, *within_entities: List[str]*, *triples: List[Tuple[str]]*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE encoding (head and tail).

Parameters

- **model** – KGE model wrapper with BPE support.
- **within_entities** – List of entities to evaluate within.
- **triples** – Test triples as list of (head, relation, tail) tuples.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **re_vocab** – Mapping (relation, entity) -> list of valid head entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.link_prediction.**evaluate_lp**(*model*, *triple_idx*, *num_entities: int*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *info: str = 'Eval Starts'*, *batch_size: int = 128*, *chunk_size: int = 1000*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*, *tie_generator: torch.Generator | None = None*, *graph_rank_plan: dicee.evaluation._graph_inference.GraphRankPlan | None = None*, *reuse_graph_queries: bool = True*) → Dict[str, float]

Evaluate link prediction with batched processing.

Memory-efficient evaluation using chunked entity scoring.

Parameters

- **model** – The KGE model to evaluate.
- **triple_idx** – Integer-indexed triples as numpy array.
- **num_entities** – Total number of entities.
- **er_vocab** – Mapping (head_idx, rel_idx) -> list of tail indices.
- **re_vocab** – Mapping (rel_idx, tail_idx) -> list of head indices.
- **info** – Description to print.
- **batch_size** – Batch size for triple processing.
- **chunk_size** – Chunk size for entity scoring.
- **tie_generator** – Optional CPU generator to continue random ties across calls. When supplied, its state takes precedence over tie_seed.
- **graph_rank_plan** – Optional whole-split plan shared across progress batches.
- **reuse_graph_queries** – Reuse deterministic ULTRA/TRIX queries. Flock keeps its original query and sampling order regardless of this option.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.link_prediction.**evaluate_bpe_lp**(*model*, *triple_idx: List[Tuple]*, *all_bpe_shaped_entities*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *info: str = 'Eval Starts'*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE-encoded entities.

Parameters

- **model** – The KGE model to evaluate.
- **triple_idx** – List of BPE-encoded triple tuples.
- **all_bpe_shaped_entities** – All entities with BPE representations.
- **er_vocab** – Mapping for tail filtering.
- **re_vocab** – Mapping for head filtering.
- **info** – Description to print.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.link_prediction.**evaluate_lp_bpe_k_vs_all**(*model*, *triples: List[List[str]]*, *er_vocab: Dict | None = None*, *batch_size: int | None = None*, *func_triple_to_bpe_representation: Callable | None = None*, *str_to_bpe_entity_to_idx: Dict | None = None*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate BPE link prediction with KvsAll scoring.

Parameters

- **model** – The KGE model wrapper.
- **triples** – List of string triples.
- **er_vocab** – Entity-relation vocabulary for filtering.
- **batch_size** – Batch size for processing.
- **func_triple_to_bpe_representation** – Function to convert triples to BPE.
- **str_to_bpe_entity_to_idx** – Mapping from string entities to BPE indices.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

Raises
ValueError – If batch_size is not provided.

dicee.evaluation.literal_prediction

Literal prediction evaluation functions.

This module provides functions for evaluating literal/attribute prediction performance of knowledge graph embedding models.

Attributes

logger

Functions

evaluate_literal_prediction(→ Optional[pandas.DataFrame])	Evaluate trained literal prediction model on a test file.

Module Contents

dicee.evaluation.literal_prediction.**logger**

dicee.evaluation.literal_prediction.**evaluate_literal_prediction**(*kge_model*, *eval_file_path: str | None = None*, *store_lit_preds: bool = True*, *eval_literals: bool = True*, *loader_backend: str = 'pandas'*, *return_attr_error_metrics: bool = False*) → pandas.DataFrame | None

Evaluate trained literal prediction model on a test file.

Evaluates the literal prediction capabilities of a KGE model by computing MAE and RMSE metrics for each attribute.

Parameters

- **kge_model** – Trained KGE model with literal prediction capability.
- **eval_file_path** – Path to the evaluation file containing test literals.
- **store_lit_preds** – If True, stores predictions to CSV file.
- **eval_literals** – If True, evaluates and prints error metrics.
- **loader_backend** – Backend for loading dataset ('pandas' or 'rdflib').
- **return_attr_error_metrics** – If True, returns the metrics DataFrame.

Returns

DataFrame with per-attribute MAE and RMSE if return_attr_error_metrics is True, otherwise None.

Raises

- **RuntimeError** – If the KGE model doesn't have a trained literal model.
- **AssertionError** – If model is invalid or test set has no valid data.

Example

```
>>> from dicee import KGE
>>> from dicee.evaluation import evaluate_literal_prediction
>>> model = KGE(path="pretrained_model")
>>> metrics = evaluate_literal_prediction(
...     model,
...     eval_file_path="test_literals.csv",
...     return_attr_error_metrics=True
... )
>>> print(metrics)
```

dicee.evaluation.utils

Utility functions for evaluation module.

This module contains shared helper functions used across different evaluation components.

Attributes

DEFAULT_HITS_RANGE
ALL_HITS_RANGE

Functions

make_iterable_verbose(→ Iterable)	Wrap an iterable with tqdm progress bar if verbose is True.
compute_metrics_from_ranks(→ Dict[str, float])	Compute standard link prediction metrics from ranks.
compute_metrics_from_ranks_simple(→ Dict[str, float])	Compute link prediction metrics without scaling factor.
update_hits(→ None)	Update hits dictionary based on rank.
create_hits_dict(→ Dict[int, List[float]])	Create an initialized hits dictionary.
efficient_zero_grad(→ None)	Efficiently zero gradients using parameter.grad = None.

Module Contents

dicee.evaluation.utils.**DEFAULT_HITS_RANGE: List[int] = [1, 3, 10]**

dicee.evaluation.utils.**ALL_HITS_RANGE: List[int]**

dicee.evaluation.utils.**make_iterable_verbose**(*iterable_object: Iterable*, *verbose: bool*, *desc: str = 'Default'*, *position: int | None = None*, *leave: bool = True*) → Iterable

Wrap an iterable with tqdm progress bar if verbose is True.

Parameters

- **iterable_object** – The iterable to potentially wrap.
- **verbose** – Whether to show progress bar.
- **desc** – Description for the progress bar.
- **position** – Position of the progress bar.

- **leave** – Whether to leave the progress bar after completion.

Returns
The original iterable or a tqdm-wrapped version.

dicee.evaluation.utils.**compute_metrics_from_ranks**(*ranks: List[int]*, *num_triples: int*, *hits_dict: Dict[int, List[float]]*, *scale_factor: int = 1*) → Dict[str, float]

Compute standard link prediction metrics from ranks.

Parameters

- **ranks** – List of ranks for each prediction.
- **num_triples** – Total number of triples evaluated.
- **hits_dict** – Dictionary mapping hit levels to lists of hits.
- **scale_factor** – Factor to scale the denominator (e.g., 2 for head+tail).

Returns
Dictionary containing H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.utils.**compute_metrics_from_ranks_simple**(*ranks: List[int]*, *num_triples: int*, *hits_dict: Dict[int, List[float]]*) → Dict[str, float]

Compute link prediction metrics without scaling factor.

Parameters

- **ranks** – List of ranks for each prediction.
- **num_triples** – Total number of triples evaluated.
- **hits_dict** – Dictionary mapping hit levels to lists of hits.

Returns
Dictionary containing H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.utils.**update_hits**(*hits: Dict[int, List[float]]*, *rank: int*, *hits_range: List[int] | None = None*) → None

Update hits dictionary based on rank.

Parameters

- **hits** – Dictionary to update in-place.
- **rank** – The rank to check against hit levels.
- **hits_range** – List of hit levels to check (default: ALL_HITS_RANGE).

dicee.evaluation.utils.**create_hits_dict**(*hits_range: List[int] | None = None*) → Dict[int, List[float]]

Create an initialized hits dictionary.

Parameters
hits_range – List of hit levels to initialize (default: ALL_HITS_RANGE).

Returns
Dictionary with empty lists for each hit level.

dicee.evaluation.utils.**efficient_zero_grad**(*model*) → None

Efficiently zero gradients using parameter.grad = None.

This is more efficient than optimizer.zero_grad() as it avoids memory operations.

See: https://pytorch.org/tutorials/recipes/recipes/tuning_guide.html

Parameters

model – PyTorch model to zero gradients for.

Classes

Evaluator	Evaluator class for KGE models in various downstream tasks.

Functions

evaluate_ensemble_link_prediction_performa	Evaluate link prediction performance of an ensemble of KGE models.
evaluate_bpe_lp(→ Dict[str, float])	Evaluate link prediction with BPE-encoded entities.
evaluate_link_prediction_performance(→ Dict[str, float])	Evaluate link prediction performance with head and tail prediction.
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with BPE encoding (head and tail).
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with BPE encoding and reciprocals.
*evaluate_link_prediction_performance_with_*	Evaluate link prediction with reciprocal relations.
evaluate_lp(→ Dict[str, float])	Evaluate link prediction with batched processing.
evaluate_lp_bpe_k_vs_all(→ Dict[str, float])	Evaluate BPE link prediction with KvsAll scoring.
evaluate_literal_prediction(→ Optional[pandas.DataFrame])	Evaluate trained literal prediction model on a test file.

Package Contents

dicee.evaluation.**evaluate_ensemble_link_prediction_performance**(*models: List*, *triples*, *er_vocab: Dict[Tuple, List]*, *weights: List[float] | None = None*, *batch_size: int = 512*, *weighted_averaging: bool = True*, *normalize_scores: bool = True*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction performance of an ensemble of KGE models.

Combines predictions from multiple models using weighted or simple averaging, with optional score normalization.

Parameters

- **models** – List of KGE models (e.g., snapshots from training).
- **triples** – Test triples as numpy array or list, shape (N, 3), with integer indices (head, relation, tail).
- **er_vocab** – Mapping (head_idx, rel_idx) -> list of tail indices for filtered evaluation.
- **weights** – Weights for model averaging. Required if weighted_averaging is True. Must sum to 1 for proper averaging.
- **batch_size** – Batch size for processing triples.
- **weighted_averaging** – If True, use weighted averaging of predictions. If False, use simple mean.
- **normalize_scores** – If True, normalize scores to [0, 1] range per sample before averaging.
- **tie_policy** – Exact prediction ties: sort (legacy), optimistic, random, or pessimistic. Applied after combining model scores.

- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

Raises

AssertionError – If weighted_averaging is True but weights are not provided or have wrong length.

Example

```
>>> from dicee.evaluation import evaluate_ensemble_link_prediction_performance
>>> models = [model1, model2, model3]
>>> weights = [0.5, 0.3, 0.2]
>>> results = evaluate_ensemble_link_prediction_performance(
...     models, test_triples, er_vocab,
...     weights=weights, weighted_averaging=True
... )
>>> print(f"MRR: {results['MRR']:.4f}")
```

class dicee.evaluation.**Evaluator**(*args*, *is_continual_training: bool = False*)

Evaluator class for KGE models in various downstream tasks.

Orchestrates link prediction evaluation with different scoring techniques including standard evaluation and byte-pair encoding based evaluation.

er_vocab

Entity-relation to tail vocabulary for filtered ranking.

re_vocab

Relation-entity (tail) to head vocabulary.

ee_vocab

Entity-entity to relation vocabulary.

num_entities

Total number of entities in the knowledge graph.

num_relations

Total number of relations in the knowledge graph.

args

Configuration arguments.

report

Dictionary storing evaluation results.

during_training

Whether evaluation is happening during training.

Example

```
>>> from dicee.evaluation import Evaluator
>>> evaluator = Evaluator(args)
>>> results = evaluator.eval(dataset, model, 'EntityPrediction')
>>> print(f"Test MRR: {results['Test']['MRR']:.4f}")
```

re_vocab: Dict | None = None

er_vocab: Dict | None = None

ee_vocab: Dict | None = None

func_triple_to_bpe_representation = None

is_continual_training = False

num_entities: int | None = None

num_relations: int | None = None

domain_constraints_per_rel = None

range_constraints_per_rel = None

args

report: Dict

during_training = False

vocab_preparation(*dataset*) → None

Prepare vocabularies from the dataset for evaluation.

Resolves any future objects and saves vocabularies to disk.

Parameters

dataset – Knowledge graph dataset with vocabulary attributes.

eval(*dataset*, *trained_model*, *form_of_labelling: str*, *during_training: bool = False*) → Dict | None

Evaluate the trained model on the dataset.

Parameters

- **dataset** – Knowledge graph dataset (KG instance).
- **trained_model** – The trained KGE model.
- **form_of_labelling** – Type of labelling ('EntityPrediction' or 'RelationPrediction').
- **during_training** – Whether evaluation is during training.

Returns

Dictionary of evaluation metrics, or None if evaluation is skipped.

eval_rank_of_head_and_tail_entity(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*) → None

Evaluate with negative sampling scoring.

eval_rank_of_head_and_tail_byte_pair_encoded_entity(*, *train_set=None*, *valid_set=None*, *test_set=None*, *ordered_bpe_entities*, *trained_model*) → None

Evaluate with BPE-encoded entities and negative sampling.

eval_with_byte(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate BytE model with generation.

eval_with_bpe_vs_all(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with BPE and KvsAll scoring.

eval_with_vs_all(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with KvsAll or 1vsAll scoring.

evaluate_lp_k_vs_all(*model*, *triple_idx*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Filtered link prediction evaluation with KvsAll scoring.

Parameters

- **model** – The trained model to evaluate.
- **triple_idx** – Integer-indexed test triples.
- **info** – Description to print.
- **form_of_labelling** – 'EntityPrediction' or 'RelationPrediction'.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp_with_byte(*model*, *triples: List[List[str]]*, *info: str | None = None*) → Dict[str, float]

Evaluate BytE model with text generation.

Parameters

- **model** – BytE model.
- **triples** – String triples.
- **info** – Description to print.

Returns

Dictionary with placeholder metrics (-1 values).

evaluate_lp_bpe_k_vs_all(*model*, *triples: List[List[str]]*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Evaluate BPE model with KvsAll scoring.

Parameters

- **model** – BPE-enabled model.
- **triples** – String triples.
- **info** – Description to print.
- **form_of_labelling** – Type of labelling.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp(*model*, *triple_idx*, *info: str*) → Dict[str, float]

Evaluate link prediction with negative sampling.

Parameters

- **model** – The model to evaluate.
- **triple_idx** – Integer-indexed triples.
- **info** – Description to print.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

dummy_eval(*trained_model*, *form_of_labelling: str*) → None

Run evaluation from saved data (for continual training).

Parameters

- **trained_model** – The trained model.
- **form_of_labelling** – Type of labelling.

eval_with_data(*dataset*, *trained_model*, *triple_idx: numpy.ndarray*, *form_of_labelling: str*) → Dict[str, float]

Evaluate a trained model on a given dataset.

Parameters

- **dataset** – Knowledge graph dataset.
- **trained_model** – The trained model.
- **triple_idx** – Integer-indexed triples to evaluate.
- **form_of_labelling** – Type of labelling.

Returns
Dictionary with evaluation metrics.

Raises
ValueError – If scoring technique is invalid.

dicee.evaluation.**evaluate_bpe_lp**(*model*, *triple_idx: List[Tuple]*, *all_bpe_shaped_entities*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *info: str = 'Eval Starts'*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE-encoded entities.

Parameters

- **model** – The KGE model to evaluate.
- **triple_idx** – List of BPE-encoded triple tuples.
- **all_bpe_shaped_entities** – All entities with BPE representations.
- **er_vocab** – Mapping for tail filtering.
- **re_vocab** – Mapping for head filtering.
- **info** – Description to print.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.**evaluate_link_prediction_performance**(*model*, *triples*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction performance with head and tail prediction.

Performs filtered evaluation where known correct answers are filtered out before computing ranks.

Parameters

- **model** – KGE model wrapper with entity/relation mappings.
- **triples** – Test triples as list of (head, relation, tail) strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **re_vocab** – Mapping (relation, entity) -> list of valid head entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.**evaluate_link_prediction_performance_with_bpe**(*model*, *within_entities: List[str]*, *triples: List[Tuple[str]]*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE encoding (head and tail).

Parameters

- **model** – KGE model wrapper with BPE support.
- **within_entities** – List of entities to evaluate within.
- **triples** – Test triples as list of (head, relation, tail) tuples.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **re_vocab** – Mapping (relation, entity) -> list of valid head entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.**evaluate_link_prediction_performance_with_bpe_reciprocals**(*model*, *within_entities: List[str]*, *triples: List[List[str]]*, *er_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE encoding and reciprocals.

Parameters

- **model** – KGE model wrapper with BPE support.
- **within_entities** – List of entities to evaluate within.
- **triples** – Test triples as list of [head, relation, tail] strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.**evaluate_link_prediction_performance_with_reciprocals**(*model*, *triples*, *er_vocab: Dict[Tuple, List]*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with reciprocal relations.

Optimized for models trained with reciprocal triples where only tail prediction is needed.

Parameters

- **model** – KGE model wrapper.
- **triples** – Test triples as list of (head, relation, tail) strings.
- **er_vocab** – Mapping (entity, relation) -> list of valid tail entities.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.**evaluate_lp**(*model*, *triple_idx*, *num_entities: int*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *info: str = 'Eval Starts'*, *batch_size: int = 128*, *chunk_size: int = 1000*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*, *tie_generator: torch.Generator | None = None*, *graph_rank_plan: dicee.evaluation._graph_inference.GraphRankPlan | None = None*, *reuse_graph_queries: bool = True*) → Dict[str, float]

Evaluate link prediction with batched processing.

Memory-efficient evaluation using chunked entity scoring.

Parameters

- **model** – The KGE model to evaluate.
- **triple_idx** – Integer-indexed triples as numpy array.
- **num_entities** – Total number of entities.
- **er_vocab** – Mapping (head_idx, rel_idx) -> list of tail indices.
- **re_vocab** – Mapping (rel_idx, tail_idx) -> list of head indices.
- **info** – Description to print.
- **batch_size** – Batch size for triple processing.
- **chunk_size** – Chunk size for entity scoring.
- **tie_generator** – Optional CPU generator to continue random ties across calls. When supplied, its state takes precedence over tie_seed.
- **graph_rank_plan** – Optional whole-split plan shared across progress batches.
- **reuse_graph_queries** – Reuse deterministic ULTRA/TRIX queries. Flock keeps its original query and sampling order regardless of this option.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.evaluation.**evaluate_lp_bpe_k_vs_all**(*model*, *triples: List[List[str]]*, *er_vocab: Dict | None = None*, *batch_size: int | None = None*, *func_triple_to_bpe_representation: Callable | None = None*, *str_to_bpe_entity_to_idx: Dict | None = None*, *, tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate BPE link prediction with KvsAll scoring.

Parameters

- **model** – The KGE model wrapper.
- **triples** – List of string triples.
- **er_vocab** – Entity-relation vocabulary for filtering.
- **batch_size** – Batch size for processing.
- **func_triple_to_bpe_representation** – Function to convert triples to BPE.
- **str_to_bpe_entity_to_idx** – Mapping from string entities to BPE indices.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

Raises

ValueError – If batch_size is not provided.

dicee.evaluation.**evaluate_literal_prediction**(*kge_model*, *eval_file_path: str | None = None*, *store_lit_preds: bool = True*, *eval_literals: bool = True*, *loader_backend: str = 'pandas'*, *return_attr_error_metrics: bool = False*) → pandas.DataFrame | None

Evaluate trained literal prediction model on a test file.

Evaluates the literal prediction capabilities of a KGE model by computing MAE and RMSE metrics for each attribute.

Parameters

- **kge_model** – Trained KGE model with literal prediction capability.
- **eval_file_path** – Path to the evaluation file containing test literals.
- **store_lit_preds** – If True, stores predictions to CSV file.
- **eval_literals** – If True, evaluates and prints error metrics.
- **loader_backend** – Backend for loading dataset ('pandas' or 'rdflib').
- **return_attr_error_metrics** – If True, returns the metrics DataFrame.

Returns

DataFrame with per-attribute MAE and RMSE if return_attr_error_metrics is True, otherwise None.

Raises

- **RuntimeError** – If the KGE model doesn't have a trained literal model.
- **AssertionError** – If model is invalid or test set has no valid data.

Example

```
>>> from dicee import KGE
>>> from dicee.evaluation import evaluate_literal_prediction
>>> model = KGE(path="pretrained_model")
>>> metrics = evaluate_literal_prediction(
...     model,
...     eval_file_path="test_literals.csv",
...     return_attr_error_metrics=True
... )
>>> print(metrics)
```

dicee.evaluator

Evaluator module for knowledge graph embedding models.

This module provides backward compatibility by re-exporting from the new dicee.evaluation module.

Deprecated since version Use: `dicee.evaluation.Evaluator` instead. This module will be removed in a future version.

Classes

Evaluator	Evaluator class for KGE models in various downstream tasks.

Module Contents

class dicee.evaluator.**Evaluator**(*args*, *is_continual_training: bool = False*)

Evaluator class for KGE models in various downstream tasks.

Orchestrates link prediction evaluation with different scoring techniques including standard evaluation and byte-pair encoding based evaluation.

er_vocab

Entity-relation to tail vocabulary for filtered ranking.

re_vocab

Relation-entity (tail) to head vocabulary.

ee_vocab

Entity-entity to relation vocabulary.

num_entities

Total number of entities in the knowledge graph.

num_relations

Total number of relations in the knowledge graph.

args

Configuration arguments.

report

Dictionary storing evaluation results.

during_training
Whether evaluation is happening during training.

Example

```
>>> from dicee.evaluation import Evaluator
>>> evaluator = Evaluator(args)
>>> results = evaluator.eval(dataset, model, 'EntityPrediction')
>>> print(f"Test MRR: {results['Test']['MRR']:.4f}")
```

re_vocab: Dict | None = None

er_vocab: Dict | None = None

ee_vocab: Dict | None = None

func_triple_to_bpe_representation = None

is_continual_training = False

num_entities: int | None = None

num_relations: int | None = None

domain_constraints_per_rel = None

range_constraints_per_rel = None

args

report: Dict

during_training = False

vocab_preparation(*dataset*) → None
Prepare vocabularies from the dataset for evaluation.

Resolves any future objects and saves vocabularies to disk.

Parameters
dataset – Knowledge graph dataset with vocabulary attributes.

eval(*dataset*, *trained_model*, *form_of_labelling: str*, *during_training: bool = False*) → Dict | None
Evaluate the trained model on the dataset.

Parameters

- **dataset** – Knowledge graph dataset (KG instance).
- **trained_model** – The trained KGE model.
- **form_of_labelling** – Type of labelling (‘EntityPrediction’ or ‘RelationPrediction’).
- **during_training** – Whether evaluation is during training.

Returns
Dictionary of evaluation metrics, or None if evaluation is skipped.

eval_rank_of_head_and_tail_entity(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*) → None
Evaluate with negative sampling scoring.

eval_rank_of_head_and_tail_byte_pair_encoded_entity(*, *train_set=None*, *valid_set=None*, *test_set=None*, *ordered_bpe_entities*, *trained_model*) → None

Evaluate with BPE-encoded entities and negative sampling.

eval_with_byte(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate BytE model with generation.

eval_with_bpe_vs_all(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with BPE and KvsAll scoring.

eval_with_vs_all(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with KvsAll or 1vsAll scoring.

evaluate_lp_k_vs_all(*model*, *triple_idx*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Filtered link prediction evaluation with KvsAll scoring.

Parameters

- **model** – The trained model to evaluate.
- **triple_idx** – Integer-indexed test triples.
- **info** – Description to print.
- **form_of_labelling** – 'EntityPrediction' or 'RelationPrediction'.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp_with_byte(*model*, *triples: List[List[str]]*, *info: str | None = None*) → Dict[str, float]

Evaluate BytE model with text generation.

Parameters

- **model** – BytE model.
- **triples** – String triples.
- **info** – Description to print.

Returns

Dictionary with placeholder metrics (-1 values).

evaluate_lp_bpe_k_vs_all(*model*, *triples: List[List[str]]*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Evaluate BPE model with KvsAll scoring.

Parameters

- **model** – BPE-enabled model.
- **triples** – String triples.
- **info** – Description to print.
- **form_of_labelling** – Type of labelling.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp(*model*, *triple_idx*, *info: str*) → Dict[str, float]

Evaluate link prediction with negative sampling.

Parameters

- **model** – The model to evaluate.
- **triple_idx** – Integer-indexed triples.
- **info** – Description to print.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dummy_eval(*trained_model*, *form_of_labelling: str*) → None

Run evaluation from saved data (for continual training).

Parameters

- **trained_model** – The trained model.
- **form_of_labelling** – Type of labelling.

eval_with_data(*dataset*, *trained_model*, *triple_idx: numpy.ndarray*, *form_of_labelling: str*) → Dict[str, float]

Evaluate a trained model on a given dataset.

Parameters

- **dataset** – Knowledge graph dataset.
- **trained_model** – The trained model.
- **triple_idx** – Integer-indexed triples to evaluate.
- **form_of_labelling** – Type of labelling.

Returns

Dictionary with evaluation metrics.

Raises

ValueError – If scoring technique is invalid.

dicee.executer

Executor module for training, retraining and evaluating KGE models.

This module provides the Execute and ContinuousExecute classes for managing the full lifecycle of knowledge graph embedding model training.

Attributes

logger	

Classes

Execute	Executor class for training, retraining and evaluating KGE models.

continues on next page

Table 26 – continued from previous page

ContinuousExecute	A subclass of Execute Class for retraining

Module Contents

dicee.executer.**logger**

class dicee.executer.**Execute**(*args*, *continuous_training: bool = False*)

Executor class for training, retraining and evaluating KGE models.

Handles the complete workflow: 1. Loading & Preprocessing & Serializing input data 2. Training & Validation & Testing 3. Storing all necessary information

args
: Processed input arguments.

distributed
: Whether distributed training is enabled.

rank
: Process rank in distributed training.

world_size
: Total number of processes.

local_rank
: Local GPU rank.

trainer
: Training handler instance.

trained_model
: The trained model after training completes.

knowledge_graph
: The loaded knowledge graph.

report
: Dictionary storing training metrics and results.

evaluator
: Model evaluation handler.

distributed

rank

world_size

local_rank

args

is_continual_training = False

trainer: *dicee.trainer.DICE_Trainer* | None = None

trained_model = None

knowledge_graph: *dicee.knowledge_graph.KG* | None = None

report: Dict

evaluator: *dicee.evaluator.Evaluator* | None = None

start_time: float | None = None

is_local_rank_zero() → bool

is_global_rank_zero() → bool

cleanup()

setup_executor() → None

Set up storage directories for the experiment.

Creates or reuses experiment directories based on configuration. Saves the configuration to a JSON file.

create_and_store_kg() → None

Create knowledge graph and store as memory-mapped file.

Only executed on global rank 0 in distributed training, so exactly one process writes the shared memmap/vocab files regardless of how many nodes are involved. Skips if memmap already exists.

load_from_memmap() → None

Load knowledge graph from memory-mapped file.

save_trained_model() → None

Save a knowledge graph embedding model

(1) Send model to eval mode and cpu.

(2) Store the memory footprint of the model.

(3) Save the model into disk.

(4) Update the stats of KG again ?

Parameter

rtype
None

end(*form_of_labelling: str*) → dict

End training

(1) Store trained model.

(2) Report runtimes.

(3) Eval model if required.

Parameter

rtype
A dict containing information about the training and/or evaluation

write_report() → None

Report training related information in a report.json file

start() → dict

Start training

(1) Loading the Data # (2) Create an evaluator object. # (3) Create a trainer object. # (4) Start the training

Parameter

rtype
A dict containing information about the training and/or evaluation

class dicee.executer.**ContinuousExecute**(*args*)

Bases: *Execute*

A subclass of Execute Class for retraining

(1) Loading & Preprocessing & Serializing input data.

(2) Training & Validation & Testing

(3) Storing all necessary info

During the continual learning we can only modify *** num_epochs *** parameter. Trained model stored in the same folder as the seed model for the training. Trained model is noted with the current time.

continual_start() → dict

Start Continual Training

(1) Initialize training.

(2) Start continual training.

(3) Save trained model.

Parameter

rtype
A dict containing information about the training and/or evaluation

dicee.knowledge_graph

Knowledge Graph module for data loading and preprocessing.

Provides the KG class for handling knowledge graph data including loading, preprocessing, and indexing operations.

Classes

KG	Knowledge Graph container and processor.

Module Contents

class dicee.knowledge_graph.**KG**(*dataset_dir: str | None = None*, *byte_pair_encoding: bool = False*, *padding: bool = False*, *add_noise_rate: float | None = None*, *sparql_endpoint: str | None = None*, *path_single_kg: str | None = None*, *path_for_deserialization: str | None = None*, *add_reciprocal: bool | None = None*, *eval_model: str | None = None*, *read_only_few: int | None = None*, *sample_triples_ratio: float | None = None*, *path_for_serialization: str | None = None*, *entity_to_idx: Dict | None = None*, *relation_to_idx: Dict | None = None*, *backend: str | None = None*, *training_technique: str | None = None*, *separator: str | None = None*)

Knowledge Graph container and processor.

Handles loading, preprocessing, and indexing of knowledge graph data from various sources including files, SPARQL endpoints, and serialized formats.

dataset_dir

Path to directory containing train/valid/test files.

num_entities

Total number of unique entities.

num_relations

Total number of unique relations.

train_set

Indexed training triples as numpy array.

valid_set

Indexed validation triples (optional).

test_set

Indexed test triples (optional).

entity_to_idx

Mapping from entity strings to indices.

relation_to_idx

Mapping from relation strings to indices.

```
dataset_dir = None

sparql_endpoint = None

path_single_kg = None

byte_pair_encoding = False

ordered_shaped_bpe_tokens = None

add_noise_rate = None

num_entities: int | None = None

num_relations: int | None = None

path_for_deserialization = None

add_reciprocal = None

eval_model = None

read_only_few = None

sample_triples_ratio = None

path_for_serialization = None

entity_to_idx = None
```

relation_to_idx = None

backend = 'pandas'

training_technique = None

separator = None

raw_train_set = None

raw_valid_set = None

raw_test_set = None

train_set = None

valid_set = None

test_set = None

idx_entity_to_bpe_shaped: Dict

enc

num_tokens

num_bpe_entities: int | None = None

padding = False

dummy_id

max_length_subword_tokens: int | None = None

train_set_target = None

target_dim: int | None = None

train_target_indices = None

ordered_bpe_entities = None

description_of_input = None

describe() → None

Generate a description string of the dataset statistics.

property entities_str: List[str]

Get list of all entity strings.

property relations_str: List[str]

Get list of all relation strings.

exists(*h: str*, *r: str*, *t: str*) → bool

Check if a triple exists in the training set.

Parameters

- **h** – Head entity string.
- **r** – Relation string.

- **t** – Tail entity string.

Returns
True if the triple exists, False otherwise.

__iter__() → Iterator[Tuple[str, str, str]]

Iterate over training triples as string tuples.

__len__() → int

Return number of triples in the raw training set.

func_triple_to_bpe_representation(*triple: List[str]*)

dicee.knowledge_graph_embeddings

Attributes

logger	

Classes

KGE	Knowledge Graph Embedding Class for interactive usage of pre-trained models

Module Contents

dicee.knowledge_graph_embeddings.**logger**

class dicee.knowledge_graph_embeddings.**KGE**(*path=None*, *url=None*, *construct_ensemble=False*, *model_name=None*)

Bases: *dicee.abstracts.BaseInteractiveKGE*, *dicee.abstracts.InteractiveQueryDecomposition*, *dicee.abstracts.BaseInteractiveTrainKGE*

Knowledge Graph Embedding Class for interactive usage of pre-trained models

__str__()

to(*device: str*) → None

get_transductive_entity_embeddings(*indices: torch.LongTensor | List[str]*, *as_pytorch=False*, *as_numpy=False*, *as_list=True*) → torch.FloatTensor | numpy.ndarray | List[float]

create_vector_database(*collection_name: str*, *distance: str*, *location: str = 'localhost'*, *port: int = 6333*)

generate(*h=''*, *r=''*)

eval_lp_performance(*dataset=List[Tuple[str, str, str]]*, *filtered=True*, ***, *tie_policy=None*, *tie_seed=None*)

Evaluate head/tail ranks, inheriting saved tie settings unless overridden.

`tie_policy` accepts sort (legacy), optimistic, random, or pessimistic. `tie_seed` seeds an independent random stream for each evaluation.

predict_missing_head_entity(*relation: List[str] | str*, *tail_entity: List[str] | str*, *within=None*, *batch_size=2*, *topk=1*, *return_indices=False*) → Tuple

Given a relation and a tail entity, return top k ranked head entity.

argmax_{e in E } f(e,r,t), where r in R, t in E.

Parameter

relation: Union[List[str], str]

String representation of selected relations.

tail_entity: Union[List[str], str]

String representation of selected entities.

k: int

Highest ranked k entities.

Returns: Tuple

Highest K scores and entities

predict_missing_relations(*head_entity: List[str] | str*, *tail_entity: List[str] | str*, *within=None*, *batch_size=2*, *topk=1*, *return_indices=False*) → Tuple

Given a head entity and a tail entity, return top k ranked relations.

argmax_{r in R } f(h,r,t), where h, t in E.

Parameter

head_entity: List[str]

String representation of selected entities.

tail_entity: List[str]

String representation of selected entities.

k: int

Highest ranked k entities.

Returns: Tuple

Highest K scores and entities

predict_missing_tail_entity(*head_entity: List[str] | str*, *relation: List[str] | str*, *within: List[str] | None = None*, *batch_size=2*, *topk=1*, *return_indices=False*) → torch.FloatTensor

Given a head entity and a relation, return top k ranked entities

argmax_{e in E } f(h,r,e), where h in E and r in R.

Parameter

head_entity: List[str]

String representation of selected entities.

tail_entity: List[str]

String representation of selected entities.

Returns: Tuple

scores

predict(*, *h: List[str] | str | None = None*, *r: List[str] | str | None = None*, *t: List[str] | str | None = None*, *within: List[str] | None = None*, *logits: bool = True*) → torch.FloatTensor

Predict scores for triples or missing triple elements.

Parameters

- **h** – Head entity/entities. None to predict heads.
- **r** – Relation/relations. None to predict relations.
- **t** – Tail entity/entities. None to predict tails.
- **within** – Optional list of entities to restrict predictions to.
- **logits** – If True, return raw scores. If False, return sigmoid scores (0-1).

Returns

- Single triple (h, r, t): scalar score
- Missing element: vector of all possible scores

Return type
torch.FloatTensor of scores. Shape depends on the query type

Raises

- **TypeError** – If inputs are not strings or lists of strings.
- **ValueError** – If a required argument for the query type is missing.

Examples

```
>>> # Score a specific triple
>>> model.predict(h="Mongolia", r="isLocatedIn", t="Asia", logits=False)
tensor(0.9523)
```

```
>>> # Get scores for all possible tail entities
>>> model.predict(h="Mongolia", r="isLocatedIn", t=None)
tensor([0.21, 0.95, 0.03, ...])  # One score per entity
```

predict_topk(*, *h: str | List[str] | None = None*, *r: str | List[str] | None = None*, *t: str | List[str] | None = None*, *topk: int = 10*, *within: List[str] | None = None*, *batch_size: int = 1024*) → List[Tuple[str, float]] | List[List[Tuple[str, float]]]

Predict top-k missing items in a given triple pattern.

Parameters

- **h** – Head entity/entities. None to predict heads.
- **r** – Relation/relations. None to predict relations.
- **t** – Tail entity/entities. None to predict tails.
- **topk** – Number of top predictions to return.
- **within** – Optional list of entities to restrict predictions to.

- **batch_size** – Batch size for processing multiple queries.

Returns
List[(item, score), …] of length topk. For batch query: List of such lists, one per query.

Return type
For single query

Raises

- **TypeError** – If h, r, or t is not a str or list of str.
- **ValueError** – If the required arguments for a query type are None.

Examples

```
>>> model.predict_topk(h=["Mongolia"], r=["isLocatedIn"], topk=3)
[('Asia', 0.99), ('Europe', 0.02), ...]
```

```
>>> model.predict_topk(r=["isLocatedIn"], t=["Asia"], topk=5)
[('Mongolia', 0.85), ('China', 0.82), ...]
```

triple_score(*h: List[str] | str | None = None*, *r: List[str] | str | None = None*, *t: List[str] | str | None = None*, *logits=False*) → torch.FloatTensor

Predict triple score

Parameter

head_entity: List[str]

String representation of selected entities.

relation: List[str]

String representation of selected relations.

tail_entity: List[str]

String representation of selected entities.

logits: bool

If logits is True, unnormalized score returned

Returns: Tuple

pytorch tensor of triple score

return_multi_hop_query_results(*aggregated_query_for_all_entities*, *k: int*, *only_scores*)

single_hop_query_answering(*query: tuple*, *only_scores: bool = True*, *k: int | None = None*, *use_logits: bool = True*)

answer_multi_hop_query(*query_type: str | None = None*, *query: Tuple[str | Tuple[str, str], ...] | None = None*, *queries: List[Tuple[str | Tuple[str, str], ...]] | None = None*, *tnorm: str = 'prod'*, *neg_norm: str = 'standard'*, *lambda_: float = 0.0*, *k: int = 10*, *only_scores: bool = False*, *use_logits: bool = True*)
→ List[Tuple[str, torch.Tensor]] | List[List[Tuple[str, torch.Tensor]]]

Answer multi-hop EPFO (Existential Positive First-Order) queries.

Supports 9 query types: 1p, 2p, 3p, 2i, 3i, ip, pi, 2u, up. See docs/guides/multi_hop_queries.md for detailed query patterns.

Parameters

- **query_type** – Query pattern name. One of: - 1p: (e, (r,)) # One-hop - 2p: (e, (r1, r2)) # Two-hop - 3p: (e, (r1, r2, r3)) # Three-hop - 2i: ((e1, (r1,)), (e2, (r2,))) # Two-way intersection - 3i: ((e1, (r1,)), (e2, (r2,)), (e3, (r3,))) # Three-way intersection - ip: (((e1, (r1,)), (e2, (r2,))), (r3,)) # Intersection + projection - pi: ((e, (r1, r2)), (r3,)) # Projection + intersection (2i meets 2p) - 2u: ((e1, (r1,)), (e2, (r2,))) # Two-way union - up: ((e, (r1, r2)), (e, (r3,))) # Union + projection
- **query** – Single query tuple matching the query_type pattern.
- **queries** – Batch of queries. If provided, query must be None.
- **tnorm** – T-norm for intersection/union. Options: “prod”, “min”.
- **neg_norm** – Negation norm. Options: “standard”, “sugeno”, “yager”.
- **lambda** – Parameter for sugeno and yager negation (0.0-1.0).
- **k** – Number of top answer entities to return.
- **only_scores** – If True, return only scores tensor. If False, return (entity, score) tuples.
- **use_logits** – If True, use raw model logits. If False, use sigmoid probabilities.

Returns

List[(entity, score), …] of top-k answers. For batch queries: List of such lists, one per query.

Return type

For single query

Raises

- **ValueError** – If query_type is not in {1p, 2p, 3p, 2i, 3i, ip, pi, 2u, up}.
- **AssertionError** – If query structure doesn’t match query_type pattern.

Examples

```
>>> # 1p: Find entities located in Asia
>>> model.answer_multi_hop_query(
...     query_type="1p",
...     query=("Asia", ("isLocatedIn",)),
...     k=5
... )
[("Mongolia", 0.92), ("China", 0.89), ...]
```

```
>>> # 2p: Two-hop query (e.g., "capital of countries in Europe")
>>> model.answer_multi_hop_query(
...     query_type="2p",
...     query=("Europe", ("isLocatedIn", "hasCapital")),
...     k=3
... )
[("Paris", 0.85), ("Berlin", 0.82), ...]
```

```
>>> # 2i: Intersection query
>>> model.answer_multi_hop_query(
...     query_type="2i",
...     query=(("Asia", ("isLocatedIn",)), ("Mountains", ("hasGeography",))),
...     k=5
... )
[("Nepal", 0.78), ("Tibet", 0.65), ...]
```

See also

- docs/guides/multi_hop_queries.md: Complete guide with all query patterns
- tests/test_answer_multi_hop_query.py: Usage examples

find_missing_triples(*confidence: float*, *entities: List[str] | None = None*, *relations: List[str] | None = None*, *topk: int = 10*, *at_most: int = sys.maxsize*) → Set

Find missing triples

Iterative over a set of entities E and a set of relation R :

orall e in E and orall r in R f(e,r,x)

Return (e,r,x)

otin G and f(e,r,x) > confidence

confidence: float

A threshold for an output of a sigmoid function given a triple.

topk: int

Highest ranked k item to select triples with f(e,r,x) > confidence .

at_most: int

Stop after finding at_most missing triples

{(e,r,x) | f(e,r,x) > confidence land (e,r,x)

otin G

predict_literals(*entity: List[str] | str | None = None*, *attribute: List[str] | str | None = None*, *denormalize_preds: bool = True*) → numpy.ndarray

Predicts literal values for given entities and attributes.

Parameters

- **entity** (*Union[List[str], str]*) – Entity or list of entities to predict literals for.
- **attribute** (*Union[List[str], str]*) – Attribute or list of attributes to predict literals for.
- **denormalize_preds** (*bool*) – If True, denormalizes the predictions.

Returns

Predictions for the given entities and attributes.

Return type

numpy ndarray

dicee.losses

Submodules

dicee.losses.custom_losses

Alternative loss functions selectable via the `loss_fn` config option.

`BaseKGE.__init__` (`dicee/models/base_model.py`) instantiates one of these in place of the framework's default loss based on `args["loss_fn"]`; see that dispatch for the exact string values and which additional config options (`label_smoothing_rate`, `label_relaxation_alpha`) feed each one's constructor.

Several of these declare a `current_epoch` parameter on `forward` so they can vary their behavior over training; `BaseKGE.loss_function` detects this once per model (`self._loss_needs_epoch`, set by inspecting `self.loss.forward`'s parameters) and forwards the current epoch only to losses that declare it.

Classes

DefaultBCELoss	Plain `BCEWithLogitsLoss`, selectable via `loss_fn="BCELoss"`.
WeightedBCELoss	BCE weighted by each prediction's own confidence.
LabelSmoothingLoss	Fixed-rate label-smoothing loss, selectable via `loss_fn="LS"`
AdaptiveLabelSmoothingLoss	KL-divergence label smoothing with a smoothing factor that self-adjusts
LabelRelaxationLoss	Label relaxation loss, selectable via `loss_fn="LRLoss"`.
AdaptiveLabelRelaxationLoss	*LabelRelaxationLoss* with an `alpha` that self-adjusts from
ConfidenceBasedAdaptiveLabelRelaxationLoss	*LabelRelaxationLoss* where `alpha` shrinks as the model's
CombinedLSandLR	Switches from *LabelSmoothingLoss* to *LabelRelaxationLoss*
CombinedAdaptiveLSandAdaptiveLR	Switches from *AdaptiveLabelSmoothingLoss* to
AggregatedLSandLR	Fixed 0.4/0.6 weighted average of *LabelSmoothingLoss* and
ACLS	Adaptive and Conditional Label Smoothing-style calibration loss,

Module Contents

class dicee.losses.custom_losses.**DefaultBCELoss**

Bases: `torch.nn.Module`

Plain `BCEWithLogitsLoss`, selectable via `loss_fn="BCELoss"`.

Equivalent to the framework's own default entity/relation-prediction loss (see `BaseKGE.__init__`); provided as an explicit `loss_fn` option for parity with the other choices in this module.

forward(*pred*, *target*)

class dicee.losses.custom_losses.**WeightedBCELoss**

Bases: `torch.nn.Module`

BCE weighted by each prediction's own confidence.

Down-weights low-confidence predictions (`sigmoid(pred)` near 0.5) and up-weights confident ones, clamped to `[0.5, 1.0]`. Weights are detached, so they don't receive gradients themselves. `current_epoch` is accepted for interface consistency with the other `loss_fn` options but is currently unused.

forward(*pred*, *target*, *current_epoch*)

class dicee.losses.custom_losses.**LabelSmoothingLoss**(*smoothness_ratio=0.0*)

Bases: `torch.nn.Module`

Fixed-rate label-smoothing loss, selectable via `loss_fn="LS"` (`smoothness_ratio` comes from the `label_smoothing_rate` config option).

Converts `pred` (logits) to log-probabilities and compares against a smoothed target distribution (mass `smoothness_ratio` redistributed uniformly over the non-target classes) via `KL(pred || smoothed_target)` — the same fixed-rate variant of *AdaptiveLabelSmoothingLoss*'s per-call smoothing, without the batch-to-batch adaptation.

This is independent of the (separately, correctly implemented) dataset-level smoothing controlled by the same `label_smoothing_rate` option (`dicee/dataset_classes/_label_based.py`, `_negative_sampling.py`), which bakes smoothing into the training targets regardless of `loss_fn`.

Parameters

smoothness_ratio (*float*) – Fraction of probability mass redistributed from the target class(es) to the non-target classes; `0` recovers a KL-divergence loss with no smoothing.

smoothness_ratio = 0.0

forward(*pred*, *target*)

class dicee.losses.custom_losses.**AdaptiveLabelSmoothingLoss**(*min_smoothing_factor=0.01*, *max_smoothing_factor=0.2*, *smoothing_factor_step=0.01*, *initial_smoothing_factor=0.1*)

Bases: `torch.nn.Module`

KL-divergence label smoothing with a smoothing factor that self-adjusts from batch to batch, selectable via `loss_fn="AdaptiveLabelSmoothingLoss"`.

Converts `logits` to log-probabilities and compares against a smoothed target distribution (mass `smoothing_factor` redistributed uniformly over the non-target classes) via `KL(pred || smoothed_target)`. After each call, `smoothing_factor` moves one `smoothing_factor_step` toward `max_smoothing_factor` if the loss increased since the previous call, or toward `min_smoothing_factor` if it decreased — i.e. it backs off smoothing while the model is improving and leans into it when training stalls. `current_epoch` is accepted for interface consistency with the other `loss_fn` options but is currently unused.

This module is stateful (`prev_loss`, `smoothing_factor` persist across calls), so a fresh instance should be used per training run.

Parameters

- **min_smoothing_factor** (*float*) – Lower bound the adaptive smoothing factor can decay to.
- **max_smoothing_factor** (*float*) – Upper bound the adaptive smoothing factor can grow to.
- **smoothing_factor_step** (*float*) – Adjustment applied to the smoothing factor after each call.
- **initial_smoothing_factor** (*float*) – Starting value before any adjustment has occurred.

min_smoothing_factor = 0.01

max_smoothing_factor = 0.2

smoothing_factor_step = 0.01

smoothing_factor = 0.1

prev_loss = None

eps = 1e-14

forward(*logits*, *target*, *current_epoch*)

class dicee.losses.custom_losses.**LabelRelaxationLoss**(*alpha=0.0*)

Bases: `torch.nn.Module`

Label relaxation loss, selectable via `loss_fn="LRLoss"`.

Based on "From Label Smoothing to Label Relaxation" (Lienen & Hullermeier, AAAI 2021): instead of smoothing toward one fixed soft target, it constructs a *credal set* of distributions that assign at least `1 - alpha` probability to the target class(es) and measures the KL divergence to the closest point in that set, rather than to a single fixed distribution. Predictions that are already confident enough (probability mass on the target above `1 - alpha`) incur zero loss.

Parameters

alpha (*float*) – Relaxation strength in `[0, 1)`; `0` recovers ordinary cross-entropy-style behavior (no slack), larger values tolerate more probability mass elsewhere before penalizing.

alpha = 0.0

gz_threshold = 0.1

eps = 1e-14

forward(*pred*, *target*)

class dicee.losses.custom_losses.**AdaptiveLabelRelaxationLoss**(*min_alpha=0.01*, *max_alpha=0.2*, *alpha_step=0.01*, *initial_alpha=0.1*)

Bases: `torch.nn.Module`

LabelRelaxationLoss with an `alpha` that self-adjusts from batch to batch, selectable via `loss_fn="AdaptiveLabelRelaxationLoss"`.

Computes the label-relaxation loss once under `torch.no_grad()` with the current `alpha` to decide the adjustment (grow toward `max_alpha` if the loss increased since the previous call, shrink toward `min_alpha` if it decreased), then recomputes the loss with the updated `alpha` for the actual gradient-carrying return value. Like *AdaptiveLabelSmoothingLoss*, this module is stateful (`prev_loss`, `alpha` persist across calls) - use a fresh instance per training run.

Parameters

- **min_alpha** (*float*) – Lower bound the adaptive `alpha` can decay to.
- **max_alpha** (*float*) – Upper bound the adaptive `alpha` can grow to.
- **alpha_step** (*float*) – Adjustment applied to `alpha` after each call.
- **initial_alpha** (*float*) – Starting value before any adjustment has occurred.

min_alpha = 0.01

max_alpha = 0.2

alpha_step = 0.01

alpha = 0.1

prev_loss = None

eps = 1e-14

gz_threshold = 0.1

forward(*pred*, *target*)

class dicee.losses.custom_losses.**ConfidenceBasedAdaptiveLabelRelaxationLoss**(*alpha=0.1*)

Bases: `torch.nn.Module`

LabelRelaxationLoss where `alpha` shrinks as the model's own average prediction confidence grows, selectable via `loss_fn="ConfidenceBasedAdaptiveLabelRelaxationLoss"`.

Each call rescales `alpha` by `(1 - mean(pred))` before computing the loss. Unlike *AdaptiveLabelRelaxationLoss*, this adjustment is one-directional and unbounded below: `alpha` only ever decays toward 0 as confidence rises, with no mechanism to grow back if confidence later drops. `current_epoch` is accepted for interface consistency with the other `loss_fn` options but is currently unused. Stateful (`alpha` persists and mutates across calls) - use a fresh instance per training run.

Parameters

alpha (*float*) – Initial relaxation strength before any confidence-based decay.

alpha = 0.1

gz_threshold = 0.1

eps = 1e-14

forward(*pred*, *target*, *current_epoch*)

class dicee.losses.custom_losses.**CombinedLSandLR**(*smoothness_ratio=0.0*, *alpha=0.0*)

Bases: `torch.nn.Module`

Switches from *LabelSmoothingLoss* to *LabelRelaxationLoss* after epoch 20, selectable via `loss_fn="CombinedLSandLR"`.

Intended to smooth early training then relax the target distribution later on.

Parameters

- **smoothness_ratio** (*float*) – Forwarded to *LabelSmoothingLoss* for the first 20 epochs.
- **alpha** (*float*) – Forwarded to *LabelRelaxationLoss* from epoch 20 onward.

smoothness_ratio = 0.0

alpha = 0.0

forward(*pred*, *target*, *current_epoch*)

class dicee.losses.custom_losses.**CombinedAdaptiveLSandAdaptiveLR**

Bases: `torch.nn.Module`

Switches from *AdaptiveLabelSmoothingLoss* to *AdaptiveLabelRelaxationLoss* after epoch 100, selectable via `loss_fn="CombinedAdaptiveLSandAdaptiveLR"`.

Both inner losses use their default hyperparameters; there is currently no way to configure them through this wrapper.

adaptive_label_smoothing

adaptive_label_relaxation

criterion = ''

forward(*pred*, *target*, *current_epoch*)

class dicee.losses.custom_losses.**AggregatedLSandLR**(*smoothness_ratio=0.1*, *alpha=0.1*)

Bases: `torch.nn.Module`

Fixed 0.4/0.6 weighted average of *LabelSmoothingLoss* and *LabelRelaxationLoss*, selectable via `loss_fn="AggregatedLSandLR"`.

Unlike *CombinedLSandLR*, both terms are computed every call and blended (`0.4 * smoothing + 0.6 * relaxation`) rather than switched between by epoch.

Parameters

- **smoothness_ratio** (*float*) – Forwarded to the inner *LabelSmoothingLoss*.
- **alpha** (*float*) – Forwarded to the inner *LabelRelaxationLoss*.

smoothness_ratio = 0.1

alpha = 0.1

forward(*pred*, *target*, *current_epoch*)

class dicee.losses.custom_losses.**ACLS**(*pos_lambda: float = 1.0*, *neg_lambda: float = 0.1*, *alpha: float = 0.1*, *margin: float = 10.0*, *num_classes: int = 200*, *ignore_index: int = -100*)

Bases: `torch.nn.Module`

Adaptive and Conditional Label Smoothing-style calibration loss, selectable via `loss_fn="ACLS"`.

Adds a margin-based confidence regularizer to cross-entropy: for the predicted (argmax) class, logits above `margin` are penalized quadratically (discourages over-confidence on the top class), and for every other class, logits within `margin` of the top logit are also penalized quadratically (discourages under-separation from the runner-up classes). The two penalties are weighted by `pos_lambda`/`neg_lambda` and added to `CrossEntropyLoss` scaled by `alpha`.

Warning

`nn.CrossEntropyLoss` expects `targets` as class indices (or a proper probability distribution per row); KGE's `KvsAll` labels are multi-hot (possibly several true tails per row), which is not an equivalent input. `num_classes` and `ignore_index` are currently unused in `forward`/`get_reg`.

Parameters

- **pos_lambda** (*float*) – Weight of the over-confidence penalty on the predicted class.

- **neg_lambda** (*float*) – Weight of the under-separation penalty on the other classes.
- **alpha** (*float*) – Weight of the regularization term relative to cross-entropy.
- **margin** (*float*) – Logit margin used by both penalty terms.
- **num_classes** (*int*) – Currently unused.
- **ignore_index** (*int*) – Currently unused.

pos_lambda = 1.0

neg_lambda = 0.1

alpha = 0.1

margin = 10.0

num_classes = 200

ignore_index = -100

cross_entropy

property names

get_reg(*inputs*, *targets*)

forward(*inputs*, *targets*)

dicee.models

Submodules

dicee.models.adopt

ADOPT Optimizer Implementation.

This module implements the ADOPT (Adaptive Optimization with Precise Tracking) algorithm, an advanced optimization method for training neural networks.

ADOPT Overview:

ADOPT is an adaptive learning rate optimization algorithm that combines the benefits of momentum-based methods with per-parameter learning rate adaptation. Unlike Adam, which applies momentum to raw gradients, ADOPT normalizes gradients first and then applies momentum, leading to more stable training dynamics.

Key Features: - Gradient normalization before momentum application - Adaptive per-parameter learning rates - Optional gradient clipping that grows with training steps - Support for decoupled weight decay (AdamW-style) - Multiple execution modes: single-tensor, multi-tensor (foreach), and fused (planned)

Algorithm Comparison:

Adam: $m = \beta_1 * m + (1-\beta_1) * g$, $\theta = \theta - \alpha * m/\sqrt{v}$ ADOPT: $m = \beta_1 * m + (1-\beta_1) * g/\sqrt{v}$, $\theta = \theta - \alpha * m$

The key difference is that ADOPT normalizes gradients before momentum, which provides better stability and can lead to improved convergence.

Classes:

- ADOPT: Main optimizer class (extends torch.optim.Optimizer)

Functions:

- adopt: Functional API for ADOPT algorithm computation
- _single_tensor_adopt: Single-tensor implementation (TorchScript compatible)
- _multi_tensor_adopt: Multi-tensor implementation using foreach operations

Performance:

- Single-tensor: Default, compatible with torch.jit.script
- Multi-tensor (foreach): 2-3x faster on GPU through vectorization
- Fused (planned): Would provide maximum performance via specialized kernels

Example:

```
>>> import torch
>>> from dicee.models.adopt import ADOPT
>>>
>>> model = torch.nn.Linear(10, 1)
>>> optimizer = ADOPT(model.parameters(), lr=0.001, weight_decay=0.01, decouple=True)
>>>
>>> # Training loop
>>> for epoch in range(num_epochs):
...     optimizer.zero_grad()
...     output = model(input)
...     loss = criterion(output, target)
...     loss.backward()
...     optimizer.step()
```

References:

Original implementation: https://github.com/iShohei220/adopt

Notes:

This implementation is based on the original ADOPT implementation and adapted to work with the PyTorch optimizer interface and the dicee framework.

Classes

ADOPT	ADOPT Optimizer.

Functions

adopt(params, grads, exp_avgs, exp_avg_sqs, state_steps)	Functional API that performs ADOPT algorithm computation.

Module Contents

class dicee.models.adopt.**ADOPT**(*params: torch.optim.optimizer.ParamsT*, *lr: float | torch.Tensor = 0.001*, *betas: Tuple[float, float] = (0.9, 0.9999)*, *eps: float = 1e-06*, *clip_lambda: Callable[[int], float] | None = lambda step: ...*, *weight_decay: float = 0.0*, *decouple: bool = False*, ***, *foreach: bool | None = None*, *maximize: bool = False*, *capturable: bool = False*, *differentiable: bool = False*, *fused: bool | None = None*)

Bases: `torch.optim.optimizer.Optimizer`

ADOPT Optimizer.

ADOPT is an adaptive learning rate optimization algorithm that combines momentum-based updates with adaptive per-parameter learning rates. It uses exponential moving averages of gradients and squared gradients, with gradient clipping for stability.

The algorithm performs the following key operations: 1. Normalizes gradients by the square root of the second moment estimate 2. Applies optional gradient clipping based on the training step 3. Updates parameters using momentum-smoothed normalized gradients 4. Supports decoupled weight decay (AdamW-style) or L2 regularization

Mathematical formulation:

$m_t = \beta_1 * m_\{t-1\} + (1 - \beta_1) * clip(g_t / \sqrt{(v_t)})$ $v_t = \beta_2 * v_\{t-1\} + (1 - \beta_2) * g_t^2$ $\theta_t = \theta_\{t-1\} - \alpha * m_t$

where:

- θ_t: parameter at step t
- g_t: gradient at step t
- m_t: first moment estimate (momentum)
- v_t: second moment estimate (variance)
- α: learning rate
- β_1, β_2: exponential decay rates
- clip(): optional gradient clipping function

Reference:

Original implementation: https://github.com/iShohei220/adopt

Parameters

- **params** (*ParamsT*) – Iterable of parameters to optimize or dicts defining parameter groups.
- **lr** (*float or Tensor, optional*) – Learning rate. Can be a float or 1-element Tensor. Default: 1e-3
- **betas** (*Tuple[float, float], optional*) – Coefficients (β_1, β_2) for computing running averages of gradient and its square. β_1 controls momentum, β_2 controls variance. Default: (0.9, 0.9999)

- **eps** (*float, optional*) – Term added to denominator to improve numerical stability. Default: 1e-6
- **clip_lambda** (*Callable[[int], float], optional*) – Function that takes the step number and returns the gradient clipping threshold. Common choices: - lambda step: step**0.25 (default, gradually increases clipping threshold) - lambda step: 1.0 (constant clipping) - None (no clipping) Default: lambda step: step**0.25
- **weight_decay** (*float, optional*) – Weight decay coefficient (L2 penalty). Default: 0.0
- **decouple** (*bool, optional*) – If True, uses decoupled weight decay (AdamW-style), applying weight decay directly to parameters. If False, adds weight decay to gradients (L2 regularization). Default: False
- **foreach** (*bool, optional*) – If True, uses the faster foreach implementation for multi-tensor operations. Default: None (auto-select)
- **maximize** (*bool, optional*) – If True, maximizes parameters instead of minimizing. Useful for reinforcement learning. Default: False
- **capturable** (*bool, optional*) – If True, the optimizer is safe to capture in a CUDA graph. Requires learning rate as Tensor. Default: False
- **differentiable** (*bool, optional*) – If True, the optimization step can be differentiated. Useful for meta-learning. Default: False
- **fused** (*bool, optional*) – If True, uses fused kernel implementation (currently not supported). Default: None

Raises

- **ValueError** – If learning rate, epsilon, betas, or weight_decay are invalid.
- **RuntimeError** – If fused is enabled (not currently supported).
- **RuntimeError** – If lr is a Tensor with foreach=True and capturable=False.

Example

```
>>> # Basic usage
>>> optimizer = ADOPT(model.parameters(), lr=0.001)
>>> optimizer.zero_grad()
>>> loss.backward()
>>> optimizer.step()
```

```
>>> # With decoupled weight decay
>>> optimizer = ADOPT(model.parameters(), lr=0.001, weight_decay=0.01,␣
↪decouple=True)
```

```
>>> # Custom gradient clipping
>>> optimizer = ADOPT(model.parameters(), clip_lambda=lambda step: max(1.0,␣
↪step**0.5))
```

Note

- For most use cases, the default hyperparameters work well
- Consider using decouple=True for better generalization (similar to AdamW)

- The clip_lambda function helps stabilize training in early steps

clip_lambda

__setstate__(*state*)

Restore optimizer state from a checkpoint.

This method handles backward compatibility when loading optimizer state from older versions. It ensures all required fields are present with default values and properly converts step counters to tensors if needed.

Key responsibilities: 1. Set default values for newly added hyperparameters 2. Convert old-style scalar step counters to tensor format 3. Place step tensors on appropriate devices based on capturable/fused modes

Parameters

state (*dict*) – Optimizer state dictionary (typically from torch.load()).

Note

- This enables loading checkpoints saved with older ADOPT versions
- Step counters are converted to appropriate device/dtype for compatibility
- Capturable and fused modes require step tensors on parameter devices

step(*closure=None*)

Perform a single optimization step.

This method executes one iteration of the ADOPT optimization algorithm across all parameter groups. It orchestrates the following workflow:

1. Optionally evaluates a closure to recompute the loss (useful for algorithms like LBFGS or when loss needs multiple evaluations)
2. For each parameter group: - Collects parameters with gradients and their associated state - Extracts hyperparameters (betas, learning rate, etc.) - Calls the functional adopt() API to perform the actual update
3. Returns the loss value if a closure was provided

The functional API (adopt()) handles three execution modes: - Single-tensor: Updates one parameter at a time (default, JIT-compatible) - Multi-tensor (foreach): Batches operations for better performance - Fused: Uses fused CUDA kernels (not yet implemented)

Gradient scaling support: This method is compatible with automatic mixed precision (AMP) training. It can access grad_scale and found_inf attributes for gradient unscaling and inf/nan detection when used with GradScaler.

Parameters

closure (*Callable, optional*) – A callable that reevaluates the model and returns the loss. The closure should: - Enable gradients (torch.enable_grad()) - Compute forward pass - Compute loss - Compute backward pass - Return the loss value Example: lambda: (loss := model(x), loss.backward(), loss)[-1] Default: None

Returns

The loss value returned by the closure, or None if no
closure was provided.

Return type
Optional[Tensor]

Example

```
>>> # Standard usage
>>> loss = criterion(model(input), target)
>>> loss.backward()
>>> optimizer.step()
```

```
>>> # With closure (e.g., for line search)
>>> def closure():
...     optimizer.zero_grad()
...     output = model(input)
...     loss = criterion(output, target)
...     loss.backward()
...     return loss
>>> loss = optimizer.step(closure)
```

Note

- Call zero_grad() before computing gradients for the next step
- CUDA graph capture is checked for safety when capturable=True
- The method is thread-safe for different parameter groups

dicee.models.adopt.**adopt**(*params: List[torch.Tensor]*, *grads: List[torch.Tensor]*, *exp_avgs: List[torch.Tensor]*, *exp_avg_sqs: List[torch.Tensor]*, *state_steps: List[torch.Tensor]*, *foreach: bool | None = None*, *capturable: bool = False*, *differentiable: bool = False*, *fused: bool | None = None*, *grad_scale: torch.Tensor | None = None*, *found_inf: torch.Tensor | None = None*, *has_complex: bool = False*, ***, *beta1: float*, *beta2: float*, *lr: float | torch.Tensor*, *clip_lambda: Callable[[int], float] | None*, *weight_decay: float*, *decouple: bool*, *eps: float*, *maximize: bool*)

Functional API that performs ADOPT algorithm computation.

This is the main functional interface for the ADOPT optimization algorithm. It dispatches to one of three implementations based on the execution mode:

1. **Single-tensor mode** (default): Updates parameters one at a time - Compatible with torch.jit.script - More flexible but slower - Used when foreach=False or automatically for small models
2. **Multi-tensor (foreach) mode**: Batches operations across tensors - 2-3x faster on GPU through vectorization - Groups tensors by device/dtype automatically - Used when foreach=True
3. **Fused mode**: Uses specialized fused kernels (not yet implemented) - Would provide maximum performance - Currently raises RuntimeError if enabled

Algorithm overview (ADOPT):

ADOPT adapts learning rates per-parameter while using momentum on normalized gradients. The key innovation is normalizing gradients before momentum, which provides more stable training than standard Adam.

Mathematical formulation:

Normalize gradient by its historical variance normed_g_t = g_t / √(v_t + ε)

Optional gradient clipping for stability normed_g_t = clip(normed_g_t, threshold(t))

Momentum on normalized gradients (key difference from Adam) m_t = β_1 * m_{t-1} + (1 - β_1) * normed_g_t

Parameter update θ_t = θ_{t-1} - α * m_t

Update variance estimate v_t = β_2 * v_{t-1} + (1 - β_2) * g_t^2

where:

- θ: parameters
- g: gradients
- m: first moment (momentum of normalized gradients)
- v: second moment (variance of raw gradients)
- α: learning rate
- β_1, β_2: exponential decay rates
- ε: numerical stability constant
- clip(): gradient clipping function based on step

Automatic mode selection:

When foreach and fused are both None (default), the function automatically selects the best implementation based on: - Parameter types and devices - Whether differentiable mode is enabled - Learning rate type (float vs Tensor) - Capturable mode requirements

param params
Parameters to optimize.

type params
List[Tensor]

param grads
Gradients for each parameter.

type grads
List[Tensor]

param exp_avgs
First moment estimates (momentum).

type exp_avgs
List[Tensor]

param exp_avg_sqs
Second moment estimates (variance).

type exp_avg_sqs
List[Tensor]

param state_steps
Step counters (must be singleton tensors).

type state_steps
List[Tensor]

param foreach
Whether to use multi-tensor implementation. None: auto-select based on configuration (default).

type foreach
Optional[bool]

param capturable
If True, ensure CUDA graph capture safety.

type capturable
bool

param differentiable
If True, allow gradients through optimization step.

type differentiable
bool

param fused
If True, use fused kernels (not implemented).

type fused
Optional[bool]

param grad_scale
Gradient scaler for AMP training.

type grad_scale
Optional[Tensor]

param found_inf
Flag for inf/nan detection in AMP.

type found_inf
Optional[Tensor]

param has_complex
Whether any parameters are complex-valued.

type has_complex
bool

param beta1
Exponential decay rate for first moment (momentum). Typical range: 0.9-0.95.

type beta1
float

param beta2
Exponential decay rate for second moment (variance). Typical range: 0.999-0.9999 (higher than Adam).

type beta2
float

param lr
Learning rate. Can be a scalar Tensor for dynamic learning rate with capturable=True.

type lr
Union[float, Tensor]

param clip_lambda
Function that maps step number to gradient clipping threshold. None disables clipping.

type clip_lambda
Optional[Callable[[int], float]]

param weight_decay
Weight decay coefficient (L2 penalty).

type weight_decay
float

param decouple
If True, use decoupled weight decay (AdamW-style). Recommended for better generalization.

type decouple
bool

param eps
Small constant for numerical stability in normalization.

type eps
float

param maximize
If True, maximize objective instead of minimize.

type maximize
bool

raises RuntimeError
If torch.jit.script is used with foreach or fused.

raises RuntimeError
If state_steps contains non-tensor elements.

raises RuntimeError
If fused=True (not yet implemented).

raises RuntimeError
If lr is Tensor with foreach=True and capturable=False.

Example

```
>>> # Typically called by ADOPT optimizer, not directly
>>> adopt(
...     params=[p1, p2],
...     grads=[g1, g2],
...     exp_avgs=[m1, m2],
...     exp_avg_sqs=[v1, v2],
...     state_steps=[step1, step2],
...     beta1=0.9,
...     beta2=0.9999,
...     lr=0.001,
...     clip_lambda=lambda s: s**0.25,
...     weight_decay=0.01,
...     decouple=True,
...     eps=1e-6,
...     maximize=False,
... )
```

Note

- For distributed training, this API is compatible with torch/distributed/optim
- The foreach mode is generally preferred for GPU training
- Complex parameters are handled transparently by viewing as real
- First optimization step only initializes variance, doesn't update parameters

See also

- ADOPT class: High-level optimizer interface
- _single_tensor_adopt: Single-tensor implementation details
- _multi_tensor_adopt: Multi-tensor implementation details

dicee.models.base_model

Attributes

logger

Classes

BaseKGELightning	Thin PyTorch Lightning wrapper shared by all KGE models.
BaseKGE	Base class for all Knowledge Graph Embedding models.
IdentityClass	No-op normalisation / dropout placeholder.

Module Contents

dicee.models.base_model.**logger**

class dicee.models.base_model.**BaseKGELightning**(**args*, ***kwargs*)

Bases: lightning.LightningModule

Thin PyTorch Lightning wrapper shared by all KGE models.

Provides the standard Lightning training loop hooks (training_step, on_train_epoch_end, configure_optimizers) as well as a helper for reporting model size. All concrete KGE models should extend *BaseKGE* rather than this class directly.

args: Dict[str, Any]

loss: torch.nn.Module

training_step_outputs = []

mem_of_model() → Dict

Size of model in MB and number of params

training_step(*batch*, *batch_idx=None*)

Execute one optimisation step for the given mini-batch.

Handles two- and three-element batches produced by the different dataset classes (`KvsAll` / `NegSample` vs. `KvsSample`).

Parameters

- **batch** (*tuple*) – `(x, y)` for standard scoring, or `(x, y_select, y)` for sample-based labelling.
- **batch_idx** (*int, optional*) – Index of the current batch (unused, kept for Lightning API compat).

Returns

Scalar loss value for this batch.

Return type

torch.FloatTensor

loss_function(*yhat_batch: torch.Tensor*, *y_batch: torch.Tensor*, *current_epoch: int | None = None*) → torch.Tensor

Compute the loss between model predictions and targets.

Delegates to `self.loss` which is configured in `BaseKGE.__init__` based on the scoring technique (`BCEWithLogitsLoss` for entity/relation prediction, `CrossEntropyLoss` for classification, `MarginRankingLoss` for `scoring_technique='NegSampleMargin'`) or on `loss_fn` (one of the classes in *dicee.losses.custom_losses*).

For `NegSampleMargin`, *yhat_batch*/*y_batch* still arrive in the flat `NegSample` layout (positives first, followed by `neg_ratio` tiled blocks of negatives, row-aligned by triple). This method splits that flat batch into positive/negative score pairs before delegating to `MarginRankingLoss`.

When `adversarial_temperature` is set (grouped/strict negative sampling), *yhat_batch*/*y_batch* instead arrive as `(batch, 1 + neg_ratio)` groups with the positive in column 0, produced by `GroupedNegativeSamplingDataset`. This delegates to *grouped_adversarial_bce()* instead of `self.loss`.

Some `loss_fn` classes (e.g. `CombinedLSandLR`, `AdaptiveLabelSmoothingLoss`) declare a `current_epoch` parameter on their own `forward` to schedule their behavior over training. `self._loss_needs_epoch` (computed once in `BaseKGE.__init__` by inspecting `self.loss.forward`) says whether that argument should be forwarded to `self.loss` here.

Parameters

- **yhat_batch** (*torch.FloatTensor*) – Model output scores, shape `(batch_size, *)`.
- **y_batch** (*torch.FloatTensor*) – Ground-truth labels of the same shape as *yhat_batch*.
- **current_epoch** (*int, optional*) – Current training epoch, forwarded to `self.loss` only when it declares a `current_epoch` parameter. Callers that never use an epoch-aware `loss_fn` may omit this.

Returns

Scalar loss value.

Return type

torch.FloatTensor

on_train_epoch_end(**args*, ***kwargs*)

Called in the training loop at the very end of the epoch.

To access all batch outputs at the end of the epoch, you can cache step outputs as an attribute of the `LightningModule` and access them in this hook:

```
class MyLightningModule(L.LightningModule):
    def __init__(self):
        super().__init__()
        self.training_step_outputs = []

    def training_step(self):
        loss = ...
        self.training_step_outputs.append(loss)
        return loss

    def on_train_epoch_end(self):
        # do something with all training_step outputs, for example:
        epoch_mean = torch.stack(self.training_step_outputs).mean()
        self.log("training_epoch_mean", epoch_mean)
        # free up the memory
        self.training_step_outputs.clear()
```

test_epoch_end(*outputs: List[Any]*)

test_dataloader() → None

An iterable or collection of iterables specifying test samples.

For more information about multiple dataloaders, see this section.

For data processing use the following pattern:

- download in `prepare_data()`
- process and split in `setup()`

However, the above are only necessary for distributed processing.

Warning

do not assign state in prepare_data

- `test()`
- `prepare_data()`
- `setup()`

Note

Lightning tries to add the correct sampler for distributed and arbitrary hardware. There is no need to set it yourself.

Note

If you don't need a test dataset and a `test_step()`, you don't need to implement this method.

val_dataloader() → None

An iterable or collection of iterables specifying validation samples.

For more information about multiple dataloaders, see this section.

The dataloader you return will not be reloaded unless you set **:paramref:`~lightning.pytorch.trainer.trainer.Trainer.reload_dataloaders_every_n_epochs`** to a positive integer.

It's recommended that all data downloads and preparation happen in `prepare_data()`.

- `fit()`
- `validate()`
- `prepare_data()`
- `setup()`

> **Note**
>
> Lightning tries to add the correct sampler for distributed and arbitrary hardware There is no need to set it yourself.

> **Note**
>
> If you don't need a validation dataset and a `validation_step()`, you don't need to implement this method.

predict_dataloader() → None

An iterable or collection of iterables specifying prediction samples.

For more information about multiple dataloaders, see this section.

It's recommended that all data downloads and preparation happen in `prepare_data()`.

- `predict()`
- `prepare_data()`
- `setup()`

> **Note**
>
> Lightning tries to add the correct sampler for distributed and arbitrary hardware There is no need to set it yourself.

Returns

A `torch.utils.data.DataLoader` or a sequence of them specifying prediction samples.

train_dataloader() → None

An iterable or collection of iterables specifying training samples.

For more information about multiple dataloaders, see this section.

The dataloader you return will not be reloaded unless you set **:paramref:`~lightning.pytorch.trainer.trainer.Trainer.reload_dataloaders_every_n_epochs`** to a positive integer.

For data processing use the following pattern:

- download in `prepare_data()`
- process and split in `setup()`

However, the above are only necessary for distributed processing.

> **Warning**
>
> do not assign state in prepare_data

- `fit()`
- `prepare_data()`
- `setup()`

> **Note**
>
> Lightning tries to add the correct sampler for distributed and arbitrary hardware. There is no need to set it yourself.

configure_optimizers(*parameters=None*)

Instantiate and return the optimiser for training.

The optimiser type is taken from `self.optimizer_name` which is set in *BaseKGE.init_params_with_sanity_checking()* from the `--optim` argument. Supported values: `'SGD'`, `'Adam'`, `'Adopt'`, `'AdamW'`, `'NAdam'`, `'Adagrad'`, `'ASGD'`, `'Muon'`.

Parameters

parameters (*iterable, optional*) – Model parameters to optimise. Defaults to `self.parameters()` when `None`.

Returns

The configured optimiser instance.

Return type

torch.optim.Optimizer

class dicee.models.base_model.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns

Shape `(batch_size, num_bpe_entities)` score matrix.

Return type

torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where k is the number of sampled target entities.

Return type
torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters
idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns
head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters
indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns
head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where T is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type
torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of T subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns

head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type

torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

class dicee.models.base_model.**IdentityClass**(*args=None*)

Bases: `torch.nn.Module`

No-op normalisation / dropout placeholder.

Used whenever no normalisation layer is requested (`--normalization None`). All inputs are returned unchanged so that the rest of the model code does not need conditional checks around normalisation calls.

args = None

__call__(*x*)

static forward(*x*)

dicee.models.clifford

Attributes

logger

Classes

Keci	Keci: Knowledge Graph Embedding via Clifford Algebra.
KeciTransformer	Keci with Transformer architecture.
TransformerBlock	A single transformer block with self-attention and MLP.
TransformerSelfAttention	Multi-head self-attention for the Keci Transformer.
TransformerMLP	MLP for the Keci Transformer.
CKeci	Without learning dimension scaling
DeCaL	Base class for all Knowledge Graph Embedding models.

Module Contents

dicee.models.clifford.**logger**

class dicee.models.clifford.**Keci**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Keci: Knowledge Graph Embedding via Clifford Algebra.

Embeds entities and relations as multi-vectors in the Clifford algebra Cl_{p,q}(R^d) and scores triples via the Clifford product. The algebra is parameterised by two non-negative integers p and q:

- `p = 0, q = 0` — reduces to a standard bilinear (DistMult-like) model.
- `p = 0, q = 1` — equivalent to ComplEx.
- Larger `p` and `q` capture higher-order geometric interactions.

The embedding dimension must satisfy `embedding_dim % (p + q + 1) == 0`; the resulting quotient is stored as `self.r`.

Parameters

args (*dict*) – Configuration dictionary. Recognised keys (beyond those in `BaseKGE`): `p` (int, default 0) and `q` (int, default 0).

References

Demir et al., *Clifford Embeddings — A Generalized Approach for Embedding in Normed Algebras*, ECML 2023.

name = 'Keci'

p

q

r

requires_grad_for_interactions = True

compute_sigma_pp(*hp*, *rp*)

Compute sigma_{pp} = sum_{i=1}^{p-1} sum_{k=i+1}^p (h_i r_k - h_k r_i) e_i e_k

sigma_{pp} captures the interactions between along p bases For instance, let p e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for i in range(p - 1):

for k in range(i + 1, p):
results.append(hp[:, :, i] * rp[:, :, k] - hp[:, :, k] * rp[:, :, i])

sigma_pp = torch.stack(results, dim=2) assert sigma_pp.shape == (b, r, int((p * (p - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

compute_sigma_qq(*hq*, *rq*)

Compute sigma_{qq} = sum_{j=1}^{p+q-1} sum_{k=j+1}^{p+q} (h_j r_k - h_k r_j) e_j e_k sigma_{q} captures the interactions between along q bases For instance, let q e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for j in range(q - 1):

for k in range(j + 1, q):
results.append(hq[:, :, j] * rq[:, :, k] - hq[:, :, k] * rq[:, :, j])

sigma_qq = torch.stack(results, dim=2) assert sigma_qq.shape == (b, r, int((q * (q - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

compute_sigma_pq(*, *hp*, *hq*, *rp*, *rq*)

sum_{i=1}^{p} sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

apply_coefficients(*hp*, *hq*, *rp*, *rq*)

Multiplying a base vector with its scalar coefficient

clifford_multiplication(*h0*, *hp*, *hq*, *r0*, *rp*, *rq*)

Compute our CL multiplication

h = h_0 + sum_{i=1}^p h_i e_i + sum_{j=p+1}^{p+q} h_j e_j r = r_0 + sum_{i=1}^p r_i e_i + sum_{j=p+1}^{p+q} r_j e_j

ei ^2 = +1 for i =< i =< p ej ^2 = -1 for p < j =< p+q ei ej = -eje1 for i

eq j

h r = sigma_0 + sigma_p + sigma_q + sigma_{pp} + sigma_{q}+ sigma_{pq} where

(1) sigma_0 = h_0 r_0 + sum_{i=1}^p (h_0 r_i) e_i - sum_{j=p+1}^{p+q} (h_j r_j) e_j

(2) sigma_p = sum_{i=1}^p (h_0 r_i + h_i r_0) e_i

(3) sigma_q = sum_{j=p+1}^{p+q} (h_0 r_j + h_j r_0) e_j

(4) sigma_{pp} = sum_{i=1}^{p-1} sum_{k=i+1}^p (h_i r_k - h_k r_i) e_i e_k

(5) sigma_{qq} = sum_{j=1}^{p+q-1} sum_{k=j+1}^{p+q} (h_j r_k - h_k r_j) e_j e_k

(6) sigma_{pq} = sum_{i=1}^{p} sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j

construct_cl_multivector(*x: torch.FloatTensor*, *r: int*, *p: int*, *q: int*)
→ tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Split a flat embedding vector into the three Clifford components.

Given an embedding `x` of dimension `d = r + r*p + r*q`, returns the scalar part `a0`, the *p*-blade part `ap`, and the *q*-blade part `aq`.

Parameters

- **x** (*torch.FloatTensor*) – Shape `(batch_size, d)`.
- **r** (*int*) – Scalar block size (`embedding_dim // (p + q + 1)`).
- **p** (*int*) – Number of positive-signature basis elements.
- **q** (*int*) – Number of negative-signature basis elements.

Returns

- **a0** (*torch.FloatTensor*) – Shape `(batch_size, r)` — scalar (grade-0) part.
- **ap** (*torch.FloatTensor*) – Shape `(batch_size, r, p)` — positive-blade part.
- **aq** (*torch.FloatTensor*) – Shape `(batch_size, r, q)` — negative-blade part.

forward_k_vs_with_explicit(*x: torch.Tensor*) → torch.FloatTensor

KvsAll scoring using an explicit loop over sigma_pp/qq/pq terms.

Functionally equivalent to `forward_k_vs_all()` but computes the higher-order interaction terms (sigma_pp, sigma_qq, sigma_pq) with explicit nested loops rather than einsum contractions. Kept for reference and correctness verification.

Parameters

x (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

k_vs_all_score(*bpe_head_ent_emb: torch.FloatTensor*, *bpe_rel_ent_emb: torch.FloatTensor*, *E: torch.FloatTensor*) → torch.FloatTensor

Compute Clifford-product scores for a head/relation batch vs. all entities.

Decomposes the head-entity and relation embeddings into Clifford multi-vectors, performs the Cl_{p,q} product, and inner-products the result against the entity embedding matrix *E*.

Parameters

- **bpe_head_ent_emb** (*torch.FloatTensor*) – Head-entity embeddings, shape `(batch_size, embedding_dim)`.
- **bpe_rel_ent_emb** (*torch.FloatTensor*) – Relation embeddings, shape `(batch_size, embedding_dim)`.
- **E** (*torch.FloatTensor*) – All entity embeddings, shape `(num_entities, embedding_dim)`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

Kvsall training

(1) Retrieve real-valued embedding vectors for heads and relations mathbb{R}^d .

(2) Construct head entity and relation embeddings according to Cl_{p,q}(mathbb{R}^d) .

(3) Perform Cl multiplication

(4) Inner product of (3) and all entity embeddings

forward_k_vs_with_explicit and this funcitons are identical Parameter ——— x: torch.LongTensor with (n,2) shape :rtype: torch.FloatTensor with (n, |E|) shape

construct_batch_selected_cl_multivector(*x: torch.FloatTensor*, *r: int*, *p: int*, *q: int*) → tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Split a batched, *k*-selected embedding tensor into Clifford components.

A variant of `construct_cl_multivector()` for tensors that have an extra *k* dimension (e.g. when scoring against *k* sampled targets).

Parameters

- **x** (*torch.FloatTensor*) – Shape `(batch_size, k, d)`.
- **r** (*int*) – Scalar block size.
- **p** (*int*) – Number of positive-signature basis elements.
- **q** (*int*) – Number of negative-signature basis elements.

Returns

- **a0** (*torch.FloatTensor*) – Shape `(batch_size, k, r)`.
- **ap** (*torch.FloatTensor*) – Shape `(batch_size, k, r, p)`.
- **aq** (*torch.FloatTensor*) – Shape `(batch_size, k, r, q)`.

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

Parameter

x: torch.LongTensor with (n,2) shape

target_entity_idx: torch.LongTensor with (n, k) shape k denotes the selected number of examples.

rtype
torch.FloatTensor with (n, k) shape

score(*h*, *r*, *t*)

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Parameter

x: torch.LongTensor with (n,3) shape

rtype
torch.FloatTensor with (n) shape

class dicee.models.clifford.**KeciTransformer**(*args*)

Bases: *Keci*

Keci with Transformer architecture.

Concatenates h0, hp, hq, r0, rp, rq into a single embedding vector and processes through transformer.

name = 'KeciTransformer'

use_clifford_mul

seq_len = 1

transformer

lm_head

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor
Kvsall training

Parameter

x: torch.LongTensor with (n,2) shape

rtype
torch.FloatTensor with (n, |E|) shape

class dicee.models.clifford.**TransformerBlock**(*n_embd*, *n_head*, *dropout=0.0*, *bias=False*)

Bases: torch.nn.Module

A single transformer block with self-attention and MLP.

ln_1

attn

ln_2

mlp

forward(*x*)

class dicee.models.clifford.**TransformerSelfAttention**(*n_embd*, *n_head*, *dropout=0.0*, *bias=False*)

Bases: torch.nn.Module

Multi-head self-attention for the Keci Transformer.

c_attn

c_proj

attn_dropout

resid_dropout

n_head

n_embd

dropout = 0.0

forward(*x*)

class dicee.models.clifford.**TransformerMLP**(*n_embd*, *dropout=0.0*, *bias=False*)

Bases: torch.nn.Module

MLP for the Keci Transformer.

c_fc

gelu

c_proj

dropout

forward(*x*)

class dicee.models.clifford.**CKeci**(*args*)

Bases: *Keci*

Without learning dimension scaling

name = 'CKeci'

requires_grad_for_interactions = False

class dicee.models.clifford.**DeCaL**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from `BaseKGELightning` and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'DeCaL'

p

q

r

re

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Parameter

x: torch.LongTensor with (n,) shape

rtype

torch.FloatTensor with (n) shape

cl_pqr(*a: torch.tensor*) → torch.tensor

Input: tensor(batch_size, emb_dim) —> output: tensor with 1+p+q+r components with size (batch_size, emb_dim/(1+p+q+r)) each.

1) takes a tensor of size (batch_size, emb_dim), split it into 1 + p + q +r components, hence 1+p+q+r must be a divisor of the emb_dim. 2) Return a list of the 1+p+q+r components vectors, each are tensors of size (batch_size, emb_dim/(1+p+q+r))

compute_sigmas_single(*list_h_emb*, *list_r_emb*, *list_t_emb*)

here we compute all the sums with no others vectors interaction taken with the scalar product with t, that is,

$$s0 = h_0 r_0 t_0 s1 = \sum_{i=1}^{p} h_i r_i t_0 s2 = \sum_{j=p+1}^{p+q} h_j r_j t_0 s3 = \sum_{i=1}^{q} (h_0 r_i t_i + h_i r_0 t_i) s4 = \sum_{i=p+1}^{p+q} (h_0 r_i t_i + h_i r_0 t_i) s5 = \sum_{i=p+q+1}^{p+q+r} (h_0 r_i t$$

and return:

$$sigma_0 t = \sigma_0 \cdot t_0 = s0 + s1 - s2 s3, s4 and s5$$

compute_sigmas_multivect (*list_h_emb*, *list_r_emb*)

Here we compute and return all the sums with vectors interaction for the same and different bases.

For same bases vectors interaction we have

$$\sigma_p p = \sum_{i=1}^{p-1} \sum_{i'=i+1}^{p} (h_i r_{i'} - h_{i'} r_i)(models the interactions between e_i and e'_i for 1 <= i, i' <= p) \sigma_q q = \sum_{j=p+1}^{p+q-1} \sum_{j'=j+1}^{p+q} (h_j r_{j'}$$

For different base vector interactions, we have

$$\sigma_p q = \sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i)(interactions n between e_i and e_j for 1 <= i <= p and p+1 <= j <= p+q) \sigma_p r = \sum_{i=}^{}$$

forward_k_vs_all (*x: torch.Tensor*) → torch.FloatTensor

Kvsall training

(1) Retrieve real-valued embedding vectors for heads and relations

(2) Construct head entity and relation embeddings according to Cl_{p,q, r}(mathbb{R}^d) .

(3) Perform Cl multiplication

(4) Inner product of (3) and all entity embeddings

forward_k_vs_with_explicit and this funcitons are identical Parameter ——— x: torch.LongTensor with (n,) shape :rtype: torch.FloatTensor with (n, |E|) shape

apply_coefficients (*h0*, *hp*, *hq*, *hk*, *r0*, *rp*, *rq*, *rk*)

Multiplying a base vector with its scalar coefficient

construct_cl_multivector (*x: torch.FloatTensor*, *re: int*, *p: int*, *q: int*, *r: int*)
→ tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Construct a batch of multivectors Cl_{p,q,r}(mathbb{R}^d)

Parameter

x: torch.FloatTensor with (n,d) shape

returns

- **a0** (*torch.FloatTensor*)
- **ap** (*torch.FloatTensor*)
- **aq** (*torch.FloatTensor*)
- **ar** (*torch.FloatTensor*)

compute_sigma_pp (*hp*, *rp*)

Compute .. math:

```
\sigma_{p,p}^* = \sum_{i=1}^{p-1}\sum_{i'=i+1}^{p}(x_iy_{i'}-x_{i'}y_i)
```

sigma_{pp} captures the interactions between along p bases For instance, let p e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for i in range(p - 1):

for k in range(i + 1, p):
results.append(hp[:, :, i] * rp[:, :, k] - hp[:, :, k] * rp[:, :, i])

sigma_pp = torch.stack(results, dim=2) assert sigma_pp.shape == (b, r, int((p * (p - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

`compute_sigma_qq`(*hq*, *rq*)

Compute

$$\sigma^*_{q,q} = \sum_{j=p+1}^{p+q-1} \sum_{j'=j+1}^{p+q} (x_j y_{j'} - x_{j'} y_j) Eq.16$$

sigma_{q} captures the interactions between along q bases For instance, let q e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for j in range(q - 1):

for k in range(j + 1, q):
results.append(hq[:, :, j] * rq[:, :, k] - hq[:, :, k] * rq[:, :, j])

sigma_qq = torch.stack(results, dim=2) assert sigma_qq.shape == (b, r, int((q * (q - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

`compute_sigma_rr`(*hk*, *rk*)

$$\sigma^*_{r,r} = \sum_{k=p+q+1}^{p+q+r-1} \sum_{k'=k+1}^{p} (x_k y_{k'} - x_{k'} y_k)$$

`compute_sigma_pq`(*, *hp*, *hq*, *rp*, *rq*)

Compute

$$\sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j$$

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

compute_sigma_pr(*, *hp*, *hk*, *rp*, *rk*)

Compute

$$\sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j$$

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

compute_sigma_qr(*, *hq*, *hk*, *rq*, *rk*)

$$\sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j$$

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

dicee.models.complex

Classes

ConEx	Convolutional ComplEx Knowledge Graph Embeddings
AConEx	Additive Convolutional ComplEx Knowledge Graph Embeddings
ComplEx	Base class for all Knowledge Graph Embedding models.
RotatE	RotatE: knowledge graph embedding by relational rotation in complex space.

Module Contents

class dicee.models.complex.**ConEx**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Convolutional ComplEx Knowledge Graph Embeddings

name = 'ConEx'

conv2d

fc_num_input

fc1

norm_fc1

bn_conv2d

feature_map_dropout

residual_convolution(*C_1: Tuple[torch.Tensor, torch.Tensor]*, *C_2: Tuple[torch.Tensor, torch.Tensor]*) → torch.FloatTensor

Compute residual score of two complex-valued embeddings. :param C_1: a tuple of two pytorch tensors that corresponds complex-valued embeddings :param C_2: a tuple of two pytorch tensors that corresponds complex-valued embeddings :return:

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.Tensor*, *target_entity_idx: torch.Tensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.complex.**AConEx**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Additive Convolutional ComplEx Knowledge Graph Embeddings

name = 'AConEx'

conv2d

fc_num_input

fc1

norm_fc1

bn_conv2d

feature_map_dropout

residual_convolution(*C_1: Tuple[torch.Tensor, torch.Tensor]*, *C_2: Tuple[torch.Tensor, torch.Tensor]*) → torch.FloatTensor

Compute residual score of two complex-valued embeddings. :param C_1: a tuple of two pytorch tensors that corresponds complex-valued embeddings :param C_2: a tuple of two pytorch tensors that corresponds complex-valued embeddings :return:

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.Tensor*, *target_entity_idx: torch.Tensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.complex.**ComplEx**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from `BaseKGELightning` and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- `forward_triples()` — score a batch of `(h, r, t)` triples.
- *`forward_k_vs_all()`* — score a `(h, r)` batch against every entity.

Parameters
args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'ComplEx'

static score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*)

static k_vs_all_score(*emb_h: torch.FloatTensor*, *emb_r: torch.FloatTensor*, *emb_E: torch.FloatTensor*)

Parameters

- **emb_h**
- **emb_r**
- **emb_E**

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.complex.**RotatE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

RotatE: knowledge graph embedding by relational rotation in complex space.

Represents each entity as a complex vector in ℂ^(d/2), obtained by splitting its `d`-dimensional embedding into a real half and an imaginary half. Each relation is represented by `d/2` phase angles θ that define a unit-modulus rotation `r_i = e^{iθ_i}`; since a phase angle needs only one real number, the relation embedding table is reinitialised at half of the entity embedding size. A true triple `(h, r, t)` should satisfy `h ∘ r ≈ t` under element-wise complex multiplication, giving the score:

```
f(h, r, t) = margin - ||h ∘ r - t||_2
```

Unlike TransE, RotatE can model symmetric, antisymmetric, inverse, and composition relation patterns.

References

Sun et al., *RotatE: Knowledge Graph Embedding by Relational Rotation in Complex Space*, ICLR 2019. https://arxiv.org/abs/1902.10197

name = 'RotatE'

margin

half_dim

relation_embeddings

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*) → torch.FloatTensor

Score a batch of triples using the RotatE margin-distance formula.

Parameters

- **head_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **tail_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **rel_ent_emb** (*torch.FloatTensor*) – Shape `(batch_size, d/2)` relation phase angles θ.

Returns

Shape `(batch_size,)` scores equal to `margin - ||h ∘ r - t||_2`.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Computes `margin - ||h ∘ r - e||_2` for every entity embedding *e*.

Parameters

x (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

forward_k_vs_sample(*x: torch.Tensor*, *target_entity_idx: torch.Tensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Computes `margin - ||h ∘ r - e||_2` for each of the *k* sampled entities *e*.

Parameters

- **x** (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.
- **target_entity_idx** (*torch.Tensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns

Shape `(batch_size, k)` score matrix.

Return type

torch.FloatTensor

dicee.models.dualE

Classes

DualE	Dual Quaternion Knowledge Graph Embeddings (https://ojs.aaai.org/index.php/AAAI/article/download/16850/16657)

Module Contents

class dicee.models.dualE.**DualE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Dual Quaternion Knowledge Graph Embeddings (https://ojs.aaai.org/index.php/AAAI/article/download/16850/16657)

name = 'DualE'

entity_embeddings

relation_embeddings

num_ent = None

kvsall_score(*e_1_h*, *e_2_h*, *e_3_h*, *e_4_h*, *e_5_h*, *e_6_h*, *e_7_h*, *e_8_h*, *e_1_t*, *e_2_t*, *e_3_t*, *e_4_t*, *e_5_t*, *e_6_t*, *e_7_t*, *e_8_t*, *r_1*, *r_2*, *r_3*, *r_4*, *r_5*, *r_6*, *r_7*, *r_8*) → torch.tensor

KvsAll scoring function

Input

x: torch.LongTensor with (n,) shape

Output

torch.FloatTensor with (n) shape

forward_triples(*idx_triple: torch.tensor*) → torch.tensor

Negative Sampling forward pass:

Input

x: torch.LongTensor with (n,) shape

Output

torch.FloatTensor with (n) shape

forward_k_vs_all(*x*)

KvsAll forward pass

Input

x: torch.LongTensor with (n,) shape

Output

torch.FloatTensor with (n) shape

T(*x: torch.tensor*) → torch.tensor

Transpose function

Input: Tensor with shape (nxm) Output: Tensor with shape (mxn)

dicee.models.ensemble

Classes

EnsembleKGE

Module Contents

class dicee.models.ensemble.**EnsembleKGE**(*models: list | None = None*, *seed_model=None*, *pretrained_models: List | None = None*)

name

train_mode = True

args

named_children()

property example_input_array

parameters()

modules()

__iter__()

__len__()

eval()

to(*device*)

state_dict()

Return the state dict of the ensemble.

load_state_dict(*state_dict*, *strict=True*)

Load the state dict into the ensemble.

mem_of_model()

__call__(*x_batch*)

step()

get_embeddings()

__str__()

dicee.models.flock

Pure PyTorch Flock, compatible with both official jw9730/flock checkpoints.

Architecture adapted from https://github.com/jw9730/flock at UPSTREAM_COMMIT. See docs/flock.md for random-walk and numerical parity details.

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Attributes

UPSTREAM_COMMIT	

Classes

RMSNorm	Base class for all neural network modules.
FeedForward	Base class for all neural network modules.
BidirectionalGRU	Base class for all neural network modules.
FlockBase	Shared graph lifecycle and entity-scoring interfaces for graph foundation models.
Flock	Entity-prediction Flock with official `flock_entity.pth` parameters.
FlockRelation	Relation-prediction Flock with official `flock_relation.pth` parameters.

Functions

consensus(x, logits, ids, num_ids)	Stable multihead softmax pooling over all occurrences of each graph ID.

Module Contents

dicee.models.flock.**UPSTREAM_COMMIT = 'f35103d25a78bdf4075de5c673a51de4979aa4d7'**

class dicee.models.flock.**RMSNorm**(*dim*, *eps=1e-05*)

Bases: torch.nn.Module

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call to(), etc.

> **Note**
>
> As per the example above, an __init__() call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

eps = 1e-05

inference_backend = 'auto'

weight

forward(*x*)

class dicee.models.flock.**FeedForward**(*dim*)

Bases: torch.nn.Module

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F
```

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```
class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

w1

w2

w3

forward(*x*)

class dicee.models.flock.**BidirectionalGRU**(*dim*, *layers*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

gru_norm

gru

gru_out

ffn_norm

feed_forward

forward(*x*)

dicee.models.flock.**consensus**(*x*, *logits*, *ids*, *num_ids*)

Stable multihead softmax pooling over all occurrences of each graph ID.

class dicee.models.flock.**FlockBase**(*args*)

Bases: *dicee.models.graph_model.GraphKGE*

Shared graph lifecycle and entity-scoring interfaces for graph foundation models.

config_prefix = 'flock'

graph_filename = 'flock_graph.pt'

relation_prediction = False

checkpoint_hint = 'official Flock requires dim=64, walk_len=128, refinements=6, num_layers=1, attention_heads=4...

emb_anon_node: torch.nn.ModuleList

emb_anon_type: torch.nn.ModuleList

emb_restart: torch.nn.ModuleList

emb_neighbor: torch.nn.ModuleList

emb_direction: torch.nn.ModuleList

emb_head_is_query: torch.nn.ModuleList

emb_tail_is_query: torch.nn.ModuleList

emb_node_is_query: torch.nn.ModuleList

emb_type_is_query: torch.nn.ModuleList

from_node: torch.nn.ModuleList

from_type: torch.nn.ModuleList

to_node: torch.nn.ModuleList

to_type: torch.nn.ModuleList

node_logit: torch.nn.ModuleList

type_logit: torch.nn.ModuleList

dim

walk_num

walk_len

refinements

test_samples

seed

prefetch_walks

compact_state

compile_sampler

pack_walks

node_init

type_init

net

head

sample_walks(*heads*, *tails=None*, *edges=None*, *generator=None*)

Sample official-format records; IDs use the attached graph vocabulary.

Returns (nodes, anonymous nodes, restarts, neighbors, relations, anonymous relations, directions), each shaped [T,B,S,L]. Anonymous names start at 1; no-relation markers are 2*R and L+1 respectively.

score_walks(*heads*, *query*, *candidates*, *records*)

Score fixed walks for parity/replay; query/candidates use internal IDs.

Entity mode: query=relations, candidates=tails. Relation mode: query=tails, candidates=relations. This scores the supplied records; callers constructing training records must first remove target edges.

class dicee.models.flock.**Flock**(*args*)

Bases: *FlockBase*

Entity-prediction Flock with official `flock_entity.pth` parameters.

name = 'Flock'

class dicee.models.flock.**FlockRelation**(*args*)

Bases: *FlockBase*, *dicee.models.graph_model.RelationGraphKGE*

Relation-prediction Flock with official `flock_relation.pth` parameters.

name = 'FlockRelation'

relation_prediction = True

dicee.models.flock_walks

Flock's non-backtracking walk and recording protocol in PyTorch.

Semantics follow jw9730/flock at f35103d25a78bdf4075de5c673a51de4979aa4d7. The RNG is PyTorch's CPU generator, not the reference C++ per-walk MT19937. Walk records can be supplied explicitly to compare neural computations exactly.

Classes

WalkGraph	CPU CSR context, kept outside a model's transferable parameters.

Functions

compiled_walk_nodes()	
anonymize(values[, missing_id, missing_name])	Assign 1-based names in order of discovery, independently in each row.

Module Contents

dicee.models.flock_walks.**compiled_walk_nodes**()

dicee.models.flock_walks.**anonymize**(*values*, *missing_id=None*, *missing_name=None*)

Assign 1-based names in order of discovery, independently in each row.

class dicee.models.flock_walks.**WalkGraph**

CPU CSR context, kept outside a model's transferable parameters.

edge_index: torch.Tensor

edge_type: torch.Tensor

num_nodes: int

num_types: int

compile_sampler: bool = True

__post_init__()

walk(*prefix*, *length*, *remove_loops*, *generator=None*, *prefix_types=None*)

Sample uniform neighbors, then a uniform typed edge in either direction.

parse_types(*walks*, *generator=None*)

sample(*heads*, *tails*, *walk_num*, *length*, *refinements*, *generator=None*)

Return seven official-format records of shape [T,B,3N or 4N,L].

dicee.models.fsdp_models

Row-wise sharded entity embeddings for multi-GPU training.

Entity embeddings are partitioned row-wise across ranks:

- dist.all_to_all_single routes index requests to the owning rank
- A custom autograd Function propagates gradients back through the all_to_all
- _LocalSparseAdam updates only the rows that received a non-zero gradient

Public API:

model.setup_fsdp_training(device, lr) — called by trainer before loop model.gather_entity_embeddings_on_rank_zero() — called by trainer after loop model.save_local_shard_checkpoint(path) — per-rank, called periodically during loop model.load_local_shard_checkpoint(path) — per-rank, called by trainer on resume create_fsdp_sharded_model_class(ModelClass) — factory used by static_funcs.py

Classes

RowWiseShardedEmbedding	Entity embedding table sharded row-wise across ranks.
FSDPShardedEntityModel	Mixin that defers entity embedding allocation and wires up row-wise sharding.

Functions

create_fsdp_sharded_model_class(...)	Return a row-wise sharded variant of *model_class*.

Module Contents

class dicee.models.fsdp_models.**RowWiseShardedEmbedding**(*num_embeddings: int*, *embedding_dim: int*, *rank: int*, *world_size: int*, *device: torch.device*, *weight_dtype: torch.dtype = torch.bfloat16*)

Bases: `torch.nn.Module`

Entity embedding table sharded row-wise across ranks.

Each rank owns rows [start_row, end_row). forward() looks like nn.Embedding so every existing scoring function works unchanged.

num_embeddings

embedding_dim

rank

world_size

device

shard_size

start_row

end_row

local_rows

weight

forward(*entity_ids: torch.LongTensor*) → torch.FloatTensor

class dicee.models.fsdp_models.**FSDPShardedEntityModel**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Mixin that defers entity embedding allocation and wires up row-wise sharding.

Lifecycle

1. __init__() — entity_embeddings is None
2. setup_fsdp_training(dev, lr) — creates RowWiseShardedEmbedding
3. gather_entity_embeddings_on_rank_zero() — called by trainer after last epoch

setup_fsdp_training(*device: torch.device*, *lr: float*, *optimizer_cls=None*, *optimizer_kwargs: dict | None = None*, *adam_device: torch.device | None = None*) → None

Create the per-rank embedding shard and its local sparse Adam.

adam_device — where exp_avg / exp_avg_sq are stored:
GPU (default) : pure GPU kernels, no PCIe, ~42.8 GiB extra GPU RAM per rank. CPU : lower GPU footprint, PCIe transfer each step.

Falls back to the embedding device (GPU) when not specified.

gather_entity_embeddings_on_rank_zero() → torch.nn.Embedding | None

Gather the full entity table on rank 0; return None on other ranks.

save_local_shard_checkpoint(*path: str*) → None

Save this rank's entity embedding shard + its local Adam state to *path*.

Cheap and O(local_rows) — unlike gather_entity_embeddings_on_rank_zero(), no collective communication is involved, so this is safe to call periodically even when the full table is too large to gather.

load_local_shard_checkpoint(*path: str*) → None

Load this rank's entity embedding shard + local Adam state from *path*.

The checkpoint must have been saved by a run with the same world_size: shard boundaries (start_row/local_rows) are derived from world_size and num_entities, so a mismatch means this rank does not own the same rows as when the checkpoint was written.

get_embeddings()

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

dicee.models.fsdp_models.**create_fsdp_sharded_model_class**(
model_class: Type[dicee.models.base_model.BaseKGE])
→ Type[*dicee.models.base_model.BaseKGE*]

Return a row-wise sharded variant of *model_class*.

The returned class inherits all scoring functions from *model_class* unchanged. Only entity_embeddings is replaced at training time.

dicee.models.function_space

Attributes

logger

Classes

FMult	Learning Knowledge Neural Graphs
GFMult	Learning Knowledge Neural Graphs
FMult2	Learning Knowledge Neural Graphs
LFMult1	Embedding with trigonometric functions. We represent all entities and relations in the complex number space as:
LFMult	Embedding with polynomial functions. We represent all entities and relations in the polynomial space as:

Module Contents

dicee.models.function_space.**logger**

class dicee.models.function_space.**FMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Learning Knowledge Neural Graphs

name = 'FMult'

entity_embeddings

relation_embeddings

k

num_sample = 50

gamma

roots

weights

compute_func(*weights: torch.FloatTensor*, *x*) → torch.FloatTensor

chain_func(*weights*, *x: torch.FloatTensor*)

forward_triples(*idx_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

class dicee.models.function_space.**GFMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Learning Knowledge Neural Graphs

name = 'GFMult'

entity_embeddings

relation_embeddings

k

num_sample = 250

roots

weights

compute_func(*weights: torch.FloatTensor*, *x*) → torch.FloatTensor

chain_func(*weights*, *x: torch.FloatTensor*)

forward_triples(*idx_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

class dicee.models.function_space.**FMult2**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Learning Knowledge Neural Graphs

name = 'FMult2'

n_layers = 3

k

n = 50

score_func = 'compositional'

discrete_points

entity_embeddings

relation_embeddings

build_func(*Vec*)

build_chain_funcs(*list_Vec*)

compute_func(*W*, *b*, *x*) → torch.FloatTensor

function(*list_W*, *list_b*)

trapezoid(*list_W*, *list_b*)

forward_triples(*idx_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

class dicee.models.function_space.**LFMult1**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Embedding with trigonometric functions. We represent all entities and relations in the complex number space as: f(x) = sum_{k=0}^{k=d-1}wk e^{kix}. and use the three differents scoring function as in the paper to evaluate the score

name = 'LFMult1'

entity_embeddings

relation_embeddings

forward_triples(*idx_triple*)

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

tri_score(*h*, *r*, *t*)

vtp_score(*h*, *r*, *t*)

class dicee.models.function_space.**LFMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Embedding with polynomial functions. We represent all entities and relations in the polynomial space as: f(x) = sum_{i=0}^{d-1} a_k x^{i%d} and use the three differents scoring function as in the paper to evaluate the score. We also consider combining with Neural Networks.

name = 'LFMult'

entity_embeddings

relation_embeddings

degree

m

x_values

forward_triples(*idx_triple*)

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

construct_multi_coeff(*x*)

poly_NN(*x*, *coefh*, *coefr*, *coeft*)

Constructing a 2 layers NN to represent the embeddings. h = sigma(wh^T x + bh), r = sigma(wr^T x + br), t = sigma(wt^T x + bt)

linear(*x*, *w*, *b*)

scalar_batch_NN(*a*, *b*, *c*)

element wise multiplication between a,b and c: Inputs : a, b, c ====> torch.tensor of size batch_size x m x d Output : a tensor of size batch_size x d

tri_score(*coeff_h*, *coeff_r*, *coeff_t*)

this part implement the trilinear scoring techniques:

score(h,r,t) = int_{0}{1} h(x)r(x)t(x) dx = sum_{i,j,k = 0}^{d-1} dfrac{a_i*b_j*c_k}{1+(i+j+k)%d}

1. generate the range for i,j and k from [0 d-1]
2. perform dfrac{a_i*b_j*c_k}{1+(i+j+k)%d} in parallel for every batch
3. take the sum over each batch

vtp_score(*h*, *r*, *t*)

this part implement the vector triple product scoring techniques:

score(h,r,t) = int_{0}{1} h(x)r(x)t(x) dx = sum_{i,j,k = 0}^{d-1} dfrac{a_i*c_j*b_k - b_i*c_j*a_k}{(1+(i+j)%d)(1+k)}

1. generate the range for i,j and k from [0 d-1]
2. Compute the first and second terms of the sum
3. Multiply with then denominator and take the sum
4. take the sum over each batch

comp_func(*h*, *r*, *t*)

this part implement the function composition scoring techniques: i.e. score = <hor, t>

polynomial(*coeff*, *x*, *degree*)

This function takes a matrix tensor of coefficients (coeff), a tensor vector of points x and range of integer [0,1,…d] and return a vector tensor (coeff[0][0] + coeff[0][1]x +…+ coeff[0][d]x^d,

coeff[1][0] + coeff[1][1]x +…+ coeff[1][d]x^d)

pop(*coeff*, *x*, *degree*)

This function allow us to evaluate the composition of two polynomes without for loops :) it takes a matrix tensor of coefficients (coeff), a matrix tensor of points x and range of integer [0,1,…d]

and return a tensor (coeff[0][0] + coeff[0][1]x +…+ coeff[0][d]x^d,

coeff[1][0] + coeff[1][1]x +…+ coeff[1][d]x^d)

dicee.models.graph_model

Shared graph-conditioned model support, extracted from the DICE ULTRA implementation.

Architecture and graph construction adapted from https://github.com/DeepGraphLearning/ULTRA at commit 427966ad8ed60420eef034063d44f3153addff90. Reference fixtures and the grouped sampling and adversarial loss implementations also follow this upstream work. Graphs are runtime context, excluded from the transferable state dictionary.

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Classes

GraphKGE	Shared graph lifecycle and entity-scoring interfaces for graph foundation models.
RelationGraphKGE	Shared relation prediction API with (head, tail) query pairs.

Module Contents

class dicee.models.graph_model.**GraphKGE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Shared graph lifecycle and entity-scoring interfaces for graph foundation models.

name = 'GraphKGE'

config_prefix = 'graph'

graph_filename = 'graph.pt'

checkpoint_hint = 'all keys and shapes must match'

deterministic_inference = False

query_batch_size

num_direct_relations = 0

clear_inference_cache()

Drop derived representations when graph, weights or device change.

set_inference_backend(*backend*)

inference_token()

train(*mode=True*)

Set the module in training mode.

This has an effect only on certain modules. See the documentation of particular modules for details of their behaviors in training/evaluation mode, i.e., whether they are affected, e.g. `Dropout`, `BatchNorm`, etc.

Parameters

mode (*bool*) – whether to set training mode (`True`) or evaluation mode (`False`). Default: `True`.

Returns

self

Return type

Module

init_entity_embeddings(*embedding_dim=None*)

Create (or re-create) the entity embedding table.

This is the single place that honours `defer_large_embeddings`: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to `embedding_dim`.

init_relation_embeddings(*embedding_dim=None*)

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to `embedding_dim`.

abstractmethod get_embeddings()

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

set_graph(*triples*, *num_entities=None*, *num_relations=None*, *inverse_relations=None*)

Attach training facts; inverse_relations maps direct DICE IDs to inverse IDs.

Without a mapping every supplied relation is treated as a direct relation. Repeated facts and explicitly supplied inverse facts are deduplicated.

save_graph(*path*)

load_graph(*path*)

load_pretrained(*path*)

forward_grouped(*triples*, *head_prediction=None*)

forward_triples(*x*)

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_sample(*x*, *target_entity_idx*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns

Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

forward_k_vs_all(*x*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_all_heads(*x*, *target_entity_idx=None*)

Score head candidates for DICE pairs (relation, tail).

forward(*x*, *y_idx=None*)

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → `forward_byte_pair_encoded_triple()`
- BPE pair tensor → `forward_byte_pair_encoded_k_vs_all()`

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

class dicee.models.graph_model.**RelationGraphKGE**(*args*)

Bases: *GraphKGE*

Shared relation prediction API with (head, tail) query pairs.

forward_grouped(*triples*, *head_prediction=None*)

forward_triples(*x*)

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_sample(*x*, *target_entity_idx*)

Score relation candidates [K] or [B,K] for (head, tail) pairs.

forward_k_vs_all(*x*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_all_relations

forward_k_vs_all_heads(*x*, *target_entity_idx=None*)

Score head candidates for DICE pairs (relation, tail).

dicee.models.literal

Classes

LiteralEmbeddings	A model for learning and predicting numerical literals using pre-trained KGE.

Module Contents

class dicee.models.literal.**LiteralEmbeddings**(*num_of_data_properties: int*, *embedding_dims: int*, *entity_embeddings: torch.tensor*, *dropout: float = 0.3*, *gate_residual=True*, *freeze_entity_embeddings=True*)

Bases: `torch.nn.Module`

A model for learning and predicting numerical literals using pre-trained KGE.

num_of_data_properties

Number of data properties (attributes).

Type
int

embedding_dims

Dimension of the embeddings.

Type
int

entity_embeddings

Pre-trained entity embeddings.

Type
torch.tensor

dropout

Dropout rate for regularization.

Type
float

gate_residual

Whether to use gated residual connections.

Type
bool

freeze_entity_embeddings

Whether to freeze the entity embeddings during training.

Type
bool

embedding_dim

num_of_data_properties

hidden_dim

gate_residual = True

freeze_entity_embeddings = True

entity_embeddings

data_property_embeddings

fc

fc_out

dropout

gated_residual_proj

layer_norm

forward(*entity_idx*, *attr_idx*)

Parameters

- **entity_idx** (*Tensor*) – Entity indices (batch).
- **attr_idx** (*Tensor*) – Attribute (Data property) indices (batch).

Returns
scalar predictions.

Return type
Tensor

property device

dicee.models.octonion

Classes

OMult	Base class for all Knowledge Graph Embedding models.
ConvO	Base class for all Knowledge Graph Embedding models.
AConvO	Additive Convolutional Octonion Knowledge Graph Embeddings

Functions

octonion_mul(*, O_1, O_2)
octonion_mul_norm(*, O_1, O_2)

Module Contents

dicee.models.octonion.**octonion_mul**(*, *O_1*, *O_2*)

dicee.models.octonion.**octonion_mul_norm**(*, *O_1*, *O_2*)

class dicee.models.octonion.**OMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from `BaseKGELightning` and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- `forward_triples()` — score a batch of `(h, r, t)` triples.
- *`forward_k_vs_all()`* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'OMult'

static octonion_normalizer(*emb_rel_e0*, *emb_rel_e1*, *emb_rel_e2*, *emb_rel_e3*, *emb_rel_e4*, *emb_rel_e5*, *emb_rel_e6*, *emb_rel_e7*)

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*)

k_vs_all_score(*bpe_head_ent_emb*, *bpe_rel_ent_emb*, *E*)

forward_k_vs_all(*x*)

Completed. Given a head entity and a relation (h,r), we compute scores for all possible triples,i.e., [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,| Entities|)

class dicee.models.octonion.**ConvO**(*args: dict*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from `BaseKGELightning` and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'ConvO'

conv2d

fc_num_input

fc1

bn_conv2d

norm_fc1

feature_map_dropout

static octonion_normalizer(*emb_rel_e0*, *emb_rel_e1*, *emb_rel_e2*, *emb_rel_e3*, *emb_rel_e4*, *emb_rel_e5*, *emb_rel_e6*, *emb_rel_e7*)

residual_convolution(*O_1*, *O_2*)

forward_triples(*x: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,| Entities|)

class dicee.models.octonion.**AConvO**(*args: dict*)

Bases: *dicee.models.base_model.BaseKGE*

Additive Convolutional Octonion Knowledge Graph Embeddings

name = 'AConvO'

conv2d

fc_num_input

fc1

bn_conv2d

norm_fc1

feature_map_dropout

static octonion_normalizer(*emb_rel_e0*, *emb_rel_e1*, *emb_rel_e2*, *emb_rel_e3*, *emb_rel_e4*, *emb_rel_e5*, *emb_rel_e6*, *emb_rel_e7*)

residual_convolution(*O_1*, *O_2*)

forward_triples(*x: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,|Entities|)

dicee.models.pykeen_models

Attributes

logger

Classes

PykeenKGE	A class for using knowledge graph embedding models implemented in Pykeen

Module Contents

dicee.models.pykeen_models.**logger**

class dicee.models.pykeen_models.**PykeenKGE**(*args: dict*)

Bases: *dicee.models.base_model.BaseKGE*

A class for using knowledge graph embedding models implemented in Pykeen

Notes: Pykeen_DistMult: C Pykeen_ComplEx: Pykeen_QuatE: Pykeen_MuRE: Pykeen_CP: Pykeen_HolE: Pykeen_HolE: Pykeen_HolE: Pykeen_TransD: Pykeen_TransE: Pykeen_TransF: Pykeen_TransH: Pykeen_TransR:

model_kwargs

name

model

loss_history = []

args

entity_embeddings = None

relation_embeddings = None

forward_k_vs_all(*x: torch.LongTensor*)

=> Explicit version by this we can apply bn and dropout

(1) Retrieve embeddings of heads and relations + apply Dropout & Normalization if given. h, r = self.get_head_relation_representation(x) # (2) Reshape (1). if self.last_dim > 0:

h = h.reshape(len(x), self.embedding_dim, self.last_dim) r = r.reshape(len(x), self.embedding_dim, self.last_dim)

(3) Reshape all entities. if self.last_dim > 0:

t = self.entity_embeddings.weight.reshape(self.num_entities, self.embedding_dim, self.last_dim)

else:

t = self.entity_embeddings.weight

(4) Call the score_t from interactions to generate triple scores. return self.interaction.score_t(h=h, r=r, all_entities=t, slice_size=1)

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

=> Explicit version by this we can apply bn and dropout

(1) Retrieve embeddings of heads, relations and tails and apply Dropout & Normalization if given. h, r, t = self.get_triple_representation(x) # (2) Reshape (1). if self.last_dim > 0:

h = h.reshape(len(x), self.embedding_dim, self.last_dim) r = r.reshape(len(x), self.embedding_dim, self.last_dim) t = t.reshape(len(x), self.embedding_dim, self.last_dim)

(3) Compute the triple score return self.interaction.score(h=h, r=r, t=t, slice_size=None, slice_dim=0)

abstractmethod forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns

Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

dicee.models.quaternion

Classes

QMult	Base class for all Knowledge Graph Embedding models.
ConvQ	Convolutional Quaternion Knowledge Graph Embeddings
AConvQ	Additive Convolutional Quaternion Knowledge Graph Embeddings

Functions

quaternion_mul_with_unit_norm(*, Q_1, Q_2)

Module Contents

dicee.models.quaternion.**quaternion_mul_with_unit_norm**(*, *Q_1*, *Q_2*)

class dicee.models.quaternion.**QMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from `BaseKGELightning` and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- `forward_triples()` — score a batch of `(h, r, t)` triples.
- *`forward_k_vs_all()`* — score a `(h, r)` batch against every entity.

Parameters
args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'QMult'

explicit = True

quaternion_multiplication_followed_by_inner_product(*h*, *r*, *t*)

Parameters

- **h** – shape: (**batch_dims*, dim) The head representations.
- **r** – shape: (**batch_dims*, dim) The head representations.
- **t** – shape: (**batch_dims*, dim) The tail representations.

Returns
Triple scores.

static quaternion_normalizer(*x: torch.FloatTensor*) → torch.FloatTensor

Normalize the length of relation vectors, if the forward constraint has not been applied yet.

Absolute value of a quaternion

$$|a + bi + cj + dk| = \sqrt{a^2 + b^2 + c^2 + d^2}$$

L2 norm of quaternion vector:

$$\|x\|^2 = \sum_{i=1}^{d} |x_i|^2 = \sum_{i=1}^{d} (x_i.re^2 + x_i.im_1^2 + x_i.im_2^2 + x_i.im_3^2)$$

Parameters

x – The vector.

Returns

The normalized vector.

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*)

k_vs_all_score(*bpe_head_ent_emb*, *bpe_rel_ent_emb*, *E*)

Parameters

- **bpe_head_ent_emb**
- **bpe_rel_ent_emb**
- **E**

forward_k_vs_all(*x*)

Parameters

x

forward_k_vs_sample(*x*, *target_entity_idx*)

Completed. Given a head entity and a relation (h,r), we compute scores for all possible triples,i.e., [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, **|Entities|**) Given a batch of head entities and relations => shape (size of batch,| Entities|)

class dicee.models.quaternion.**ConvQ**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Convolutional Quaternion Knowledge Graph Embeddings

name = 'ConvQ'

entity_embeddings

relation_embeddings

conv2d

fc_num_input

fc1

bn_conv1

bn_conv2

feature_map_dropout

residual_convolution(*Q_1*, *Q_2*)

forward_triples(*indexed_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,|Entities|)

class dicee.models.quaternion.**AConvQ**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Additive Convolutional Quaternion Knowledge Graph Embeddings

name = 'AConvQ'

entity_embeddings

relation_embeddings

conv2d

fc_num_input

fc1

bn_conv1

bn_conv2

feature_map_dropout

residual_convolution(*Q_1*, *Q_2*)

forward_triples(*indexed_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,|Entities|)

dicee.models.real

Classes

DistMult	DistMult: bilinear diagonal knowledge graph embedding.
TransE	TransE: translation-based knowledge graph embedding.
TransH	TransH: translation-based knowledge graph embedding on relation-specific hyperplanes.
MuRE	MuRE: multi-relational graph embedding with relation-specific diagonal scaling, translation, and entity biases.
Shallom	Shallom: shallow neural model for relation prediction.
Pyke	Pyke: Physical Embedding Model for Knowledge Graphs.
CoKEConfig	Configuration for the CoKE (Contextualized Knowledge Graph Embedding) model.
CoKE	Contextualized Knowledge Graph Embedding (CoKE) model.

Module Contents

class dicee.models.real.**DistMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

DistMult: bilinear diagonal knowledge graph embedding.

Scores a triple `(h, r, t)` as the element-wise product of the head, relation, and tail embeddings summed over the embedding dimension:

```
f(h, r, t) = \sum_i  h_i  \cdot  r_i  \cdot  t_i
```

Simple yet effective baseline; incapable of modelling asymmetric relations.

References

Yang et al., *Embedding Entities and Relations for Learning and Inference in Knowledge Bases*, ICLR 2015. https://arxiv.org/abs/1412.6575

name = 'DistMult'

k_vs_all_score(*emb_h: torch.FloatTensor*, *emb_r: torch.FloatTensor*, *emb_E: torch.FloatTensor*) → torch.FloatTensor

Score a head/relation batch against all entity embeddings.

Computes `(h * r) @ E^T` after applying hidden dropout and normalisation to the element-wise product.

Parameters

- **emb_h** (*torch.FloatTensor*) – Head entity embeddings, shape `(batch_size, embedding_dim)`.
- **emb_r** (*torch.FloatTensor*) – Relation embeddings, shape `(batch_size, embedding_dim)`.

- **emb_E** (*torch.FloatTensor*) – All entity embeddings, shape `(num_entities, embedding_dim)`.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Parameters

- **x** (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.
- **target_entity_idx** (*torch.LongTensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns
Shape `(batch_size, k)` score matrix.

Return type
torch.FloatTensor

score(*h: torch.FloatTensor*, *r: torch.FloatTensor*, *t: torch.FloatTensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` embedding triples.

Parameters

- **h** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **r** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **t** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.real.**TransE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

TransE: translation-based knowledge graph embedding.

Models a relation *r* as a translation in embedding space such that `h + r ≈ t` for a true triple `(h, r, t)`. The score function is defined as:

```
f(h, r, t) = margin - ||h + r - t||_2
```

TransE is effective for 1-to-1 relations but struggles with reflexive, one-to-many, and many-to-one patterns.

References

Bordes et al., *Translating Embeddings for Modeling Multi-relational Data*, NeurIPS 2013. https://proceedings.neurips.cc/paper/2013/file/1cecc7a77928ca8133fa24680a88d2f9-Paper.pdf

name = 'TransE'

margin

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*) → torch.FloatTensor

Score a batch of triples using the TransE margin-distance formula.

Parameters

- **head_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **rel_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **tail_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.

Returns

Shape `(batch_size,)` scores equal to `margin - ||h + r - t||_2`.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Computes `margin - ||h + r - e||_2` for every entity embedding *e*.

Parameters

x (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

class dicee.models.real.**TransH**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

TransH: translation-based knowledge graph embedding on relation-specific hyperplanes.

Addresses TransE's inability to model one-to-many, many-to-one, and many-to-many relations by letting each relation *r* define its own hyperplane. An entity is first projected onto that hyperplane before the translation is applied, so the same entity can be positioned differently depending on the relation. Concretely, for a triple `(h, r, t)` with unit-norm hyperplane normal `r_w` and in-hyperplane translation vector `r_d`:

```
h_r = h - (r_w^T h) * r_w
t_r = t - (r_w^T t) * r_w
f(h, r, t) = margin - ||h_r + r_d - t_r||_2^2
```

`r_w` reuses the inherited `relation_embeddings` table and is normalised to unit norm before every use, since only a unit-norm normal yields a true orthogonal projection onto the hyperplane. `r_d` is a second, relation-specific embedding table analogous to TransE's relation vector.

References

Wang et al., *Knowledge Graph Embedding by Translating on Hyperplanes*, AAAI 2014. https://aaai.org/papers/8870-knowledge-graph-embedding-by-translating-on-hyperplanes/

name = 'TransH'

margin

relation_normal_translations

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Every candidate tail entity must be projected onto the batch row's relation-specific hyperplane, and the hyperplane normal differs per batch row, so unlike TransE this materialises a full `(batch_size, num_entities, embedding_dim)` tensor of projected tail candidates – O(B * E * D) memory, versus TransE's O(B * E).

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Same hyperplane-projection logic as *forward_k_vs_all()*, but restricted to the *k* sampled tail candidates instead of every entity, so it costs `O(B * k * D)` memory instead of `O(B * E * D)`.

Parameters

- **x** (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

- **target_entity_idx** (*torch.LongTensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns
Shape `(batch_size, k)` score matrix.

Return type
torch.FloatTensor

class dicee.models.real.**MuRE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

MuRE: multi-relational graph embedding with relation-specific diagonal scaling, translation, and entity biases.

Scores a triple `(h, r, t)` as:

```
f(h, r, t) = -||R_r \odot h + t_r - t||_2 + b_h + b_t
```

`R_r` is a relation-specific diagonal matrix applied to the head via an element-wise (Hadamard) product; it reuses the inherited `relation_embeddings` table directly, shape `(num_relations, embedding_dim)`. `t_r` is a second, relation-specific translation vector (own embedding table, same shape convention: `(num_relations, embedding_dim)`) analogous to TransE's relation vector. `b_h` and `b_t` are learnable scalar biases indexed per entity (head and tail respectively), not per relation.

References

Balažević et al., *Multi-relational Poincaré Graph Embeddings*, NeurIPS 2019. https://arxiv.org/abs/1905.09791

name = 'MuRE'

relation_translations

b_h

b_t

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Computes `-||R_r \odot h + t_r - e||_2 + b_h + b_e` for every entity embedding *e*.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Parameters

- **x** (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.
- **target_entity_idx** (*torch.LongTensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns
Shape `(batch_size, k)` score matrix.

Return type
torch.FloatTensor

class dicee.models.real.**Shallom**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Shallom: shallow neural model for relation prediction.

Represents each triple as the concatenation of head and tail entity embeddings and feeds it through a two-layer MLP to predict the relation. Designed for the `RelationPrediction` labelling form.

References

Demir et al., *A Shallow Neural Model for Relation Prediction*, ISWC 2021. https://arxiv.org/abs/2101.09090

name = 'Shallom'

shallom

get_embeddings() → Tuple[numpy.ndarray, None]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

forward_k_vs_all(*x*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_triples(*x*) → torch.FloatTensor

Score a batch of triples by looking up relation scores from `forward_k_vs_all`.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.real.**Pyke**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Pyke: Physical Embedding Model for Knowledge Graphs.

Scores a triple `(h, r, t)` based on the average pairwise distance between head-to-relation and relation-to-tail in embedding space:

```
f(h, r, t) = margin - (||h - r||_2 + ||r - t||_2) / 2
```

The model encodes geometric proximity between entities and the relations that connect them.

name = 'Pyke'

dist_func

margin

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

Score a batch of triples using the Pyke distance formula.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.real.**CoKEConfig**

Configuration for the CoKE (Contextualized Knowledge Graph Embedding) model.

block_size
Sequence length for transformer (3 for triples: head, relation, tail)

vocab_size
Total vocabulary size (num_entities + num_relations)

n_layer
Number of transformer layers

n_head
Number of attention heads per layer

n_embd
Embedding dimension (set to match model embedding_dim)

dropout
Dropout rate applied throughout the model

bias
Whether to use bias in linear layers

causal

Whether to use causal masking (False for bidirectional attention)

block_size: int = 3

vocab_size: int = None

n_layer: int = 6

n_head: int = 8

n_embd: int = None

dropout: float = 0.3

bias: bool = True

causal: bool = False

class dicee.models.real.**CoKE**(*args*, *config: CoKEConfig = CoKEConfig()*)

Bases: *dicee.models.base_model.BaseKGE*

Contextualized Knowledge Graph Embedding (CoKE) model. Based on: https://arxiv.org/pdf/1911.02168.

CoKE uses a transformer encoder to learn contextualized representations of entities and relations. For link prediction, it predicts masked elements in (head, relation, tail) triples using bidirectional attention, similar to BERT's masked language modeling approach.

The model creates a sequence [head_emb, relation_emb, mask_emb], adds positional embeddings, and processes it through transformer layers to predict the tail entity.

name = 'CoKE'

config

pos_emb

mask_emb

blocks

ln_f

coke_dropout

forward_k_vs_all(*x: torch.Tensor*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

score(*emb_h*, *emb_r*, *emb_t*)

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

dicee.models.sampled_loss

Reusable loss for positive-first groups of negative samples.

Functions

grouped_adversarial_bce(→ torch.Tensor)	Balance the positive against a weighted negative group (ULTRA convention).

Module Contents

dicee.models.sampled_loss.**grouped_adversarial_bce**(*logits: torch.Tensor*, *targets: torch.Tensor*, *temperature: float*) → torch.Tensor

Balance the positive against a weighted negative group (ULTRA convention).

Positive temperature uses detached softmax(logits / temperature); zero uses uniform negative weights. This intentionally differs from ordinary mean BCE.

dicee.models.static_funcs

Functions

quaternion_mul(→ Tuple[torch.Tensor, torch.Tensor, ...)	Perform quaternion multiplication

Module Contents

dicee.models.static_funcs.**quaternion_mul**(*, *Q_1*, *Q_2*)
→ Tuple[torch.Tensor, torch.Tensor, torch.Tensor, torch.Tensor]

Perform quaternion multiplication :param Q_1: :param Q_2: :return:

dicee.models.transformers

Full definition of a GPT Language Model, all of it in this single file. References: 1) the official GPT-2 TensorFlow implementation released by OpenAI: https://github.com/openai/gpt-2/blob/master/src/model.py 2) huggingface/transformers PyTorch implementation: https://github.com/huggingface/transformers/blob/main/src/transformers/models/gpt2/modeling_gpt2.py

Attributes

logger

Classes

BytE	Base class for all Knowledge Graph Embedding models.
LayerNorm	LayerNorm but with an optional bias. PyTorch doesn't support simply bias=False
SelfAttention	Base class for all neural network modules.
MLP	Base class for all neural network modules.
Block	Base class for all neural network modules.
GPTConfig	
GPT	Base class for all neural network modules.

Module Contents

dicee.models.transformers.**logger**

class dicee.models.transformers.**BytE**(**args*, ***kwargs*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from `BaseKGELightning` and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- `forward_triples()` — score a batch of `(h, r, t)` triples.
- `forward_k_vs_all()` — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'BytE'

config

temperature = 0.5

topk = 2

transformer

lm_head

loss_function(*yhat_batch*, *y_batch*)

Compute the loss between model predictions and targets.

Delegates to `self.loss` which is configured in `BaseKGE.__init__` based on the scoring technique (`BCEWithLogitsLoss` for entity/relation prediction, `CrossEntropyLoss` for classification, `MarginRankingLoss` for `scoring_technique='NegSampleMargin'`) or on `loss_fn` (one of the classes in *dicee.losses.custom_losses*).

For `NegSampleMargin`, *yhat_batch*/*y_batch* still arrive in the flat `NegSample` layout (positives first, followed by `neg_ratio` tiled blocks of negatives, row-aligned by triple). This method splits that flat batch into positive/negative score pairs before delegating to `MarginRankingLoss`.

When `adversarial_temperature` is set (grouped/strict negative sampling), *yhat_batch*/*y_batch* instead arrive as `(batch, 1 + neg_ratio)` groups with the positive in column 0, produced by `GroupedNegativeSamplingDataset`. This delegates to *grouped_adversarial_bce()* instead of `self.loss`.

Some `loss_fn` classes (e.g. `CombinedLSandLR`, `AdaptiveLabelSmoothingLoss`) declare a `current_epoch` parameter on their own `forward` to schedule their behavior over training. `self._loss_needs_epoch` (computed once in `BaseKGE.__init__` by inspecting `self.loss.forward`) says whether that argument should be forwarded to `self.loss` here.

Parameters

- **yhat_batch** (*torch.FloatTensor*) – Model output scores, shape `(batch_size, *)`.
- **y_batch** (*torch.FloatTensor*) – Ground-truth labels of the same shape as *yhat_batch*.
- **current_epoch** (*int, optional*) – Current training epoch, forwarded to `self.loss` only when it declares a `current_epoch` parameter. Callers that never use an epoch-aware `loss_fn` may omit this.

Returns
Scalar loss value.

Return type
torch.FloatTensor

forward(*x: torch.LongTensor*)

Parameters
x (*B by T tensor*)

generate(*idx*, *max_new_tokens*, *temperature=1.0*, *top_k=None*)

Take a conditioning sequence of indices idx (LongTensor of shape (b,t)) and complete the sequence max_new_tokens times, feeding the predictions back into the model each time. Most likely you'll want to make sure to be in model.eval() mode of operation for this.

training_step(*batch*, *batch_idx=None*)

Execute one optimisation step for the given mini-batch.

Handles two- and three-element batches produced by the different dataset classes (`KvsAll` / `NegSample` vs. `KvsSample`).

Parameters

- **batch** (*tuple*) – `(x, y)` for standard scoring, or `(x, y_select, y)` for sample-based labelling.
- **batch_idx** (*int, optional*) – Index of the current batch (unused, kept for Lightning API compat).

Returns
Scalar loss value for this batch.

Return type
torch.FloatTensor

class dicee.models.transformers.**LayerNorm**(*ndim*, *bias*)

Bases: torch.nn.Module

LayerNorm but with an optional bias. PyTorch doesn't support simply bias=False

weight

bias

forward(*input*)

class dicee.models.transformers.**SelfAttention**(*config*)

Bases: torch.nn.Module

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

c_attn

c_proj

attn_dropout

resid_dropout

n_head

n_embd

dropout

causal

flash = True

forward(*x*)

class dicee.models.transformers.**MLP**(*config*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

c_fc

gelu

c_proj

dropout

forward(*x*)

class dicee.models.transformers.**Block**(*config*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

ln_1

attn

ln_2

mlp

forward(*x*)

class dicee.models.transformers.**GPTConfig**

block_size: int = 1024

vocab_size: int = 50304

n_layer: int = 12

n_head: int = 12

n_embd: int = 768

dropout: float = 0.0

bias: bool = False

causal: bool = True

class dicee.models.transformers.**GPT**(*config*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

config

transformer

lm_head

get_num_params(*non_embedding=True*)

Return the number of parameters in the model. For non-embedding count (default), the position embeddings get subtracted. The token embeddings would too, except due to the parameter sharing these params are actually used as weights in the final layer, so we include them.

forward(*idx*, *targets=None*)

crop_block_size(*block_size*)

classmethod from_pretrained(*model_type*, *override_args=None*)

configure_optimizers(*weight_decay*, *learning_rate*, *betas*, *device_type*)

estimate_mfu(*fwdbwd_per_iter*, *dt*)

estimate model flops utilization (MFU) in units of A100 bfloat16 peak FLOPS

dicee.models.trix

Pure PyTorch TRIX entity and relation prediction.

Reimplementation of Zhang et al., https://arxiv.org/abs/2502.19512, verified against https://github.com/yuchengz99/TRIX at UPSTREAM_COMMIT. The module names, update schedule, binary entity-labelled relation edges, and fused convolution direction follow the released code/checkpoints (see docs/trix.md).

Attributes

UPSTREAM_COMMIT	
INTERACTIONS	

Classes

EntityReasoner	Base class for all neural network modules.
RelationReasoner	Base class for all neural network modules.
TRIXBase	Shared graph lifecycle and entity-scoring interfaces for graph foundation models.
TRIX	Entity predictor compatible with official `entity_prediction.pth`.
TRIXRelation	Relation predictor compatible with official `relation_prediction.pth`.

Functions

build_relation_graph(edge_index, edge_type, ...)	Sparse (relation, relation, shared entity) edges in official role order.

Module Contents

dicee.models.trix.**UPSTREAM_COMMIT = '7596e14eefefe89e61396205a0550172cadeddb0'**

dicee.models.trix.**INTERACTIONS = ('hh', 'ht', 'th', 'tt')**

dicee.models.trix.**build_relation_graph**(*edge_index*, *edge_type*, *num_entities*, *num_relations*)

Sparse (relation, relation, shared entity) edges in official role order.

Repeated incidences have binary support. hh/tt exclude equal relations; ht/th include them. Distinct shared entities remain distinct edges. No dense entity-by-relation-by-relation tensor is materialized.

class dicee.models.trix.**EntityReasoner**(*dim*, *num_layers*, *output_dim=1*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

layers

mlp

features(*edges*, *num_entities*, *relations*, *heads*, *rels*, *states=None*, *tails=None*, *candidates=None*, *split=False*)

class dicee.models.trix.**RelationReasoner**(*dim*, *num_layers*, *entity_feedback=False*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F
```

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```
class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

step(*index*, *states*, *boundary*, *graph*, *entities*, *constant_relations=None*)

class dicee.models.trix.**TRIXBase**(*args*)

Bases: *dicee.models.graph_model.GraphKGE*

Shared graph lifecycle and entity-scoring interfaces for graph foundation models.

config_prefix = 'trix'

graph_filename = 'trix_graph.pt'

checkpoint_hint = 'use the matching entity/relation checkpoint with trix_dim=32; all keys and shapes must match'

dim

property relation_graph

class dicee.models.trix.**TRIX**(*args*)

Bases: *TRIXBase*

Entity predictor compatible with official `entity_prediction.pth`.

DICE order is (head, relation, tail). Head corruption retains the original relation seed, then uses the inverse relation for both entity reasoners. Explicit reciprocal DICE queries use that same convention.

name = 'TRIX'

deterministic_inference = True

relation_model

entity_model_1

entity_model_2

relation_cache_mb

clear_inference_cache()

Drop derived representations when graph, weights or device change.

class dicee.models.trix.**TRIXRelation**(*args*)

Bases: *TRIXBase*, *dicee.models.graph_model.RelationGraphKGE*

Relation predictor compatible with official `relation_prediction.pth`.

Pairs are (head, tail); triples and grouped relation corruptions use DICE's (head, relation, tail) order. One reasoning pass scores every relation for a given pair, including external inverse IDs when present in the vocabulary.

name = 'TRIXRelation'

relation_model

entity_model

mlp

dicee.models.ultra

Pure PyTorch ULTRA, compatible with DeepGraphLearning/ULTRA link predictors.

Architecture and graph construction adapted from https://github.com/DeepGraphLearning/ULTRA at commit 427966ad8ed60420eef034063d44f3153addff90. Reference fixtures and the grouped sampling and adversarial loss implementations also follow this upstream work. Graphs are runtime context, excluded from the transferable state dictionary.

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Attributes

UPSTREAM_COMMIT

Classes

RelationalConv	ULTRA's DistMult/sum convolution, including the boundary message.
RelNBFNet	Base class for all neural network modules.
EntityNBFNet	Base class for all neural network modules.
ULTRA	Pure PyTorch ULTRA with the official state dictionary and DICE triple order.

Functions

build_relation_graph(edge_index, edge_type, ...)	Binary support of the hh, tt, ht, th incidence products, in upstream order.

Module Contents

dicee.models.ultra.**UPSTREAM_COMMIT = '427966ad8ed60420eef034063d44f3153addff90'**

class dicee.models.ultra.**RelationalConv**(*dim*, *project_relations=False*)

Bases: torch.nn.Module

ULTRA's DistMult/sum convolution, including the boundary message.

inference_backend = 'auto'

inference_compile = False

layer_norm

linear

forward(*states*, *boundary*, *edge_index*, *edge_type*, *relations=None*, *, *projected=None*, *residual=False*, *constant_relations=None*)

class dicee.models.ultra.**RelNBFNet**(*dim=64*, *num_layers=6*)

Bases: torch.nn.Module

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

layers

dim = 64

forward(*edge_index*, *edge_type*, *num_relations*, *query*)

class dicee.models.ultra.**EntityNBFNet**(*dim=64*, *num_layers=6*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

layers

mlp

forward(*edge_index*, *edge_type*, *num_entities*, *relations*, *heads*, *rels*, *tails*, *projections=None*)

dicee.models.ultra.**build_relation_graph**(*edge_index*, *edge_type*, *num_entities*, *num_relations*)

Binary support of the hh, tt, ht, th incidence products, in upstream order.

class dicee.models.ultra.**ULTRA**(*args*)

Bases: *dicee.models.graph_model.GraphKGE*

Pure PyTorch ULTRA with the official state dictionary and DICE triple order.

name = 'ULTRA'

config_prefix = 'ultra'

graph_filename = 'ultra_graph.pt'

checkpoint_hint = 'official ULTRA requires ultra_dim=64, ultra_num_layers=6; all keys and shapes must match'

deterministic_inference = True

dim

num_layers

relation_model

entity_model

relation_cache_mb

projection_cache_mb

clear_inference_cache()

Drop derived representations when graph, weights or device change.

Attributes

logger
logger
logger

Classes

ACLS	Adaptive and Conditional Label Smoothing-style calibration loss,
AdaptiveLabelRelaxationLoss	*LabelRelaxationLoss* with an `alpha` that self-adjusts from
AdaptiveLabelSmoothingLoss	KL-divergence label smoothing with a smoothing factor that self-adjusts
AggregatedLSandLR	Fixed 0.4/0.6 weighted average of *LabelSmoothingLoss* and

continues on next page

Table 69 – continued from previous page

CombinedAdaptiveLSandAdaptiveLR	Switches from *AdaptiveLabelSmoothingLoss* to
CombinedLSandLR	Switches from *LabelSmoothingLoss* to *LabelRelaxationLoss*
ConfidenceBasedAdaptiveLabelRelaxationLoss	*LabelRelaxationLoss* where `alpha` shrinks as the model's
DefaultBCELoss	Plain `BCEWithLogitsLoss`, selectable via `loss_fn="BCELoss"`.
LabelRelaxationLoss	Label relaxation loss, selectable via `loss_fn="LRLoss"`.
LabelSmoothingLoss	Fixed-rate label-smoothing loss, selectable via `loss_fn="LS"`
WeightedBCELoss	BCE weighted by each prediction's own confidence.
ADOPT	ADOPT Optimizer.
BaseKGELightning	Thin PyTorch Lightning wrapper shared by all KGE models.
BaseKGE	Base class for all Knowledge Graph Embedding models.
IdentityClass	No-op normalisation / dropout placeholder.
Block	Base class for all neural network modules.
BaseKGE	Base class for all Knowledge Graph Embedding models.
DistMult	DistMult: bilinear diagonal knowledge graph embedding.
TransE	TransE: translation-based knowledge graph embedding.
TransH	TransH: translation-based knowledge graph embedding on relation-specific hyperplanes.
MuRE	MuRE: multi-relational graph embedding with relation-specific diagonal scaling, translation, and entity biases.
Shallom	Shallom: shallow neural model for relation prediction.
Pyke	Pyke: Physical Embedding Model for Knowledge Graphs.
CoKEConfig	Configuration for the CoKE (Contextualized Knowledge Graph Embedding) model.
CoKE	Contextualized Knowledge Graph Embedding (CoKE) model.
BaseKGE	Base class for all Knowledge Graph Embedding models.
ConEx	Convolutional ComplEx Knowledge Graph Embeddings
AConEx	Additive Convolutional ComplEx Knowledge Graph Embeddings
ComplEx	Base class for all Knowledge Graph Embedding models.
RotatE	RotatE: knowledge graph embedding by relational rotation in complex space.
BaseKGE	Base class for all Knowledge Graph Embedding models.
IdentityClass	No-op normalisation / dropout placeholder.
QMult	Base class for all Knowledge Graph Embedding models.
ConvQ	Convolutional Quaternion Knowledge Graph Embeddings
AConvQ	Additive Convolutional Quaternion Knowledge Graph Embeddings
BaseKGE	Base class for all Knowledge Graph Embedding models.
IdentityClass	No-op normalisation / dropout placeholder.
OMult	Base class for all Knowledge Graph Embedding models.
ConvO	Base class for all Knowledge Graph Embedding models.
AConvO	Additive Convolutional Octonion Knowledge Graph Embeddings
Keci	Keci: Knowledge Graph Embedding via Clifford Algebra.

continues on next page

Table 69 – continued from previous page

CKeci	Without learning dimension scaling
DeCaL	Base class for all Knowledge Graph Embedding models.
KeciTransformer	Keci with Transformer architecture.
BaseKGE	Base class for all Knowledge Graph Embedding models.
PykeenKGE	A class for using knowledge graph embedding models implemented in Pykeen
BaseKGE	Base class for all Knowledge Graph Embedding models.
FMult	Learning Knowledge Neural Graphs
GFMult	Learning Knowledge Neural Graphs
FMult2	Learning Knowledge Neural Graphs
LFMult1	Embedding with trigonometric functions. We represent all entities and relations in the complex number space as:
LFMult	Embedding with polynomial functions. We represent all entities and relations in the polynomial space as:
DualE	Dual Quaternion Knowledge Graph Embeddings (https://ojs.aaai.org/index.php/AAAI/article/download/16850/16657)
ULTRA	Pure PyTorch ULTRA with the official state dictionary and DICE triple order.
TRIX	Entity predictor compatible with official `entity_prediction.pth`.
TRIXRelation	Relation predictor compatible with official `relation_prediction.pth`.
Flock	Entity-prediction Flock with official `flock_entity.pth` parameters.
FlockRelation	Relation-prediction Flock with official `flock_relation.pth` parameters.

Functions

quaternion_mul(→ Tuple[torch.Tensor, torch.Tensor, ...)	Perform quaternion multiplication
quaternion_mul_with_unit_norm(*, Q_1, Q_2)	
octonion_mul(*, O_1, O_2)	
octonion_mul_norm(*, O_1, O_2)	

Package Contents

class dicee.models.**ACLS**(*pos_lambda: float = 1.0*, *neg_lambda: float = 0.1*, *alpha: float = 0.1*, *margin: float = 10.0*, *num_classes: int = 200*, *ignore_index: int = -100*)

Bases: `torch.nn.Module`

Adaptive and Conditional Label Smoothing-style calibration loss, selectable via `loss_fn="ACLS"`.

Adds a margin-based confidence regularizer to cross-entropy: for the predicted (argmax) class, logits above `margin` are penalized quadratically (discourages over-confidence on the top class), and for every other class, logits within `margin` of the top logit are also penalized quadratically (discourages under-separation from the runner-up classes). The two penalties are weighted by `pos_lambda`/`neg_lambda` and added to `CrossEntropyLoss` scaled by `alpha`.

Warning

`nn.CrossEntropyLoss` expects `targets` as class indices (or a proper probability distribution per row); KGE's `KvsAll` labels are multi-hot (possibly several true tails per row), which is not an equivalent input. `num_classes` and `ignore_index` are currently unused in `forward`/`get_reg`.

Parameters

- **pos_lambda** (*float*) – Weight of the over-confidence penalty on the predicted class.
- **neg_lambda** (*float*) – Weight of the under-separation penalty on the other classes.
- **alpha** (*float*) – Weight of the regularization term relative to cross-entropy.
- **margin** (*float*) – Logit margin used by both penalty terms.
- **num_classes** (*int*) – Currently unused.
- **ignore_index** (*int*) – Currently unused.

pos_lambda = 1.0

neg_lambda = 0.1

alpha = 0.1

margin = 10.0

num_classes = 200

ignore_index = -100

cross_entropy

property names

get_reg(*inputs*, *targets*)

forward(*inputs*, *targets*)

class dicee.models.**AdaptiveLabelRelaxationLoss**(*min_alpha=0.01*, *max_alpha=0.2*, *alpha_step=0.01*, *initial_alpha=0.1*)

Bases: `torch.nn.Module`

LabelRelaxationLoss with an `alpha` that self-adjusts from batch to batch, selectable via `loss_fn="AdaptiveLabelRelaxationLoss"`.

Computes the label-relaxation loss once under `torch.no_grad()` with the current `alpha` to decide the adjustment (grow toward `max_alpha` if the loss increased since the previous call, shrink toward `min_alpha` if it decreased), then recomputes the loss with the updated `alpha` for the actual gradient-carrying return value. Like *AdaptiveLabelSmoothingLoss*, this module is stateful (`prev_loss`, `alpha` persist across calls) - use a fresh instance per training run.

Parameters

- **min_alpha** (*float*) – Lower bound the adaptive `alpha` can decay to.
- **max_alpha** (*float*) – Upper bound the adaptive `alpha` can grow to.
- **alpha_step** (*float*) – Adjustment applied to `alpha` after each call.
- **initial_alpha** (*float*) – Starting value before any adjustment has occurred.

min_alpha = 0.01

max_alpha = 0.2

alpha_step = 0.01

alpha = 0.1

prev_loss = None

eps = 1e-14

gz_threshold = 0.1

forward(*pred*, *target*)

class dicee.models.**AdaptiveLabelSmoothingLoss**(*min_smoothing_factor=0.01*, *max_smoothing_factor=0.2*, *smoothing_factor_step=0.01*, *initial_smoothing_factor=0.1*)

Bases: `torch.nn.Module`

KL-divergence label smoothing with a smoothing factor that self-adjusts from batch to batch, selectable via `loss_fn="AdaptiveLabelSmoothingLoss"`.

Converts `logits` to log-probabilities and compares against a smoothed target distribution (mass `smoothing_factor` redistributed uniformly over the non-target classes) via `KL(pred || smoothed_target)`. After each call, `smoothing_factor` moves one `smoothing_factor_step` toward `max_smoothing_factor` if the loss increased since the previous call, or toward `min_smoothing_factor` if it decreased — i.e. it backs off smoothing while the model is improving and leans into it when training stalls. `current_epoch` is accepted for interface consistency with the other `loss_fn` options but is currently unused.

This module is stateful (`prev_loss`, `smoothing_factor` persist across calls), so a fresh instance should be used per training run.

Parameters

- **min_smoothing_factor** (*float*) – Lower bound the adaptive smoothing factor can decay to.
- **max_smoothing_factor** (*float*) – Upper bound the adaptive smoothing factor can grow to.
- **smoothing_factor_step** (*float*) – Adjustment applied to the smoothing factor after each call.
- **initial_smoothing_factor** (*float*) – Starting value before any adjustment has occurred.

min_smoothing_factor = 0.01

max_smoothing_factor = 0.2

smoothing_factor_step = 0.01

smoothing_factor = 0.1

prev_loss = None

eps = 1e-14

forward(*logits*, *target*, *current_epoch*)

class dicee.models.**AggregatedLSandLR**(*smoothness_ratio=0.1*, *alpha=0.1*)

Bases: `torch.nn.Module`

Fixed 0.4/0.6 weighted average of *LabelSmoothingLoss* and *LabelRelaxationLoss*, selectable via `loss_fn="AggregatedLSandLR"`.

Unlike *CombinedLSandLR*, both terms are computed every call and blended (`0.4 * smoothing + 0.6 * relaxation`) rather than switched between by epoch.

Parameters

- **smoothness_ratio** (*float*) – Forwarded to the inner *LabelSmoothingLoss*.
- **alpha** (*float*) – Forwarded to the inner *LabelRelaxationLoss*.

smoothness_ratio = 0.1

alpha = 0.1

forward(*pred*, *target*, *current_epoch*)

class dicee.models.**CombinedAdaptiveLSandAdaptiveLR**

Bases: `torch.nn.Module`

Switches from *AdaptiveLabelSmoothingLoss* to *AdaptiveLabelRelaxationLoss* after epoch 100, selectable via `loss_fn="CombinedAdaptiveLSandAdaptiveLR"`.

Both inner losses use their default hyperparameters; there is currently no way to configure them through this wrapper.

adaptive_label_smoothing

adaptive_label_relaxation

criterion = ''

forward(*pred*, *target*, *current_epoch*)

class dicee.models.**CombinedLSandLR**(*smoothness_ratio=0.0*, *alpha=0.0*)

Bases: `torch.nn.Module`

Switches from *LabelSmoothingLoss* to *LabelRelaxationLoss* after epoch 20, selectable via `loss_fn="CombinedLSandLR"`.

Intended to smooth early training then relax the target distribution later on.

Parameters

- **smoothness_ratio** (*float*) – Forwarded to *LabelSmoothingLoss* for the first 20 epochs.
- **alpha** (*float*) – Forwarded to *LabelRelaxationLoss* from epoch 20 onward.

smoothness_ratio = 0.0

alpha = 0.0

forward(*pred*, *target*, *current_epoch*)

class dicee.models.**ConfidenceBasedAdaptiveLabelRelaxationLoss**(*alpha=0.1*)

Bases: torch.nn.Module

LabelRelaxationLoss where alpha shrinks as the model's own average prediction confidence grows, selectable via loss_fn="ConfidenceBasedAdaptiveLabelRelaxationLoss".

Each call rescales alpha by (1 - mean(pred)) before computing the loss. Unlike *AdaptiveLabelRelaxationLoss*, this adjustment is one-directional and unbounded below: alpha only ever decays toward 0 as confidence rises, with no mechanism to grow back if confidence later drops. current_epoch is accepted for interface consistency with the other loss_fn options but is currently unused. Stateful (alpha persists and mutates across calls) - use a fresh instance per training run.

Parameters

alpha (*float*) – Initial relaxation strength before any confidence-based decay.

alpha = 0.1

gz_threshold = 0.1

eps = 1e-14

forward(*pred*, *target*, *current_epoch*)

class dicee.models.**DefaultBCELoss**

Bases: torch.nn.Module

Plain BCEWithLogitsLoss, selectable via loss_fn="BCELoss".

Equivalent to the framework's own default entity/relation-prediction loss (see BaseKGE.__init__); provided as an explicit loss_fn option for parity with the other choices in this module.

forward(*pred*, *target*)

class dicee.models.**LabelRelaxationLoss**(*alpha=0.0*)

Bases: torch.nn.Module

Label relaxation loss, selectable via loss_fn="LRLoss".

Based on "From Label Smoothing to Label Relaxation" (Lienen & Hullermeier, AAAI 2021): instead of smoothing toward one fixed soft target, it constructs a *credal set* of distributions that assign at least 1 - alpha probability to the target class(es) and measures the KL divergence to the closest point in that set, rather than to a single fixed distribution. Predictions that are already confident enough (probability mass on the target above 1 - alpha) incur zero loss.

Parameters

alpha (*float*) – Relaxation strength in [0, 1); 0 recovers ordinary cross-entropy-style behavior (no slack), larger values tolerate more probability mass elsewhere before penalizing.

alpha = 0.0

gz_threshold = 0.1

eps = 1e-14

forward(*pred*, *target*)

class dicee.models.**LabelSmoothingLoss**(*smoothness_ratio=0.0*)

Bases: torch.nn.Module

Fixed-rate label-smoothing loss, selectable via loss_fn="LS" (smoothness_ratio comes from the label_smoothing_rate config option).

Converts `pred` (logits) to log-probabilities and compares against a smoothed target distribution (mass `smoothness_ratio` redistributed uniformly over the non-target classes) via `KL(pred || smoothed_target)` — the same fixed-rate variant of *AdaptiveLabelSmoothingLoss*'s per-call smoothing, without the batch-to-batch adaptation.

This is independent of the (separately, correctly implemented) dataset-level smoothing controlled by the same `label_smoothing_rate` option (`dicee/dataset_classes/_label_based.py`, `_negative_sampling.py`), which bakes smoothing into the training targets regardless of `loss_fn`.

Parameters

smoothness_ratio (*float*) – Fraction of probability mass redistributed from the target class(es) to the non-target classes; `0` recovers a KL-divergence loss with no smoothing.

smoothness_ratio = 0.0

forward(*pred*, *target*)

class dicee.models.**WeightedBCELoss**

Bases: `torch.nn.Module`

BCE weighted by each prediction's own confidence.

Down-weights low-confidence predictions (`sigmoid(pred)` near 0.5) and up-weights confident ones, clamped to `[0.5, 1.0]`. Weights are detached, so they don't receive gradients themselves. `current_epoch` is accepted for interface consistency with the other `loss_fn` options but is currently unused.

forward(*pred*, *target*, *current_epoch*)

class dicee.models.**ADOPT**(*params: torch.optim.optimizer.ParamsT*, *lr: float | torch.Tensor = 0.001*, *betas: Tuple[float, float] = (0.9, 0.9999)*, *eps: float = 1e-06*, *clip_lambda: Callable[[int], float] | None = lambda step: ...*, *weight_decay: float = 0.0*, *decouple: bool = False*, ***, *foreach: bool | None = None*, *maximize: bool = False*, *capturable: bool = False*, *differentiable: bool = False*, *fused: bool | None = None*)

Bases: `torch.optim.optimizer.Optimizer`

ADOPT Optimizer.

ADOPT is an adaptive learning rate optimization algorithm that combines momentum-based updates with adaptive per-parameter learning rates. It uses exponential moving averages of gradients and squared gradients, with gradient clipping for stability.

The algorithm performs the following key operations: 1. Normalizes gradients by the square root of the second moment estimate 2. Applies optional gradient clipping based on the training step 3. Updates parameters using momentum-smoothed normalized gradients 4. Supports decoupled weight decay (AdamW-style) or L2 regularization

Mathematical formulation:

$m_t = \beta_1 * m_\{t-1\} + (1 - \beta_1) * clip(g_t / \sqrt{(v_t)})$ $v_t = \beta_2 * v_\{t-1\} + (1 - \beta_2) * g_t^2$ $\theta_t = \theta_\{t-1\} - \alpha * m_t$

where:

- θ_t: parameter at step t
- g_t: gradient at step t
- m_t: first moment estimate (momentum)
- v_t: second moment estimate (variance)
- α: learning rate
- β_1, β_2: exponential decay rates

- clip(): optional gradient clipping function

Reference:

Original implementation: https://github.com/iShohei220/adopt

Parameters

- **params** (*ParamsT*) – Iterable of parameters to optimize or dicts defining parameter groups.
- **lr** (*float or Tensor, optional*) – Learning rate. Can be a float or 1-element Tensor. Default: 1e-3
- **betas** (*Tuple[float, float], optional*) – Coefficients (β_1, β_2) for computing running averages of gradient and its square. β_1 controls momentum, β_2 controls variance. Default: (0.9, 0.9999)
- **eps** (*float, optional*) – Term added to denominator to improve numerical stability. Default: 1e-6
- **clip_lambda** (*Callable[[int], float], optional*) – Function that takes the step number and returns the gradient clipping threshold. Common choices: - lambda step: step**0.25 (default, gradually increases clipping threshold) - lambda step: 1.0 (constant clipping) - None (no clipping) Default: lambda step: step**0.25
- **weight_decay** (*float, optional*) – Weight decay coefficient (L2 penalty). Default: 0.0
- **decouple** (*bool, optional*) – If True, uses decoupled weight decay (AdamW-style), applying weight decay directly to parameters. If False, adds weight decay to gradients (L2 regularization). Default: False
- **foreach** (*bool, optional*) – If True, uses the faster foreach implementation for multi-tensor operations. Default: None (auto-select)
- **maximize** (*bool, optional*) – If True, maximizes parameters instead of minimizing. Useful for reinforcement learning. Default: False
- **capturable** (*bool, optional*) – If True, the optimizer is safe to capture in a CUDA graph. Requires learning rate as Tensor. Default: False
- **differentiable** (*bool, optional*) – If True, the optimization step can be differentiated. Useful for meta-learning. Default: False
- **fused** (*bool, optional*) – If True, uses fused kernel implementation (currently not supported). Default: None

Raises

- **ValueError** – If learning rate, epsilon, betas, or weight_decay are invalid.
- **RuntimeError** – If fused is enabled (not currently supported).
- **RuntimeError** – If lr is a Tensor with foreach=True and capturable=False.

Example

```
>>> # Basic usage
>>> optimizer = ADOPT(model.parameters(), lr=0.001)
>>> optimizer.zero_grad()
>>> loss.backward()
>>> optimizer.step()
```

```
>>> # With decoupled weight decay
>>> optimizer = ADOPT(model.parameters(), lr=0.001, weight_decay=0.01,␣
↪decouple=True)
```

```
>>> # Custom gradient clipping
>>> optimizer = ADOPT(model.parameters(), clip_lambda=lambda step: max(1.0,␣
↪step**0.5))
```

Note

- For most use cases, the default hyperparameters work well
- Consider using decouple=True for better generalization (similar to AdamW)
- The clip_lambda function helps stabilize training in early steps

clip_lambda

__setstate__(*state*)

Restore optimizer state from a checkpoint.

This method handles backward compatibility when loading optimizer state from older versions. It ensures all required fields are present with default values and properly converts step counters to tensors if needed.

Key responsibilities: 1. Set default values for newly added hyperparameters 2. Convert old-style scalar step counters to tensor format 3. Place step tensors on appropriate devices based on capturable/fused modes

Parameters

state (*dict*) – Optimizer state dictionary (typically from torch.load()).

Note

- This enables loading checkpoints saved with older ADOPT versions
- Step counters are converted to appropriate device/dtype for compatibility
- Capturable and fused modes require step tensors on parameter devices

step(*closure=None*)

Perform a single optimization step.

This method executes one iteration of the ADOPT optimization algorithm across all parameter groups. It orchestrates the following workflow:

1. Optionally evaluates a closure to recompute the loss (useful for algorithms like LBFGS or when loss needs multiple evaluations)
2. For each parameter group: - Collects parameters with gradients and their associated state - Extracts hyperparameters (betas, learning rate, etc.) - Calls the functional adopt() API to perform the actual update
3. Returns the loss value if a closure was provided

The functional API (adopt()) handles three execution modes: - Single-tensor: Updates one parameter at a time (default, JIT-compatible) - Multi-tensor (foreach): Batches operations for better performance - Fused: Uses fused CUDA kernels (not yet implemented)

Gradient scaling support: This method is compatible with automatic mixed precision (AMP) training. It can access grad_scale and found_inf attributes for gradient unscaling and inf/nan detection when used with GradScaler.

Parameters

closure (*Callable, optional*) – A callable that reevaluates the model and returns the loss. The closure should: - Enable gradients (torch.enable_grad()) - Compute forward pass - Compute loss - Compute backward pass - Return the loss value Example: lambda: (loss := model(x), loss.backward(), loss)[-1] Default: None

Returns

The loss value returned by the closure, or None if no
closure was provided.

Return type
Optional[Tensor]

Example

```
>>> # Standard usage
>>> loss = criterion(model(input), target)
>>> loss.backward()
>>> optimizer.step()
```

```
>>> # With closure (e.g., for line search)
>>> def closure():
...     optimizer.zero_grad()
...     output = model(input)
...     loss = criterion(output, target)
...     loss.backward()
...     return loss
>>> loss = optimizer.step(closure)
```

Note

- Call zero_grad() before computing gradients for the next step
- CUDA graph capture is checked for safety when capturable=True
- The method is thread-safe for different parameter groups

dicee.models.**logger**

class dicee.models.**BaseKGELightning**(**args*, ***kwargs*)

Bases: lightning.LightningModule

Thin PyTorch Lightning wrapper shared by all KGE models.

Provides the standard Lightning training loop hooks (training_step, on_train_epoch_end, configure_optimizers) as well as a helper for reporting model size. All concrete KGE models should extend *BaseKGE* rather than this class directly.

args: Dict[str, Any]

loss: torch.nn.Module

training_step_outputs = []

mem_of_model() → Dict

Size of model in MB and number of params

training_step(*batch*, *batch_idx=None*)

Execute one optimisation step for the given mini-batch.

Handles two- and three-element batches produced by the different dataset classes (`KvsAll` / `NegSample` vs. `KvsSample`).

Parameters

- **batch** (*tuple*) – `(x, y)` for standard scoring, or `(x, y_select, y)` for sample-based labelling.
- **batch_idx** (*int, optional*) – Index of the current batch (unused, kept for Lightning API compat).

Returns

Scalar loss value for this batch.

Return type

torch.FloatTensor

loss_function(*yhat_batch: torch.Tensor*, *y_batch: torch.Tensor*, *current_epoch: int | None = None*) → torch.Tensor

Compute the loss between model predictions and targets.

Delegates to `self.loss` which is configured in `BaseKGE.__init__` based on the scoring technique (`BCEWithLogitsLoss` for entity/relation prediction, `CrossEntropyLoss` for classification, `MarginRankingLoss` for `scoring_technique='NegSampleMargin'`) or on `loss_fn` (one of the classes in *dicee.losses.custom_losses*).

For `NegSampleMargin`, *yhat_batch*/*y_batch* still arrive in the flat `NegSample` layout (positives first, followed by `neg_ratio` tiled blocks of negatives, row-aligned by triple). This method splits that flat batch into positive/negative score pairs before delegating to `MarginRankingLoss`.

When `adversarial_temperature` is set (grouped/strict negative sampling), *yhat_batch*/*y_batch* instead arrive as `(batch, 1 + neg_ratio)` groups with the positive in column 0, produced by `GroupedNegativeSamplingDataset`. This delegates to *grouped_adversarial_bce()* instead of `self.loss`.

Some `loss_fn` classes (e.g. `CombinedLSandLR`, `AdaptiveLabelSmoothingLoss`) declare a `current_epoch` parameter on their own `forward` to schedule their behavior over training. `self._loss_needs_epoch` (computed once in `BaseKGE.__init__` by inspecting `self.loss.forward`) says whether that argument should be forwarded to `self.loss` here.

Parameters

- **yhat_batch** (*torch.FloatTensor*) – Model output scores, shape `(batch_size, *)`.
- **y_batch** (*torch.FloatTensor*) – Ground-truth labels of the same shape as *yhat_batch*.
- **current_epoch** (*int, optional*) – Current training epoch, forwarded to `self.loss` only when it declares a `current_epoch` parameter. Callers that never use an epoch-aware `loss_fn` may omit this.

Returns

Scalar loss value.

Return type
torch.FloatTensor

on_train_epoch_end(**args*, ***kwargs*)

Called in the training loop at the very end of the epoch.

To access all batch outputs at the end of the epoch, you can cache step outputs as an attribute of the `LightningModule` and access them in this hook:

```
class MyLightningModule(L.LightningModule):
    def __init__(self):
        super().__init__()
        self.training_step_outputs = []

    def training_step(self):
        loss = ...
        self.training_step_outputs.append(loss)
        return loss

    def on_train_epoch_end(self):
        # do something with all training_step outputs, for example:
        epoch_mean = torch.stack(self.training_step_outputs).mean()
        self.log("training_epoch_mean", epoch_mean)
        # free up the memory
        self.training_step_outputs.clear()
```

test_epoch_end(*outputs: List[Any]*)

test_dataloader() → None

An iterable or collection of iterables specifying test samples.

For more information about multiple dataloaders, see this section.

For data processing use the following pattern:

- download in `prepare_data()`
- process and split in `setup()`

However, the above are only necessary for distributed processing.

> **Warning**
>
> do not assign state in prepare_data

- `test()`
- `prepare_data()`
- `setup()`

> **Note**
>
> Lightning tries to add the correct sampler for distributed and arbitrary hardware. There is no need to set it yourself.

> **Note**
>
> If you don't need a test dataset and a `test_step()`, you don't need to implement this method.

`val_dataloader``()` → None

An iterable or collection of iterables specifying validation samples.

For more information about multiple dataloaders, see this section.

The dataloader you return will not be reloaded unless you set **:paramref:`~lightning.pytorch.trainer.trainer.Trainer.reload_dataloaders_every_n_epochs`** to a positive integer.

It's recommended that all data downloads and preparation happen in `prepare_data()`.

- `fit()`
- `validate()`
- `prepare_data()`
- `setup()`

> **Note**
>
> Lightning tries to add the correct sampler for distributed and arbitrary hardware There is no need to set it yourself.

> **Note**
>
> If you don't need a validation dataset and a `validation_step()`, you don't need to implement this method.

`predict_dataloader``()` → None

An iterable or collection of iterables specifying prediction samples.

For more information about multiple dataloaders, see this section.

It's recommended that all data downloads and preparation happen in `prepare_data()`.

- `predict()`
- `prepare_data()`
- `setup()`

> **Note**
>
> Lightning tries to add the correct sampler for distributed and arbitrary hardware There is no need to set it yourself.

Returns

A `torch.utils.data.DataLoader` or a sequence of them specifying prediction samples.

train_dataloader() → None

An iterable or collection of iterables specifying training samples.

For more information about multiple dataloaders, see this section.

The dataloader you return will not be reloaded unless you set **:paramref:`~lightning.pytorch.trainer.trainer.Trainer.reload_dataloaders_every_n_epochs`** to a positive integer.

For data processing use the following pattern:

- download in `prepare_data()`
- process and split in `setup()`

However, the above are only necessary for distributed processing.

> **Warning**
>
> do not assign state in prepare_data

- `fit()`
- `prepare_data()`
- `setup()`

> **Note**
>
> Lightning tries to add the correct sampler for distributed and arbitrary hardware. There is no need to set it yourself.

configure_optimizers(*parameters=None*)

Instantiate and return the optimiser for training.

The optimiser type is taken from `self.optimizer_name` which is set in *BaseKGE.init_params_with_sanity_checking()* from the `--optim` argument. Supported values: `'SGD'`, `'Adam'`, `'Adopt'`, `'AdamW'`, `'NAdam'`, `'Adagrad'`, `'ASGD'`, `'Muon'`.

Parameters

parameters (*iterable, optional*) – Model parameters to optimise. Defaults to `self.parameters()` when `None`.

Returns

The configured optimiser instance.

Return type

torch.optim.Optimizer

class dicee.models.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns

Shape `(batch_size, num_bpe_entities)` score matrix.

Return type

torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where k is the number of sampled target entities.

Return type
torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters
idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns
head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters
indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns
head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where T is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type
torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of T subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns

head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type

torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

class dicee.models.**IdentityClass**(*args=None*)

Bases: `torch.nn.Module`

No-op normalisation / dropout placeholder.

Used whenever no normalisation layer is requested (`--normalization None`). All inputs are returned unchanged so that the rest of the model code does not need conditional checks around normalisation calls.

args = None

__call__(*x*)

static forward(*x*)

class dicee.models.**Block**(*config*)

Bases: `torch.nn.Module`

Base class for all neural network modules.

Your models should also subclass this class.

Modules can also contain other Modules, allowing them to be nested in a tree structure. You can assign the submodules as regular attributes:

```
import torch.nn as nn
import torch.nn.functional as F


class Model(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.conv1 = nn.Conv2d(1, 20, 5)
        self.conv2 = nn.Conv2d(20, 20, 5)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        return F.relu(self.conv2(x))
```

Submodules assigned in this way will be registered, and will also have their parameters converted when you call `to()`, etc.

> **Note**
>
> As per the example above, an `__init__()` call to the parent class must be made before assignment on the child.

Variables

training (*bool*) – Boolean represents whether this module is in training or evaluation mode.

ln_1

attn

ln_2

mlp

forward(*x*)

class dicee.models.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and *T* is `max_length_subword_tokens`.

Returns
Shape `(batch_size, num_bpe_entities)` score matrix.

Return type
torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters
- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns

Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type

torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters

idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns

head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type

torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters

indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns

head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type

torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*) → Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where *T* is `max_length_subword_tokens`.

Returns

head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type

torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*) → Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of *T* subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and *T* is `max_length_subword_tokens`.

Returns

head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type

torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

class dicee.models.**DistMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

DistMult: bilinear diagonal knowledge graph embedding.

Scores a triple `(h, r, t)` as the element-wise product of the head, relation, and tail embeddings summed over the embedding dimension:

```
f(h, r, t) = \sum_i  h_i  \cdot  r_i  \cdot  t_i
```

Simple yet effective baseline; incapable of modelling asymmetric relations.

References

Yang et al., *Embedding Entities and Relations for Learning and Inference in Knowledge Bases*, ICLR 2015. https://arxiv.org/abs/1412.6575

name = 'DistMult'

k_vs_all_score(*emb_h: torch.FloatTensor*, *emb_r: torch.FloatTensor*, *emb_E: torch.FloatTensor*) → torch.FloatTensor

Score a head/relation batch against all entity embeddings.

Computes `(h * r) @ E^T` after applying hidden dropout and normalisation to the element-wise product.

Parameters

- **emb_h** (*torch.FloatTensor*) – Head entity embeddings, shape `(batch_size, embedding_dim)`.
- **emb_r** (*torch.FloatTensor*) – Relation embeddings, shape `(batch_size, embedding_dim)`.
- **emb_E** (*torch.FloatTensor*) – All entity embeddings, shape `(num_entities, embedding_dim)`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Parameters

- **x** (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.
- **target_entity_idx** (*torch.LongTensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns

Shape `(batch_size, k)` score matrix.

Return type

torch.FloatTensor

score(*h: torch.FloatTensor*, *r: torch.FloatTensor*, *t: torch.FloatTensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` embedding triples.

Parameters

- **h** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **r** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **t** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.**TransE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

TransE: translation-based knowledge graph embedding.

Models a relation *r* as a translation in embedding space such that `h + r ≈ t` for a true triple `(h, r, t)`. The score function is defined as:

```
f(h, r, t) = margin - ||h + r - t||_2
```

TransE is effective for 1-to-1 relations but struggles with reflexive, one-to-many, and many-to-one patterns.

References

Bordes et al., *Translating Embeddings for Modeling Multi-relational Data*, NeurIPS 2013. https://proceedings.neurips.cc/paper/2013/file/1cecc7a77928ca8133fa24680a88d2f9-Paper.pdf

name = 'TransE'

margin

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*) → torch.FloatTensor

Score a batch of triples using the TransE margin-distance formula.

Parameters

- **head_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **rel_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **tail_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.

Returns
Shape `(batch_size,)` scores equal to `margin - ||h + r - t||_2`.

Return type
torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Computes `margin - ||h + r - e||_2` for every entity embedding *e*.

Parameters
x (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

class dicee.models.**TransH**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

TransH: translation-based knowledge graph embedding on relation-specific hyperplanes.

Addresses TransE's inability to model one-to-many, many-to-one, and many-to-many relations by letting each relation *r* define its own hyperplane. An entity is first projected onto that hyperplane before the translation is applied, so the same entity can be positioned differently depending on the relation. Concretely, for a triple `(h, r, t)` with unit-norm hyperplane normal `r_w` and in-hyperplane translation vector `r_d`:

```
h_r = h - (r_w^T h) * r_w
t_r = t - (r_w^T t) * r_w
f(h, r, t) = margin - ||h_r + r_d - t_r||_2^2
```

`r_w` reuses the inherited `relation_embeddings` table and is normalised to unit norm before every use, since only a unit-norm normal yields a true orthogonal projection onto the hyperplane. `r_d` is a second, relation-specific embedding table analogous to TransE's relation vector.

References

Wang et al., *Knowledge Graph Embedding by Translating on Hyperplanes*, AAAI 2014. https://aaai.org/papers/8870-knowledge-graph-embedding-by-translating-on-hyperplanes/

name = 'TransH'

margin

relation_normal_translations

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Every candidate tail entity must be projected onto the batch row's relation-specific hyperplane, and the hyperplane normal differs per batch row, so unlike TransE this materialises a full `(batch_size, num_entities, embedding_dim)` tensor of projected tail candidates – O(B * E * D) memory, versus TransE's O(B * E).

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Same hyperplane-projection logic as *forward_k_vs_all()*, but restricted to the *k* sampled tail candidates instead of every entity, so it costs `O(B * k * D)` memory instead of `O(B * E * D)`.

Parameters

- **x** (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.
- **target_entity_idx** (*torch.LongTensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns
Shape `(batch_size, k)` score matrix.

Return type
torch.FloatTensor

class dicee.models.**MuRE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

MuRE: multi-relational graph embedding with relation-specific diagonal scaling, translation, and entity biases.

Scores a triple `(h, r, t)` as:

```
f(h, r, t) = -||R_r \odot h + t_r - t||_2 + b_h + b_t
```

`R_r` is a relation-specific diagonal matrix applied to the head via an element-wise (Hadamard) product; it reuses the inherited `relation_embeddings` table directly, shape `(num_relations, embedding_dim)`. `t_r` is a second, relation-specific translation vector (own embedding table, same shape convention: `(num_relations, embedding_dim)`) analogous to TransE's relation vector. `b_h` and `b_t` are learnable scalar biases indexed per entity (head and tail respectively), not per relation.

References

Balažević et al., *Multi-relational Poincaré Graph Embeddings*, NeurIPS 2019. https://arxiv.org/abs/1905.09791

name = 'MuRE'

relation_translations

b_h

b_t

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Computes `-||R_r \odot h + t_r - e||_2 + b_h + b_e` for every entity embedding *e*.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Parameters

- **x** (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.
- **target_entity_idx** (*torch.LongTensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns
Shape `(batch_size, k)` score matrix.

Return type
torch.FloatTensor

class dicee.models.**Shallom**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Shallom: shallow neural model for relation prediction.

Represents each triple as the concatenation of head and tail entity embeddings and feeds it through a two-layer MLP to predict the relation. Designed for the `RelationPrediction` labelling form.

References

Demir et al., *A Shallow Neural Model for Relation Prediction*, ISWC 2021. https://arxiv.org/abs/2101.09090

name = 'Shallom'

shallom

get_embeddings() → Tuple[numpy.ndarray, None]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

forward_k_vs_all(*x*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_triples(*x*) → torch.FloatTensor

Score a batch of triples by looking up relation scores from `forward_k_vs_all`.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.**Pyke**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Pyke: Physical Embedding Model for Knowledge Graphs.

Scores a triple `(h, r, t)` based on the average pairwise distance between head-to-relation and relation-to-tail in embedding space:

```
f(h, r, t) = margin - (||h - r||_2 + ||r - t||_2) / 2
```

The model encodes geometric proximity between entities and the relations that connect them.

name = 'Pyke'

dist_func

margin

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

Score a batch of triples using the Pyke distance formula.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.**CoKEConfig**

Configuration for the CoKE (Contextualized Knowledge Graph Embedding) model.

block_size

Sequence length for transformer (3 for triples: head, relation, tail)

vocab_size
: Total vocabulary size (num_entities + num_relations)

n_layer
: Number of transformer layers

n_head
: Number of attention heads per layer

n_embd
: Embedding dimension (set to match model embedding_dim)

dropout
: Dropout rate applied throughout the model

bias
: Whether to use bias in linear layers

causal
: Whether to use causal masking (False for bidirectional attention)

block_size: int = 3

vocab_size: int = None

n_layer: int = 6

n_head: int = 8

n_embd: int = None

dropout: float = 0.3

bias: bool = True

causal: bool = False

class dicee.models.**CoKE**(*args*, *config: CoKEConfig = CoKEConfig()*)

Bases: *dicee.models.base_model.BaseKGE*

Contextualized Knowledge Graph Embedding (CoKE) model. Based on: https://arxiv.org/pdf/1911.02168.

CoKE uses a transformer encoder to learn contextualized representations of entities and relations. For link prediction, it predicts masked elements in (head, relation, tail) triples using bidirectional attention, similar to BERT's masked language modeling approach.

The model creates a sequence [head_emb, relation_emb, mask_emb], adds positional embeddings, and processes it through transformer layers to predict the tail entity.

name = 'CoKE'

config

pos_emb

mask_emb

blocks

ln_f

coke_dropout

forward_k_vs_all(*x: torch.Tensor*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

score(*emb_h*, *emb_r*, *emb_t*)

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters
args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and *T* is `max_length_subword_tokens`.

Returns

Shape `(batch_size, num_bpe_entities)` score matrix.

Return type

torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters
idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns
head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters
indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns
head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where T is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type
torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of T subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type
torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

class dicee.models.**ConEx**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Convolutional ComplEx Knowledge Graph Embeddings

name = 'ConEx'

conv2d

fc_num_input

fc1

norm_fc1

bn_conv2d

feature_map_dropout

residual_convolution(*C_1: Tuple[torch.Tensor, torch.Tensor]*, *C_2: Tuple[torch.Tensor, torch.Tensor]*) → torch.FloatTensor

Compute residual score of two complex-valued embeddings. :param C_1: a tuple of two pytorch tensors that corresponds complex-valued embeddings :param C_2: a tuple of two pytorch tensors that corresponds complex-valued embeddings :return:

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.Tensor*, *target_entity_idx: torch.Tensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.**AConEx**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Additive Convolutional ComplEx Knowledge Graph Embeddings

name = 'AConEx'

conv2d

fc_num_input

fc1

norm_fc1

bn_conv2d

feature_map_dropout

residual_convolution(*C_1: Tuple[torch.Tensor, torch.Tensor]*, *C_2: Tuple[torch.Tensor, torch.Tensor]*) → torch.FloatTensor

Compute residual score of two complex-valued embeddings. :param C_1: a tuple of two pytorch tensors that corresponds complex-valued embeddings :param C_2: a tuple of two pytorch tensors that corresponds complex-valued embeddings :return:

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.Tensor*, *target_entity_idx: torch.Tensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.**ComplEx**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- `forward_triples()` — score a batch of `(h, r, t)` triples.
- *`forward_k_vs_all()`* — score a `(h, r)` batch against every entity.

Parameters
args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'ComplEx'

static score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*)

static k_vs_all_score(*emb_h: torch.FloatTensor*, *emb_r: torch.FloatTensor*, *emb_E: torch.FloatTensor*)

Parameters
- **emb_h**
- **emb_r**
- **emb_E**

forward_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.**RotatE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

RotatE: knowledge graph embedding by relational rotation in complex space.

Represents each entity as a complex vector in ℂ^(d/2), obtained by splitting its `d`-dimensional embedding into a real half and an imaginary half. Each relation is represented by `d/2` phase angles θ that define a unit-modulus rotation `r_i = e^{iθ_i}`; since a phase angle needs only one real number, the relation embedding table is reinitialised at half of the entity embedding size. A true triple `(h, r, t)` should satisfy `h ∘ r ≈ t` under element-wise complex multiplication, giving the score:

```
f(h, r, t) = margin - ||h ∘ r - t||_2
```

Unlike TransE, RotatE can model symmetric, antisymmetric, inverse, and composition relation patterns.

References

Sun et al., *RotatE: Knowledge Graph Embedding by Relational Rotation in Complex Space*, ICLR 2019. https://arxiv.org/abs/1902.10197

name = 'RotatE'

margin

half_dim

relation_embeddings

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*) → torch.FloatTensor

Score a batch of triples using the RotatE margin-distance formula.

Parameters

- **head_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **tail_ent_emb** (*torch.FloatTensor*) – Each has shape `(batch_size, embedding_dim)`.
- **rel_ent_emb** (*torch.FloatTensor*) – Shape `(batch_size, d/2)` relation phase angles θ.

Returns

Shape `(batch_size,)` scores equal to `margin - ||h ∘ r - t||_2`.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

KvsAll forward pass: score head/relation against all entities.

Computes `margin - ||h ∘ r - e||_2` for every entity embedding *e*.

Parameters

x (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

forward_k_vs_sample(*x: torch.Tensor*, *target_entity_idx: torch.Tensor*) → torch.FloatTensor

KvsSample forward pass: score head/relation against a sampled entity subset.

Computes `margin - ||h ∘ r - e||_2` for each of the *k* sampled entities *e*.

Parameters

- **x** (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.
- **target_entity_idx** (*torch.Tensor*) – Shape `(batch_size, k)` indices of the *k* target entities per sample.

Returns
Shape `(batch_size, k)` score matrix.

Return type
torch.FloatTensor

class dicee.models.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters
args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and *T* is `max_length_subword_tokens`.

Returns

Shape `(batch_size, num_bpe_entities)` score matrix.

Return type

torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where k is the number of sampled target entities.

Return type
torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters
idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns
head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters
indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns
head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where T is `max_length_subword_tokens`.

Returns

head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type

torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*) → Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of T subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns

head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type

torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

class dicee.models.**IdentityClass**(*args=None*)

Bases: `torch.nn.Module`

No-op normalisation / dropout placeholder.

Used whenever no normalisation layer is requested (`--normalization None`). All inputs are returned unchanged so that the rest of the model code does not need conditional checks around normalisation calls.

args = None

__call__(*x*)

static forward(*x*)

dicee.models.**quaternion_mul**(*, *Q_1*, *Q_2*) → Tuple[torch.Tensor, torch.Tensor, torch.Tensor, torch.Tensor]

Perform quaternion multiplication :param Q_1: :param Q_2: :return:

dicee.models.**quaternion_mul_with_unit_norm**(*, *Q_1*, *Q_2*)

class dicee.models.**QMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- `forward_triples()` — score a batch of `(h, r, t)` triples.
- *`forward_k_vs_all()`* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'QMult'

explicit = True

quaternion_multiplication_followed_by_inner_product(*h*, *r*, *t*)

Parameters

- **h** – shape: (**batch_dims*, dim) The head representations.
- **r** – shape: (**batch_dims*, dim) The head representations.
- **t** – shape: (**batch_dims*, dim) The tail representations.

Returns

Triple scores.

static quaternion_normalizer(*x: torch.FloatTensor*) → torch.FloatTensor

Normalize the length of relation vectors, if the forward constraint has not been applied yet.

Absolute value of a quaternion

$$|a + bi + cj + dk| = \sqrt{a^2 + b^2 + c^2 + d^2}$$

L2 norm of quaternion vector:

$$\|x\|^2 = \sum_{i=1}^{d} |x_i|^2 = \sum_{i=1}^{d} (x_i.re^2 + x_i.im_1^2 + x_i.im_2^2 + x_i.im_3^2)$$

Parameters

x – The vector.

Returns

The normalized vector.

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*)

k_vs_all_score(*bpe_head_ent_emb*, *bpe_rel_ent_emb*, *E*)

Parameters

- **bpe_head_ent_emb**
- **bpe_rel_ent_emb**
- **E**

forward_k_vs_all(*x*)

Parameters

x

forward_k_vs_sample(*x*, *target_entity_idx*)

Completed. Given a head entity and a relation (h,r), we compute scores for all possible triples,i.e., [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,| Entities|)

class dicee.models.**ConvQ**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Convolutional Quaternion Knowledge Graph Embeddings

name = 'ConvQ'

entity_embeddings

relation_embeddings

conv2d

fc_num_input

fc1

bn_conv1

bn_conv2

feature_map_dropout

residual_convolution(*Q_1*, *Q_2*)

forward_triples(*indexed_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,| Entities|)

class dicee.models.**AConvQ**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Additive Convolutional Quaternion Knowledge Graph Embeddings

name = 'AConvQ'

entity_embeddings

relation_embeddings

conv2d

fc_num_input

fc1

bn_conv1

bn_conv2

feature_map_dropout

residual_convolution(*Q_1*, *Q_2*)

forward_triples(*indexed_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,|Entities|)

class dicee.models.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and *T* is `max_length_subword_tokens`.

Returns

Shape `(batch_size, num_bpe_entities)` score matrix.

Return type

torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters
idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns
head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters
indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns
head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where T is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type
torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of T subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type
torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

class dicee.models.**IdentityClass**(*args=None*)

Bases: torch.nn.Module

No-op normalisation / dropout placeholder.

Used whenever no normalisation layer is requested (`--normalization None`). All inputs are returned unchanged so that the rest of the model code does not need conditional checks around normalisation calls.

args = None

__call__(*x*)

static forward(*x*)

dicee.models.**octonion_mul**(*, *O_1*, *O_2*)

dicee.models.**octonion_mul_norm**(*, *O_1*, *O_2*)

class dicee.models.**OMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- `forward_triples()` — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'OMult'

static octonion_normalizer(*emb_rel_e0*, *emb_rel_e1*, *emb_rel_e2*, *emb_rel_e3*, *emb_rel_e4*, *emb_rel_e5*, *emb_rel_e6*, *emb_rel_e7*)

score(*head_ent_emb: torch.FloatTensor*, *rel_ent_emb: torch.FloatTensor*, *tail_ent_emb: torch.FloatTensor*)

k_vs_all_score(*bpe_head_ent_emb*, *bpe_rel_ent_emb*, *E*)

forward_k_vs_all(*x*)

Completed. Given a head entity and a relation (h,r), we compute scores for all possible triples,i.e., [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,| Entities|)

class dicee.models.**ConvO**(*args: dict*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'ConvO'

conv2d

fc_num_input

fc1

bn_conv2d

norm_fc1

feature_map_dropout

static octonion_normalizer(*emb_rel_e0*, *emb_rel_e1*, *emb_rel_e2*, *emb_rel_e3*, *emb_rel_e4*, *emb_rel_e5*, *emb_rel_e6*, *emb_rel_e7*)

residual_convolution(*O_1*, *O_2*)

forward_triples(*x: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,|Entities|)

class dicee.models.**AConvO**(*args: dict*)

Bases: *dicee.models.base_model.BaseKGE*

Additive Convolutional Octonion Knowledge Graph Embeddings

name = 'AConvO'

conv2d

fc_num_input

fc1

bn_conv2d

norm_fc1

feature_map_dropout

static octonion_normalizer(*emb_rel_e0*, *emb_rel_e1*, *emb_rel_e2*, *emb_rel_e3*, *emb_rel_e4*, *emb_rel_e5*, *emb_rel_e6*, *emb_rel_e7*)

residual_convolution(*O_1*, *O_2*)

forward_triples(*x: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*)

Given a head entity and a relation (h,r), we compute scores for all entities. [score(h,r,x)|x in Entities] => [0.0,0.1,…,0.8], shape=> (1, |Entities|) Given a batch of head entities and relations => shape (size of batch,|Entities|)

class dicee.models.**Keci**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Keci: Knowledge Graph Embedding via Clifford Algebra.

Embeds entities and relations as multi-vectors in the Clifford algebra Cl_{p,q}(R^d) and scores triples via the Clifford product. The algebra is parameterised by two non-negative integers *p* and *q*:

- `p = 0, q = 0` — reduces to a standard bilinear (DistMult-like) model.
- `p = 0, q = 1` — equivalent to ComplEx.
- Larger `p` and `q` capture higher-order geometric interactions.

The embedding dimension must satisfy `embedding_dim % (p + q + 1) == 0`; the resulting quotient is stored as `self.r`.

Parameters

args (*dict*) – Configuration dictionary. Recognised keys (beyond those in *BaseKGE*): `p` (int, default 0) and `q` (int, default 0).

References

Demir et al., *Clifford Embeddings — A Generalized Approach for Embedding in Normed Algebras*, ECML 2023.

name = 'Keci'

p

q

r

requires_grad_for_interactions = True

compute_sigma_pp(*hp*, *rp*)

Compute sigma_{pp} = sum_{i=1}^{p-1} sum_{k=i+1}^p (h_i r_k - h_k r_i) e_i e_k

sigma_{pp} captures the interactions between along p bases For instance, let p e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for i in range(p - 1):

for k in range(i + 1, p):
results.append(hp[:, :, i] * rp[:, :, k] - hp[:, :, k] * rp[:, :, i])

sigma_pp = torch.stack(results, dim=2) assert sigma_pp.shape == (b, r, int((p * (p - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

compute_sigma_qq(*hq*, *rq*)

Compute sigma_{qq} = sum_{j=1}^{p+q-1} sum_{k=j+1}^{p+q} (h_j r_k - h_k r_j) e_j e_k sigma_{q} captures the interactions between along q bases For instance, let q e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for j in range(q - 1):

for k in range(j + 1, q):
results.append(hq[:, :, j] * rq[:, :, k] - hq[:, :, k] * rq[:, :, j])

sigma_qq = torch.stack(results, dim=2) assert sigma_qq.shape == (b, r, int((q * (q - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

compute_sigma_pq(*, *hp*, *hq*, *rp*, *rq*)

sum_{i=1}^{p} sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

apply_coefficients(*hp*, *hq*, *rp*, *rq*)

Multiplying a base vector with its scalar coefficient

clifford_multiplication(*h0*, *hp*, *hq*, *r0*, *rp*, *rq*)

Compute our CL multiplication

h = h_0 + sum_{i=1}^p h_i e_i + sum_{j=p+1}^{p+q} h_j e_j r = r_0 + sum_{i=1}^p r_i e_i + sum_{j=p+1}^{p+q} r_j e_j

ei ^2 = +1 for i =< i =< p ej ^2 = -1 for p < j =< p+q ei ej = -eje1 for i

eq j

h r = sigma_0 + sigma_p + sigma_q + sigma_{pp} + sigma_{q}+ sigma_{pq} where

(1) sigma_0 = h_0 r_0 + sum_{i=1}^p (h_0 r_i) e_i - sum_{j=p+1}^{p+q} (h_j r_j) e_j

(2) sigma_p = sum_{i=1}^p (h_0 r_i + h_i r_0) e_i

(3) sigma_q = sum_{j=p+1}^{p+q} (h_0 r_j + h_j r_0) e_j

(4) sigma_{pp} = sum_{i=1}^{p-1} sum_{k=i+1}^p (h_i r_k - h_k r_i) e_i e_k

(5) sigma_{qq} = sum_{j=1}^{p+q-1} sum_{k=j+1}^{p+q} (h_j r_k - h_k r_j) e_j e_k

(6) sigma_{pq} = sum_{i=1}^{p} sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j

construct_cl_multivector(*x: torch.FloatTensor*, *r: int*, *p: int*, *q: int*) → tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Split a flat embedding vector into the three Clifford components.

Given an embedding `x` of dimension `d = r + r*p + r*q`, returns the scalar part `a0`, the *p*-blade part `ap`, and the *q*-blade part `aq`.

Parameters

- **x** (*torch.FloatTensor*) – Shape `(batch_size, d)`.
- **r** (*int*) – Scalar block size (`embedding_dim // (p + q + 1)`).
- **p** (*int*) – Number of positive-signature basis elements.
- **q** (*int*) – Number of negative-signature basis elements.

Returns

- **a0** (*torch.FloatTensor*) – Shape `(batch_size, r)` — scalar (grade-0) part.
- **ap** (*torch.FloatTensor*) – Shape `(batch_size, r, p)` — positive-blade part.
- **aq** (*torch.FloatTensor*) – Shape `(batch_size, r, q)` — negative-blade part.

forward_k_vs_with_explicit(*x: torch.Tensor*) → torch.FloatTensor

KvsAll scoring using an explicit loop over sigma_pp/qq/pq terms.

Functionally equivalent to *forward_k_vs_all()* but computes the higher-order interaction terms (sigma_pp, sigma_qq, sigma_pq) with explicit nested loops rather than einsum contractions. Kept for reference and correctness verification.

Parameters

x (*torch.Tensor*) – Shape `(batch_size, 2)` integer tensor `[head_idx, relation_idx]`.

Returns

Shape `(batch_size, num_entities)` score matrix.

Return type

torch.FloatTensor

k_vs_all_score(*bpe_head_ent_emb: torch.FloatTensor*, *bpe_rel_ent_emb: torch.FloatTensor*, *E: torch.FloatTensor*) → torch.FloatTensor

Compute Clifford-product scores for a head/relation batch vs. all entities.

Decomposes the head-entity and relation embeddings into Clifford multi-vectors, performs the Cl_{p,q} product, and inner-products the result against the entity embedding matrix *E*.

Parameters

- **bpe_head_ent_emb** (*torch.FloatTensor*) – Head-entity embeddings, shape `(batch_size, embedding_dim)`.
- **bpe_rel_ent_emb** (*torch.FloatTensor*) – Relation embeddings, shape `(batch_size, embedding_dim)`.

- **E** (*torch.FloatTensor*) – All entity embeddings, shape `(num_entities, embedding_dim)`.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

Kvsall training

(1) Retrieve real-valued embedding vectors for heads and relations mathbb{R}^d .

(2) Construct head entity and relation embeddings according to Cl_{p,q}(mathbb{R}^d) .

(3) Perform Cl multiplication

(4) Inner product of (3) and all entity embeddings

forward_k_vs_with_explicit and this funcitons are identical Parameter ——— x: torch.LongTensor with (n,2) shape :rtype: torch.FloatTensor with (n, |E|) shape

construct_batch_selected_cl_multivector(*x: torch.FloatTensor*, *r: int*, *p: int*, *q: int*) → tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Split a batched, *k*-selected embedding tensor into Clifford components.

A variant of *construct_cl_multivector()* for tensors that have an extra *k* dimension (e.g. when scoring against *k* sampled targets).

Parameters

- **x** (*torch.FloatTensor*) – Shape `(batch_size, k, d)`.
- **r** (*int*) – Scalar block size.
- **p** (*int*) – Number of positive-signature basis elements.
- **q** (*int*) – Number of negative-signature basis elements.

Returns

- **a0** (*torch.FloatTensor*) – Shape `(batch_size, k, r)`.
- **ap** (*torch.FloatTensor*) – Shape `(batch_size, k, r, p)`.
- **aq** (*torch.FloatTensor*) – Shape `(batch_size, k, r, q)`.

forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx: torch.LongTensor*) → torch.FloatTensor

Parameter

x: torch.LongTensor with (n,2) shape

target_entity_idx: torch.LongTensor with (n, k) shape k denotes the selected number of examples.

rtype
torch.FloatTensor with (n, k) shape

score(*h*, *r*, *t*)

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Parameter

x: torch.LongTensor with (n,3) shape

rtype
torch.FloatTensor with (n) shape

class dicee.models.**CKeci**(*args*)

Bases: *Keci*

Without learning dimension scaling

name = 'CKeci'

requires_grad_for_interactions = False

class dicee.models.**DeCaL**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters
args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

name = 'DeCaL'

p

q

r

re

forward_triples(*x: torch.Tensor*) → torch.FloatTensor

Parameter

x: torch.LongTensor with (n,) shape

rtype
torch.FloatTensor with (n) shape

cl_pqr(*a: torch.tensor*) → torch.tensor

Input: tensor(batch_size, emb_dim) —> output: tensor with 1+p+q+r components with size (batch_size, emb_dim/(1+p+q+r)) each.

1) takes a tensor of size (batch_size, emb_dim), split it into 1 + p + q +r components, hence 1+p+q+r must be a divisor of the emb_dim. 2) Return a list of the 1+p+q+r components vectors, each are tensors of size (batch_size, emb_dim/(1+p+q+r))

compute_sigmas_single (*list_h_emb*, *list_r_emb*, *list_t_emb*)

here we compute all the sums with no others vectors interaction taken with the scalar product with t, that is,

$$s0 = h_0 r_0 t_0 s1 = \sum_{i=1}^{p} h_i r_i t_0 s2 = \sum_{j=p+1}^{p+q} h_j r_j t_0 s3 = \sum_{i=1}^{q} (h_0 r_i t_i + h_i r_0 t_i) s4 = \sum_{i=p+1}^{p+q} (h_0 r_i t_i + h_i r_0 t_i) s5 = \sum_{i=p+q+1}^{p+q+r} (h_0 r_i t$$

and return:

$$sigma_0 t = \sigma_0 \cdot t_0 = s0 + s1 - s2 s3, s4 and s5$$

compute_sigmas_multivect (*list_h_emb*, *list_r_emb*)

Here we compute and return all the sums with vectors interaction for the same and different bases.

For same bases vectors interaction we have

$$\sigma_p p = \sum_{i=1}^{p-1} \sum_{i'=i+1}^{p} (h_i r_{i'} - h_{i'} r_i)(models the interactions between e_i and e_i' for 1 <= i, i' <= p) \sigma_q q = \sum_{j=p+1}^{p+q-1} \sum_{j'=j+1}^{p+q} (h_j r_{j'}$$

For different base vector interactions, we have

$$\sigma_p q = \sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i)(interactions n between e_i and e_j for 1 <= i <= p and p+1 <= j <= p+q) \sigma_p r = \sum_{i=}^{}$$

forward_k_vs_all (*x: torch.Tensor*) → torch.FloatTensor

Kvsall training

(1) Retrieve real-valued embedding vectors for heads and relations

(2) Construct head entity and relation embeddings according to Cl_{p,q, r}(mathbb{R}^d) .

(3) Perform Cl multiplication

(4) Inner product of (3) and all entity embeddings

forward_k_vs_with_explicit and this funcitons are identical Parameter ——— x: torch.LongTensor with (n,) shape :rtype: torch.FloatTensor with (n, |E|) shape

apply_coefficients (*h0*, *hp*, *hq*, *hk*, *r0*, *rp*, *rq*, *rk*)

Multiplying a base vector with its scalar coefficient

construct_cl_multivector (*x: torch.FloatTensor*, *re: int*, *p: int*, *q: int*, *r: int*)
→ tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Construct a batch of multivectors Cl_{p,q,r}(mathbb{R}^d)

Parameter

x: torch.FloatTensor with (n,d) shape

returns

- **a0** (*torch.FloatTensor*)
- **ap** (*torch.FloatTensor*)
- **aq** (*torch.FloatTensor*)
- **ar** (*torch.FloatTensor*)

compute_sigma_pp(*hp*, *rp*)

Compute .. math:

```
\sigma_{p,p}^* = \sum_{i=1}^{p-1}\sum_{i'=i+1}^{p}(x_iy_{i'}-x_{i'}y_i)
```

sigma_{pp} captures the interactions between along p bases For instance, let p e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for i in range(p - 1):

for k in range(i + 1, p):
results.append(hp[:, :, i] * rp[:, :, k] - hp[:, :, k] * rp[:, :, i])

sigma_pp = torch.stack(results, dim=2) assert sigma_pp.shape == (b, r, int((p * (p - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

compute_sigma_qq(*hq*, *rq*)

Compute

$$\sigma_{q,q}^* = \sum_{j=p+1}^{p+q-1} \sum_{j'=j+1}^{p+q} (x_j y_{j'} - x_{j'} y_j) Eq.16$$

sigma_{q} captures the interactions between along q bases For instance, let q e_1, e_2, e_3, we compute interactions between e_1 e_2, e_1 e_3 , and e_2 e_3 This can be implemented with a nested two for loops

results = [] for j in range(q - 1):

for k in range(j + 1, q):
results.append(hq[:, :, j] * rq[:, :, k] - hq[:, :, k] * rq[:, :, j])

sigma_qq = torch.stack(results, dim=2) assert sigma_qq.shape == (b, r, int((q * (q - 1)) / 2))

Yet, this computation would be quite inefficient. Instead, we compute interactions along all p, e.g., e1e1, e1e2, e1e3,

e2e1, e2e2, e2e3, e3e1, e3e2, e3e3

Then select the triangular matrix without diagonals: e1e2, e1e3, e2e3.

compute_sigma_rr(*hk*, *rk*)

$$\sigma_{r,r}^* = \sum_{k=p+q+1}^{p+q+r-1} \sum_{k'=k+1}^{p} (x_k y_{k'} - x_{k'} y_k)$$

compute_sigma_pq(*, *hp*, *hq*, *rp*, *rq*)

Compute

$$\sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j$$

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

compute_sigma_pr(*, *hp*, *hk*, *rp*, *rk*)

Compute

$$\sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j$$

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

compute_sigma_qr(*, *hq*, *hk*, *rq*, *rk*)

$$\sum_{i=1}^{p} \sum_{j=p+1}^{p+q} (h_i r_j - h_j r_i) e_i e_j$$

results = [] sigma_pq = torch.zeros(b, r, p, q) for i in range(p):

for j in range(q):
sigma_pq[:, :, i, j] = hp[:, :, i] * rq[:, :, j] - hq[:, :, j] * rp[:, :, i]

print(sigma_pq.shape)

class dicee.models.**KeciTransformer**(*args*)

Bases: *Keci*

Keci with Transformer architecture.

Concatenates h0, hp, hq, r0, rp, rq into a single embedding vector and processes through transformer.

name = 'KeciTransformer'

use_clifford_mul

seq_len = 1

transformer

lm_head

forward_k_vs_all(*x: torch.Tensor*) → torch.FloatTensor

Kvsall training

Parameter

x: torch.LongTensor with (n,2) shape

rtype
torch.FloatTensor with (n, |E|) shape

class dicee.models.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters

args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns

Shape `(batch_size, num_bpe_entities)` score matrix.

Return type

torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where k is the number of sampled target entities.

Return type
torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters
idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns
head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters
indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns
head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where T is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type
torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of T subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and T is `max_length_subword_tokens`.

Returns

head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type

torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

dicee.models.**logger**

class dicee.models.**PykeenKGE**(*args: dict*)

Bases: *dicee.models.base_model.BaseKGE*

A class for using knowledge graph embedding models implemented in Pykeen

Notes: Pykeen_DistMult: C Pykeen_ComplEx: Pykeen_QuatE: Pykeen_MuRE: Pykeen_CP: Pykeen_HolE: Pykeen_HolE: Pykeen_HolE: Pykeen_TransD: Pykeen_TransE: Pykeen_TransF: Pykeen_TransH: Pykeen_TransR:

model_kwargs

name

model

loss_history = []

args

entity_embeddings = None

relation_embeddings = None

forward_k_vs_all(*x: torch.LongTensor*)

=> Explicit version by this we can apply bn and dropout

(1) Retrieve embeddings of heads and relations + apply Dropout & Normalization if given. h, r = self.get_head_relation_representation(x) # (2) Reshape (1). if self.last_dim > 0:

h = h.reshape(len(x), self.embedding_dim, self.last_dim) r = r.reshape(len(x), self.embedding_dim, self.last_dim)

(3) Reshape all entities. if self.last_dim > 0:

t = self.entity_embeddings.weight.reshape(self.num_entities, self.embedding_dim, self.last_dim)

else:

t = self.entity_embeddings.weight

(4) Call the score_t from interactions to generate triple scores. return self.interaction.score_t(h=h, r=r, all_entities=t, slice_size=1)

forward_triples(*x: torch.LongTensor*) → torch.FloatTensor

=> Explicit version by this we can apply bn and dropout

(1) Retrieve embeddings of heads, relations and tails and apply Dropout & Normalization if given. h, r, t = self.get_triple_representation(x) # (2) Reshape (1). if self.last_dim > 0:

h = h.reshape(len(x), self.embedding_dim, self.last_dim) r = r.reshape(len(x), self.embedding_dim, self.last_dim) t = t.reshape(len(x), self.embedding_dim, self.last_dim)

(3) Compute the triple score return self.interaction.score(h=h, r=r, t=t, slice_size=None, slice_dim=0)

abstractmethod forward_k_vs_sample(*x: torch.LongTensor*, *target_entity_idx*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

class dicee.models.**BaseKGE**(*args: dict*)

Bases: *BaseKGELightning*

Base class for all Knowledge Graph Embedding models.

Inherits the Lightning training loop from *BaseKGELightning* and adds the embedding tables, normalisation / dropout layers, and the routing logic that dispatches `forward()` calls to the appropriate scoring method.

Sub-classes must implement at minimum:

- *forward_triples()* — score a batch of `(h, r, t)` triples.
- *forward_k_vs_all()* — score a `(h, r)` batch against every entity.

Parameters
args (*dict*) – Flat configuration dictionary produced by `vars(argparse.Namespace)`. Required keys: `embedding_dim`, `num_entities`, `num_relations`, `learning_rate` (or `lr`), `optim`, `scoring_technique`.

args

embedding_dim = None

num_entities: int | None = None

num_relations: int | None = None

num_tokens = None

learning_rate = None

apply_unit_norm = None

input_dropout_rate = None

hidden_dropout_rate = None

optimizer_name = None

feature_map_dropout_rate = None

kernel_size = None

num_of_output_channels = None

weight_decay = None

selected_optimizer = None

normalizer_class = None

normalize_head_entity_embeddings

normalize_relation_embeddings

normalize_tail_entity_embeddings

hidden_normalizer

param_init

input_dp_ent_real

input_dp_rel_real

hidden_dropout

loss_history = []

byte_pair_encoding

max_length_subword_tokens

block_size

defer_large_embeddings

init_entity_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the entity embedding table.

This is the single place that honours *defer_large_embeddings*: when entity rows are sharded across FSDP ranks the table must stay `None` until the trainer allocates the sharded adapter. Subclasses that need an entity table of a non-default width must call this method instead of assigning `self.entity_embeddings` directly, so the deferral is never silently undone.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

init_relation_embeddings(*embedding_dim: int | None = None*) → None

Create (or re-create) the relation embedding table.

Relation tables are never deferred - they are small enough to be replicated on every rank.

Parameters

embedding_dim (*Optional[int]*) – Width of the table. Defaults to *embedding_dim*.

forward_byte_pair_encoded_k_vs_all(*x: torch.LongTensor*) → torch.FloatTensor

KvsAll scoring for BPE-encoded head entities and relations.

Retrieves subword-unit embeddings for the head entity and relation, reduces them to fixed-size vectors via a linear projection, then scores against all BPE entity embeddings.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` BPE token indices where dim 1 indexes `[head, relation]` and *T* is `max_length_subword_tokens`.

Returns

Shape `(batch_size, num_bpe_entities)` score matrix.

Return type

torch.FloatTensor

forward_byte_pair_encoded_triple(*x: Tuple[torch.LongTensor, torch.LongTensor]*) → torch.FloatTensor

NegSample scoring for BPE-encoded `(head, relation, tail)` triples.

Retrieves subword-unit embeddings for all three elements and reduces them to fixed-size vectors via a linear projection before computing the triple score.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` BPE token indices.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

init_params_with_sanity_checking() → None

Populate model hyper-parameters from `self.args` with safe defaults.

Reads embedding dimension, learning rate, dropout rates, normalisation strategy, optimizer name, and parameter initialisation scheme from the `args` dict. Falls back to sensible defaults for any missing key so that minimal `args` dicts (e.g. for unit tests) are still valid.

forward(*x: torch.LongTensor | Tuple[torch.LongTensor, torch.LongTensor]*, *y_idx: torch.LongTensor | None = None*) → torch.Tensor

Route the forward pass to the appropriate scoring method.

Inspects the shape and type of *x* to decide which low-level scorer to call:

- Tuple `(x, y_idx)` → *forward_k_vs_sample()*
- `(batch, 3)` tensor → *forward_triples()*
- `(batch, 2)` tensor → *forward_k_vs_all()*
- BPE triple tensor → *forward_byte_pair_encoded_triple()*
- BPE pair tensor → *forward_byte_pair_encoded_k_vs_all()*

Parameters

- **x** (*torch.LongTensor or Tuple[torch.LongTensor, torch.LongTensor]*) – Either a plain index tensor or a `(triple_idx, target_idx)` tuple for sample-based labelling.
- **y_idx** (*torch.LongTensor, optional*) – Target entity indices used by *forward_k_vs_sample()*. Ignored when *x* is a plain tensor.

Returns
Score tensor whose shape depends on the selected scorer.

Return type
torch.FloatTensor

forward_triples(*x: torch.LongTensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

forward_k_vs_all(**args*, ***kwargs*)

Score a `(head, relation)` batch against every entity.

Sub-classes must override this method. The default implementation raises `ValueError` to make missing overrides obvious at runtime.

Returns
Shape `(batch_size, num_entities)` score matrix.

Return type
torch.FloatTensor

forward_k_vs_sample(**args*, ***kwargs*)

Score a `(head, relation)` batch against a sampled subset of entities.

Used by `KvsSample` and `1vsSample` datasets. Sub-classes that support sample-based labelling must override this method.

Returns
Shape `(batch_size, k)` score matrix where *k* is the number of sampled target entities.

Return type
torch.FloatTensor

get_triple_representation(*idx_hrt*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for a triple index batch.

Parameters
idx_hrt (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor with columns `[head_idx, relation_idx, tail_idx]`.

Returns
head_ent_emb, rel_ent_emb, tail_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_head_relation_representation(*indexed_triple*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve and normalise embedding vectors for head entities and relations.

Parameters
indexed_triple (*torch.LongTensor*) – Shape `(batch_size, 2)` integer tensor with columns `[head_idx, relation_idx]`.

Returns
head_ent_emb, rel_ent_emb – Each has shape `(batch_size, embedding_dim)` after applying the configured dropout and normalisation.

Return type
torch.FloatTensor

get_sentence_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor, torch.FloatTensor]

Retrieve BPE subword-unit embeddings for a batch of triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3, T)` where *T* is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb, tail_emb – Each has shape `(batch_size, T, embedding_dim)`.

Return type
torch.FloatTensor

get_bpe_head_and_relation_representation(*x: torch.LongTensor*)
→ Tuple[torch.FloatTensor, torch.FloatTensor]

Retrieve unit-normalised BPE embeddings for head entities and relations.

Each entity/relation is represented as a sequence of *T* subword tokens. Their token embeddings are L2-normalised across the sequence dimension so that the resulting matrix has unit Frobenius norm.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 2, T)` where dim 1 indexes `[head, relation]` and *T* is `max_length_subword_tokens`.

Returns
head_ent_emb, rel_emb – Each has shape `(batch_size, T, embedding_dim)`, L2-normalised over the `(T, D)` dimensions.

Return type
torch.FloatTensor

get_embeddings() → Tuple[numpy.ndarray, numpy.ndarray]

Return the entity and relation embedding matrices as numpy arrays.

Returns

- **entity_embeddings** (*numpy.ndarray*) – Shape `(num_entities, embedding_dim)`.
- **relation_embeddings** (*numpy.ndarray*) – Shape `(num_relations, embedding_dim)`.

dicee.models.**logger**

class dicee.models.**FMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Learning Knowledge Neural Graphs

name = 'FMult'

entity_embeddings

relation_embeddings

k

num_sample = 50

gamma

roots

weights

compute_func(*weights: torch.FloatTensor*, *x*) → torch.FloatTensor

chain_func(*weights*, *x: torch.FloatTensor*)

forward_triples(*idx_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns

Shape `(batch_size,)` triple scores.

Return type

torch.FloatTensor

class dicee.models.**GFMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Learning Knowledge Neural Graphs

name = 'GFMult'

entity_embeddings

relation_embeddings

k

num_sample = 250

roots

weights

compute_func(*weights: torch.FloatTensor*, *x*) → torch.FloatTensor

chain_func(*weights*, *x: torch.FloatTensor*)

forward_triples(*idx_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters

x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.**FMult2**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Learning Knowledge Neural Graphs

name = 'FMult2'

n_layers = 3

k

n = 50

score_func = 'compositional'

discrete_points

entity_embeddings

relation_embeddings

build_func(*Vec*)

build_chain_funcs(*list_Vec*)

compute_func(*W*, *b*, *x*) → torch.FloatTensor

function(*list_W*, *list_b*)

trapezoid(*list_W*, *list_b*)

forward_triples(*idx_triple: torch.Tensor*) → torch.Tensor

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

class dicee.models.**LFMult1**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Embedding with trigonometric functions. We represent all entities and relations in the complex number space as: f(x) = sum_{k=0}^{k=d-1}wk e^{kix}. and use the three differents scoring function as in the paper to evaluate the score

name = 'LFMult1'

entity_embeddings

relation_embeddings

forward_triples(*idx_triple*)

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

tri_score(*h*, *r*, *t*)

vtp_score(*h*, *r*, *t*)

class dicee.models.**LFMult**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Embedding with polynomial functions. We represent all entities and relations in the polynomial space as: f(x) = sum_{i=0}^{d-1} a_k x^{i%d} and use the three differents scoring function as in the paper to evaluate the score. We also consider combining with Neural Networks.

name = 'LFMult'

entity_embeddings

relation_embeddings

degree

m

x_values

forward_triples(*idx_triple*)

Score a batch of `(head, relation, tail)` index triples.

Parameters
x (*torch.LongTensor*) – Shape `(batch_size, 3)` integer tensor where each row is `[head_idx, relation_idx, tail_idx]`.

Returns
Shape `(batch_size,)` triple scores.

Return type
torch.FloatTensor

construct_multi_coeff(*x*)

poly_NN(*x*, *coefh*, *coefr*, *coeft*)

Constructing a 2 layers NN to represent the embeddings. h = sigma(wh^T x + bh), r = sigma(wr^T x + br), t = sigma(wt^T x + bt)

linear(*x*, *w*, *b*)

scalar_batch_NN(*a*, *b*, *c*)

element wise multiplication between a,b and c: Inputs : a, b, c ====> torch.tensor of size batch_size x m x d Output : a tensor of size batch_size x d

tri_score(*coeff_h*, *coeff_r*, *coeff_t*)

this part implement the trilinear scoring techniques:

score(h,r,t) = int_{0}{1} h(x)r(x)t(x) dx = sum_{i,j,k = 0}^{d-1} dfrac{a_i*b_j*c_k}{1+(i+j+k)%d}

1. generate the range for i,j and k from [0 d-1]
2. perform dfrac{a_i*b_j*c_k}{1+(i+j+k)%d} in parallel for every batch
3. take the sum over each batch

vtp_score(*h*, *r*, *t*)

this part implement the vector triple product scoring techniques:

score(h,r,t) = int_{0}{1} h(x)r(x)t(x) dx = sum_{i,j,k = 0}^{d-1} dfrac{a_i*c_j*b_k - b_i*c_j*a_k}{(1+(i+j)%d)(1+k)}

1. generate the range for i,j and k from [0 d-1]
2. Compute the first and second terms of the sum
3. Multiply with then denominator and take the sum
4. take the sum over each batch

comp_func(*h*, *r*, *t*)

this part implement the function composition scoring techniques: i.e. score = <hor, t>

polynomial(*coeff*, *x*, *degree*)

This function takes a matrix tensor of coefficients (coeff), a tensor vector of points x and range of integer [0,1,…d] and return a vector tensor (coeff[0][0] + coeff[0][1]x +…+ coeff[0][d]x^d,

coeff[1][0] + coeff[1][1]x +…+ coeff[1][d]x^d)

pop(*coeff*, *x*, *degree*)

This function allow us to evaluate the composition of two polynomes without for loops :) it takes a matrix tensor of coefficients (coeff), a matrix tensor of points x and range of integer [0,1,…d]

and return a tensor (coeff[0][0] + coeff[0][1]x +…+ coeff[0][d]x^d,

coeff[1][0] + coeff[1][1]x +…+ coeff[1][d]x^d)

class dicee.models.**DualE**(*args*)

Bases: *dicee.models.base_model.BaseKGE*

Dual Quaternion Knowledge Graph Embeddings (https://ojs.aaai.org/index.php/AAAI/article/download/16850/16657)

name = 'DualE'

entity_embeddings

relation_embeddings

num_ent = None

kvsall_score(*e_1_h*, *e_2_h*, *e_3_h*, *e_4_h*, *e_5_h*, *e_6_h*, *e_7_h*, *e_8_h*, *e_1_t*, *e_2_t*, *e_3_t*, *e_4_t*, *e_5_t*, *e_6_t*, *e_7_t*, *e_8_t*, *r_1*, *r_2*, *r_3*, *r_4*, *r_5*, *r_6*, *r_7*, *r_8*) → torch.tensor

KvsAll scoring function

Input

x: torch.LongTensor with (n,) shape

Output

torch.FloatTensor with (n) shape

forward_triples(*idx_triple: torch.tensor*) → torch.tensor

Negative Sampling forward pass:

Input

x: torch.LongTensor with (n,) shape

Output

torch.FloatTensor with (n) shape

forward_k_vs_all(*x*)

KvsAll forward pass

Input

x: torch.LongTensor with (n,) shape

Output

torch.FloatTensor with (n) shape

T(*x: torch.tensor*) → torch.tensor

Transpose function

Input: Tensor with shape (nxm) Output: Tensor with shape (mxn)

class dicee.models.**ULTRA**(*args*)

Bases: *dicee.models.graph_model.GraphKGE*

Pure PyTorch ULTRA with the official state dictionary and DICE triple order.

name = 'ULTRA'

config_prefix = 'ultra'

graph_filename = 'ultra_graph.pt'

checkpoint_hint = 'official ULTRA requires ultra_dim=64, ultra_num_layers=6; all keys and shapes must match'

deterministic_inference = True

dim

num_layers

relation_model

entity_model

relation_cache_mb

projection_cache_mb

clear_inference_cache()

Drop derived representations when graph, weights or device change.

class dicee.models.**TRIX**(*args*)

Bases: TRIXBase

Entity predictor compatible with official entity_prediction.pth.

DICE order is (head, relation, tail). Head corruption retains the original relation seed, then uses the inverse relation for both entity reasoners. Explicit reciprocal DICE queries use that same convention.

name = 'TRIX'

deterministic_inference = True

relation_model

entity_model_1

entity_model_2

relation_cache_mb

clear_inference_cache()

Drop derived representations when graph, weights or device change.

class dicee.models.**TRIXRelation**(*args*)

Bases: TRIXBase, *dicee.models.graph_model.RelationGraphKGE*

Relation predictor compatible with official relation_prediction.pth.

Pairs are (head, tail); triples and grouped relation corruptions use DICE's (head, relation, tail) order. One reasoning pass scores every relation for a given pair, including external inverse IDs when present in the vocabulary.

name = 'TRIXRelation'

relation_model

entity_model

mlp

class dicee.models.**Flock**(*args*)

Bases: FlockBase

Entity-prediction Flock with official flock_entity.pth parameters.

name = 'Flock'

class dicee.models.**FlockRelation**(*args*)

Bases: FlockBase, *dicee.models.graph_model.RelationGraphKGE*

Relation-prediction Flock with official flock_relation.pth parameters.

name = 'FlockRelation'

relation_prediction = True

dicee.query_generator

Attributes

logger

Classes

QueryGenerator

Module Contents

dicee.query_generator.**logger**

class dicee.query_generator.**QueryGenerator**(*train_path*, *val_path: str*, *test_path: str*, *ent2id: Dict | None = None*, *rel2id: Dict | None = None*, *seed: int = 1*, *gen_valid: bool = False*, *gen_test: bool = True*)

train_path

val_path

test_path

gen_valid = False

gen_test = True

seed = 1

max_ans_num = 1000000.0

mode

ent2id = None

rel2id: Dict = None

ent_in: Dict

ent_out: Dict

query_name_to_struct

list2tuple(*list_data*)

tuple2list(*x: List | Tuple*) → List | Tuple

Convert a nested tuple to a nested list.

set_global_seed(*seed: int*)

Set seed

construct_graph(*paths: List[str]*) → Tuple[Dict, Dict]

Construct graph from triples Returns dicts with incoming and outgoing edges

fill_query(*query_structure: List[str | List]*, *ent_in: Dict*, *ent_out: Dict*, *answer: int*) → bool

Private method for fill_query logic.

achieve_answer(*query: List[str | List]*, *ent_in: Dict*, *ent_out: Dict*) → set

Private method for achieve_answer logic. @TODO: Document the code

write_links(*ent_out*, *small_ent_out*)

ground_queries(*query_structure: List[str | List]*, *ent_in: Dict*, *ent_out: Dict*, *small_ent_in: Dict*, *small_ent_out: Dict*, *gen_num: int*, *query_name: str*)

Generating queries and achieving answers

unmap(*query_type*, *queries*, *tp_answers*, *fp_answers*, *fn_answers*)

unmap_query(*query_structure*, *query*, *id2ent*, *id2rel*)

generate_queries(*query_struct: List*, *gen_num: int*, *query_type: str*)

Passing incoming and outgoing edges to ground queries depending on mode [train valid or text] and getting queries and answers in return @ TODO: create a class for each single query struct

save_queries(*query_type: str*, *gen_num: int*, *save_path: str*)

abstractmethod load_queries(*path*)

get_queries(*query_type: str*, *gen_num: int*)

static save_queries_and_answers(*path: str*, *data: List[Tuple[str, Tuple[collections.defaultdict]]]*) → None

Save Queries into Disk

static load_queries_and_answers(*path: str*) → List[Tuple[str, Tuple[collections.defaultdict]]]

Load Queries from Disk to Memory

dicee.read_preprocess_save_load_kg

Submodules

dicee.read_preprocess_save_load_kg.preprocess

Attributes

logger

Classes

PreprocessKG	Preprocess the data in memory

Module Contents

dicee.read_preprocess_save_load_kg.preprocess.**logger**

class dicee.read_preprocess_save_load_kg.preprocess.**PreprocessKG**(*kg*)

Preprocess the data in memory

kg

start() → None

Preprocess train, valid and test datasets stored in knowledge graph instance

Parameter

rtype
None

preprocess_with_byte_pair_encoding()

preprocess_with_byte_pair_encoding_with_padding() → None

Preprocess with byte pair encoding and add padding

preprocess_with_pandas() → None

Preprocess with pandas: add reciprocal triples, construct vocabulary, and index datasets

preprocess_with_polars() → None

Preprocess with polars: add reciprocal triples and create indexed datasets

sequential_vocabulary_construction() → None

(1) Read input data into memory

(2) Remove triples with a condition

(3) **Serialize vocabularies in a pandas dataframe where**
=> the index is integer and => a single column is string (e.g. URI)

dicee.read_preprocess_save_load_kg.read_from_disk

Attributes

logger	

Classes

ReadFromDisk	Read the data from disk into memory

Module Contents

dicee.read_preprocess_save_load_kg.read_from_disk.**logger**

class dicee.read_preprocess_save_load_kg.read_from_disk.**ReadFromDisk**(*kg*)

Read the data from disk into memory

kg

start() → None

Read a knowledge graph from disk into memory

Data will be available at the train_set, test_set, valid_set attributes.

Parameter

None

rtype
None

add_noisy_triples_into_training()

dicee.read_preprocess_save_load_kg.save_load_disk

Attributes

logger

Classes

LoadSaveToDisk

Module Contents

dicee.read_preprocess_save_load_kg.save_load_disk.**logger**

class dicee.read_preprocess_save_load_kg.save_load_disk.**LoadSaveToDisk**(*kg*)

kg

save()

load()

dicee.read_preprocess_save_load_kg.util

Attributes

logger

Functions

polars_dataframe_indexer(→ polars.DataFrame)	Replaces 'subject', 'relation', and 'object' columns in the input Polars DataFrame with their corresponding index values
pandas_dataframe_indexer(→ pandas.DataFrame)	Replaces 'subject', 'relation', and 'object' columns in the input Pandas DataFrame with their corresponding index values
apply_reciprocal_or_noise(add_reciprocal, eval_model)	Add reciprocal triples if conditions are met
timeit(func)	
read_with_polars(→ polars.DataFrame)	Load and Preprocess via Polars

continues on next page

Table 80 – continued from previous page

read_with_pandas(data_path[, read_only_few, ...])	Load and Preprocess via Pandas
read_from_disk(→ Tuple[polars.DataFrame, pandas.DataFrame])	
count_triples(→ int)	Returns the total number of triples in the triple store.
fetch_worker(endpoint, offsets, chunk_size, ...)	Worker process: fetch assigned chunks and save to disk with per-worker tqdm.
read_from_triple_store_with_polars(endpoint[, ...])	Main function to read all triples in parallel, save as Parquet, and load into Polars dataframe.
read_from_triple_store_with_pandas([endpoint]	Read triples from triple store into pandas dataframe
get_filter_vocabs(data[, directory])	Build all evaluation filters in one pass without copying the input splits.
get_er_vocab(data[, file_path])	
get_re_vocab(data[, file_path])	
get_ee_vocab(data[, file_path])	
create_constraints(triples[, file_path])	
load_with_pandas(→ None)	Deserialize data
save_numpy_ndarray(*, data, file_path)	
load_numpy_ndarray(*, file_path)	
save_pickle(*, data[, file_path])	
load_pickle(*[, file_path])	
create_recipriocal_triples(x)	Add inverse triples into dask dataframe
dataset_sanity_checking(→ None)	

Module Contents

dicee.read_preprocess_save_load_kg.util.**logger**

dicee.read_preprocess_save_load_kg.util.**polars_dataframe_indexer**(*df_polars: polars.DataFrame*, *idx_entity: polars.DataFrame*, *idx_relation: polars.DataFrame*) → polars.DataFrame

Replaces ‘subject’, ‘relation’, and ‘object’ columns in the input Polars DataFrame with their corresponding index values from the entity and relation index DataFrames.

This function processes the DataFrame in three main steps: 1. Replace the ‘relation’ values with the corresponding index from *idx_relation*. 2. Replace the ‘subject’ values with the corresponding index from *idx_entity*. 3. Replace the ‘object’ values with the corresponding index from *idx_entity*.

Parameters:

df_polars
: [polars.DataFrame] The input Polars DataFrame containing columns: ‘subject’, ‘relation’, and ‘object’.

idx_entity
: [polars.DataFrame] A Polars DataFrame that contains the mapping between entity names and their corresponding indices. Must have columns: ‘entity’ and ‘index’.

idx_relation
: [polars.DataFrame] A Polars DataFrame that contains the mapping between relation names and their corresponding indices. Must have columns: ‘relation’ and ‘index’.

Returns:

polars.DataFrame
A DataFrame with the 'subject', 'relation', and 'object' columns replaced by their corresponding indices.

Example Usage:

```
>>> df_polars = pl.DataFrame({
        "subject": ["Alice", "Bob", "Charlie"],
        "relation": ["knows", "works_with", "lives_in"],
        "object": ["Dave", "Eve", "Frank"]
    })
>>> idx_entity = pl.DataFrame({
        "entity": ["Alice", "Bob", "Charlie", "Dave", "Eve", "Frank"],
        "index": [0, 1, 2, 3, 4, 5]
    })
>>> idx_relation = pl.DataFrame({
        "relation": ["knows", "works_with", "lives_in"],
        "index": [0, 1, 2]
    })
>>> polars_dataframe_indexer(df_polars, idx_entity, idx_relation)
```

Steps:

1. Join the input DataFrame *df_polars* on the 'relation' column with *idx_relation* to replace the relations with their indices.
2. Join on 'subject' to replace it with the corresponding entity index using a left join on *idx_entity*.
3. Join on 'object' to replace it with the corresponding entity index using a left join on *idx_entity*.
4. Select only the 'subject', 'relation', and 'object' columns to return the final result.

dicee.read_preprocess_save_load_kg.util.**pandas_dataframe_indexer**(
df_pandas: pandas.DataFrame, *idx_entity: pandas.DataFrame*, *idx_relation: pandas.DataFrame*)
→ pandas.DataFrame

Replaces 'subject', 'relation', and 'object' columns in the input Pandas DataFrame with their corresponding index values from the entity and relation index DataFrames.

Parameters:

df_pandas
[pd.DataFrame] The input Pandas DataFrame containing columns: 'subject', 'relation', and 'object'.

idx_entity
[pd.DataFrame] A Pandas DataFrame that contains the mapping between entity names and their corresponding indices. Must have columns: 'entity' and 'index'.

idx_relation
[pd.DataFrame] A Pandas DataFrame that contains the mapping between relation names and their corresponding indices. Must have columns: 'relation' and 'index'.

Returns:

pd.DataFrame
A DataFrame with the 'subject', 'relation', and 'object' columns replaced by their corresponding indices.

dicee.read_preprocess_save_load_kg.util.**apply_reciprocal_or_noise**(*add_reciprocal: bool*, *eval_model: str*, *df: object | None = None*, *info: str | None = None*)

Add reciprocal triples if conditions are met

dicee.read_preprocess_save_load_kg.util.**timeit**(*func*)

dicee.read_preprocess_save_load_kg.util.**read_with_polars**(*data_path*, *read_only_few: int | None = None*, *sample_triples_ratio: float | None = None*, *separator: str | None = None*) → polars.DataFrame

Load and Preprocess via Polars

dicee.read_preprocess_save_load_kg.util.**read_with_pandas**(*data_path*, *read_only_few: int | None = None*, *sample_triples_ratio: float | None = None*, *separator: str | None = None*)

Load and Preprocess via Pandas

dicee.read_preprocess_save_load_kg.util.**read_from_disk**(*data_path: str*, *read_only_few: int | None = None*, *sample_triples_ratio: float | None = None*, *backend: str | None = None*, *separator: str | None = None*)
→ Tuple[polars.DataFrame, pandas.DataFrame]

dicee.read_preprocess_save_load_kg.util.**count_triples**(*endpoint: str*) → int

Returns the total number of triples in the triple store.

dicee.read_preprocess_save_load_kg.util.**fetch_worker**(*endpoint: str*, *offsets: list[int]*, *chunk_size: int*, *output_dir: str*, *worker_id: int*)

Worker process: fetch assigned chunks and save to disk with per-worker tqdm.

dicee.read_preprocess_save_load_kg.util.**read_from_triple_store_with_polars**(*endpoint: str*, *chunk_size: int = 500000*, *output_dir: str = 'triples_parquet'*)

Main function to read all triples in parallel, save as Parquet, and load into Polars dataframe.

dicee.read_preprocess_save_load_kg.util.**read_from_triple_store_with_pandas**(*endpoint: str | None = None*)

Read triples from triple store into pandas dataframe

dicee.read_preprocess_save_load_kg.util.**get_filter_vocabs**(*data*, *directory=None*)

Build all evaluation filters in one pass without copying the input splits.

Keep the existing defaultdict/list format and pickle filenames so saved checkpoints and evaluation APIs remain compatible.

dicee.read_preprocess_save_load_kg.util.**get_er_vocab**(*data*, *file_path: str | None = None*)

dicee.read_preprocess_save_load_kg.util.**get_re_vocab**(*data*, *file_path: str | None = None*)

dicee.read_preprocess_save_load_kg.util.**get_ee_vocab**(*data*, *file_path: str | None = None*)

dicee.read_preprocess_save_load_kg.util.**create_constraints**(*triples*, *file_path: str | None = None*)

(1) Extract domains and ranges of relations

(2) Store a mapping from relations to entities that are outside of the domain and range. Crete constrainted entities based on the range of relations :param triples: :return: Tuple[dict, dict]

dicee.read_preprocess_save_load_kg.util.**load_with_pandas**(*self*) → None

Deserialize data

dicee.read_preprocess_save_load_kg.util.**save_numpy_ndarray**(*, *data: numpy.ndarray*, *file_path: str*)

dicee.read_preprocess_save_load_kg.util.**load_numpy_ndarray**(*, *file_path: str*)

dicee.read_preprocess_save_load_kg.util.**save_pickle**(*, *data: object*, *file_path=str*)

dicee.read_preprocess_save_load_kg.util.**load_pickle**(*, *file_path=str*)

dicee.read_preprocess_save_load_kg.util.**create_recipriocal_triples**(*x*)

Add inverse triples into dask dataframe :param x: :return:

dicee.read_preprocess_save_load_kg.util.**dataset_sanity_checking**(*train_set: numpy.ndarray*, *num_entities: int*, *num_relations: int*) → None

Parameters

- **train_set**
- **num_entities**
- **num_relations**

Returns

Classes

PreprocessKG	Preprocess the data in memory
LoadSaveToDisk	
ReadFromDisk	Read the data from disk into memory

Package Contents

class dicee.read_preprocess_save_load_kg.**PreprocessKG**(*kg*)

Preprocess the data in memory

kg

start() → None

Preprocess train, valid and test datasets stored in knowledge graph instance

Parameter

rtype
None

preprocess_with_byte_pair_encoding()

preprocess_with_byte_pair_encoding_with_padding() → None

Preprocess with byte pair encoding and add padding

preprocess_with_pandas() → None

Preprocess with pandas: add reciprocal triples, construct vocabulary, and index datasets

preprocess_with_polars() → None

Preprocess with polars: add reciprocal triples and create indexed datasets

sequential_vocabulary_construction() → None

(1) Read input data into memory

(2) Remove triples with a condition

(3) **Serialize vocabularies in a pandas dataframe where**
=> the index is integer and => a single column is string (e.g. URI)

class dicee.read_preprocess_save_load_kg.**LoadSaveToDisk**(*kg*)

kg

save()

load()

class dicee.read_preprocess_save_load_kg.**ReadFromDisk**(*kg*)

Read the data from disk into memory

kg

start() → None

Read a knowledge graph from disk into memory

Data will be available at the train_set, test_set, valid_set attributes.

Parameter

None

rtype
None

add_noisy_triples_into_training()

dicee.sanity_checkers

Attributes

logger

Functions

is_sparql_endpoint_alive([sparql_endpoint])	
validate_knowledge_graph(args)	Validating the source of knowledge graph
sanity_checking_with_arguments(args)	
sanity_check_callback_args(args)	Perform sanity checks on callback-related arguments.

Module Contents

dicee.sanity_checkers.**logger**

dicee.sanity_checkers.**is_sparql_endpoint_alive**(*sparql_endpoint: str | None = None*)

dicee.sanity_checkers.**validate_knowledge_graph**(*args*)

Validating the source of knowledge graph

dicee.sanity_checkers.**sanity_checking_with_arguments**(*args*)

dicee.sanity_checkers.**sanity_check_callback_args**(*args*)

Perform sanity checks on callback-related arguments.

dicee.scripts

Submodules

dicee.scripts.index_serve

$ docker pull qdrant/qdrant && docker run -p 6333:6333 -p 6334:6334 -v $(pwd)/qdrant_storage:/qdrant/storage:z qdrant/qdrant $ dicee_vector_db –index –serve –path CountryEmbeddings –collection “countries_vdb”

Attributes

logger
app
neural_searcher

Classes

NeuralSearcher
StringListRequest

Functions

get_default_arguments()
index(args)
root()
search_embeddings(q)
retrieve_embeddings(q)
search_embeddings_batch(request)
serve(args)
main()

Module Contents

dicee.scripts.index_serve.**logger**

dicee.scripts.index_serve.**get_default_arguments**()

dicee.scripts.index_serve.**index**(*args*)

dicee.scripts.index_serve.**app**

dicee.scripts.index_serve.**neural_searcher = None**

class dicee.scripts.index_serve.**NeuralSearcher**(*args*)

collection_name

entity_to_idx

qdrant_client

topk = 5

retrieve_embedding(*entity: str | None = None*, *entities: List[str] | None = None*) → List

search(*entity: str*)

async dicee.scripts.index_serve.**root**()

async dicee.scripts.index_serve.**search_embeddings**(*q: str*)

async dicee.scripts.index_serve.**retrieve_embeddings**(*q: str*)

class dicee.scripts.index_serve.**StringListRequest**

Bases: pydantic.BaseModel

queries: List[str]

reducer: str | None = None

async dicee.scripts.index_serve.**search_embeddings_batch**(*request: StringListRequest*)

dicee.scripts.index_serve.**serve**(*args*)

dicee.scripts.index_serve.**main**()

dicee.scripts.run

Functions

get_default_arguments([description])	Extends lightning Trainer's arguments with ours
main()	

Module Contents

dicee.scripts.run.**get_default_arguments**(*description=None*)

Extends lightning Trainer's arguments with ours

dicee.scripts.run.**main**()

dicee.static_funcs

Static utility functions for DICE embeddings.

This module provides utility functions for model initialization, data loading, serialization, and various helper operations.

Attributes

logger
MODEL_REGISTRY

Functions

create_recipriocal_triples(→ pandas.DataFrame)	Add inverse triples to a DataFrame.
get_er_vocab(→ Dict[Tuple[int, int], List[int]])	Build entity-relation to tail vocabulary.
get_re_vocab(→ Dict[Tuple[int, int], List[int]])	Build relation-entity (tail) to head vocabulary.
get_ee_vocab(→ Dict[Tuple[int, int], List[int]])	Build entity-entity to relation vocabulary.
timeit(→ Callable)	Decorator to measure and print execution time and memory usage.
setup_distributed_training(→ Dict[str, Union[bool, int]])	Resolve distributed-training state and initialize custom DDP when needed.
save_pickle(→ None)	Save data to a pickle file.
load_pickle(→ object)	Load data from a pickle file.
load_term_mapping(→ Union[dict, polars.DataFrame])	Load term-to-index mapping from pickle or CSV file.
select_model(args, is_continual_training, storage_path)	
load_model(→ Tuple[object, Tuple[dict, dict]])	Load weights and initialize pytorch module from namespace arguments
load_model_ensemble(...)	Construct Ensemble Of weights and initialize pytorch module from namespace arguments
save_numpy_ndarray(*, data, file_path)	
numpy_data_type_changer(→ numpy.ndarray)	Detect most efficient data type for a given triples
save_checkpoint_model(→ None)	Store Pytorch model into disk
store(→ None)	
add_noisy_triples(→ pandas.DataFrame)	Add randomly constructed triples
read_or_load_kg(args, cls)	
intialize_model(...)	Initialize a knowledge graph embedding model.
load_json(→ Dict)	Load JSON file into a dictionary.
save_embeddings(→ None)	Save embeddings to a CSV file.
vocab_to_parquet(vocab_to_idx, name, ...)	
create_experiment_folder(→ str)	Create a timestamped experiment folder.
continual_training_setup_executor(→ None)	
exponential_function(→ torch.FloatTensor)	
load_numpy(→ numpy.ndarray)	
evaluate(entity_to_idx, scores, easy_answers, hard_answers)	# @TODO: CD: Renamed this function
download_file(url[, destination_folder])	
download_files_from_url(→ None)	
download_pretrained_model(→ str)	
write_csv_from_model_parallel(path)	Create
from_pretrained_model_write_embeddings_int(None)	

Module Contents

dicee.static_funcs.**logger**

dicee.static_funcs.**MODEL_REGISTRY: Dict[str, Tuple[Type, str]]**

dicee.static_funcs.**create_recipriocal_triples**(*df: pandas.DataFrame*) → pandas.DataFrame

Add inverse triples to a DataFrame.

For each triple (s, p, o), creates an inverse triple (o, p_inverse, s).

Parameters

df – DataFrame with ‘subject’, ‘relation’, and ‘object’ columns.

Returns

DataFrame with original and inverse triples concatenated.

dicee.static_funcs.**get_er_vocab**(*data: numpy.ndarray*, *file_path: str | None = None*) → Dict[Tuple[int, int], List[int]]

Build entity-relation to tail vocabulary.

Parameters

- **data** – Array of triples with shape (n, 3) where columns are (head, relation, tail).
- **file_path** – Optional path to save the vocabulary as pickle.

Returns

Dictionary mapping (head, relation) pairs to list of tail entities.

dicee.static_funcs.**get_re_vocab**(*data: numpy.ndarray*, *file_path: str | None = None*) → Dict[Tuple[int, int], List[int]]

Build relation-entity (tail) to head vocabulary.

Parameters

- **data** – Array of triples with shape (n, 3) where columns are (head, relation, tail).
- **file_path** – Optional path to save the vocabulary as pickle.

Returns

Dictionary mapping (relation, tail) pairs to list of head entities.

dicee.static_funcs.**get_ee_vocab**(*data: numpy.ndarray*, *file_path: str | None = None*) → Dict[Tuple[int, int], List[int]]

Build entity-entity to relation vocabulary.

Parameters

- **data** – Array of triples with shape (n, 3) where columns are (head, relation, tail).
- **file_path** – Optional path to save the vocabulary as pickle.

Returns

Dictionary mapping (head, tail) pairs to list of relations.

dicee.static_funcs.**timeit**(*func: Callable*) → Callable

Decorator to measure and print execution time and memory usage.

Parameters

func – Function to be timed.

Returns

Wrapped function that prints timing information.

dicee.static_funcs.**setup_distributed_training**(*args*) → Dict[str, bool | int]

Resolve distributed-training state and initialize custom DDP when needed.

dicee.static_funcs.**save_pickle**(*, *data: object | None = None*, *file_path: str*) → None

Save data to a pickle file.

Note: Consider using more portable formats (JSON, Parquet) for new code.

Parameters

- **data** – Object to serialize. If None, nothing is saved.
- **file_path** – Path where the pickle file will be saved.

dicee.static_funcs.**load_pickle**(*file_path: str*) → object

Load data from a pickle file.

Note: Consider using more portable formats (JSON, Parquet) for new code.

Parameters

file_path – Path to the pickle file.

Returns

Deserialized object from the pickle file.

dicee.static_funcs.**load_term_mapping**(*file_path: str*) → dict | polars.DataFrame

Load term-to-index mapping from pickle or CSV file.

Attempts to load from pickle first, falls back to CSV if not found.

Parameters

file_path – Base path without extension.

Returns

Dictionary or Polars DataFrame containing the mapping.

dicee.static_funcs.**select_model**(*args: dict*, *is_continual_training: bool*, *storage_path: str*)

dicee.static_funcs.**load_model**(*path_of_experiment_folder: str*, *model_name='model.pt'*, *verbose=0*)
→ Tuple[object, Tuple[dict, dict]]

Load weights and initialize pytorch module from namespace arguments

dicee.static_funcs.**load_model_ensemble**(*path_of_experiment_folder: str*)
→ Tuple[*dicee.models.base_model.BaseKGE*, Tuple[pandas.DataFrame, pandas.DataFrame]]

Construct Ensemble Of weights and initialize pytorch module from namespace arguments

(1) Detect models under given path

(2) Accumulate parameters of detected models

(3) Normalize parameters

(4) Insert (3) into model.

dicee.static_funcs.**save_numpy_ndarray**(*, *data: numpy.ndarray*, *file_path: str*)

dicee.static_funcs.**numpy_data_type_changer**(*train_set: numpy.ndarray*, *num: int*)
→ numpy.ndarray

Detect most efficient data type for a given triples :param train_set: :param num: :return:

dicee.static_funcs.**save_checkpoint_model**(*model*, *path: str*) → None

Store Pytorch model into disk

dicee.static_funcs.**store**(*trained_model*, *full_storage_path: str*, *model_name: str = 'model'*, *save_embeddings_as_csv=False*) → None

dicee.static_funcs.**add_noisy_triples**(*train_set: pandas.DataFrame*, *add_noise_rate: float*) → pandas.DataFrame

Add randomly constructed triples :param train_set: :param add_noise_rate: :return:

dicee.static_funcs.**read_or_load_kg**(*args*, *cls*)

dicee.static_funcs.**intialize_model**(*args: Dict*, *verbose: int = 0*, *for_inference: bool = False*) → Tuple[*dicee.models.base_model.BaseKGE*, str]

Initialize a knowledge graph embedding model.

Parameters

- **args** – Dictionary containing model configuration including 'model' key.
- **verbose** – Verbosity level. If > 0, prints initialization message.
- **for_inference** – If True, never build the FSDP row-wise-sharded shell, even if args["trainer"] == "torchFSDP". A completed run's model.pt (written by TorchFSDP-Trainer._materialize_model()) already has a plain, full entity_embeddings.weight — loading it into the sharded shell (which defers entity_embeddings until setup_fsdp_training()) fails with "Unexpected key(s): entity_embeddings.weight". Callers loading a finished model for inference (load_model, load_model_ensemble) must set this; callers building a fresh model to train/resume with (select_model) must not.

Returns

Tuple of (initialized model, form of labelling string).

Raises

ValueError – If the model name is not recognized.

dicee.static_funcs.**load_json**(*path: str*) → Dict

Load JSON file into a dictionary.

Parameters

path – Path to the JSON file.

Returns

Dictionary containing the JSON data.

Raises

- **FileNotFoundError** – If the file does not exist.
- **json.JSONDecodeError** – If the file contains invalid JSON.

dicee.static_funcs.**save_embeddings**(*embeddings: numpy.ndarray*, *indexes: List*, *path: str*) → None

Save embeddings to a CSV file.

Parameters

- **embeddings** – NumPy array of embeddings with shape (n_items, embedding_dim).
- **indexes** – List of index labels (entity/relation names).
- **path** – Output file path.

dicee.static_funcs.**vocab_to_parquet**(*vocab_to_idx*, *name*, *path_for_serialization*, *print_into*)

dicee.static_funcs.**create_experiment_folder**(*folder_name: str = 'Experiments'*) → str

Create a timestamped experiment folder.

Parameters

folder_name – Base directory name for experiments.

Returns

Full path to the created experiment folder.

dicee.static_funcs.**continual_training_setup_executor**(*executor*) → None

dicee.static_funcs.**exponential_function**(*x: numpy.ndarray*, *lam: float*, *ascending_order=True*) → torch.FloatTensor

dicee.static_funcs.**load_numpy**(*path*) → numpy.ndarray

dicee.static_funcs.**evaluate**(*entity_to_idx*, *scores*, *easy_answers*, *hard_answers*)

@TODO: CD: Renamed this function Evaluate multi hop query answering on different query types

dicee.static_funcs.**download_file**(*url*, *destination_folder='.'*)

dicee.static_funcs.**download_files_from_url**(*base_url: str*, *destination_folder='.'*) → None

Parameters

- **base_url** (e.g. "https://files.dice-research.org/projects/DiceEmbeddings/KINSHIP-Keci-dim128-epoch256-KvsAll")
- **destination_folder** (*e.g. "KINSHIP-Keci-dim128-epoch256-KvsAll"*)

dicee.static_funcs.**download_pretrained_model**(*url: str*) → str

dicee.static_funcs.**write_csv_from_model_parallel**(*path: str*)

Create

dicee.static_funcs.**from_pretrained_model_write_embeddings_into_csv**(*path: str*) → None

dicee.static_funcs_training

Training-related static functions.

This module provides backward compatibility by re-exporting evaluation functions from the new dicee.evaluation module, along with training utilities.

Deprecated since version Evaluation: functions have moved to `dicee.evaluation`. Use that module for new code. This module will continue to export training utilities.

Functions

evaluate_bpe_lp(→ Dict[str, float])	Evaluate link prediction with BPE-encoded entities.
evaluate_lp(→ Dict[str, float])	Evaluate link prediction with batched processing.
efficient_zero_grad(→ None)	Efficiently zero gradients using parameter.grad = None.
make_iterable_verbose(→ Iterable)	Wrap an iterable with tqdm progress bar if verbose is True.

Module Contents

dicee.static_funcs_training.**evaluate_bpe_lp**(*model*, *triple_idx: List[Tuple]*, *all_bpe_shaped_entities*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *info: str = 'Eval Starts'*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*) → Dict[str, float]

Evaluate link prediction with BPE-encoded entities.

Parameters

- **model** – The KGE model to evaluate.
- **triple_idx** – List of BPE-encoded triple tuples.
- **all_bpe_shaped_entities** – All entities with BPE representations.
- **er_vocab** – Mapping for tail filtering.
- **re_vocab** – Mapping for head filtering.
- **info** – Description to print.
- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.static_funcs_training.**evaluate_lp**(*model*, *triple_idx*, *num_entities: int*, *er_vocab: Dict[Tuple, List]*, *re_vocab: Dict[Tuple, List]*, *info: str = 'Eval Starts'*, *batch_size: int = 128*, *chunk_size: int = 1000*, ***, *tie_policy: str = 'sort'*, *tie_seed: int = 0*, *tie_generator: torch.Generator | None = None*, *graph_rank_plan: dicee.evaluation._graph_inference.GraphRankPlan | None = None*, *reuse_graph_queries: bool = True*) → Dict[str, float]

Evaluate link prediction with batched processing.

Memory-efficient evaluation using chunked entity scoring.

Parameters

- **model** – The KGE model to evaluate.
- **triple_idx** – Integer-indexed triples as numpy array.
- **num_entities** – Total number of entities.
- **er_vocab** – Mapping (head_idx, rel_idx) -> list of tail indices.
- **re_vocab** – Mapping (rel_idx, tail_idx) -> list of head indices.
- **info** – Description to print.
- **batch_size** – Batch size for triple processing.
- **chunk_size** – Chunk size for entity scoring.
- **tie_generator** – Optional CPU generator to continue random ties across calls. When supplied, its state takes precedence over tie_seed.
- **graph_rank_plan** – Optional whole-split plan shared across progress batches.
- **reuse_graph_queries** – Reuse deterministic ULTRA/TRIX queries. Flock keeps its original query and sampling order regardless of this option.

- **tie_policy** – How exact prediction ties are ranked: sort (legacy), optimistic, random, or pessimistic.
- **tie_seed** – Independent random tie seed; reset on each evaluation call.

Returns
Dictionary with H@1, H@3, H@10, and MRR metrics.

dicee.static_funcs_training.**efficient_zero_grad**(*model*) → None

Efficiently zero gradients using parameter.grad = None.

This is more efficient than optimizer.zero_grad() as it avoids memory operations.

See: https://pytorch.org/tutorials/recipes/recipes/tuning_guide.html

Parameters
model – PyTorch model to zero gradients for.

dicee.static_funcs_training.**make_iterable_verbose**(*iterable_object: Iterable*, *verbose: bool*, *desc: str = 'Default'*, *position: int | None = None*, *leave: bool = True*) → Iterable

Wrap an iterable with tqdm progress bar if verbose is True.

Parameters
- **iterable_object** – The iterable to potentially wrap.
- **verbose** – Whether to show progress bar.
- **desc** – Description for the progress bar.
- **position** – Position of the progress bar.
- **leave** – Whether to leave the progress bar after completion.

Returns
The original iterable or a tqdm-wrapped version.

dicee.static_preprocess_funcs

Attributes

logger
enable_log

Functions

timeit(func)	
preprocesses_input_args(args)	Sanity Checking in input arguments
create_constraints(→ Tuple[dict, dict, dict, dict])	
get_er_vocab(data)	
get_re_vocab(data)	
get_ee_vocab(data)	
mapping_from_first_two_cols_to_third(train_se	

Module Contents

dicee.static_preprocess_funcs.**logger**

dicee.static_preprocess_funcs.**enable_log = False**

dicee.static_preprocess_funcs.**timeit**(*func*)

dicee.static_preprocess_funcs.**preprocesses_input_args**(*args*)

Sanity Checking in input arguments

dicee.static_preprocess_funcs.**create_constraints**(*triples: numpy.ndarray*)
→ Tuple[dict, dict, dict, dict]

(1) Extract domains and ranges of relations

(2) Store a mapping from relations to entities that are outside of the domain and range. Create constraints entities based on the range of relations :param triples: :return:

dicee.static_preprocess_funcs.**get_er_vocab**(*data*)

dicee.static_preprocess_funcs.**get_re_vocab**(*data*)

dicee.static_preprocess_funcs.**get_ee_vocab**(*data*)

dicee.static_preprocess_funcs.**mapping_from_first_two_cols_to_third**(*train_set_idx*)

dicee.trainer

Submodules

dicee.trainer.auto_batch_finder

Attributes

logger

Functions

find_good_batch_size(→ Tuple[int, Optional[float]])	Find a batch size that uses GPU memory efficiently.

Module Contents

dicee.trainer.auto_batch_finder.**logger**

dicee.trainer.auto_batch_finder.**find_good_batch_size**(
train_loader: torch.utils.data.DataLoader, *training_step_fn: Callable*, *device*)
→ Tuple[int, float | None]

Find a batch size that uses GPU memory efficiently.

Progressively increases batch size (with tunable delta) until GPU memory exceeds 90% or a CUDA OOM error is raised, then backs off to the last safe batch size. Only supported on CUDA devices; returns the initial batch size unchanged on CPU.

Parameters

- **train_loader** – DataLoader wrapping the training dataset.
- **training_step_fn** – Callable[[batch], float] — runs one forward+backward pass and returns the scalar loss. Each trainer passes its own implementation so this function stays trainer-agnostic.
- **device** – torch.device (or anything accepted by torch.device()) used to monitor GPU memory.

Returns

(batch_size, runtime) where runtime is the wall-clock seconds of the last successful batch, or None when batch finding was skipped.

dicee.trainer.dice_trainer

DICE Trainer module for knowledge graph embedding training.

Provides the DICE_Trainer class which supports multiple training backends including PyTorch Lightning, DDP, and custom CPU/GPU trainers.

Attributes

logger	

Classes

DICE_Trainer	DICE_Trainer implement

Functions

load_term_mapping(→ polars.DataFrame)	Load term-to-index mapping from CSV file.
initialize_trainer(...)	Initialize the appropriate trainer based on configuration.
get_callbacks(→ List)	Create list of callbacks based on configuration.

Module Contents

dicee.trainer.dice_trainer.**logger**

dicee.trainer.dice_trainer.**load_term_mapping**(*file_path: str*) → polars.DataFrame

Load term-to-index mapping from CSV file.

Parameters

file_path – Base path without extension.

Returns

Polars DataFrame containing the mapping.

dicee.trainer.dice_trainer.**initialize_trainer**(*args*, *callbacks: List*)
→ *dicee.trainer.torch_trainer.TorchTrainer* | *dicee.trainer.model_parallelism.TensorParallel* | *dicee.trainer.torch_trainer_ddp.*

Initialize the appropriate trainer based on configuration.

Parameters

- **args** – Configuration arguments containing trainer type.
- **callbacks** – List of training callbacks.

Returns
Initialized trainer instance.

Raises
AssertionError – If trainer is None after initialization.

dicee.trainer.dice_trainer.**get_callbacks**(*args*) → List

Create list of callbacks based on configuration.

Parameters
args – Configuration arguments.

Returns
List of callback instances.

class dicee.trainer.dice_trainer.**DICE_Trainer**(*args*, *is_continual_training: bool*, *storage_path*, *evaluator=None*)

DICE_Trainer implement
1- Pytorch Lightning trainer (https://pytorch-lightning.readthedocs.io/en/stable/common/trainer.html) 2- Multi-GPU Trainer(https://pytorch.org/docs/stable/generated/torch.nn.parallel.DistributedDataParallel.html) 3- CPU Trainer

args

is_continual_training:bool

storage_path:str

evaluator:

report:dict

report

args

trainer = None

is_continual_training

storage_path

evaluator = None

form_of_labelling = None

continual_start(*knowledge_graph*)

(1) Initialize training.

(2) Load model

(3) Load trainer (3) Fit model

Parameter

returns

- *model*
- **form_of_labelling** (*str*)

initialize_trainer(*callbacks: List*)
→ lightning.Trainer | *dicee.trainer.model_parallelism.TensorParallel* | *dicee.trainer.torch_trainer.TorchTrainer* | *dicee.t*

Initialize Trainer from input arguments

initialize_or_load_model()

init_dataloader(*dataset: torch.utils.data.Dataset*) → torch.utils.data.DataLoader

init_dataset() → torch.utils.data.Dataset

prepare_graph_model(*model*, *knowledge_graph*)

Attach only training facts, preserving the indexed external vocabulary.

prepare_ultra

start(*knowledge_graph: dicee.knowledge_graph.KG | numpy.memmap*)
→ Tuple[*dicee.models.base_model.BaseKGE*, str]

Start the training

(1) Initialize Trainer

(2) Initialize or load a pretrained KGE model

in DDP setup, we need to load the memory map of already read/index KG.

k_fold_cross_validation(*dataset*) → Tuple[*dicee.models.base_model.BaseKGE*, str]

Perform K-fold Cross-Validation

1. Obtain K train and test splits.
2. **For each split,**
   2.1 initialize trainer and model 2.2. Train model with configuration provided in args. 2.3. Compute the mean reciprocal rank (MRR) score of the model on the test respective split.
3. Report the mean and average MRR .

Parameters

- **self**
- **dataset**

Returns
model

dicee.trainer.model_parallelism

Tensor Parallelism trainer for ensemble knowledge graph embeddings.

This module implements the tensor parallelism training strategy described in:

"Multiple Run Ensemble Learning with Low-Dimensional Knowledge Graph Embeddings"
https://arxiv.org/abs/2104.05003

The TensorParallel trainer creates an ensemble of models, each trained on a separate GPU, allowing efficient utilization of multi-GPU systems through model parallelism.

Attributes

logger

Classes

TensorParallel	Abstract base class for KGE model trainers.

Functions

extract_input_outputs(z[, device])
forward_backward_update_loss(→ float)

Module Contents

dicee.trainer.model_parallelism.**logger**

dicee.trainer.model_parallelism.**extract_input_outputs**(*z: list*, *device=None*)

dicee.trainer.model_parallelism.**forward_backward_update_loss**(*z: Tuple*, *ensemble_model*) → float

class dicee.trainer.model_parallelism.**TensorParallel**(*args*, *callbacks*)

Bases: *dicee.abstracts.AbstractTrainer*

Abstract base class for KGE model trainers.

Provides the callback dispatch mechanism shared by all concrete trainer implementations (TorchTrainer, TorchDDPTrainer, etc.). Sub-classes call the `on_*` hooks at the appropriate points in the training loop so that any registered `AbstractCallback` can react.

Parameters

- **args** (*argparse.Namespace or similar*) – Processed configuration object. Must expose at least `random_seed` (int).
- **callbacks** (*list of AbstractCallback*) – Ordered list of callback instances to invoke at each lifecycle hook.

fit(**args*, ***kwargs*)

Train model

dicee.trainer.torch_trainer

Attributes

logger

Classes

TorchTrainer	TorchTrainer for using single GPU or multi CPUs on a single node

Module Contents

dicee.trainer.torch_trainer.**logger**

class dicee.trainer.torch_trainer.**TorchTrainer**(*args*, *callbacks*)

Bases: *dicee.abstracts.AbstractTrainer*

TorchTrainer for using single GPU or multi CPUs on a single node

Arguments

callbacks: list of Abstract callback instances

loss_function = None

optimizer = None

model = None

train_dataloaders = None

training_step = None

process

fit(**args*, *train_dataloaders*, ***kwargs*) → None

Training starts

Arguments

kwargs:Tuple
empty dictionary

Return type
batch loss (float)

forward_backward_update(*x_batch: torch.Tensor*, *y_batch: torch.Tensor*, ***, *loss_divisor: int = 1*, *update_weights: bool = True*) → float

Compute forward, loss, backward, and parameter update

Arguments

Return type
batch loss (float)

extract_input_outputs_set_device(*batch: list*) → Tuple

Construct inputs and outputs from a batch of inputs with outputs From a batch of inputs and put

Arguments

Return type
(tuple) mini-batch on select device

dicee.trainer.torch_trainer_ddp

Attributes

logger

Classes

TorchDDPTrainer	Abstract base class for KGE model trainers.
NodeTrainer	

Functions

make_iterable_verbose(→ Iterable)

Module Contents

dicee.trainer.torch_trainer_ddp.**logger**

dicee.trainer.torch_trainer_ddp.**make_iterable_verbose**(*iterable_object*, *verbose*, *desc='Default'*, *position=None*, *leave=True*) → Iterable

class dicee.trainer.torch_trainer_ddp.**TorchDDPTrainer**(*args*, *callbacks*)

Bases: *dicee.abstracts.AbstractTrainer*

Abstract base class for KGE model trainers.

Provides the callback dispatch mechanism shared by all concrete trainer implementations (TorchTrainer, TorchDDPTrainer, etc.). Sub-classes call the `on_*` hooks at the appropriate points in the training loop so that any registered `AbstractCallback` can react.

Parameters

- **args** (*argparse.Namespace or similar*) – Processed configuration object. Must expose at least `random_seed` (int).
- **callbacks** (*list of AbstractCallback*) – Ordered list of callback instances to invoke at each lifecycle hook.

fit(**args*, ***kwargs*)

class dicee.trainer.torch_trainer_ddp.**NodeTrainer**(*trainer*, *model: torch.nn.Module*, *train_dataset_loader: torch.utils.data.DataLoader*, *callbacks*, *num_epochs: int*)

trainer

local_rank

global_rank

optimizer

train_dataset_loader

loss_func

callbacks

model

num_epochs

loss_history = []

ctx

scaler

extract_input_outputs(*z: list*)

train()

dicee.trainer.torch_trainer_fsdp

Multi-GPU trainer: row-wise sharded entity embeddings + FSDP dense parameters.

Entity embeddings are sharded row-wise across ranks via dist.all_to_all_single. Each rank owns a contiguous slice of entity rows; forward and backward use all_to_all to route index requests and gradient returns to the owning rank. A per-rank _LocalSparseAdam updates only the rows that received a non-zero gradient each batch.

Dense parameters (relation embeddings, Clifford coefficients, …) are wrapped with FSDP and updated by the standard dense optimizer.

Attributes

logger	
OptimizedModule	

Classes

TorchFSDPTrainer	Single-node multi-GPU trainer: row-wise sharded entity embeddings + FSDP.

Module Contents

dicee.trainer.torch_trainer_fsdp.**logger**

dicee.trainer.torch_trainer_fsdp.**OptimizedModule = None**

class dicee.trainer.torch_trainer_fsdp.**TorchFSDPTrainer**(*args*, *callbacks*)

Bases: *dicee.abstracts.AbstractTrainer*

Single-node multi-GPU trainer: row-wise sharded entity embeddings + FSDP.

Entity embeddings

Sharded row-wise across GPUs. all_to_all routes each index request to its owning rank; embeddings and gradients travel back the same way. _LocalSparseAdam updates only the rows accessed in each batch — O(active_rows) not O(shard_size). Each entity receives gradient from every rank's batches.

Dense parameters (relation embeddings, Clifford coefficients, ...)

Wrapped with FSDP; updated by the standard dense optimizer step().

Usage

Launched with torchrun:

torchrun –nproc_per_node=4 -u dicee –trainer torchFSDP …

local_rank

global_rank

is_global_zero

device

model = None

raw_model = None

optimizer = None

loss_func = None

train_dataset_loader = None

ptdtype

ctx

scaler

sharding_strategy

gradient_clip_val

use_compile

num_workers

prefetch_factor

entity_optim_device

checkpoint_every_n_epochs

fit(**args*, ***kwargs*)

extract_input_outputs(*z*)

Classes

DICE_Trainer	DICE_Trainer implement

Package Contents

class dicee.trainer.**DICE_Trainer**(*args*, *is_continual_training: bool*, *storage_path*, *evaluator=None*)

DICE_Trainer implement

1- Pytorch Lightning trainer (https://pytorch-lightning.readthedocs.io/en/stable/common/trainer.html) 2- Multi-GPU Trainer(https://pytorch.org/docs/stable/generated/torch.nn.parallel.DistributedDataParallel.html) 3- CPU Trainer

args

is_continual_training:bool

storage_path:str

evaluator:

report:dict

report

args

trainer = None

is_continual_training

storage_path

evaluator = None

form_of_labelling = None

continual_start(*knowledge_graph*)

(1) Initialize training.

(2) Load model

(3) Load trainer (3) Fit model

Parameter

returns

- *model*
- **form_of_labelling** (*str*)

initialize_trainer(*callbacks: List*)
→ lightning.Trainer | *dicee.trainer.model_parallelism.TensorParallel* | *dicee.trainer.torch_trainer.TorchTrainer* | *dicee.t*

Initialize Trainer from input arguments

initialize_or_load_model()

init_dataloader(*dataset: torch.utils.data.Dataset*) → torch.utils.data.DataLoader

init_dataset() → torch.utils.data.Dataset

prepare_graph_model(*model*, *knowledge_graph*)

Attach only training facts, preserving the indexed external vocabulary.

prepare_ultra

start(*knowledge_graph: dicee.knowledge_graph.KG | numpy.memmap*)
→ Tuple[*dicee.models.base_model.BaseKGE*, str]

Start the training

(1) Initialize Trainer

(2) Initialize or load a pretrained KGE model

in DDP setup, we need to load the memory map of already read/index KG.

k_fold_cross_validation(*dataset*) → Tuple[*dicee.models.base_model.BaseKGE*, str]

Perform K-fold Cross-Validation

1. Obtain K train and test splits.
2. **For each split,**
   2.1 initialize trainer and model 2.2. Train model with configuration provided in args. 2.3. Compute the mean reciprocal rank (MRR) score of the model on the test respective split.
3. Report the mean and average MRR .

Parameters

- **self**
- **dataset**

Returns
model

dicee.weight_averaging

Classes

AMWA	Adaptive Momentum Weight Averaging (AMWA).
ASWA	Adaptive stochastic weight averaging
SWA	Stochastic Weight Averaging callback.
SWAG	Stochastic Weight Averaging - Gaussian (SWAG).
EMA	Exponential Moving Average (EMA) callback.
TWA	Train with Weight Averaging (TWA) using subspace projection + averaging.

Module Contents

class dicee.weight_averaging.**AMWA**(*amwa_start_epoch: int = 0*, *amwa_c_epochs: int = 1*, *path: str | None = None*, *monitor: str = 'MRR'*, *maximize: bool = True*, *beta: float | None = None*, *beta_window: int = 10*, *beta_init: float = 1.0*, *beta_floor: float = 1e-08*)

Bases: *dicee.abstracts.AbstractCallback*

Adaptive Momentum Weight Averaging (AMWA).

Paper: https://www.sciencedirect.com/science/article/pii/S0031320325009586

BaseNet is the ordinary model optimized during training, while StableNet is updated once per epoch via weight averaging,

w_stable <- m * w_stable + (1 - m) * w_base

where the adaptive momentum is computed from the validation-performance difference between StableNet and BaseNet,

m(Delta) = 1 / (1 + exp(-max(0, Delta) / beta_n))

with Delta defined so that Delta > 0 means StableNet performs better than BaseNet. This yields the exact behavior described in the paper: StableNet stays close to its current solution when it already validates better, and it takes a larger step toward BaseNet when BaseNet validates better. The paper sets beta_n to the standard deviation of Delta over the past 10 epochs; this implementation follows that default and also allows a fixed beta for the ablation setting.

param amwa_start_epoch
Epoch at which AMWA starts tracking StableNet.

type amwa_start_epoch
int

param amwa_c_epochs
Update interval in epochs.

type amwa_c_epochs
int

param path
Directory used to persist `amwa.pt` and `amwa_history.json`.

type path
str, optional

param monitor
Validation metric to monitor, e.g. `"MRR"`.

type monitor
str

param maximize
Whether larger values of `monitor` are better.

type maximize
bool

param beta
If provided, use a fixed beta instead of the adaptive beta_n from the paper. This matches the paper's fixed-beta ablation.

type beta
float, optional

param beta_window
Number of past Delta values used to estimate beta_n.

type beta_window
int

param beta_init
Beta used before enough Delta history exists.

type beta_init
float

param beta_floor
Numerical floor used when the empirical standard deviation is zero.

type beta_floor
float

amwa_start_epoch = 0

amwa_c_epochs = 1

path = None

monitor = 'MRR'

maximize = True

beta = None

beta_window

beta_init

beta_floor

current_epoch = -1

initial_eval_setting: str | None = None

stable_state_dict: Dict[str, torch.Tensor] | None = None

delta_history: List[float] = []

momentum_history: List[float] = []

beta_history: List[float] = []

base_history: List[float] = []

stable_history: List[float] = []

history: List[Dict[str, Any]] = []

on_train_epoch_start(*trainer*, *model*)
Called when the train epoch begins.

on_train_epoch_end(*trainer*, *model*)
Called at the end of each training epoch.

Parameters

- **trainer** (AbstractTrainer *or pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model being trained. `model.loss_history` contains the per-epoch average losses accumulated so far.

on_fit_end(*trainer*, *model*)
Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

class dicee.weight_averaging.**ASWA**(*num_epochs*, *path*)

Bases: *dicee.abstracts.AbstractCallback*

Adaptive stochastic weight averaging ASWE keeps track of the validation performance and update s the ensemble model accordingly.

path

num_epochs

initial_eval_setting = None

epoch_count = 0

alphas = []

val_aswa = -1

on_fit_end(*trainer*, *model*)

Called once after the final training epoch completes.

Override to perform post-training actions such as saving the final model state, computing evaluation metrics, or cleaning up resources.

static compute_mrr(*trainer*, *model*) → float

get_aswa_state_dict(*model*)

decide(*running_model_state_dict*, *ensemble_state_dict*, *val_running_model*, *mrr_updated_ensemble_model*)

Perform Hard Update, software or rejection

Parameters

- **running_model_state_dict**
- **ensemble_state_dict**
- **val_running_model**
- **mrr_updated_ensemble_model**

on_train_epoch_end(*trainer*, *model*)

Called at the end of each training epoch.

Parameters

- **trainer** (AbstractTrainer *or pl.Trainer*) – The active trainer instance.
- **model** (BaseKGE) – The model being trained. `model.loss_history` contains the per-epoch average losses accumulated so far.

class dicee.weight_averaging.**SWA**(*swa_start_epoch*, *swa_c_epochs: int = 1*, *lr_init: float = 0.1*, *swa_lr: float = 0.05*, *max_epochs: int | None = None*)

Bases: *dicee.abstracts.AbstractCallback*

Stochastic Weight Averaging callback.

Initialize SWA callback.

swa_start_epoch: int
The epoch at which to start SWA.

swa_c_epochs: int
The number of epochs to use for SWA.

lr_init: float
The initial learning rate.

swa_lr: float
The learning rate to use during SWA.

max_epochs: int
The maximum number of epochs. args.num_epochs

swa_start_epoch

swa_c_epochs = 1

swa_lr = 0.05

lr_init = 0.1

max_epochs = None

swa_model = None

swa_n = 0

current_epoch = -1

static moving_average(*swa_model*, *running_model*, *alpha*)

Update SWA model with moving average of current model. Math: # SWA update: # θ_swa ← (1 - alpha) * θ_swa + alpha * θ # alpha = 1 / (n + 1), where n = number of models already averaged # alpha is tracked via self.swa_n in code

on_train_epoch_start(*trainer*, *model*)

Update learning rate according to SWA schedule.

on_train_epoch_end(*trainer*, *model*)

Apply SWA averaging if conditions are met.

on_fit_end(*trainer*, *model*)

Replace main model with SWA model at the end of training.

class dicee.weight_averaging.**SWAG**(*swa_start_epoch*, *swa_c_epochs: int = 1*, *lr_init: float = 0.1*, *swa_lr: float = 0.05*, *max_epochs: int | None = None*, *max_num_models: int = 20*, *var_clamp: float = 1e-30*)

Bases: *dicee.abstracts.AbstractCallback*

Stochastic Weight Averaging - Gaussian (SWAG). Parameters

swa_start_epoch
[int] Epoch at which to start collecting weights.

swa_c_epochs
[int] Interval of epochs between updates.

lr_init
[float] Initial LR.

swa_lr
[float] LR in SWA / GSWA phase.

max_epochs
[int] Total number of epochs.

max_num_models
[int] Number of models to keep for low-rank covariance approx.

var_clamp
[float] Clamp low variance for stability.

swa_start_epoch

swa_c_epochs = 1

swa_lr = 0.05

lr_init = 0.1

max_epochs = None

max_num_models = 20

var_clamp = 1e-30

mean = None

sq_mean = None

deviations = []

gswa_n = 0

current_epoch = -1

get_mean_and_var()
Return mean + variance (diagonal part).

sample(*base_model*, *scale=0.5*)
Sample new model from SWAG posterior distribution.

Math: # From SWAG, posterior is approximated as: # θ ~ N(mean, Σ) # where Σ ≈ diag(var) + (1/(K-1)) * D D^T # - mean = running average of weights # - var = elementwise variance (sq_mean - mean^2) # - D = [dev_1, dev_2, …, dev_K], deviations from mean (low-rank approx) # - K = number of collected models

Sampling step: # 1. θ_diag = mean + scale * std ⊙ ε, where ε ~ N(0, I) # 2. θ_lowrank = θ_diag + (D z) / sqrt(K-1), where z ~ N(0, I_K) # Final sample = θ_lowrank

on_train_epoch_start(*trainer*, *model*)
Update LR schedule (same as SWA).

on_train_epoch_end(*trainer*, *model*)
Collect Gaussian stats at the end of epochs after swa_start.

on_fit_end(*trainer*, *model*)
Set model weights to the collected SWAG mean at the end of training.

class dicee.weight_averaging.**EMA**(*ema_start_epoch: int*, *decay: float = 0.999*, *max_epochs: int | None = None*, *ema_c_epochs: int = 1*)

Bases: *dicee.abstracts.AbstractCallback*

Exponential Moving Average (EMA) callback.

Parameters

- **ema_start_epoch** (*int*) – Epoch to start EMA.
- **decay** (*float*) – EMA decay rate (typical: 0.99 - 0.9999) Math: θ_ema <- decay * θ_ema + (1 - decay) * θ
- **max_epochs** (*int*) – Maximum number of epochs.

ema_start_epoch

decay = 0.999

max_epochs = None

ema_c_epochs = 1

ema_model = None

current_epoch = -1

static ema_update(*ema_model*, *running_model*, *decay: float*)

Update EMA model with exponential moving average of current model. Math: # EMA update: # θ_ema ← (1 - alpha) * θ_ema + alpha * θ # alpha = 1 - decay, where decay is the EMA smoothing factor (typical 0.99 - 0.999) # alpha controls how much of the current model θ contributes to the EMA # decay is fixed in code –> can be extended to scheduled

on_train_epoch_start(*trainer*, *model*)

Track current epoch.

on_train_epoch_end(*trainer*, *model*)

Update EMA if past start epoch.

on_fit_end(*trainer*, *model*)

Replace main model with EMA model at the end of training.

class dicee.weight_averaging.**TWA**(*twa_start_epoch: int*, *lr_init: float*, *num_samples: int = 5*, *reg_lambda: float = 0.0*, *max_epochs: int | None = None*, *twa_c_epochs: int = 1*)

Bases: *dicee.abstracts.AbstractCallback*

Train with Weight Averaging (TWA) using subspace projection + averaging.

Parameters

twa_start_epoch
[int] Epoch to start TWA.

lr_init
[float] Learning rate used for β updates.

num_samples
[int] Number of sampled weight snapshots to build projection subspace.

reg_lambda
[float] Regularization coefficient for β updates.

max_epochs
[int] Total number of training epochs.

twa_c_epochs
[int] Interval of epochs between TWA updates.

twa_start_epoch

num_samples = 5

reg_lambda = 0.0

max_epochs = None

lr_init

twa_c_epochs = 1

current_epoch = -1

weight_samples = []

twa_model = None

base_weights = None

P = None

beta = None

sample_weights(*model*)

Collect sampled weights from the current model and maintain rolling buffer.

build_projection(*weight_samples*, *k=None*)

Build projection subspace from collected weight samples. :param weight_samples: list of flat weight tensors [(D,), …] :param k: number of basis vectors to keep. Defaults to min(N, D).

Returns

(D,) base weight vector (average) P: (D, k) projection matrix with top-k basis directions

Return type

mean_w

on_train_epoch_start(*trainer*, *model*)

Track epoch.

on_train_epoch_end(*trainer*, *model*)

Main TWA logic: build subspace and update in β space.

Math: # TWA weight update: # w_twa = mean_w + P * beta # mean_w = (1/n) * sum_i w_i (SWA baseline) # beta <- (1 - eta * lambda) * beta - eta * P^T * g # g = gradient of training loss w.r.t. full model weights # eta = learning rate, lambda = ridge regularization # P = orthonormal basis spanning sampled checkpoints {w_i}

on_fit_end(*trainer*, *model*)

Replace with TWA model at the end.

14.2 Attributes

__version__

14.3 Classes

Execute	Executor class for training, retraining and evaluating KGE models.
KGE	Knowledge Graph Embedding Class for interactive usage of pre-trained models
QueryGenerator	
DICE_Trainer	DICE_Trainer implement
Evaluator	Evaluator class for KGE models in various downstream tasks.

14.4 Package Contents

class dicee.**Execute**(*args*, *continuous_training: bool = False*)

Executor class for training, retraining and evaluating KGE models.

Handles the complete workflow: 1. Loading & Preprocessing & Serializing input data 2. Training & Validation & Testing 3. Storing all necessary information

args

Processed input arguments.

distributed

Whether distributed training is enabled.

rank

Process rank in distributed training.

world_size

Total number of processes.

local_rank

Local GPU rank.

trainer

Training handler instance.

trained_model

The trained model after training completes.

knowledge_graph

The loaded knowledge graph.

report

Dictionary storing training metrics and results.

evaluator

Model evaluation handler.

distributed

rank

world_size

local_rank

args

is_continual_training = False

trainer: *dicee.trainer.DICE_Trainer* **| None = None**

trained_model = None

knowledge_graph: *dicee.knowledge_graph.KG* **| None = None**

report: Dict

evaluator: *dicee.evaluator.Evaluator* **| None = None**

start_time: float | None = None

is_local_rank_zero() → bool

is_global_rank_zero() → bool

cleanup()

setup_executor() → None

Set up storage directories for the experiment.

Creates or reuses experiment directories based on configuration. Saves the configuration to a JSON file.

create_and_store_kg() → None

Create knowledge graph and store as memory-mapped file.

Only executed on global rank 0 in distributed training, so exactly one process writes the shared memmap/vocab files regardless of how many nodes are involved. Skips if memmap already exists.

load_from_memmap() → None

Load knowledge graph from memory-mapped file.

save_trained_model() → None

Save a knowledge graph embedding model

(1) Send model to eval mode and cpu.

(2) Store the memory footprint of the model.

(3) Save the model into disk.

(4) Update the stats of KG again ?

Parameter

rtype
None

end(*form_of_labelling: str*) → dict

End training

(1) Store trained model.

(2) Report runtimes.

(3) Eval model if required.

Parameter

rtype
A dict containing information about the training and/or evaluation

write_report() → None
Report training related information in a report.json file

start() → dict
Start training

(1) Loading the Data # (2) Create an evaluator object. # (3) Create a trainer object. # (4) Start the training

Parameter

rtype
A dict containing information about the training and/or evaluation

class dicee.**KGE**(*path=None*, *url=None*, *construct_ensemble=False*, *model_name=None*)
Bases: *dicee.abstracts.BaseInteractiveKGE*, *dicee.abstracts.InteractiveQueryDecomposition*, *dicee.abstracts.BaseInteractiveTrainKGE*

Knowledge Graph Embedding Class for interactive usage of pre-trained models

__str__()

to(*device: str*) → None

get_transductive_entity_embeddings(*indices: torch.LongTensor | List[str]*, *as_pytorch=False*, *as_numpy=False*, *as_list=True*) → torch.FloatTensor | numpy.ndarray | List[float]

create_vector_database(*collection_name: str*, *distance: str*, *location: str = 'localhost'*, *port: int = 6333*)

generate(*h=''*, *r=''*)

eval_lp_performance(*dataset=List[Tuple[str, str, str]]*, *filtered=True*, ***, *tie_policy=None*, *tie_seed=None*)
Evaluate head/tail ranks, inheriting saved tie settings unless overridden.

`tie_policy` accepts sort (legacy), optimistic, random, or pessimistic. `tie_seed` seeds an independent random stream for each evaluation.

predict_missing_head_entity(*relation: List[str] | str*, *tail_entity: List[str] | str*, *within=None*, *batch_size=2*, *topk=1*, *return_indices=False*) → Tuple
Given a relation and a tail entity, return top k ranked head entity.

argmax_{e in E } f(e,r,t), where r in R, t in E.

Parameter

relation: Union[List[str], str]

String representation of selected relations.

tail_entity: Union[List[str], str]

String representation of selected entities.

k: int

Highest ranked k entities.

Returns: Tuple

Highest K scores and entities

predict_missing_relations(*head_entity: List[str] | str*, *tail_entity: List[str] | str*, *within=None*, *batch_size=2*, *topk=1*, *return_indices=False*) → Tuple

Given a head entity and a tail entity, return top k ranked relations.

argmax_{r in R } f(h,r,t), where h, t in E.

Parameter

head_entity: List[str]

String representation of selected entities.

tail_entity: List[str]

String representation of selected entities.

k: int

Highest ranked k entities.

Returns: Tuple

Highest K scores and entities

predict_missing_tail_entity(*head_entity: List[str] | str*, *relation: List[str] | str*, *within: List[str] | None = None*, *batch_size=2*, *topk=1*, *return_indices=False*) → torch.FloatTensor

Given a head entity and a relation, return top k ranked entities

argmax_{e in E } f(h,r,e), where h in E and r in R.

Parameter

head_entity: List[str]

String representation of selected entities.

tail_entity: List[str]

String representation of selected entities.

Returns: Tuple

scores

predict(*, *h: List[str] | str | None = None*, *r: List[str] | str | None = None*, *t: List[str] | str | None = None*, *within: List[str] | None = None*, *logits: bool = True*) → torch.FloatTensor

Predict scores for triples or missing triple elements.

Parameters

- **h** – Head entity/entities. None to predict heads.
- **r** – Relation/relations. None to predict relations.
- **t** – Tail entity/entities. None to predict tails.
- **within** – Optional list of entities to restrict predictions to.
- **logits** – If True, return raw scores. If False, return sigmoid scores (0-1).

Returns

- Single triple (h, r, t): scalar score
- Missing element: vector of all possible scores

Return type
torch.FloatTensor of scores. Shape depends on the query type

Raises

- **`TypeError`** – If inputs are not strings or lists of strings.
- **`ValueError`** – If a required argument for the query type is missing.

Examples

```
>>> # Score a specific triple
>>> model.predict(h="Mongolia", r="isLocatedIn", t="Asia", logits=False)
tensor(0.9523)
```

```
>>> # Get scores for all possible tail entities
>>> model.predict(h="Mongolia", r="isLocatedIn", t=None)
tensor([0.21, 0.95, 0.03, ...])  # One score per entity
```

`predict_topk`(*, *h: str | List[str] | None = None*, *r: str | List[str] | None = None*, *t: str | List[str] | None = None*, *topk: int = 10*, *within: List[str] | None = None*, *batch_size: int = 1024*) → List[Tuple[str, float]] | List[List[Tuple[str, float]]]

Predict top-k missing items in a given triple pattern.

Parameters

- **`h`** – Head entity/entities. None to predict heads.
- **`r`** – Relation/relations. None to predict relations.
- **`t`** – Tail entity/entities. None to predict tails.
- **`topk`** – Number of top predictions to return.
- **`within`** – Optional list of entities to restrict predictions to.
- **`batch_size`** – Batch size for processing multiple queries.

Returns
List[(item, score), …] of length topk. For batch query: List of such lists, one per query.

Return type
For single query

Raises

- **`TypeError`** – If h, r, or t is not a str or list of str.
- **`ValueError`** – If the required arguments for a query type are None.

Examples

```
>>> model.predict_topk(h=["Mongolia"], r=["isLocatedIn"], topk=3)
[('Asia', 0.99), ('Europe', 0.02), ...]
```

```
>>> model.predict_topk(r=["isLocatedIn"], t=["Asia"], topk=5)
[('Mongolia', 0.85), ('China', 0.82), ...]
```

triple_score(*h: List[str] | str | None = None*, *r: List[str] | str | None = None*, *t: List[str] | str | None = None*, *logits=False*) → torch.FloatTensor

Predict triple score

Parameter

head_entity: List[str]

String representation of selected entities.

relation: List[str]

String representation of selected relations.

tail_entity: List[str]

String representation of selected entities.

logits: bool

If logits is True, unnormalized score returned

Returns: Tuple

pytorch tensor of triple score

return_multi_hop_query_results(*aggregated_query_for_all_entities*, *k: int*, *only_scores*)

single_hop_query_answering(*query: tuple*, *only_scores: bool = True*, *k: int | None = None*, *use_logits: bool = True*)

answer_multi_hop_query(*query_type: str | None = None*, *query: Tuple[str | Tuple[str, str], ...] | None = None*, *queries: List[Tuple[str | Tuple[str, str], ...]] | None = None*, *tnorm: str = 'prod'*, *neg_norm: str = 'standard'*, *lambda_: float = 0.0*, *k: int = 10*, *only_scores: bool = False*, *use_logits: bool = True*) → List[Tuple[str, torch.Tensor]] | List[List[Tuple[str, torch.Tensor]]]

Answer multi-hop EPFO (Existential Positive First-Order) queries.

Supports 9 query types: 1p, 2p, 3p, 2i, 3i, ip, pi, 2u, up. See docs/guides/multi_hop_queries.md for detailed query patterns.

Parameters

- **query_type** – Query pattern name. One of: - 1p: (e, (r,)) # One-hop - 2p: (e, (r1, r2)) # Two-hop - 3p: (e, (r1, r2, r3)) # Three-hop - 2i: ((e1, (r1,)), (e2, (r2,))) # Two-way intersection - 3i: ((e1, (r1,)), (e2, (r2,)), (e3, (r3,))) # Three-way intersection - ip: (((e1, (r1,)), (e2, (r2,))), (r3,)) # Intersection + projection - pi: ((e, (r1, r2)), (r3,)) # Projection + intersection (2i meets 2p) - 2u: ((e1, (r1,)), (e2, (r2,))) # Two-way union - up: ((e, (r1, r2)), (e, (r3,))) # Union + projection
- **query** – Single query tuple matching the query_type pattern.
- **queries** – Batch of queries. If provided, query must be None.
- **tnorm** – T-norm for intersection/union. Options: “prod”, “min”.
- **neg_norm** – Negation norm. Options: “standard”, “sugeno”, “yager”.

- **lambda** – Parameter for sugeno and yager negation (0.0-1.0).
- **k** – Number of top answer entities to return.
- **only_scores** – If True, return only scores tensor. If False, return (entity, score) tuples.
- **use_logits** – If True, use raw model logits. If False, use sigmoid probabilities.

Returns
List[(entity, score), …] of top-k answers. For batch queries: List of such lists, one per query.

Return type
For single query

Raises

- **ValueError** – If query_type is not in {1p, 2p, 3p, 2i, 3i, ip, pi, 2u, up}.
- **AssertionError** – If query structure doesn't match query_type pattern.

Examples

```
>>> # 1p: Find entities located in Asia
>>> model.answer_multi_hop_query(
...     query_type="1p",
...     query=("Asia", ("isLocatedIn",)),
...     k=5
... )
[("Mongolia", 0.92), ("China", 0.89), ...]
```

```
>>> # 2p: Two-hop query (e.g., "capital of countries in Europe")
>>> model.answer_multi_hop_query(
...     query_type="2p",
...     query=("Europe", ("isLocatedIn", "hasCapital")),
...     k=3
... )
[("Paris", 0.85), ("Berlin", 0.82), ...]
```

```
>>> # 2i: Intersection query
>>> model.answer_multi_hop_query(
...     query_type="2i",
...     query=(("Asia", ("isLocatedIn",)), ("Mountains", ("hasGeography",))),
...     k=5
... )
[("Nepal", 0.78), ("Tibet", 0.65), ...]
```

See also

- docs/guides/multi_hop_queries.md: Complete guide with all query patterns
- tests/test_answer_multi_hop_query.py: Usage examples

find_missing_triples(*confidence: float*, *entities: List[str] | None = None*, *relations: List[str] | None = None*, *topk: int = 10*, *at_most: int = sys.maxsize*) → Set

Find missing triples

Iterative over a set of entities E and a set of relation R :

orall e in E and orall r in R f(e,r,x)

Return (e,r,x)

otin G and f(e,r,x) > confidence

confidence: float

A threshold for an output of a sigmoid function given a triple.

topk: int

Highest ranked k item to select triples with f(e,r,x) > confidence .

at_most: int

Stop after finding at_most missing triples

{(e,r,x) | f(e,r,x) > confidence land (e,r,x)

otin G

predict_literals(*entity: List[str] | str | None = None*, *attribute: List[str] | str | None = None*, *denormalize_preds: bool = True*) → numpy.ndarray

Predicts literal values for given entities and attributes.

Parameters

- **entity** (*Union[List[str], str]*) – Entity or list of entities to predict literals for.
- **attribute** (*Union[List[str], str]*) – Attribute or list of attributes to predict literals for.
- **denormalize_preds** (*bool*) – If True, denormalizes the predictions.

Returns

Predictions for the given entities and attributes.

Return type

numpy ndarray

class dicee.**QueryGenerator**(*train_path*, *val_path: str*, *test_path: str*, *ent2id: Dict | None = None*, *rel2id: Dict | None = None*, *seed: int = 1*, *gen_valid: bool = False*, *gen_test: bool = True*)

train_path

val_path

test_path

gen_valid = False

gen_test = True

seed = 1

max_ans_num = 1000000.0

mode

ent2id = None

rel2id: Dict = None

ent_in: Dict

ent_out: Dict

query_name_to_struct

list2tuple(*list_data*)

tuple2list(*x: List | Tuple*) → List | Tuple

Convert a nested tuple to a nested list.

set_global_seed(*seed: int*)

Set seed

construct_graph(*paths: List[str]*) → Tuple[Dict, Dict]

Construct graph from triples Returns dicts with incoming and outgoing edges

fill_query(*query_structure: List[str | List]*, *ent_in: Dict*, *ent_out: Dict*, *answer: int*) → bool

Private method for fill_query logic.

achieve_answer(*query: List[str | List]*, *ent_in: Dict*, *ent_out: Dict*) → set

Private method for achieve_answer logic. @TODO: Document the code

write_links(*ent_out*, *small_ent_out*)

ground_queries(*query_structure: List[str | List]*, *ent_in: Dict*, *ent_out: Dict*, *small_ent_in: Dict*, *small_ent_out: Dict*, *gen_num: int*, *query_name: str*)

Generating queries and achieving answers

unmap(*query_type*, *queries*, *tp_answers*, *fp_answers*, *fn_answers*)

unmap_query(*query_structure*, *query*, *id2ent*, *id2rel*)

generate_queries(*query_struct: List*, *gen_num: int*, *query_type: str*)

Passing incoming and outgoing edges to ground queries depending on mode [train valid or text] and getting queries and answers in return @ TODO: create a class for each single query struct

save_queries(*query_type: str*, *gen_num: int*, *save_path: str*)

abstractmethod load_queries(*path*)

get_queries(*query_type: str*, *gen_num: int*)

static save_queries_and_answers(*path: str*, *data: List[Tuple[str, Tuple[collections.defaultdict]]]*) → None

Save Queries into Disk

static load_queries_and_answers(*path: str*) → List[Tuple[str, Tuple[collections.defaultdict]]]

Load Queries from Disk to Memory

class dicee.**DICE_Trainer**(*args*, *is_continual_training: bool*, *storage_path*, *evaluator=None*)

DICE_Trainer implement

1- Pytorch Lightning trainer (https://pytorch-lightning.readthedocs.io/en/stable/common/trainer.html) 2- Multi-GPU Trainer(https://pytorch.org/docs/stable/generated/torch.nn.parallel.DistributedDataParallel.html) 3- CPU Trainer

args

is_continual_training:bool

storage_path:str

evaluator:

report:dict

report

args

trainer = None

is_continual_training

storage_path

evaluator = None

form_of_labelling = None

continual_start(*knowledge_graph*)

(1) Initialize training.

(2) Load model

(3) Load trainer (3) Fit model

Parameter

returns

- *model*
- **form_of_labelling** (*str*)

initialize_trainer(*callbacks: List*)
→ lightning.Trainer | *dicee.trainer.model_parallelism.TensorParallel* | *dicee.trainer.torch_trainer.TorchTrainer* | *dicee.t*

Initialize Trainer from input arguments

initialize_or_load_model()

init_dataloader(*dataset: torch.utils.data.Dataset*) → torch.utils.data.DataLoader

init_dataset() → torch.utils.data.Dataset

prepare_graph_model(*model*, *knowledge_graph*)

Attach only training facts, preserving the indexed external vocabulary.

prepare_ultra

start(*knowledge_graph: dicee.knowledge_graph.KG | numpy.memmap*)
→ Tuple[*dicee.models.base_model.BaseKGE*, str]

Start the training

(1) Initialize Trainer

(2) Initialize or load a pretrained KGE model

in DDP setup, we need to load the memory map of already read/index KG.

k_fold_cross_validation(*dataset*) → Tuple[*dicee.models.base_model.BaseKGE*, str]

Perform K-fold Cross-Validation

1. Obtain K train and test splits.
2. **For each split,**
   2.1 initialize trainer and model 2.2. Train model with configuration provided in args. 2.3. Compute the mean reciprocal rank (MRR) score of the model on the test respective split.
3. Report the mean and average MRR .

Parameters

- **self**
- **dataset**

Returns

model

class dicee.**Evaluator**(*args*, *is_continual_training: bool = False*)

Evaluator class for KGE models in various downstream tasks.

Orchestrates link prediction evaluation with different scoring techniques including standard evaluation and byte-pair encoding based evaluation.

er_vocab

Entity-relation to tail vocabulary for filtered ranking.

re_vocab

Relation-entity (tail) to head vocabulary.

ee_vocab

Entity-entity to relation vocabulary.

num_entities

Total number of entities in the knowledge graph.

num_relations

Total number of relations in the knowledge graph.

args

Configuration arguments.

report

Dictionary storing evaluation results.

during_training

Whether evaluation is happening during training.

Example

```
>>> from dicee.evaluation import Evaluator
>>> evaluator = Evaluator(args)
>>> results = evaluator.eval(dataset, model, 'EntityPrediction')
>>> print(f"Test MRR: {results['Test']['MRR']:.4f}")
```

re_vocab: Dict | None = None

er_vocab: Dict | None = None

ee_vocab: Dict | None = None

func_triple_to_bpe_representation = None

is_continual_training = False

num_entities: int | None = None

num_relations: int | None = None

domain_constraints_per_rel = None

range_constraints_per_rel = None

args

report: Dict

during_training = False

vocab_preparation(*dataset*) → None

Prepare vocabularies from the dataset for evaluation.

Resolves any future objects and saves vocabularies to disk.

Parameters

dataset – Knowledge graph dataset with vocabulary attributes.

eval(*dataset*, *trained_model*, *form_of_labelling: str*, *during_training: bool = False*) → Dict | None

Evaluate the trained model on the dataset.

Parameters

- **dataset** – Knowledge graph dataset (KG instance).
- **trained_model** – The trained KGE model.
- **form_of_labelling** – Type of labelling ('EntityPrediction' or 'RelationPrediction').
- **during_training** – Whether evaluation is during training.

Returns

Dictionary of evaluation metrics, or None if evaluation is skipped.

eval_rank_of_head_and_tail_entity(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*) → None

Evaluate with negative sampling scoring.

eval_rank_of_head_and_tail_byte_pair_encoded_entity(*, *train_set=None*, *valid_set=None*, *test_set=None*, *ordered_bpe_entities*, *trained_model*) → None

Evaluate with BPE-encoded entities and negative sampling.

eval_with_byte(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate BytE model with generation.

eval_with_bpe_vs_all(*, *raw_train_set*, *raw_valid_set=None*, *raw_test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with BPE and KvsAll scoring.

eval_with_vs_all(*, *train_set*, *valid_set=None*, *test_set=None*, *trained_model*, *form_of_labelling*) → None

Evaluate with KvsAll or 1vsAll scoring.

evaluate_lp_k_vs_all(*model*, *triple_idx*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Filtered link prediction evaluation with KvsAll scoring.

Parameters

- **model** – The trained model to evaluate.
- **triple_idx** – Integer-indexed test triples.
- **info** – Description to print.
- **form_of_labelling** – 'EntityPrediction' or 'RelationPrediction'.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp_with_byte(*model*, *triples: List[List[str]]*, *info: str | None = None*) → Dict[str, float]

Evaluate BytE model with text generation.

Parameters

- **model** – BytE model.
- **triples** – String triples.
- **info** – Description to print.

Returns

Dictionary with placeholder metrics (-1 values).

evaluate_lp_bpe_k_vs_all(*model*, *triples: List[List[str]]*, *info: str | None = None*, *form_of_labelling: str | None = None*) → Dict[str, float]

Evaluate BPE model with KvsAll scoring.

Parameters

- **model** – BPE-enabled model.
- **triples** – String triples.
- **info** – Description to print.
- **form_of_labelling** – Type of labelling.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

evaluate_lp(*model*, *triple_idx*, *info: str*) → Dict[str, float]

Evaluate link prediction with negative sampling.

Parameters

- **model** – The model to evaluate.
- **triple_idx** – Integer-indexed triples.
- **info** – Description to print.

Returns

Dictionary with H@1, H@3, H@10, and MRR metrics.

dummy_eval(*trained_model*, *form_of_labelling: str*) → None

Run evaluation from saved data (for continual training).

Parameters

- **trained_model** – The trained model.
- **form_of_labelling** – Type of labelling.

eval_with_data(*dataset*, *trained_model*, *triple_idx: numpy.ndarray*, *form_of_labelling: str*) → Dict[str, float]

Evaluate a trained model on a given dataset.

Parameters

- **dataset** – Knowledge graph dataset.
- **trained_model** – The trained model.
- **triple_idx** – Integer-indexed triples to evaluate.
- **form_of_labelling** – Type of labelling.

Returns

Dictionary with evaluation metrics.

Raises

ValueError – If scoring technique is invalid.

dicee.**__version__ = '0.3.3'**

Python Module Index

d

dicee, 12
dicee.__main__, 12
dicee.abstracts, 12
dicee.analyse_experiments, 20
dicee.callbacks, 22
dicee.config, 28
dicee.dataset_classes, 34
dicee.eval_static_funcs, 43
dicee.evaluation, 47
dicee.evaluation.ensemble, 47
dicee.evaluation.evaluator, 48
dicee.evaluation.link_prediction, 52
dicee.evaluation.literal_prediction, 56
dicee.evaluation.utils, 57
dicee.evaluator, 67
dicee.executer, 70
dicee.knowledge_graph, 73
dicee.knowledge_graph_embeddings, 76
dicee.losses, 82
dicee.losses.custom_losses, 82
dicee.models, 87
dicee.models.adopt, 87
dicee.models.base_model, 96
dicee.models.clifford, 105
dicee.models.complex, 114
dicee.models.dualE, 119
dicee.models.ensemble, 120
dicee.models.flock, 121
dicee.models.flock_walks, 126
dicee.models.fsdp_models, 127
dicee.models.function_space, 129
dicee.models.graph_model, 133
dicee.models.literal, 137
dicee.models.octonion, 139
dicee.models.pykeen_models, 141
dicee.models.quaternion, 143
dicee.models.real, 146
dicee.models.sampled_loss, 154
dicee.models.static_funcs, 154
dicee.models.transformers, 154
dicee.models.trix, 161
dicee.models.ultra, 164
dicee.query_generator, 246
dicee.read_preprocess_save_load_kg, 247
dicee.read_preprocess_save_load_kg.preprocess, 247
dicee.read_preprocess_save_load_kg.read_from_disk, 248
dicee.read_preprocess_save_load_kg.save_load_disk, 249
dicee.read_preprocess_save_load_kg.util, 249
dicee.sanity_checkers, 254
dicee.scripts, 255
dicee.scripts.index_serve, 255
dicee.scripts.run, 256
dicee.static_funcs, 256
dicee.static_funcs_training, 261
dicee.static_preprocess_funcs, 263
dicee.trainer, 264
dicee.trainer.auto_batch_finder, 264
dicee.trainer.dice_trainer, 265
dicee.trainer.model_parallelism, 267
dicee.trainer.torch_trainer, 268
dicee.trainer.torch_trainer_ddp, 270
dicee.trainer.torch_trainer_fsdp, 271
dicee.weight_averaging, 274

Index

Non-alphabetical

__call__() (*dicee.models.base_model.IdentityClass method*), 105
__call__() (*dicee.models.ensemble.EnsembleKGE method*), 120
__call__() (*dicee.models.IdentityClass method*), 186, 212, 220
__getitem__() (*dicee.dataset_classes.AllvsAll method*), 37
__getitem__() (*dicee.dataset_classes.BPE_NegativeSamplingDataset method*), 35
__getitem__() (*dicee.dataset_classes.FixedNegSampleDataset method*), 41
__getitem__() (*dicee.dataset_classes.FSDP1vsSampleDataset method*), 38
__getitem__() (*dicee.dataset_classes.KvsAll method*), 38
__getitem__() (*dicee.dataset_classes.KvsSampleDataset method*), 39
__getitem__() (*dicee.dataset_classes.LiteralDataset method*), 40
__getitem__() (*dicee.dataset_classes.MultiClassClassificationDataset method*), 36
__getitem__() (*dicee.dataset_classes.MultiLabelDataset method*), 36
__getitem__() (*dicee.dataset_classes.OnevsAllDataset method*), 39
__getitem__() (*dicee.dataset_classes.OnevsSample method*), 42
__getitem__() (*dicee.dataset_classes.TriplePredictionDataset method*), 43
__iter__() (*dicee.config.Namespace method*), 34
__iter__() (*dicee.knowledge_graph.KG method*), 76
__iter__() (*dicee.models.ensemble.EnsembleKGE method*), 120
__len__() (*dicee.dataset_classes.AllvsAll method*), 37
__len__() (*dicee.dataset_classes.BPE_NegativeSamplingDataset method*), 35
__len__() (*dicee.dataset_classes.FixedNegSampleDataset method*), 41
__len__() (*dicee.dataset_classes.FSDP1vsSampleDataset method*), 38
__len__() (*dicee.dataset_classes.KvsAll method*), 38
__len__() (*dicee.dataset_classes.KvsSampleDataset method*), 39
__len__() (*dicee.dataset_classes.LiteralDataset method*), 40
__len__() (*dicee.dataset_classes.MultiClassClassificationDataset method*), 36
__len__() (*dicee.dataset_classes.MultiLabelDataset method*), 36
__len__() (*dicee.dataset_classes.OnevsAllDataset method*), 39
__len__() (*dicee.dataset_classes.OnevsSample method*), 42
__len__() (*dicee.dataset_classes.TriplePredictionDataset method*), 43
__len__() (*dicee.knowledge_graph.KG method*), 76
__len__() (*dicee.models.ensemble.EnsembleKGE method*), 120
__post_init__() (*dicee.models.flock_walks.WalkGraph method*), 126
__setstate__() (*dicee.models.ADOPT method*), 176
__setstate__() (*dicee.models.adopt.ADOPT method*), 91
__str__() (*dicee.KGE method*), 284
__str__() (*dicee.knowledge_graph_embeddings.KGE method*), 76
__str__() (*dicee.models.ensemble.EnsembleKGE method*), 120
__version__ (*in module dicee*), 295

A

AbstractCallback (*class in dicee.abstracts*), 18
AbstractPPECallback (*class in dicee.abstracts*), 19
AbstractTrainer (*class in dicee.abstracts*), 13
AccumulateEpochLossCallback (*class in dicee.callbacks*), 23
achieve_answer() (*dicee.query_generator.QueryGenerator method*), 247
achieve_answer() (*dicee.QueryGenerator method*), 290
ACLS (*class in dicee.losses.custom_losses*), 86
ACLS (*class in dicee.models*), 169
AConEx (*class in dicee.models*), 204
AConEx (*class in dicee.models.complex*), 115
AConvO (*class in dicee.models*), 221
AConvO (*class in dicee.models.octonion*), 140
AConvQ (*class in dicee.models*), 214
AConvQ (*class in dicee.models.quaternion*), 145
adaptive_label_relaxation (*dicee.losses.custom_losses.CombinedAdaptiveLSandAdaptiveLR attribute*), 86
adaptive_label_relaxation (*dicee.models.CombinedAdaptiveLSandAdaptiveLR attribute*), 172
adaptive_label_smoothing (*dicee.losses.custom_losses.CombinedAdaptiveLSandAdaptiveLR attribute*), 86
adaptive_label_smoothing (*dicee.models.CombinedAdaptiveLSandAdaptiveLR attribute*), 172
adaptive_lr (*dicee.config.Namespace attribute*), 33
adaptive_swa (*dicee.config.Namespace attribute*), 32
AdaptiveLabelRelaxationLoss (*class in dicee.losses.custom_losses*), 84
AdaptiveLabelRelaxationLoss (*class in dicee.models*), 170
AdaptiveLabelSmoothingLoss (*class in dicee.losses.custom_losses*), 83

AdaptiveLabelSmoothingLoss (*class in dicee.models*), 171
add_new_entity_embeddings() (*dicee.abstracts.BaseInteractiveKGE method*), 16
add_noise_rate (*dicee.config.Namespace attribute*), 30
add_noise_rate (*dicee.knowledge_graph.KG attribute*), 74
add_noisy_triples() (*in module dicee.static_funcs*), 260
add_noisy_triples_into_training() (*dicee.read_preprocess_save_load_kg.read_from_disk.ReadFromDisk method*), 249
add_noisy_triples_into_training() (*dicee.read_preprocess_save_load_kg.ReadFromDisk method*), 254
add_reciprocal (*dicee.knowledge_graph.KG attribute*), 74
ADOPT (*class in dicee.models*), 174
ADOPT (*class in dicee.models.adopt*), 89
adopt() (*in module dicee.models.adopt*), 92
adversarial_temperature (*dicee.config.Namespace attribute*), 30
AggregatedLSandLR (*class in dicee.losses.custom_losses*), 86
AggregatedLSandLR (*class in dicee.models*), 171
ALL_HITS_RANGE (*in module dicee.evaluation.utils*), 57
AllvsAll (*class in dicee.dataset_classes*), 37
alpha (*dicee.losses.custom_losses.ACLS attribute*), 87
alpha (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss attribute*), 85
alpha (*dicee.losses.custom_losses.AggregatedLSandLR attribute*), 86
alpha (*dicee.losses.custom_losses.CombinedLSandLR attribute*), 85
alpha (*dicee.losses.custom_losses.ConfidenceBasedAdaptiveLabelRelaxationLoss attribute*), 85
alpha (*dicee.losses.custom_losses.LabelRelaxationLoss attribute*), 84
alpha (*dicee.models.ACLS attribute*), 170
alpha (*dicee.models.AdaptiveLabelRelaxationLoss attribute*), 171
alpha (*dicee.models.AggregatedLSandLR attribute*), 172
alpha (*dicee.models.CombinedLSandLR attribute*), 172
alpha (*dicee.models.ConfidenceBasedAdaptiveLabelRelaxationLoss attribute*), 173
alpha (*dicee.models.LabelRelaxationLoss attribute*), 173
alpha_step (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss attribute*), 84
alpha_step (*dicee.models.AdaptiveLabelRelaxationLoss attribute*), 171
alphas (*dicee.abstracts.AbstractPPECallback attribute*), 19
alphas (*dicee.weight_averaging.ASWA attribute*), 277
AMWA (*class in dicee.weight_averaging*), 274
amwa (*dicee.config.Namespace attribute*), 33
amwa_beta (*dicee.config.Namespace attribute*), 33
amwa_beta_floor (*dicee.config.Namespace attribute*), 34
amwa_beta_init (*dicee.config.Namespace attribute*), 34
amwa_beta_window (*dicee.config.Namespace attribute*), 34
amwa_c_epochs (*dicee.config.Namespace attribute*), 33
amwa_c_epochs (*dicee.weight_averaging.AMWA attribute*), 276
amwa_maximize (*dicee.config.Namespace attribute*), 33
amwa_monitor (*dicee.config.Namespace attribute*), 33
amwa_start_epoch (*dicee.config.Namespace attribute*), 33
amwa_start_epoch (*dicee.weight_averaging.AMWA attribute*), 276
analyse() (*in module dicee.analyse_experiments*), 22
anonymize() (*in module dicee.models.flock_walks*), 126
answer_multi_hop_query() (*dicee.KGE method*), 287
answer_multi_hop_query() (*dicee.knowledge_graph_embeddings.KGE method*), 79
app (*in module dicee.scripts.index_serve*), 256
apply_coefficients() (*dicee.models.clifford.DeCaL method*), 112
apply_coefficients() (*dicee.models.clifford.Keci method*), 107
apply_coefficients() (*dicee.models.DeCaL method*), 227
apply_coefficients() (*dicee.models.Keci method*), 223
apply_reciprocal_or_noise() (*in module dicee.read_preprocess_save_load_kg.util*), 252
apply_semantic_constraint (*dicee.abstracts.BaseInteractiveKGE attribute*), 14
apply_unit_norm (*dicee.models.base_model.BaseKGE attribute*), 101
apply_unit_norm (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 215, 230, 235
args (*dicee.DICE_Trainer attribute*), 291
args (*dicee.evaluation.Evaluator attribute*), 60, 61
args (*dicee.evaluation.evaluator.Evaluator attribute*), 49, 50
args (*dicee.Evaluator attribute*), 292, 293
args (*dicee.evaluator.Evaluator attribute*), 67, 68
args (*dicee.Execute attribute*), 282
args (*dicee.executer.Execute attribute*), 71
args (*dicee.models.base_model.BaseKGE attribute*), 101
args (*dicee.models.base_model.BaseKGELightning attribute*), 96
args (*dicee.models.base_model.IdentityClass attribute*), 105
args (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 215, 230, 235

args (*dicee.models.BaseKGELightning attribute*), 177
args (*dicee.models.ensemble.EnsembleKGE attribute*), 120
args (*dicee.models.IdentityClass attribute*), 186, 212, 220
args (*dicee.models.pykeen_models.PykeenKGE attribute*), 142
args (*dicee.models.PykeenKGE attribute*), 234
args (*dicee.trainer.DICE_Trainer attribute*), 273
args (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 266
ASWA (*class in dicee.weight_averaging*), 276
aswa (*dicee.analyse_experiments.Experiment attribute*), 21
attn (*dicee.models.Block attribute*), 187
attn (*dicee.models.clifford.TransformerBlock attribute*), 110
attn (*dicee.models.transformers.Block attribute*), 159
attn_dropout (*dicee.models.clifford.TransformerSelfAttention attribute*), 110
attn_dropout (*dicee.models.transformers.SelfAttention attribute*), 157
attributes (*dicee.abstracts.AbstractTrainer attribute*), 13
auto_batch_finding (*dicee.config.Namespace attribute*), 33

B

b_h (*dicee.models.MuRE attribute*), 195
b_h (*dicee.models.real.MuRE attribute*), 150
b_t (*dicee.models.MuRE attribute*), 195
b_t (*dicee.models.real.MuRE attribute*), 150
backend (*dicee.config.Namespace attribute*), 30
backend (*dicee.knowledge_graph.KG attribute*), 75
base_history (*dicee.weight_averaging.AMWA attribute*), 276
base_weights (*dicee.weight_averaging.TWA attribute*), 281
BaseInteractiveKGE (*class in dicee.abstracts*), 14
BaseInteractiveTrainKGE (*class in dicee.abstracts*), 20
BaseKGE (*class in dicee.models*), 181, 187, 199, 208, 215, 229, 235
BaseKGE (*class in dicee.models.base_model*), 100
BaseKGELightning (*class in dicee.models*), 177
BaseKGELightning (*class in dicee.models.base_model*), 96
batch_kronecker_product() (*dicee.callbacks.KronE static method*), 26
batch_size (*dicee.analyse_experiments.Experiment attribute*), 21
batch_size (*dicee.callbacks.PseudoLabellingCallback attribute*), 25
batch_size (*dicee.config.Namespace attribute*), 30
batches_per_epoch (*dicee.callbacks.LRScheduler attribute*), 27
beta (*dicee.weight_averaging.AMWA attribute*), 276
beta (*dicee.weight_averaging.TWA attribute*), 281
beta_floor (*dicee.weight_averaging.AMWA attribute*), 276
beta_history (*dicee.weight_averaging.AMWA attribute*), 276
beta_init (*dicee.weight_averaging.AMWA attribute*), 276
beta_window (*dicee.weight_averaging.AMWA attribute*), 276
bias (*dicee.models.CoKEConfig attribute*), 198
bias (*dicee.models.real.CoKEConfig attribute*), 152, 153
bias (*dicee.models.transformers.GPTConfig attribute*), 160
bias (*dicee.models.transformers.LayerNorm attribute*), 157
BidirectionalGRU (*class in dicee.models.flock*), 123
Block (*class in dicee.models*), 186
Block (*class in dicee.models.transformers*), 158
block_size (*dicee.config.Namespace attribute*), 32
block_size (*dicee.dataset_classes.MultiClassClassificationDataset attribute*), 35
block_size (*dicee.models.base_model.BaseKGE attribute*), 101
block_size (*dicee.models.BaseKGE attribute*), 183, 188, 200, 209, 216, 231, 236
block_size (*dicee.models.CoKEConfig attribute*), 197, 198
block_size (*dicee.models.real.CoKEConfig attribute*), 152, 153
block_size (*dicee.models.transformers.GPTConfig attribute*), 159
blocks (*dicee.models.CoKE attribute*), 198
blocks (*dicee.models.real.CoKE attribute*), 153
bn_conv1 (*dicee.models.AConvQ attribute*), 215
bn_conv1 (*dicee.models.ConvQ attribute*), 214
bn_conv1 (*dicee.models.quaternion.AConvQ attribute*), 145
bn_conv1 (*dicee.models.quaternion.ConvQ attribute*), 144
bn_conv2 (*dicee.models.AConvQ attribute*), 215
bn_conv2 (*dicee.models.ConvQ attribute*), 214
bn_conv2 (*dicee.models.quaternion.AConvQ attribute*), 145
bn_conv2 (*dicee.models.quaternion.ConvQ attribute*), 144

bn_conv2d (*dicee.models.AConEx attribute*), 205
bn_conv2d (*dicee.models.AConvO attribute*), 221
bn_conv2d (*dicee.models.complex.AConEx attribute*), 115
bn_conv2d (*dicee.models.complex.ConEx attribute*), 114
bn_conv2d (*dicee.models.ConEx attribute*), 204
bn_conv2d (*dicee.models.ConvO attribute*), 221
bn_conv2d (*dicee.models.octonion.AConvO attribute*), 141
bn_conv2d (*dicee.models.octonion.ConvO attribute*), 140
BPE_NegativeSamplingDataset (*class in dicee.dataset_classes*), 35
build_chain_funcs() (*dicee.models.FMult2 method*), 241
build_chain_funcs() (*dicee.models.function_space.FMult2 method*), 131
build_func() (*dicee.models.FMult2 method*), 241
build_func() (*dicee.models.function_space.FMult2 method*), 131
build_projection() (*dicee.weight_averaging.TWA method*), 281
build_relation_graph() (*in module dicee.models.trix*), 161
build_relation_graph() (*in module dicee.models.ultra*), 167
BytE (*class in dicee.models.transformers*), 155
byte_pair_encoding (*dicee.analyse_experiments.Experiment attribute*), 21
byte_pair_encoding (*dicee.config.Namespace attribute*), 32
byte_pair_encoding (*dicee.knowledge_graph.KG attribute*), 74
byte_pair_encoding (*dicee.models.base_model.BaseKGE attribute*), 101
byte_pair_encoding (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236

C

c_attn (*dicee.models.clifford.TransformerSelfAttention attribute*), 110
c_attn (*dicee.models.transformers.SelfAttention attribute*), 157
c_fc (*dicee.models.clifford.TransformerMLP attribute*), 110
c_fc (*dicee.models.transformers.MLP attribute*), 158
c_proj (*dicee.models.clifford.TransformerMLP attribute*), 110
c_proj (*dicee.models.clifford.TransformerSelfAttention attribute*), 110
c_proj (*dicee.models.transformers.MLP attribute*), 158
c_proj (*dicee.models.transformers.SelfAttention attribute*), 157
callbacks (*dicee.abstracts.AbstractTrainer attribute*), 13
callbacks (*dicee.analyse_experiments.Experiment attribute*), 21
callbacks (*dicee.config.Namespace attribute*), 30
callbacks (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
causal (*dicee.models.CoKEConfig attribute*), 198
causal (*dicee.models.real.CoKEConfig attribute*), 152, 153
causal (*dicee.models.transformers.GPTConfig attribute*), 160
causal (*dicee.models.transformers.SelfAttention attribute*), 158
chain_func() (*dicee.models.FMult method*), 240
chain_func() (*dicee.models.function_space.FMult method*), 130
chain_func() (*dicee.models.function_space.GFMult method*), 130
chain_func() (*dicee.models.GFMult method*), 240
checkpoint_every_n_epochs (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
checkpoint_hint (*dicee.models.flock.FlockBase attribute*), 124
checkpoint_hint (*dicee.models.graph_model.GraphKGE attribute*), 134
checkpoint_hint (*dicee.models.trix.TRIXBase attribute*), 163
checkpoint_hint (*dicee.models.ULTRA attribute*), 244
checkpoint_hint (*dicee.models.ultra.ULTRA attribute*), 167
CKeci (*class in dicee.models*), 226
CKeci (*class in dicee.models.clifford*), 110
cl_pqr() (*dicee.models.clifford.DeCaL method*), 111
cl_pqr() (*dicee.models.DeCaL method*), 226
cleanup() (*dicee.Execute method*), 283
cleanup() (*dicee.executer.Execute method*), 72
clear_inference_cache() (*dicee.models.graph_model.GraphKGE method*), 134
clear_inference_cache() (*dicee.models.TRIX method*), 245
clear_inference_cache() (*dicee.models.trix.TRIX method*), 164
clear_inference_cache() (*dicee.models.ULTRA method*), 245
clear_inference_cache() (*dicee.models.ultra.ULTRA method*), 167
clifford_multiplication() (*dicee.models.clifford.Keci method*), 107
clifford_multiplication() (*dicee.models.Keci method*), 223
clip_lambda (*dicee.models.ADOPT attribute*), 176
clip_lambda (*dicee.models.adopt.ADOPT attribute*), 91
CoKE (*class in dicee.models*), 198
CoKE (*class in dicee.models.real*), 153

coke_dropout (*dicee.models.CoKE attribute*), 198
coke_dropout (*dicee.models.real.CoKE attribute*), 153
CoKEConfig (*class in dicee.models*), 197
CoKEConfig (*class in dicee.models.real*), 152
collate_fn (*dicee.dataset_classes.AllvsAll attribute*), 37
collate_fn (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
collate_fn (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
collate_fn (*dicee.dataset_classes.KvsAll attribute*), 38
collate_fn (*dicee.dataset_classes.KvsSampleDataset attribute*), 39
collate_fn (*dicee.dataset_classes.MultiClassClassificationDataset attribute*), 36
collate_fn (*dicee.dataset_classes.MultiLabelDataset attribute*), 36
collate_fn (*dicee.dataset_classes.OnevsAllDataset attribute*), 39
collate_fn (*dicee.dataset_classes.OnevsSample attribute*), 42
collate_fn() (*dicee.dataset_classes.BPE_NegativeSamplingDataset method*), 35
collate_fn() (*dicee.dataset_classes.TriplePredictionDataset method*), 43
collection_name (*dicee.scripts.index_serve.NeuralSearcher attribute*), 256
CombinedAdaptiveLSandAdaptiveLR (*class in dicee.losses.custom_losses*), 85
CombinedAdaptiveLSandAdaptiveLR (*class in dicee.models*), 172
CombinedLSandLR (*class in dicee.losses.custom_losses*), 85
CombinedLSandLR (*class in dicee.models*), 172
comp_func() (*dicee.models.function_space.LFMult method*), 133
comp_func() (*dicee.models.LFMult method*), 243
compact_state (*dicee.models.flock.FlockBase attribute*), 125
compile_sampler (*dicee.models.flock_walks.WalkGraph attribute*), 126
compile_sampler (*dicee.models.flock.FlockBase attribute*), 125
compiled_walk_nodes() (*in module dicee.models.flock_walks*), 126
ComplEx (*class in dicee.models*), 205
ComplEx (*class in dicee.models.complex*), 116
compute_convergence() (*in module dicee.callbacks*), 25
compute_func() (*dicee.models.FMult method*), 240
compute_func() (*dicee.models.FMult2 method*), 241
compute_func() (*dicee.models.function_space.FMult method*), 130
compute_func() (*dicee.models.function_space.FMult2 method*), 131
compute_func() (*dicee.models.function_space.GFMult method*), 130
compute_func() (*dicee.models.GFMult method*), 240
compute_metrics_from_ranks() (*in module dicee.evaluation.utils*), 58
compute_metrics_from_ranks_simple() (*in module dicee.evaluation.utils*), 58
compute_mrr() (*dicee.weight_averaging.ASWA static method*), 277
compute_sigma_pp() (*dicee.models.clifford.DeCaL method*), 112
compute_sigma_pp() (*dicee.models.clifford.Keci method*), 106
compute_sigma_pp() (*dicee.models.DeCaL method*), 227
compute_sigma_pp() (*dicee.models.Keci method*), 222
compute_sigma_pq() (*dicee.models.clifford.DeCaL method*), 113
compute_sigma_pq() (*dicee.models.clifford.Keci method*), 107
compute_sigma_pq() (*dicee.models.DeCaL method*), 228
compute_sigma_pq() (*dicee.models.Keci method*), 223
compute_sigma_pr() (*dicee.models.clifford.DeCaL method*), 113
compute_sigma_pr() (*dicee.models.DeCaL method*), 229
compute_sigma_qq() (*dicee.models.clifford.DeCaL method*), 113
compute_sigma_qq() (*dicee.models.clifford.Keci method*), 106
compute_sigma_qq() (*dicee.models.DeCaL method*), 228
compute_sigma_qq() (*dicee.models.Keci method*), 223
compute_sigma_qr() (*dicee.models.clifford.DeCaL method*), 114
compute_sigma_qr() (*dicee.models.DeCaL method*), 229
compute_sigma_rr() (*dicee.models.clifford.DeCaL method*), 113
compute_sigma_rr() (*dicee.models.DeCaL method*), 228
compute_sigmas_multivect() (*dicee.models.clifford.DeCaL method*), 112
compute_sigmas_multivect() (*dicee.models.DeCaL method*), 227
compute_sigmas_single() (*dicee.models.clifford.DeCaL method*), 111
compute_sigmas_single() (*dicee.models.DeCaL method*), 226
ConEx (*class in dicee.models*), 203
ConEx (*class in dicee.models.complex*), 114
ConfidenceBasedAdaptiveLabelRelaxationLoss (*class in dicee.losses.custom_losses*), 85
ConfidenceBasedAdaptiveLabelRelaxationLoss (*class in dicee.models*), 172
config (*dicee.models.CoKE attribute*), 198
config (*dicee.models.real.CoKE attribute*), 153
config (*dicee.models.transformers.BytE attribute*), 155
config (*dicee.models.transformers.GPT attribute*), 160

config_prefix (*dicee.models.flock.FlockBase attribute*), 124
config_prefix (*dicee.models.graph_model.GraphKGE attribute*), 134
config_prefix (*dicee.models.trix.TRIXBase attribute*), 163
config_prefix (*dicee.models.ULTRA attribute*), 244
config_prefix (*dicee.models.ultra.ULTRA attribute*), 167
configs (*dicee.abstracts.BaseInteractiveKGE attribute*), 14
configure_optimizers() (*dicee.models.base_model.BaseKGELightning method*), 100
configure_optimizers() (*dicee.models.BaseKGELightning method*), 181
configure_optimizers() (*dicee.models.transformers.GPT method*), 161
consensus() (*in module dicee.models.flock*), 124
construct_batch_selected_cl_multivector() (*dicee.models.clifford.Keci method*), 108
construct_batch_selected_cl_multivector() (*dicee.models.Keci method*), 225
construct_cl_multivector() (*dicee.models.clifford.DeCaL method*), 112
construct_cl_multivector() (*dicee.models.clifford.Keci method*), 107
construct_cl_multivector() (*dicee.models.DeCaL method*), 227
construct_cl_multivector() (*dicee.models.Keci method*), 224
construct_dataset() (*in module dicee.dataset_classes*), 36
construct_ensemble (*dicee.abstracts.BaseInteractiveKGE attribute*), 14
construct_graph() (*dicee.query_generator.QueryGenerator method*), 246
construct_graph() (*dicee.QueryGenerator method*), 290
construct_input_and_output() (*dicee.abstracts.BaseInteractiveKGE method*), 17
construct_multi_coeff() (*dicee.models.function_space.LFMult method*), 132
construct_multi_coeff() (*dicee.models.LFMult method*), 242
continual_learning (*dicee.config.Namespace attribute*), 32
continual_start() (*dicee.DICE_Trainer method*), 291
continual_start() (*dicee.executer.ContinuousExecute method*), 73
continual_start() (*dicee.trainer.DICE_Trainer method*), 273
continual_start() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 266
continual_training_setup_executor() (*in module dicee.static_funcs*), 261
ContinuousExecute (*class in dicee.executer*), 73
conv2d (*dicee.models.AConEx attribute*), 204
conv2d (*dicee.models.AConvO attribute*), 221
conv2d (*dicee.models.AConvQ attribute*), 214
conv2d (*dicee.models.complex.AConEx attribute*), 115
conv2d (*dicee.models.complex.ConEx attribute*), 114
conv2d (*dicee.models.ConEx attribute*), 203
conv2d (*dicee.models.ConvO attribute*), 221
conv2d (*dicee.models.ConvQ attribute*), 214
conv2d (*dicee.models.octonion.AConvO attribute*), 141
conv2d (*dicee.models.octonion.ConvO attribute*), 140
conv2d (*dicee.models.quaternion.AConvQ attribute*), 145
conv2d (*dicee.models.quaternion.ConvQ attribute*), 144
ConvO (*class in dicee.models*), 220
ConvO (*class in dicee.models.octonion*), 139
ConvQ (*class in dicee.models*), 214
ConvQ (*class in dicee.models.quaternion*), 144
count_triples() (*in module dicee.read_preprocess_save_load_kg.util*), 252
create_and_store_kg() (*dicee.Execute method*), 283
create_and_store_kg() (*dicee.executer.Execute method*), 72
create_constraints() (*in module dicee.read_preprocess_save_load_kg.util*), 252
create_constraints() (*in module dicee.static_preprocess_funcs*), 264
create_experiment_folder() (*in module dicee.static_funcs*), 260
create_fsdp_sharded_model_class() (*in module dicee.models.fsdp_models*), 129
create_hits_dict() (*in module dicee.evaluation.utils*), 58
create_random_data() (*dicee.callbacks.PseudoLabellingCallback method*), 25
create_recipriocal_triples() (*in module dicee.read_preprocess_save_load_kg.util*), 253
create_recipriocal_triples() (*in module dicee.static_funcs*), 258
create_vector_database() (*dicee.KGE method*), 284
create_vector_database() (*dicee.knowledge_graph_embeddings.KGE method*), 76
criterion (*dicee.losses.custom_losses.CombinedAdaptiveLSandAdaptiveLR attribute*), 86
criterion (*dicee.models.CombinedAdaptiveLSandAdaptiveLR attribute*), 172
crop_block_size() (*dicee.models.transformers.GPT method*), 160
cross_entropy (*dicee.losses.custom_losses.ACLS attribute*), 87
cross_entropy (*dicee.models.ACLS attribute*), 170
ctx (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
ctx (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
current_epoch (*dicee.weight_averaging.AMWA attribute*), 276
current_epoch (*dicee.weight_averaging.EMA attribute*), 280

current_epoch (*dicee.weight_averaging.SWA attribute*), 278
current_epoch (*dicee.weight_averaging.SWAG attribute*), 279
current_epoch (*dicee.weight_averaging.TWA attribute*), 281
cycle_length (*dicee.callbacks.LRScheduler attribute*), 27

D

data_module (*dicee.callbacks.PseudoLabellingCallback attribute*), 25
data_property_embeddings (*dicee.models.literal.LiteralEmbeddings attribute*), 138
dataset_dir (*dicee.config.Namespace attribute*), 28
dataset_dir (*dicee.knowledge_graph.KG attribute*), 74
dataset_sanity_checking() (*in module dicee.read_preprocess_save_load_kg.util*), 253
DeCaL (*class in dicee.models*), 226
DeCaL (*class in dicee.models.clifford*), 111
decay (*dicee.weight_averaging.EMA attribute*), 280
decide() (*dicee.weight_averaging.ASWA method*), 277
default_eval_model (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
DEFAULT_HITS_RANGE (*in module dicee.evaluation.utils*), 57
DefaultBCELoss (*class in dicee.losses.custom_losses*), 82
DefaultBCELoss (*class in dicee.models*), 173
defer_large_embeddings (*dicee.models.base_model.BaseKGE attribute*), 101
defer_large_embeddings (*dicee.models.BaseKGE attribute*), 183, 188, 200, 209, 216, 231, 236
degree (*dicee.models.function_space.LFMult attribute*), 132
degree (*dicee.models.LFMult attribute*), 242
delta_history (*dicee.weight_averaging.AMWA attribute*), 276
denormalize() (*dicee.dataset_classes.LiteralDataset static method*), 40
describe() (*dicee.knowledge_graph.KG method*), 75
description_of_input (*dicee.knowledge_graph.KG attribute*), 75
deterministic_inference (*dicee.models.graph_model.GraphKGE attribute*), 134
deterministic_inference (*dicee.models.TRIX attribute*), 245
deterministic_inference (*dicee.models.trix.TRIX attribute*), 163
deterministic_inference (*dicee.models.ULTRA attribute*), 244
deterministic_inference (*dicee.models.ultra.ULTRA attribute*), 167
deviations (*dicee.weight_averaging.SWAG attribute*), 279
device (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 127
device (*dicee.models.literal.LiteralEmbeddings property*), 138
device (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
DICE_Trainer (*class in dicee*), 290
DICE_Trainer (*class in dicee.trainer*), 273
DICE_Trainer (*class in dicee.trainer.dice_trainer*), 266
dicee
    module, 12
dicee.__main__
    module, 12
dicee.abstracts
    module, 12
dicee.analyse_experiments
    module, 20
dicee.callbacks
    module, 22
dicee.config
    module, 28
dicee.dataset_classes
    module, 34
dicee.eval_static_funcs
    module, 43
dicee.evaluation
    module, 47
dicee.evaluation.ensemble
    module, 47
dicee.evaluation.evaluator
    module, 48
dicee.evaluation.link_prediction
    module, 52
dicee.evaluation.literal_prediction
    module, 56
dicee.evaluation.utils
    module, 57

dicee.evaluator
    module, 67
dicee.executer
    module, 70
dicee.knowledge_graph
    module, 73
dicee.knowledge_graph_embeddings
    module, 76
dicee.losses
    module, 82
dicee.losses.custom_losses
    module, 82
dicee.models
    module, 87
dicee.models.adopt
    module, 87
dicee.models.base_model
    module, 96
dicee.models.clifford
    module, 105
dicee.models.complex
    module, 114
dicee.models.dualE
    module, 119
dicee.models.ensemble
    module, 120
dicee.models.flock
    module, 121
dicee.models.flock_walks
    module, 126
dicee.models.fsdp_models
    module, 127
dicee.models.function_space
    module, 129
dicee.models.graph_model
    module, 133
dicee.models.literal
    module, 137
dicee.models.octonion
    module, 139
dicee.models.pykeen_models
    module, 141
dicee.models.quaternion
    module, 143
dicee.models.real
    module, 146
dicee.models.sampled_loss
    module, 154
dicee.models.static_funcs
    module, 154
dicee.models.transformers
    module, 154
dicee.models.trix
    module, 161
dicee.models.ultra
    module, 164
dicee.query_generator
    module, 246
dicee.read_preprocess_save_load_kg
    module, 247
dicee.read_preprocess_save_load_kg.preprocess
    module, 247
dicee.read_preprocess_save_load_kg.read_from_disk
    module, 248
dicee.read_preprocess_save_load_kg.save_load_disk
    module, 249
dicee.read_preprocess_save_load_kg.util
    module, 249

dicee.sanity_checkers
    module, 254
dicee.scripts
    module, 255
dicee.scripts.index_serve
    module, 255
dicee.scripts.run
    module, 256
dicee.static_funcs
    module, 256
dicee.static_funcs_training
    module, 261
dicee.static_preprocess_funcs
    module, 263
dicee.trainer
    module, 264
dicee.trainer.auto_batch_finder
    module, 264
dicee.trainer.dice_trainer
    module, 265
dicee.trainer.model_parallelism
    module, 267
dicee.trainer.torch_trainer
    module, 268
dicee.trainer.torch_trainer_ddp
    module, 270
dicee.trainer.torch_trainer_fsdp
    module, 271
dicee.weight_averaging
    module, 274
dim (*dicee.models.flock.FlockBase attribute*), 125
dim (*dicee.models.trix.TRIXBase attribute*), 163
dim (*dicee.models.ULTRA attribute*), 244
dim (*dicee.models.ultra.RelNBFNet attribute*), 166
dim (*dicee.models.ultra.ULTRA attribute*), 167
discrete_points (*dicee.models.FMult2 attribute*), 241
discrete_points (*dicee.models.function_space.FMult2 attribute*), 131
dist_func (*dicee.models.Pyke attribute*), 197
dist_func (*dicee.models.real.Pyke attribute*), 152
DistMult (*class in dicee.models*), 191
DistMult (*class in dicee.models.real*), 146
distributed (*dicee.Execute attribute*), 282
distributed (*dicee.executer.Execute attribute*), 71
domain_constraints_per_rel (*dicee.evaluation.Evaluator attribute*), 61
domain_constraints_per_rel (*dicee.evaluation.evaluator.Evaluator attribute*), 50
domain_constraints_per_rel (*dicee.Evaluator attribute*), 293
domain_constraints_per_rel (*dicee.evaluator.Evaluator attribute*), 68
download_file() (*in module dicee.static_funcs*), 261
download_files_from_url() (*in module dicee.static_funcs*), 261
download_pretrained_model() (*in module dicee.static_funcs*), 261
dropout (*dicee.models.clifford.TransformerMLP attribute*), 110
dropout (*dicee.models.clifford.TransformerSelfAttention attribute*), 110
dropout (*dicee.models.CoKEConfig attribute*), 198
dropout (*dicee.models.literal.LiteralEmbeddings attribute*), 138
dropout (*dicee.models.real.CoKEConfig attribute*), 152, 153
dropout (*dicee.models.transformers.GPTConfig attribute*), 159
dropout (*dicee.models.transformers.MLP attribute*), 158
dropout (*dicee.models.transformers.SelfAttention attribute*), 158
DualE (*class in dicee.models*), 243
DualE (*class in dicee.models.dualE*), 119
dummy_eval() (*dicee.evaluation.Evaluator method*), 63
dummy_eval() (*dicee.evaluation.evaluator.Evaluator method*), 51
dummy_eval() (*dicee.Evaluator method*), 294
dummy_eval() (*dicee.evaluator.Evaluator method*), 70
dummy_id (*dicee.knowledge_graph.KG attribute*), 75
during_training (*dicee.evaluation.Evaluator attribute*), 60, 61
during_training (*dicee.evaluation.evaluator.Evaluator attribute*), 49, 50
during_training (*dicee.Evaluator attribute*), 292, 293

during_training (*dicee.evaluator.Evaluator attribute*), 67, 68

E

edge_index (*dicee.models.flock_walks.WalkGraph attribute*), 126
edge_type (*dicee.models.flock_walks.WalkGraph attribute*), 126
ee_vocab (*dicee.evaluation.Evaluator attribute*), 60, 61
ee_vocab (*dicee.evaluation.evaluator.Evaluator attribute*), 49
ee_vocab (*dicee.Evaluator attribute*), 292, 293
ee_vocab (*dicee.evaluator.Evaluator attribute*), 67, 68
efficient_zero_grad() (*in module dicee.evaluation.utils*), 58
efficient_zero_grad() (*in module dicee.static_funcs_training*), 263
EMA (*class in dicee.weight_averaging*), 279
ema (*dicee.config.Namespace attribute*), 32
ema_c_epochs (*dicee.weight_averaging.EMA attribute*), 280
ema_model (*dicee.weight_averaging.EMA attribute*), 280
ema_start_epoch (*dicee.weight_averaging.EMA attribute*), 280
ema_update() (*dicee.weight_averaging.EMA static method*), 280
emb_anon_node (*dicee.models.flock.FlockBase attribute*), 124
emb_anon_type (*dicee.models.flock.FlockBase attribute*), 124
emb_direction (*dicee.models.flock.FlockBase attribute*), 124
emb_head_is_query (*dicee.models.flock.FlockBase attribute*), 124
emb_neighbor (*dicee.models.flock.FlockBase attribute*), 124
emb_node_is_query (*dicee.models.flock.FlockBase attribute*), 124
emb_restart (*dicee.models.flock.FlockBase attribute*), 124
emb_tail_is_query (*dicee.models.flock.FlockBase attribute*), 124
emb_type_is_query (*dicee.models.flock.FlockBase attribute*), 124
embedding_dim (*dicee.analyse_experiments.Experiment attribute*), 21
embedding_dim (*dicee.config.Namespace attribute*), 30
embedding_dim (*dicee.models.base_model.BaseKGE attribute*), 101
embedding_dim (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 215, 230, 235
embedding_dim (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 127
embedding_dim (*dicee.models.literal.LiteralEmbeddings attribute*), 138
embedding_dims (*dicee.models.literal.LiteralEmbeddings attribute*), 137
enable_log (*in module dicee.static_preprocess_funcs*), 264
enc (*dicee.knowledge_graph.KG attribute*), 75
end() (*dicee.Execute method*), 283
end() (*dicee.executer.Execute method*), 72
end_row (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 128
EnsembleKGE (*class in dicee.models.ensemble*), 120
ent2id (*dicee.query_generator.QueryGenerator attribute*), 246
ent2id (*dicee.QueryGenerator attribute*), 289
ent_in (*dicee.query_generator.QueryGenerator attribute*), 246
ent_in (*dicee.QueryGenerator attribute*), 289
ent_out (*dicee.query_generator.QueryGenerator attribute*), 246
ent_out (*dicee.QueryGenerator attribute*), 290
entities_str (*dicee.knowledge_graph.KG property*), 75
entity_embeddings (*dicee.models.AConvQ attribute*), 214
entity_embeddings (*dicee.models.ConvQ attribute*), 214
entity_embeddings (*dicee.models.DualE attribute*), 243
entity_embeddings (*dicee.models.dualE.DualE attribute*), 119
entity_embeddings (*dicee.models.FMult attribute*), 239
entity_embeddings (*dicee.models.FMult2 attribute*), 241
entity_embeddings (*dicee.models.function_space.FMult attribute*), 129
entity_embeddings (*dicee.models.function_space.FMult2 attribute*), 131
entity_embeddings (*dicee.models.function_space.GFMult attribute*), 130
entity_embeddings (*dicee.models.function_space.LFMult attribute*), 132
entity_embeddings (*dicee.models.function_space.LFMult1 attribute*), 131
entity_embeddings (*dicee.models.GFMult attribute*), 240
entity_embeddings (*dicee.models.LFMult attribute*), 242
entity_embeddings (*dicee.models.LFMult1 attribute*), 241
entity_embeddings (*dicee.models.literal.LiteralEmbeddings attribute*), 137, 138
entity_embeddings (*dicee.models.pykeen_models.PykeenKGE attribute*), 142
entity_embeddings (*dicee.models.PykeenKGE attribute*), 234
entity_embeddings (*dicee.models.quaternion.AConvQ attribute*), 145
entity_embeddings (*dicee.models.quaternion.ConvQ attribute*), 144
entity_model (*dicee.models.TRIXRelation attribute*), 245
entity_model (*dicee.models.trix.TRIXRelation attribute*), 164

entity_model (*dicee.models.ULTRA attribute*), 244
entity_model (*dicee.models.ultra.ULTRA attribute*), 167
entity_model_1 (*dicee.models.TRIX attribute*), 245
entity_model_1 (*dicee.models.trix.TRIX attribute*), 163
entity_model_2 (*dicee.models.TRIX attribute*), 245
entity_model_2 (*dicee.models.trix.TRIX attribute*), 163
entity_optim_device (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
entity_to_idx (*dicee.dataset_classes.LiteralDataset attribute*), 40
entity_to_idx (*dicee.knowledge_graph.KG attribute*), 74
entity_to_idx (*dicee.scripts.index_serve.NeuralSearcher attribute*), 256
EntityNBFNet (*class in dicee.models.ultra*), 166
EntityReasoner (*class in dicee.models.trix*), 161
epoch_count (*dicee.abstracts.AbstractPPECallback attribute*), 19
epoch_count (*dicee.weight_averaging.ASWA attribute*), 277
epoch_counter (*dicee.callbacks.Eval attribute*), 25
epoch_counter (*dicee.callbacks.KGESaveCallback attribute*), 24
epoch_counter (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
epoch_ratio (*dicee.callbacks.Eval attribute*), 25
eps (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss attribute*), 85
eps (*dicee.losses.custom_losses.AdaptiveLabelSmoothingLoss attribute*), 84
eps (*dicee.losses.custom_losses.ConfidenceBasedAdaptiveLabelRelaxationLoss attribute*), 85
eps (*dicee.losses.custom_losses.LabelRelaxationLoss attribute*), 84
eps (*dicee.models.AdaptiveLabelRelaxationLoss attribute*), 171
eps (*dicee.models.AdaptiveLabelSmoothingLoss attribute*), 171
eps (*dicee.models.ConfidenceBasedAdaptiveLabelRelaxationLoss attribute*), 173
eps (*dicee.models.flock.RMSNorm attribute*), 122
eps (*dicee.models.LabelRelaxationLoss attribute*), 173
er_vocab (*dicee.evaluation.Evaluator attribute*), 60, 61
er_vocab (*dicee.evaluation.evaluator.Evaluator attribute*), 49
er_vocab (*dicee.Evaluator attribute*), 292
er_vocab (*dicee.evaluator.Evaluator attribute*), 67, 68
estimate_mfu() (*dicee.models.transformers.GPT method*), 161
estimate_q() (*in module dicee.callbacks*), 25
Eval (*class in dicee.callbacks*), 25
eval() (*dicee.evaluation.Evaluator method*), 61
eval() (*dicee.evaluation.evaluator.Evaluator method*), 50
eval() (*dicee.Evaluator method*), 293
eval() (*dicee.evaluator.Evaluator method*), 68
eval() (*dicee.models.ensemble.EnsembleKGE method*), 120
eval_at_epochs (*dicee.config.Namespace attribute*), 33
eval_epochs (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
eval_every_n_epochs (*dicee.config.Namespace attribute*), 33
eval_lp_performance() (*dicee.KGE method*), 284
eval_lp_performance() (*dicee.knowledge_graph_embeddings.KGE method*), 76
eval_model (*dicee.config.Namespace attribute*), 31
eval_model (*dicee.knowledge_graph.KG attribute*), 74
eval_rank_of_head_and_tail_byte_pair_encoded_entity() (*dicee.evaluation.Evaluator method*), 61
eval_rank_of_head_and_tail_byte_pair_encoded_entity() (*dicee.evaluation.evaluator.Evaluator method*), 50
eval_rank_of_head_and_tail_byte_pair_encoded_entity() (*dicee.Evaluator method*), 293
eval_rank_of_head_and_tail_byte_pair_encoded_entity() (*dicee.evaluator.Evaluator method*), 68
eval_rank_of_head_and_tail_entity() (*dicee.evaluation.Evaluator method*), 61
eval_rank_of_head_and_tail_entity() (*dicee.evaluation.evaluator.Evaluator method*), 50
eval_rank_of_head_and_tail_entity() (*dicee.Evaluator method*), 293
eval_rank_of_head_and_tail_entity() (*dicee.evaluator.Evaluator method*), 68
eval_tie_policy (*dicee.config.Namespace attribute*), 31
eval_tie_seed (*dicee.config.Namespace attribute*), 31
eval_with_bpe_vs_all() (*dicee.evaluation.Evaluator method*), 61
eval_with_bpe_vs_all() (*dicee.evaluation.evaluator.Evaluator method*), 50
eval_with_bpe_vs_all() (*dicee.Evaluator method*), 293
eval_with_bpe_vs_all() (*dicee.evaluator.Evaluator method*), 69
eval_with_byte() (*dicee.evaluation.Evaluator method*), 61
eval_with_byte() (*dicee.evaluation.evaluator.Evaluator method*), 50
eval_with_byte() (*dicee.Evaluator method*), 293
eval_with_byte() (*dicee.evaluator.Evaluator method*), 69
eval_with_data() (*dicee.evaluation.Evaluator method*), 63
eval_with_data() (*dicee.evaluation.evaluator.Evaluator method*), 52
eval_with_data() (*dicee.Evaluator method*), 295
eval_with_data() (*dicee.evaluator.Evaluator method*), 70

eval_with_vs_all() (*dicee.evaluation.Evaluator method*), 62
eval_with_vs_all() (*dicee.evaluation.evaluator.Evaluator method*), 50
eval_with_vs_all() (*dicee.Evaluator method*), 293
eval_with_vs_all() (*dicee.evaluator.Evaluator method*), 69
evaluate() (*in module dicee.static_funcs*), 261
evaluate_bpe_lp() (*in module dicee.evaluation*), 63
evaluate_bpe_lp() (*in module dicee.evaluation.link_prediction*), 55
evaluate_bpe_lp() (*in module dicee.static_funcs_training*), 262
evaluate_ensemble_link_prediction_performance() (*in module dicee.eval_static_funcs*), 43
evaluate_ensemble_link_prediction_performance() (*in module dicee.evaluation*), 59
evaluate_ensemble_link_prediction_performance() (*in module dicee.evaluation.ensemble*), 47
evaluate_link_prediction_performance() (*in module dicee.eval_static_funcs*), 44
evaluate_link_prediction_performance() (*in module dicee.evaluation*), 63
evaluate_link_prediction_performance() (*in module dicee.evaluation.link_prediction*), 52
evaluate_link_prediction_performance_with_bpe() (*in module dicee.eval_static_funcs*), 44
evaluate_link_prediction_performance_with_bpe() (*in module dicee.evaluation*), 64
evaluate_link_prediction_performance_with_bpe() (*in module dicee.evaluation.link_prediction*), 53
evaluate_link_prediction_performance_with_bpe_reciprocals() (*in module dicee.eval_static_funcs*), 45
evaluate_link_prediction_performance_with_bpe_reciprocals() (*in module dicee.evaluation*), 64
evaluate_link_prediction_performance_with_bpe_reciprocals() (*in module dicee.evaluation.link_prediction*), 53
evaluate_link_prediction_performance_with_reciprocals() (*in module dicee.eval_static_funcs*), 45
evaluate_link_prediction_performance_with_reciprocals() (*in module dicee.evaluation*), 64
evaluate_link_prediction_performance_with_reciprocals() (*in module dicee.evaluation.link_prediction*), 53
evaluate_literal_prediction() (*in module dicee.eval_static_funcs*), 46
evaluate_literal_prediction() (*in module dicee.evaluation*), 66
evaluate_literal_prediction() (*in module dicee.evaluation.literal_prediction*), 56
evaluate_lp() (*dicee.evaluation.Evaluator method*), 62
evaluate_lp() (*dicee.evaluation.evaluator.Evaluator method*), 51
evaluate_lp() (*dicee.Evaluator method*), 294
evaluate_lp() (*dicee.evaluator.Evaluator method*), 69
evaluate_lp() (*in module dicee.evaluation*), 65
evaluate_lp() (*in module dicee.evaluation.link_prediction*), 54
evaluate_lp() (*in module dicee.static_funcs_training*), 262
evaluate_lp_bpe_k_vs_all() (*dicee.evaluation.Evaluator method*), 62
evaluate_lp_bpe_k_vs_all() (*dicee.evaluation.evaluator.Evaluator method*), 51
evaluate_lp_bpe_k_vs_all() (*dicee.Evaluator method*), 294
evaluate_lp_bpe_k_vs_all() (*dicee.evaluator.Evaluator method*), 69
evaluate_lp_bpe_k_vs_all() (*in module dicee.eval_static_funcs*), 46
evaluate_lp_bpe_k_vs_all() (*in module dicee.evaluation*), 65
evaluate_lp_bpe_k_vs_all() (*in module dicee.evaluation.link_prediction*), 55
evaluate_lp_k_vs_all() (*dicee.evaluation.Evaluator method*), 62
evaluate_lp_k_vs_all() (*dicee.evaluation.evaluator.Evaluator method*), 50
evaluate_lp_k_vs_all() (*dicee.Evaluator method*), 294
evaluate_lp_k_vs_all() (*dicee.evaluator.Evaluator method*), 69
evaluate_lp_with_byte() (*dicee.evaluation.Evaluator method*), 62
evaluate_lp_with_byte() (*dicee.evaluation.evaluator.Evaluator method*), 51
evaluate_lp_with_byte() (*dicee.Evaluator method*), 294
evaluate_lp_with_byte() (*dicee.evaluator.Evaluator method*), 69
Evaluator (*class in dicee*), 292
Evaluator (*class in dicee.evaluation*), 60
Evaluator (*class in dicee.evaluation.evaluator*), 49
Evaluator (*class in dicee.evaluator*), 67
evaluator (*dicee.DICE_Trainer attribute*), 291
evaluator (*dicee.Execute attribute*), 282, 283
evaluator (*dicee.executer.Execute attribute*), 71, 72
evaluator (*dicee.trainer.DICE_Trainer attribute*), 273
evaluator (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 266
every_x_epoch (*dicee.callbacks.KGESaveCallback attribute*), 24
example_input_array (*dicee.models.ensemble.EnsembleKGE property*), 120
Execute (*class in dicee*), 282
Execute (*class in dicee.executer*), 71
exists() (*dicee.knowledge_graph.KG method*), 75
Experiment (*class in dicee.analyse_experiments*), 21
experiment_dir (*dicee.callbacks.LRScheduler attribute*), 27
experiment_dir (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
explicit (*dicee.models.QMult attribute*), 213
explicit (*dicee.models.quaternion.QMult attribute*), 143
exponential_function() (*in module dicee.static_funcs*), 261

extract_input_outputs() (*dicee.trainer.torch_trainer_ddp.NodeTrainer method*), 271
extract_input_outputs() (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer method*), 272
extract_input_outputs() (*in module dicee.trainer.model_parallelism*), 268
extract_input_outputs_set_device() (*dicee.trainer.torch_trainer.TorchTrainer method*), 269

F

f (*dicee.callbacks.KronE attribute*), 26
fc (*dicee.models.literal.LiteralEmbeddings attribute*), 138
fc1 (*dicee.models.AConEx attribute*), 205
fc1 (*dicee.models.AConvO attribute*), 221
fc1 (*dicee.models.AConvQ attribute*), 215
fc1 (*dicee.models.complex.AConEx attribute*), 115
fc1 (*dicee.models.complex.ConEx attribute*), 114
fc1 (*dicee.models.ConEx attribute*), 204
fc1 (*dicee.models.ConvO attribute*), 221
fc1 (*dicee.models.ConvQ attribute*), 214
fc1 (*dicee.models.octonion.AConvO attribute*), 141
fc1 (*dicee.models.octonion.ConvO attribute*), 140
fc1 (*dicee.models.quaternion.AConvQ attribute*), 145
fc1 (*dicee.models.quaternion.ConvQ attribute*), 144
fc_num_input (*dicee.models.AConEx attribute*), 204
fc_num_input (*dicee.models.AConvO attribute*), 221
fc_num_input (*dicee.models.AConvQ attribute*), 214
fc_num_input (*dicee.models.complex.AConEx attribute*), 115
fc_num_input (*dicee.models.complex.ConEx attribute*), 114
fc_num_input (*dicee.models.ConEx attribute*), 203
fc_num_input (*dicee.models.ConvO attribute*), 221
fc_num_input (*dicee.models.ConvQ attribute*), 214
fc_num_input (*dicee.models.octonion.AConvO attribute*), 141
fc_num_input (*dicee.models.octonion.ConvO attribute*), 140
fc_num_input (*dicee.models.quaternion.AConvQ attribute*), 145
fc_num_input (*dicee.models.quaternion.ConvQ attribute*), 144
fc_out (*dicee.models.literal.LiteralEmbeddings attribute*), 138
feature_map_dropout (*dicee.models.AConEx attribute*), 205
feature_map_dropout (*dicee.models.AConvO attribute*), 222
feature_map_dropout (*dicee.models.AConvQ attribute*), 215
feature_map_dropout (*dicee.models.complex.AConEx attribute*), 115
feature_map_dropout (*dicee.models.complex.ConEx attribute*), 114
feature_map_dropout (*dicee.models.ConEx attribute*), 204
feature_map_dropout (*dicee.models.ConvO attribute*), 221
feature_map_dropout (*dicee.models.ConvQ attribute*), 214
feature_map_dropout (*dicee.models.octonion.AConvO attribute*), 141
feature_map_dropout (*dicee.models.octonion.ConvO attribute*), 140
feature_map_dropout (*dicee.models.quaternion.AConvQ attribute*), 145
feature_map_dropout (*dicee.models.quaternion.ConvQ attribute*), 144
feature_map_dropout_rate (*dicee.config.Namespace attribute*), 32
feature_map_dropout_rate (*dicee.models.base_model.BaseKGE attribute*), 101
feature_map_dropout_rate (*dicee.models.BaseKGE attribute*), 182, 187, 200, 208, 216, 230, 235
features() (*dicee.models.trix.EntityReasoner method*), 162
feed_forward (*dicee.models.flock.BidirectionalGRU attribute*), 124
FeedForward (*class in dicee.models.flock*), 122
fetch_worker() (*in module dicee.read_preprocess_save_load_kg.util*), 252
ffn_norm (*dicee.models.flock.BidirectionalGRU attribute*), 124
fill_query() (*dicee.query_generator.QueryGenerator method*), 246
fill_query() (*dicee.QueryGenerator method*), 290
find_good_batch_size() (*in module dicee.trainer.auto_batch_finder*), 264
find_missing_triples() (*dicee.KGE method*), 288
find_missing_triples() (*dicee.knowledge_graph_embeddings.KGE method*), 81
fit() (*dicee.trainer.model_parallelism.TensorParallel method*), 268
fit() (*dicee.trainer.torch_trainer_ddp.TorchDDPTrainer method*), 270
fit() (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer method*), 272
fit() (*dicee.trainer.torch_trainer.TorchTrainer method*), 269
FixedNegSampleDataset (*class in dicee.dataset_classes*), 41
flash (*dicee.models.transformers.SelfAttention attribute*), 158
Flock (*class in dicee.models*), 245
Flock (*class in dicee.models.flock*), 125
flock_attention_heads (*dicee.config.Namespace attribute*), 29

flock_checkpoint (*dicee.config.Namespace attribute*), 29
flock_compact_state (*dicee.config.Namespace attribute*), 29
flock_compile_sampler (*dicee.config.Namespace attribute*), 29
flock_dim (*dicee.config.Namespace attribute*), 29
flock_num_layers (*dicee.config.Namespace attribute*), 29
flock_pack_walks (*dicee.config.Namespace attribute*), 29
flock_prefetch_walks (*dicee.config.Namespace attribute*), 29
flock_query_batch_size (*dicee.config.Namespace attribute*), 29
flock_refinements (*dicee.config.Namespace attribute*), 29
flock_seed (*dicee.config.Namespace attribute*), 29
flock_test_samples (*dicee.config.Namespace attribute*), 29
flock_walk_len (*dicee.config.Namespace attribute*), 29
flock_walk_num (*dicee.config.Namespace attribute*), 29
FlockBase (*class in dicee.models.flock*), 124
FlockRelation (*class in dicee.models*), 245
FlockRelation (*class in dicee.models.flock*), 125
FMult (*class in dicee.models*), 239
FMult (*class in dicee.models.function_space*), 129
FMult2 (*class in dicee.models*), 241
FMult2 (*class in dicee.models.function_space*), 130
form_of_labelling (*dicee.DICE_Trainer attribute*), 291
form_of_labelling (*dicee.trainer.DICE_Trainer attribute*), 273
form_of_labelling (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 266
forward() (*dicee.losses.custom_losses.ACLS method*), 87
forward() (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss method*), 85
forward() (*dicee.losses.custom_losses.AdaptiveLabelSmoothingLoss method*), 84
forward() (*dicee.losses.custom_losses.AggregatedLSandLR method*), 86
forward() (*dicee.losses.custom_losses.CombinedAdaptiveLSandAdaptiveLR method*), 86
forward() (*dicee.losses.custom_losses.CombinedLSandLR method*), 85
forward() (*dicee.losses.custom_losses.ConfidenceBasedAdaptiveLabelRelaxationLoss method*), 85
forward() (*dicee.losses.custom_losses.DefaultBCELoss method*), 82
forward() (*dicee.losses.custom_losses.LabelRelaxationLoss method*), 84
forward() (*dicee.losses.custom_losses.LabelSmoothingLoss method*), 83
forward() (*dicee.losses.custom_losses.WeightedBCELoss method*), 83
forward() (*dicee.models.ACLS method*), 170
forward() (*dicee.models.AdaptiveLabelRelaxationLoss method*), 171
forward() (*dicee.models.AdaptiveLabelSmoothingLoss method*), 171
forward() (*dicee.models.AggregatedLSandLR method*), 172
forward() (*dicee.models.base_model.BaseKGE method*), 102
forward() (*dicee.models.base_model.IdentityClass static method*), 105
forward() (*dicee.models.BaseKGE method*), 184, 189, 201, 210, 217, 232, 237
forward() (*dicee.models.Block method*), 187
forward() (*dicee.models.clifford.TransformerBlock method*), 110
forward() (*dicee.models.clifford.TransformerMLP method*), 110
forward() (*dicee.models.clifford.TransformerSelfAttention method*), 110
forward() (*dicee.models.CombinedAdaptiveLSandAdaptiveLR method*), 172
forward() (*dicee.models.CombinedLSandLR method*), 172
forward() (*dicee.models.ConfidenceBasedAdaptiveLabelRelaxationLoss method*), 173
forward() (*dicee.models.DefaultBCELoss method*), 173
forward() (*dicee.models.flock.BidirectionalGRU method*), 124
forward() (*dicee.models.flock.FeedForward method*), 123
forward() (*dicee.models.flock.RMSNorm method*), 122
forward() (*dicee.models.fsdp_models.RowWiseShardedEmbedding method*), 128
forward() (*dicee.models.graph_model.GraphKGE method*), 136
forward() (*dicee.models.IdentityClass static method*), 186, 212, 220
forward() (*dicee.models.LabelRelaxationLoss method*), 173
forward() (*dicee.models.LabelSmoothingLoss method*), 174
forward() (*dicee.models.literal.LiteralEmbeddings method*), 138
forward() (*dicee.models.transformers.Block method*), 159
forward() (*dicee.models.transformers.BytE method*), 156
forward() (*dicee.models.transformers.GPT method*), 160
forward() (*dicee.models.transformers.LayerNorm method*), 157
forward() (*dicee.models.transformers.MLP method*), 158
forward() (*dicee.models.transformers.SelfAttention method*), 158
forward() (*dicee.models.ultra.EntityNBFNet method*), 167
forward() (*dicee.models.ultra.RelationalConv method*), 165
forward() (*dicee.models.ultra.RelNBFNet method*), 166
forward() (*dicee.models.WeightedBCELoss method*), 174

forward_backward_update() (*dicee.trainer.torch_trainer.TorchTrainer method*), 269
forward_backward_update_loss() (*in module dicee.trainer.model_parallelism*), 268
forward_byte_pair_encoded_k_vs_all() (*dicee.models.base_model.BaseKGE method*), 102
forward_byte_pair_encoded_k_vs_all() (*dicee.models.BaseKGE method*), 183, 188, 200, 209, 217, 231, 236
forward_byte_pair_encoded_triple() (*dicee.models.base_model.BaseKGE method*), 102
forward_byte_pair_encoded_triple() (*dicee.models.BaseKGE method*), 183, 189, 201, 209, 217, 231, 237
forward_grouped() (*dicee.models.graph_model.GraphKGE method*), 135
forward_grouped() (*dicee.models.graph_model.RelationGraphKGE method*), 136
forward_k_vs_all() (*dicee.models.AConEx method*), 205
forward_k_vs_all() (*dicee.models.AConvO method*), 222
forward_k_vs_all() (*dicee.models.AConvQ method*), 215
forward_k_vs_all() (*dicee.models.base_model.BaseKGE method*), 103
forward_k_vs_all() (*dicee.models.BaseKGE method*), 184, 190, 202, 210, 218, 232, 238
forward_k_vs_all() (*dicee.models.clifford.DeCaL method*), 112
forward_k_vs_all() (*dicee.models.clifford.Keci method*), 108
forward_k_vs_all() (*dicee.models.clifford.KeciTransformer method*), 109
forward_k_vs_all() (*dicee.models.CoKE method*), 199
forward_k_vs_all() (*dicee.models.ComplEx method*), 206
forward_k_vs_all() (*dicee.models.complex.AConEx method*), 116
forward_k_vs_all() (*dicee.models.complex.ComplEx method*), 117
forward_k_vs_all() (*dicee.models.complex.ConEx method*), 115
forward_k_vs_all() (*dicee.models.complex.RotatE method*), 118
forward_k_vs_all() (*dicee.models.ConEx method*), 204
forward_k_vs_all() (*dicee.models.ConvO method*), 221
forward_k_vs_all() (*dicee.models.ConvQ method*), 214
forward_k_vs_all() (*dicee.models.DeCaL method*), 227
forward_k_vs_all() (*dicee.models.DistMult method*), 192
forward_k_vs_all() (*dicee.models.DualE method*), 244
forward_k_vs_all() (*dicee.models.dualE.DualE method*), 119
forward_k_vs_all() (*dicee.models.fsdp_models.FSDPShardedEntityModel method*), 128
forward_k_vs_all() (*dicee.models.graph_model.GraphKGE method*), 136
forward_k_vs_all() (*dicee.models.graph_model.RelationGraphKGE method*), 137
forward_k_vs_all() (*dicee.models.Keci method*), 225
forward_k_vs_all() (*dicee.models.KeciTransformer method*), 229
forward_k_vs_all() (*dicee.models.MuRE method*), 196
forward_k_vs_all() (*dicee.models.octonion.AConvO method*), 141
forward_k_vs_all() (*dicee.models.octonion.ConvO method*), 140
forward_k_vs_all() (*dicee.models.octonion.OMult method*), 139
forward_k_vs_all() (*dicee.models.OMult method*), 220
forward_k_vs_all() (*dicee.models.pykeen_models.PykeenKGE method*), 142
forward_k_vs_all() (*dicee.models.PykeenKGE method*), 234
forward_k_vs_all() (*dicee.models.QMult method*), 213
forward_k_vs_all() (*dicee.models.quaternion.AConvQ method*), 145
forward_k_vs_all() (*dicee.models.quaternion.ConvQ method*), 145
forward_k_vs_all() (*dicee.models.quaternion.QMult method*), 144
forward_k_vs_all() (*dicee.models.real.CoKE method*), 153
forward_k_vs_all() (*dicee.models.real.DistMult method*), 147
forward_k_vs_all() (*dicee.models.real.MuRE method*), 150
forward_k_vs_all() (*dicee.models.real.Shallom method*), 151
forward_k_vs_all() (*dicee.models.real.TransE method*), 148
forward_k_vs_all() (*dicee.models.real.TransH method*), 149
forward_k_vs_all() (*dicee.models.RotatE method*), 207
forward_k_vs_all() (*dicee.models.Shallom method*), 196
forward_k_vs_all() (*dicee.models.TransE method*), 193
forward_k_vs_all() (*dicee.models.TransH method*), 194
forward_k_vs_all_heads() (*dicee.models.graph_model.GraphKGE method*), 136
forward_k_vs_all_heads() (*dicee.models.graph_model.RelationGraphKGE method*), 137
forward_k_vs_all_relations (*dicee.models.graph_model.RelationGraphKGE attribute*), 137
forward_k_vs_sample() (*dicee.models.AConEx method*), 205
forward_k_vs_sample() (*dicee.models.base_model.BaseKGE method*), 103
forward_k_vs_sample() (*dicee.models.BaseKGE method*), 184, 190, 202, 211, 218, 232, 238
forward_k_vs_sample() (*dicee.models.clifford.Keci method*), 109
forward_k_vs_sample() (*dicee.models.CoKE method*), 199
forward_k_vs_sample() (*dicee.models.ComplEx method*), 206
forward_k_vs_sample() (*dicee.models.complex.AConEx method*), 116
forward_k_vs_sample() (*dicee.models.complex.ComplEx method*), 117
forward_k_vs_sample() (*dicee.models.complex.ConEx method*), 115
forward_k_vs_sample() (*dicee.models.complex.RotatE method*), 118

forward_k_vs_sample() (*dicee.models.ConEx method*), 204
forward_k_vs_sample() (*dicee.models.DistMult method*), 192
forward_k_vs_sample() (*dicee.models.graph_model.GraphKGE method*), 135
forward_k_vs_sample() (*dicee.models.graph_model.RelationGraphKGE method*), 137
forward_k_vs_sample() (*dicee.models.Keci method*), 225
forward_k_vs_sample() (*dicee.models.MuRE method*), 196
forward_k_vs_sample() (*dicee.models.pykeen_models.PykeenKGE method*), 142
forward_k_vs_sample() (*dicee.models.PykeenKGE method*), 235
forward_k_vs_sample() (*dicee.models.QMult method*), 213
forward_k_vs_sample() (*dicee.models.quaternion.QMult method*), 144
forward_k_vs_sample() (*dicee.models.real.CoKE method*), 153
forward_k_vs_sample() (*dicee.models.real.DistMult method*), 147
forward_k_vs_sample() (*dicee.models.real.MuRE method*), 151
forward_k_vs_sample() (*dicee.models.real.TransH method*), 149
forward_k_vs_sample() (*dicee.models.RotatE method*), 207
forward_k_vs_sample() (*dicee.models.TransH method*), 195
forward_k_vs_with_explicit() (*dicee.models.clifford.Keci method*), 107
forward_k_vs_with_explicit() (*dicee.models.Keci method*), 224
forward_triples() (*dicee.models.AConEx method*), 205
forward_triples() (*dicee.models.AConvO method*), 222
forward_triples() (*dicee.models.AConvQ method*), 215
forward_triples() (*dicee.models.base_model.BaseKGE method*), 103
forward_triples() (*dicee.models.BaseKGE method*), 184, 189, 202, 210, 218, 232, 238
forward_triples() (*dicee.models.clifford.DeCaL method*), 111
forward_triples() (*dicee.models.clifford.Keci method*), 109
forward_triples() (*dicee.models.complex.AConEx method*), 116
forward_triples() (*dicee.models.complex.ConEx method*), 115
forward_triples() (*dicee.models.ConEx method*), 204
forward_triples() (*dicee.models.ConvO method*), 221
forward_triples() (*dicee.models.ConvQ method*), 214
forward_triples() (*dicee.models.DeCaL method*), 226
forward_triples() (*dicee.models.DualE method*), 244
forward_triples() (*dicee.models.dualE.DualE method*), 119
forward_triples() (*dicee.models.FMult method*), 240
forward_triples() (*dicee.models.FMult2 method*), 241
forward_triples() (*dicee.models.function_space.FMult method*), 130
forward_triples() (*dicee.models.function_space.FMult2 method*), 131
forward_triples() (*dicee.models.function_space.GFMult method*), 130
forward_triples() (*dicee.models.function_space.LFMult method*), 132
forward_triples() (*dicee.models.function_space.LFMult1 method*), 131
forward_triples() (*dicee.models.GFMult method*), 240
forward_triples() (*dicee.models.graph_model.GraphKGE method*), 135
forward_triples() (*dicee.models.graph_model.RelationGraphKGE method*), 136
forward_triples() (*dicee.models.Keci method*), 225
forward_triples() (*dicee.models.LFMult method*), 242
forward_triples() (*dicee.models.LFMult1 method*), 242
forward_triples() (*dicee.models.MuRE method*), 195
forward_triples() (*dicee.models.octonion.AConvO method*), 141
forward_triples() (*dicee.models.octonion.ConvO method*), 140
forward_triples() (*dicee.models.Pyke method*), 197
forward_triples() (*dicee.models.pykeen_models.PykeenKGE method*), 142
forward_triples() (*dicee.models.PykeenKGE method*), 234
forward_triples() (*dicee.models.quaternion.AConvQ method*), 145
forward_triples() (*dicee.models.quaternion.ConvQ method*), 145
forward_triples() (*dicee.models.real.MuRE method*), 150
forward_triples() (*dicee.models.real.Pyke method*), 152
forward_triples() (*dicee.models.real.Shallom method*), 151
forward_triples() (*dicee.models.real.TransH method*), 149
forward_triples() (*dicee.models.Shallom method*), 197
forward_triples() (*dicee.models.TransH method*), 194
freeze_entity_embeddings (*dicee.models.literal.LiteralEmbeddings attribute*), 138
frequency (*dicee.callbacks.Perturb attribute*), 27
from_node (*dicee.models.flock.FlockBase attribute*), 124
from_pretrained() (*dicee.models.transformers.GPT class method*), 160
from_pretrained_model_write_embeddings_into_csv() (*in module dicee.static_funcs*), 261
from_type (*dicee.models.flock.FlockBase attribute*), 125
FSDP1vsSampleDataset (*class in dicee.dataset_classes*), 37
FSDPShardedEntityModel (*class in dicee.models.fsdp_models*), 128

full_storage_path (*dicee.analyse_experiments.Experiment attribute*), 21
func_triple_to_bpe_representation (*dicee.evaluation.Evaluator attribute*), 61
func_triple_to_bpe_representation (*dicee.evaluation.evaluator.Evaluator attribute*), 49
func_triple_to_bpe_representation (*dicee.Evaluator attribute*), 293
func_triple_to_bpe_representation (*dicee.evaluator.Evaluator attribute*), 68
func_triple_to_bpe_representation() (*dicee.knowledge_graph.KG method*), 76
function() (*dicee.models.FMult2 method*), 241
function() (*dicee.models.function_space.FMult2 method*), 131

G

gamma (*dicee.models.FMult attribute*), 240
gamma (*dicee.models.function_space.FMult attribute*), 129
gate_residual (*dicee.models.literal.LiteralEmbeddings attribute*), 138
gated_residual_proj (*dicee.models.literal.LiteralEmbeddings attribute*), 138
gather_entity_embeddings_on_rank_zero() (*dicee.models.fsdp_models.FSDPShardedEntityModel method*), 128
gelu (*dicee.models.clifford.TransformerMLP attribute*), 110
gelu (*dicee.models.transformers.MLP attribute*), 158
gen_test (*dicee.query_generator.QueryGenerator attribute*), 246
gen_test (*dicee.QueryGenerator attribute*), 289
gen_valid (*dicee.query_generator.QueryGenerator attribute*), 246
gen_valid (*dicee.QueryGenerator attribute*), 289
generate() (*dicee.KGE method*), 284
generate() (*dicee.knowledge_graph_embeddings.KGE method*), 76
generate() (*dicee.models.transformers.BytE method*), 156
generate_queries() (*dicee.query_generator.QueryGenerator method*), 247
generate_queries() (*dicee.QueryGenerator method*), 290
get_aswa_state_dict() (*dicee.weight_averaging.ASWA method*), 277
get_bpe_head_and_relation_representation() (*dicee.models.base_model.BaseKGE method*), 104
get_bpe_head_and_relation_representation() (*dicee.models.BaseKGE method*), 185, 191, 203, 212, 219, 233, 239
get_bpe_token_representation() (*dicee.abstracts.BaseInteractiveKGE method*), 14
get_callbacks() (*in module dicee.trainer.dice_trainer*), 266
get_default_arguments() (*in module dicee.analyse_experiments*), 21
get_default_arguments() (*in module dicee.scripts.index_serve*), 255
get_default_arguments() (*in module dicee.scripts.run*), 256
get_ee_vocab() (*in module dicee.read_preprocess_save_load_kg.util*), 252
get_ee_vocab() (*in module dicee.static_funcs*), 258
get_ee_vocab() (*in module dicee.static_preprocess_funcs*), 264
get_embeddings() (*dicee.models.base_model.BaseKGE method*), 105
get_embeddings() (*dicee.models.BaseKGE method*), 186, 191, 203, 212, 219, 234, 239
get_embeddings() (*dicee.models.ensemble.EnsembleKGE method*), 120
get_embeddings() (*dicee.models.fsdp_models.FSDPShardedEntityModel method*), 128
get_embeddings() (*dicee.models.graph_model.GraphKGE method*), 135
get_embeddings() (*dicee.models.real.Shallom method*), 151
get_embeddings() (*dicee.models.Shallom method*), 196
get_entity_embeddings() (*dicee.abstracts.BaseInteractiveKGE method*), 16
get_entity_index() (*dicee.abstracts.BaseInteractiveKGE method*), 15
get_er_vocab() (*in module dicee.read_preprocess_save_load_kg.util*), 252
get_er_vocab() (*in module dicee.static_funcs*), 258
get_er_vocab() (*in module dicee.static_preprocess_funcs*), 264
get_eval_report() (*dicee.abstracts.BaseInteractiveKGE method*), 14
get_filter_vocabs() (*in module dicee.read_preprocess_save_load_kg.util*), 252
get_head_relation_representation() (*dicee.models.base_model.BaseKGE method*), 104
get_head_relation_representation() (*dicee.models.BaseKGE method*), 185, 190, 202, 211, 219, 233, 238
get_kronecker_triple_representation() (*dicee.callbacks.KronE method*), 26
get_mean_and_var() (*dicee.weight_averaging.SWAG method*), 279
get_num_params() (*dicee.models.transformers.GPT method*), 160
get_padded_bpe_triple_representation() (*dicee.abstracts.BaseInteractiveKGE method*), 14
get_queries() (*dicee.query_generator.QueryGenerator method*), 247
get_queries() (*dicee.QueryGenerator method*), 290
get_re_vocab() (*in module dicee.read_preprocess_save_load_kg.util*), 252
get_re_vocab() (*in module dicee.static_funcs*), 258
get_re_vocab() (*in module dicee.static_preprocess_funcs*), 264
get_reg() (*dicee.losses.custom_losses.ACLS method*), 87
get_reg() (*dicee.models.ACLS method*), 170
get_relation_embeddings() (*dicee.abstracts.BaseInteractiveKGE method*), 16
get_relation_index() (*dicee.abstracts.BaseInteractiveKGE method*), 15
get_sentence_representation() (*dicee.models.base_model.BaseKGE method*), 104

get_sentence_representation() (*dicee.models.BaseKGE method*), 185, 191, 203, 211, 219, 233, 239
get_transductive_entity_embeddings() (*dicee.KGE method*), 284
get_transductive_entity_embeddings() (*dicee.knowledge_graph_embeddings.KGE method*), 76
get_triple_representation() (*dicee.models.base_model.BaseKGE method*), 104
get_triple_representation() (*dicee.models.BaseKGE method*), 185, 190, 202, 211, 218, 233, 238
GFMult (*class in dicee.models*), 240
GFMult (*class in dicee.models.function_space*), 130
global_rank (*dicee.abstracts.AbstractTrainer attribute*), 13
global_rank (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 270
global_rank (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
GPT (*class in dicee.models.transformers*), 160
GPTConfig (*class in dicee.models.transformers*), 159
gpus (*dicee.config.Namespace attribute*), 30
gradient_accumulation_steps (*dicee.config.Namespace attribute*), 31
gradient_clip_val (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
graph_filename (*dicee.models.flock.FlockBase attribute*), 124
graph_filename (*dicee.models.graph_model.GraphKGE attribute*), 134
graph_filename (*dicee.models.trix.TRIXBase attribute*), 163
graph_filename (*dicee.models.ULTRA attribute*), 244
graph_filename (*dicee.models.ultra.ULTRA attribute*), 167
graph_inference_backend (*dicee.config.Namespace attribute*), 29
graph_inference_compile (*dicee.config.Namespace attribute*), 29
graph_projection_cache_mb (*dicee.config.Namespace attribute*), 29
graph_relation_cache_mb (*dicee.config.Namespace attribute*), 29
GraphKGE (*class in dicee.models.graph_model*), 134
ground_queries() (*dicee.query_generator.QueryGenerator method*), 247
ground_queries() (*dicee.QueryGenerator method*), 290
grouped_adversarial_bce() (*in module dicee.models.sampled_loss*), 154
grouped_negative_sampling (*dicee.config.Namespace attribute*), 29
gru (*dicee.models.flock.BidirectionalGRU attribute*), 124
gru_norm (*dicee.models.flock.BidirectionalGRU attribute*), 124
gru_out (*dicee.models.flock.BidirectionalGRU attribute*), 124
gswa_n (*dicee.weight_averaging.SWAG attribute*), 279
gz_threshold (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss attribute*), 85
gz_threshold (*dicee.losses.custom_losses.ConfidenceBasedAdaptiveLabelRelaxationLoss attribute*), 85
gz_threshold (*dicee.losses.custom_losses.LabelRelaxationLoss attribute*), 84
gz_threshold (*dicee.models.AdaptiveLabelRelaxationLoss attribute*), 171
gz_threshold (*dicee.models.ConfidenceBasedAdaptiveLabelRelaxationLoss attribute*), 173
gz_threshold (*dicee.models.LabelRelaxationLoss attribute*), 173

H

half_dim (*dicee.models.complex.RotatE attribute*), 118
half_dim (*dicee.models.RotatE attribute*), 207
head (*dicee.models.flock.FlockBase attribute*), 125
hidden_dim (*dicee.models.literal.LiteralEmbeddings attribute*), 138
hidden_dropout (*dicee.models.base_model.BaseKGE attribute*), 101
hidden_dropout (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236
hidden_dropout_rate (*dicee.config.Namespace attribute*), 32
hidden_dropout_rate (*dicee.models.base_model.BaseKGE attribute*), 101
hidden_dropout_rate (*dicee.models.BaseKGE attribute*), 182, 187, 200, 208, 216, 230, 235
hidden_normalizer (*dicee.models.base_model.BaseKGE attribute*), 101
hidden_normalizer (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236
history (*dicee.weight_averaging.AMWA attribute*), 276

I

IdentityClass (*class in dicee.models*), 186, 212, 219
IdentityClass (*class in dicee.models.base_model*), 105
idx_entity_to_bpe_shaped (*dicee.knowledge_graph.KG attribute*), 75
ignore_index (*dicee.losses.custom_losses.ACLS attribute*), 87
ignore_index (*dicee.models.ACLS attribute*), 170
index() (*in module dicee.scripts.index_serve*), 255
index_triple() (*dicee.abstracts.BaseInteractiveKGE method*), 16
inference_backend (*dicee.models.flock.RMSNorm attribute*), 122
inference_backend (*dicee.models.ultra.RelationalConv attribute*), 165
inference_compile (*dicee.models.ultra.RelationalConv attribute*), 165
inference_token() (*dicee.models.graph_model.GraphKGE method*), 134
init_dataloader() (*dicee.DICE_Trainer method*), 291

init_dataloader() (*dicee.trainer.DICE_Trainer method*), 273
init_dataloader() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 267
init_dataset() (*dicee.DICE_Trainer method*), 291
init_dataset() (*dicee.trainer.DICE_Trainer method*), 274
init_dataset() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 267
init_entity_embeddings() (*dicee.models.base_model.BaseKGE method*), 102
init_entity_embeddings() (*dicee.models.BaseKGE method*), 183, 188, 200, 209, 216, 231, 236
init_entity_embeddings() (*dicee.models.graph_model.GraphKGE method*), 134
init_param (*dicee.config.Namespace attribute*), 31
init_params_with_sanity_checking() (*dicee.models.base_model.BaseKGE method*), 102
init_params_with_sanity_checking() (*dicee.models.BaseKGE method*), 183, 189, 201, 210, 217, 231, 237
init_relation_embeddings() (*dicee.models.base_model.BaseKGE method*), 102
init_relation_embeddings() (*dicee.models.BaseKGE method*), 183, 188, 200, 209, 216, 231, 236
init_relation_embeddings() (*dicee.models.graph_model.GraphKGE method*), 135
initial_eval_setting (*dicee.weight_averaging.AMWA attribute*), 276
initial_eval_setting (*dicee.weight_averaging.ASWA attribute*), 277
initialize_or_load_model() (*dicee.DICE_Trainer method*), 291
initialize_or_load_model() (*dicee.trainer.DICE_Trainer method*), 273
initialize_or_load_model() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 267
initialize_trainer() (*dicee.DICE_Trainer method*), 291
initialize_trainer() (*dicee.trainer.DICE_Trainer method*), 273
initialize_trainer() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 267
initialize_trainer() (*in module dicee.trainer.dice_trainer*), 265
input_dp_ent_real (*dicee.models.base_model.BaseKGE attribute*), 101
input_dp_ent_real (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236
input_dp_rel_real (*dicee.models.base_model.BaseKGE attribute*), 101
input_dp_rel_real (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236
input_dropout_rate (*dicee.config.Namespace attribute*), 32
input_dropout_rate (*dicee.models.base_model.BaseKGE attribute*), 101
input_dropout_rate (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 216, 230, 235
INTERACTIONS (*in module dicee.models.trix*), 161
InteractiveQueryDecomposition (*class in dicee.abstracts*), 17
intialize_model() (*in module dicee.static_funcs*), 260
is_continual_training (*dicee.DICE_Trainer attribute*), 291
is_continual_training (*dicee.evaluation.Evaluator attribute*), 61
is_continual_training (*dicee.evaluation.evaluator.Evaluator attribute*), 49
is_continual_training (*dicee.Evaluator attribute*), 293
is_continual_training (*dicee.evaluator.Evaluator attribute*), 68
is_continual_training (*dicee.Execute attribute*), 282
is_continual_training (*dicee.executer.Execute attribute*), 71
is_continual_training (*dicee.trainer.DICE_Trainer attribute*), 273
is_continual_training (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 266
is_global_rank_zero() (*dicee.Execute method*), 283
is_global_rank_zero() (*dicee.executer.Execute method*), 72
is_global_zero (*dicee.abstracts.AbstractTrainer attribute*), 13
is_global_zero (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
is_local_rank_zero() (*dicee.Execute method*), 283
is_local_rank_zero() (*dicee.executer.Execute method*), 72
is_seen() (*dicee.abstracts.BaseInteractiveKGE method*), 15
is_sparql_endpoint_alive() (*in module dicee.sanity_checkers*), 255

K

k (*dicee.models.FMult attribute*), 240
k (*dicee.models.FMult2 attribute*), 241
k (*dicee.models.function_space.FMult attribute*), 129
k (*dicee.models.function_space.FMult2 attribute*), 131
k (*dicee.models.function_space.GFMult attribute*), 130
k (*dicee.models.GFMult attribute*), 240
k_fold_cross_validation() (*dicee.DICE_Trainer method*), 291
k_fold_cross_validation() (*dicee.trainer.DICE_Trainer method*), 274
k_fold_cross_validation() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 267
k_vs_all_score() (*dicee.models.clifford.Keci method*), 108
k_vs_all_score() (*dicee.models.ComplEx static method*), 206
k_vs_all_score() (*dicee.models.complex.ComplEx static method*), 117
k_vs_all_score() (*dicee.models.DistMult method*), 192
k_vs_all_score() (*dicee.models.Keci method*), 224
k_vs_all_score() (*dicee.models.octonion.OMult method*), 139

k_vs_all_score() (*dicee.models.OMult method*), 220
k_vs_all_score() (*dicee.models.QMult method*), 213
k_vs_all_score() (*dicee.models.quaternion.QMult method*), 144
k_vs_all_score() (*dicee.models.real.DistMult method*), 146
Keci (*class in dicee.models*), 222
Keci (*class in dicee.models.clifford*), 105
KeciTransformer (*class in dicee.models*), 229
KeciTransformer (*class in dicee.models.clifford*), 109
kernel_size (*dicee.config.Namespace attribute*), 32
kernel_size (*dicee.models.base_model.BaseKGE attribute*), 101
kernel_size (*dicee.models.BaseKGE attribute*), 182, 187, 200, 208, 216, 230, 236
KG (*class in dicee.knowledge_graph*), 73
kg (*dicee.callbacks.PseudoLabellingCallback attribute*), 25
kg (*dicee.read_preprocess_save_load_kg.LoadSaveToDisk attribute*), 254
kg (*dicee.read_preprocess_save_load_kg.PreprocessKG attribute*), 253
kg (*dicee.read_preprocess_save_load_kg.preprocess.PreprocessKG attribute*), 247
kg (*dicee.read_preprocess_save_load_kg.read_from_disk.ReadFromDisk attribute*), 248
kg (*dicee.read_preprocess_save_load_kg.ReadFromDisk attribute*), 254
kg (*dicee.read_preprocess_save_load_kg.save_load_disk.LoadSaveToDisk attribute*), 249
KGE (*class in dicee*), 284
KGE (*class in dicee.knowledge_graph_embeddings*), 76
KGESaveCallback (*class in dicee.callbacks*), 24
knowledge_graph (*dicee.Execute attribute*), 282, 283
knowledge_graph (*dicee.executer.Execute attribute*), 71
KronE (*class in dicee.callbacks*), 26
KvsAll (*class in dicee.dataset_classes*), 38
kvsall_score() (*dicee.models.DualE method*), 243
kvsall_score() (*dicee.models.dualE.DualE method*), 119
KvsSampleDataset (*class in dicee.dataset_classes*), 38

L

label_relaxation_alpha (*dicee.config.Namespace attribute*), 33
label_smoothing_rate (*dicee.config.Namespace attribute*), 31
label_smoothing_rate (*dicee.dataset_classes.AllvsAll attribute*), 37
label_smoothing_rate (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
label_smoothing_rate (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
label_smoothing_rate (*dicee.dataset_classes.KvsAll attribute*), 38
label_smoothing_rate (*dicee.dataset_classes.KvsSampleDataset attribute*), 39
label_smoothing_rate (*dicee.dataset_classes.OnevsSample attribute*), 42
label_smoothing_rate (*dicee.dataset_classes.TriplePredictionDataset attribute*), 42
LabelRelaxationLoss (*class in dicee.losses.custom_losses*), 84
LabelRelaxationLoss (*class in dicee.models*), 173
LabelSmoothingLoss (*class in dicee.losses.custom_losses*), 83
LabelSmoothingLoss (*class in dicee.models*), 173
layer_norm (*dicee.models.literal.LiteralEmbeddings attribute*), 138
layer_norm (*dicee.models.ultra.RelationalConv attribute*), 165
LayerNorm (*class in dicee.models.transformers*), 156
layers (*dicee.models.trix.EntityReasoner attribute*), 162
layers (*dicee.models.ultra.EntityNBFNet attribute*), 166
layers (*dicee.models.ultra.RelNBFNet attribute*), 166
learning_rate (*dicee.models.base_model.BaseKGE attribute*), 101
learning_rate (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 215, 230, 235
length (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
length (*dicee.dataset_classes.TriplePredictionDataset attribute*), 43
level (*dicee.callbacks.Perturb attribute*), 26
LFMult (*class in dicee.models*), 242
LFMult (*class in dicee.models.function_space*), 132
LFMult1 (*class in dicee.models*), 241
LFMult1 (*class in dicee.models.function_space*), 131
linear (*dicee.models.ultra.RelationalConv attribute*), 165
linear() (*dicee.models.function_space.LFMult method*), 132
linear() (*dicee.models.LFMult method*), 242
list2tuple() (*dicee.query_generator.QueryGenerator method*), 246
list2tuple() (*dicee.QueryGenerator method*), 290
LiteralDataset (*class in dicee.dataset_classes*), 40
LiteralEmbeddings (*class in dicee.models.literal*), 137
lm_head (*dicee.models.clifford.KeciTransformer attribute*), 109

lm_head (*dicee.models.KeciTransformer attribute*), 229
lm_head (*dicee.models.transformers.BytE attribute*), 155
lm_head (*dicee.models.transformers.GPT attribute*), 160
ln_1 (*dicee.models.Block attribute*), 187
ln_1 (*dicee.models.clifford.TransformerBlock attribute*), 110
ln_1 (*dicee.models.transformers.Block attribute*), 159
ln_2 (*dicee.models.Block attribute*), 187
ln_2 (*dicee.models.clifford.TransformerBlock attribute*), 110
ln_2 (*dicee.models.transformers.Block attribute*), 159
ln_f (*dicee.models.CoKE attribute*), 198
ln_f (*dicee.models.real.CoKE attribute*), 153
load() (*dicee.read_preprocess_save_load_kg.LoadSaveToDisk method*), 254
load() (*dicee.read_preprocess_save_load_kg.save_load_disk.LoadSaveToDisk method*), 249
load_and_validate_literal_data() (*dicee.dataset_classes.LiteralDataset static method*), 40
load_from_memmap() (*dicee.Execute method*), 283
load_from_memmap() (*dicee.executer.Execute method*), 72
load_graph() (*dicee.models.graph_model.GraphKGE method*), 135
load_json() (*in module dicee.static_funcs*), 260
load_local_shard_checkpoint() (*dicee.models.fsdp_models.FSDPShardedEntityModel method*), 128
load_model() (*in module dicee.static_funcs*), 259
load_model_ensemble() (*in module dicee.static_funcs*), 259
load_numpy() (*in module dicee.static_funcs*), 261
load_numpy_ndarray() (*in module dicee.read_preprocess_save_load_kg.util*), 253
load_pickle() (*in module dicee.read_preprocess_save_load_kg.util*), 253
load_pickle() (*in module dicee.static_funcs*), 259
load_pretrained() (*dicee.models.graph_model.GraphKGE method*), 135
load_queries() (*dicee.query_generator.QueryGenerator method*), 247
load_queries() (*dicee.QueryGenerator method*), 290
load_queries_and_answers() (*dicee.query_generator.QueryGenerator static method*), 247
load_queries_and_answers() (*dicee.QueryGenerator static method*), 290
load_state_dict() (*dicee.models.ensemble.EnsembleKGE method*), 120
load_term_mapping() (*in module dicee.static_funcs*), 259
load_term_mapping() (*in module dicee.trainer.dice_trainer*), 265
load_with_pandas() (*in module dicee.read_preprocess_save_load_kg.util*), 253
loader_backend (*dicee.dataset_classes.LiteralDataset attribute*), 40
LoadSaveToDisk (*class in dicee.read_preprocess_save_load_kg*), 254
LoadSaveToDisk (*class in dicee.read_preprocess_save_load_kg.save_load_disk*), 249
local_rank (*dicee.abstracts.AbstractTrainer attribute*), 13
local_rank (*dicee.Execute attribute*), 282
local_rank (*dicee.executer.Execute attribute*), 71
local_rank (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 270
local_rank (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
local_rows (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 128
log_level (*dicee.config.Namespace attribute*), 31
logger (*in module dicee.abstracts*), 13
logger (*in module dicee.analyse_experiments*), 21
logger (*in module dicee.callbacks*), 23
logger (*in module dicee.evaluation.evaluator*), 49
logger (*in module dicee.evaluation.link_prediction*), 52
logger (*in module dicee.evaluation.literal_prediction*), 56
logger (*in module dicee.executer*), 71
logger (*in module dicee.knowledge_graph_embeddings*), 76
logger (*in module dicee.models*), 177, 234, 239
logger (*in module dicee.models.base_model*), 96
logger (*in module dicee.models.clifford*), 105
logger (*in module dicee.models.function_space*), 129
logger (*in module dicee.models.pykeen_models*), 141
logger (*in module dicee.models.transformers*), 155
logger (*in module dicee.query_generator*), 246
logger (*in module dicee.read_preprocess_save_load_kg.preprocess*), 247
logger (*in module dicee.read_preprocess_save_load_kg.read_from_disk*), 248
logger (*in module dicee.read_preprocess_save_load_kg.save_load_disk*), 249
logger (*in module dicee.read_preprocess_save_load_kg.util*), 250
logger (*in module dicee.sanity_checkers*), 255
logger (*in module dicee.scripts.index_serve*), 255
logger (*in module dicee.static_funcs*), 258
logger (*in module dicee.static_preprocess_funcs*), 264
logger (*in module dicee.trainer.auto_batch_finder*), 264

logger (*in module dicee.trainer.dice_trainer*), 265
logger (*in module dicee.trainer.model_parallelism*), 268
logger (*in module dicee.trainer.torch_trainer*), 269
logger (*in module dicee.trainer.torch_trainer_ddp*), 270
logger (*in module dicee.trainer.torch_trainer_fsdp*), 271
loss (*dicee.models.base_model.BaseKGELightning attribute*), 96
loss (*dicee.models.BaseKGELightning attribute*), 177
loss_fn (*dicee.config.Namespace attribute*), 33
loss_func (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
loss_func (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
loss_function (*dicee.trainer.torch_trainer.TorchTrainer attribute*), 269
loss_function() (*dicee.models.base_model.BaseKGELightning method*), 97
loss_function() (*dicee.models.BaseKGELightning method*), 178
loss_function() (*dicee.models.transformers.BytE method*), 155
loss_history (*dicee.models.base_model.BaseKGE attribute*), 101
loss_history (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236
loss_history (*dicee.models.pykeen_models.PykeenKGE attribute*), 142
loss_history (*dicee.models.PykeenKGE attribute*), 234
loss_history (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
lr (*dicee.analyse_experiments.Experiment attribute*), 21
lr (*dicee.config.Namespace attribute*), 30
lr_init (*dicee.weight_averaging.SWA attribute*), 278
lr_init (*dicee.weight_averaging.SWAG attribute*), 279
lr_init (*dicee.weight_averaging.TWA attribute*), 281
lr_lambda (*dicee.callbacks.LRScheduler attribute*), 28
LRScheduler (*class in dicee.callbacks*), 27

M

m (*dicee.models.function_space.LFMult attribute*), 132
m (*dicee.models.LFMult attribute*), 242
main() (*in module dicee.scripts.index_serve*), 256
main() (*in module dicee.scripts.run*), 256
make_iterable_verbose() (*in module dicee.evaluation.utils*), 57
make_iterable_verbose() (*in module dicee.static_funcs_training*), 263
make_iterable_verbose() (*in module dicee.trainer.torch_trainer_ddp*), 270
mapping_from_first_two_cols_to_third() (*in module dicee.static_preprocess_funcs*), 264
margin (*dicee.config.Namespace attribute*), 32
margin (*dicee.losses.custom_losses.ACLS attribute*), 87
margin (*dicee.models.ACLS attribute*), 170
margin (*dicee.models.complex.RotatE attribute*), 117
margin (*dicee.models.Pyke attribute*), 197
margin (*dicee.models.real.Pyke attribute*), 152
margin (*dicee.models.real.TransE attribute*), 148
margin (*dicee.models.real.TransH attribute*), 149
margin (*dicee.models.RotatE attribute*), 207
margin (*dicee.models.TransE attribute*), 193
margin (*dicee.models.TransH attribute*), 194
mask_emb (*dicee.models.CoKE attribute*), 198
mask_emb (*dicee.models.real.CoKE attribute*), 153
max_alpha (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss attribute*), 84
max_alpha (*dicee.models.AdaptiveLabelRelaxationLoss attribute*), 171
max_ans_num (*dicee.query_generator.QueryGenerator attribute*), 246
max_ans_num (*dicee.QueryGenerator attribute*), 289
max_epochs (*dicee.callbacks.KGESaveCallback attribute*), 24
max_epochs (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
max_epochs (*dicee.weight_averaging.EMA attribute*), 280
max_epochs (*dicee.weight_averaging.SWA attribute*), 278
max_epochs (*dicee.weight_averaging.SWAG attribute*), 279
max_epochs (*dicee.weight_averaging.TWA attribute*), 281
max_length_subword_tokens (*dicee.knowledge_graph.KG attribute*), 75
max_length_subword_tokens (*dicee.models.base_model.BaseKGE attribute*), 101
max_length_subword_tokens (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236
max_num_models (*dicee.weight_averaging.SWAG attribute*), 279
max_num_of_classes (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
max_num_of_classes (*dicee.dataset_classes.KvsSampleDataset attribute*), 39
max_smoothing_factor (*dicee.losses.custom_losses.AdaptiveLabelSmoothingLoss attribute*), 83
max_smoothing_factor (*dicee.models.AdaptiveLabelSmoothingLoss attribute*), 171

maximize (*dicee.weight_averaging.AMWA attribute*), 276
mean (*dicee.weight_averaging.SWAG attribute*), 279
mem_of_model() (*dicee.models.base_model.BaseKGELightning method*), 96
mem_of_model() (*dicee.models.BaseKGELightning method*), 178
mem_of_model() (*dicee.models.ensemble.EnsembleKGE method*), 120
method (*dicee.callbacks.Perturb attribute*), 26
min_alpha (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss attribute*), 84
min_alpha (*dicee.models.AdaptiveLabelRelaxationLoss attribute*), 171
min_smoothing_factor (*dicee.losses.custom_losses.AdaptiveLabelSmoothingLoss attribute*), 83
min_smoothing_factor (*dicee.models.AdaptiveLabelSmoothingLoss attribute*), 171
MLP (*class in dicee.models.transformers*), 158
mlp (*dicee.models.Block attribute*), 187
mlp (*dicee.models.clifford.TransformerBlock attribute*), 110
mlp (*dicee.models.transformers.Block attribute*), 159
mlp (*dicee.models.trix.EntityReasoner attribute*), 162
mlp (*dicee.models.TRIXRelation attribute*), 245
mlp (*dicee.models.trix.TRIXRelation attribute*), 164
mlp (*dicee.models.ultra.EntityNBFNet attribute*), 166
mode (*dicee.query_generator.QueryGenerator attribute*), 246
mode (*dicee.QueryGenerator attribute*), 289
model (*dicee.config.Namespace attribute*), 30
model (*dicee.models.pykeen_models.PykeenKGE attribute*), 142
model (*dicee.models.PykeenKGE attribute*), 234
model (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
model (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
model (*dicee.trainer.torch_trainer.TorchTrainer attribute*), 269
model_kwargs (*dicee.models.pykeen_models.PykeenKGE attribute*), 142
model_kwargs (*dicee.models.PykeenKGE attribute*), 234
model_name (*dicee.analyse_experiments.Experiment attribute*), 21
MODEL_REGISTRY (*in module dicee.static_funcs*), 258
module
    dicee, 12
    dicee.__main__, 12
    dicee.abstracts, 12
    dicee.analyse_experiments, 20
    dicee.callbacks, 22
    dicee.config, 28
    dicee.dataset_classes, 34
    dicee.eval_static_funcs, 43
    dicee.evaluation, 47
    dicee.evaluation.ensemble, 47
    dicee.evaluation.evaluator, 48
    dicee.evaluation.link_prediction, 52
    dicee.evaluation.literal_prediction, 56
    dicee.evaluation.utils, 57
    dicee.evaluator, 67
    dicee.executer, 70
    dicee.knowledge_graph, 73
    dicee.knowledge_graph_embeddings, 76
    dicee.losses, 82
    dicee.losses.custom_losses, 82
    dicee.models, 87
    dicee.models.adopt, 87
    dicee.models.base_model, 96
    dicee.models.clifford, 105
    dicee.models.complex, 114
    dicee.models.dualE, 119
    dicee.models.ensemble, 120
    dicee.models.flock, 121
    dicee.models.flock_walks, 126
    dicee.models.fsdp_models, 127
    dicee.models.function_space, 129
    dicee.models.graph_model, 133
    dicee.models.literal, 137
    dicee.models.octonion, 139
    dicee.models.pykeen_models, 141
    dicee.models.quaternion, 143
    dicee.models.real, 146

dicee.models.sampled_loss, 154
dicee.models.static_funcs, 154
dicee.models.transformers, 154
dicee.models.trix, 161
dicee.models.ultra, 164
dicee.query_generator, 246
dicee.read_preprocess_save_load_kg, 247
dicee.read_preprocess_save_load_kg.preprocess, 247
dicee.read_preprocess_save_load_kg.read_from_disk, 248
dicee.read_preprocess_save_load_kg.save_load_disk, 249
dicee.read_preprocess_save_load_kg.util, 249
dicee.sanity_checkers, 254
dicee.scripts, 255
dicee.scripts.index_serve, 255
dicee.scripts.run, 256
dicee.static_funcs, 256
dicee.static_funcs_training, 261
dicee.static_preprocess_funcs, 263
dicee.trainer, 264
dicee.trainer.auto_batch_finder, 264
dicee.trainer.dice_trainer, 265
dicee.trainer.model_parallelism, 267
dicee.trainer.torch_trainer, 268
dicee.trainer.torch_trainer_ddp, 270
dicee.trainer.torch_trainer_fsdp, 271
dicee.weight_averaging, 274
modules() (*dicee.models.ensemble.EnsembleKGE method*), 120
momentum_history (*dicee.weight_averaging.AMWA attribute*), 276
monitor (*dicee.weight_averaging.AMWA attribute*), 276
moving_average() (*dicee.weight_averaging.SWA static method*), 278
MultiClassClassificationDataset (*class in dicee.dataset_classes*), 35
MultiLabelDataset (*class in dicee.dataset_classes*), 36
MuRE (*class in dicee.models*), 195
MuRE (*class in dicee.models.real*), 150

N

n (*dicee.models.FMult2 attribute*), 241
n (*dicee.models.function_space.FMult2 attribute*), 131
n_embd (*dicee.models.clifford.TransformerSelfAttention attribute*), 110
n_embd (*dicee.models.CoKEConfig attribute*), 198
n_embd (*dicee.models.real.CoKEConfig attribute*), 152, 153
n_embd (*dicee.models.transformers.GPTConfig attribute*), 159
n_embd (*dicee.models.transformers.SelfAttention attribute*), 157
n_epochs_eval_model (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
n_epochs_eval_model (*dicee.config.Namespace attribute*), 33
n_head (*dicee.models.clifford.TransformerSelfAttention attribute*), 110
n_head (*dicee.models.CoKEConfig attribute*), 198
n_head (*dicee.models.real.CoKEConfig attribute*), 152, 153
n_head (*dicee.models.transformers.GPTConfig attribute*), 159
n_head (*dicee.models.transformers.SelfAttention attribute*), 157
n_layer (*dicee.models.CoKEConfig attribute*), 198
n_layer (*dicee.models.real.CoKEConfig attribute*), 152, 153
n_layer (*dicee.models.transformers.GPTConfig attribute*), 159
n_layers (*dicee.models.FMult2 attribute*), 241
n_layers (*dicee.models.function_space.FMult2 attribute*), 131
name (*dicee.abstracts.BaseInteractiveKGE property*), 14
name (*dicee.models.AConEx attribute*), 204
name (*dicee.models.AConvO attribute*), 221
name (*dicee.models.AConvQ attribute*), 214
name (*dicee.models.CKeci attribute*), 226
name (*dicee.models.clifford.CKeci attribute*), 111
name (*dicee.models.clifford.DeCaL attribute*), 111
name (*dicee.models.clifford.Keci attribute*), 106
name (*dicee.models.clifford.KeciTransformer attribute*), 109
name (*dicee.models.CoKE attribute*), 198
name (*dicee.models.ComplEx attribute*), 206
name (*dicee.models.complex.AConEx attribute*), 115

name (*dicee.models.complex.ComplEx attribute*), 116
name (*dicee.models.complex.ConEx attribute*), 114
name (*dicee.models.complex.RotatE attribute*), 117
name (*dicee.models.ConEx attribute*), 203
name (*dicee.models.ConvO attribute*), 221
name (*dicee.models.ConvQ attribute*), 214
name (*dicee.models.DeCaL attribute*), 226
name (*dicee.models.DistMult attribute*), 192
name (*dicee.models.DualE attribute*), 243
name (*dicee.models.dualE.DualE attribute*), 119
name (*dicee.models.ensemble.EnsembleKGE attribute*), 120
name (*dicee.models.Flock attribute*), 245
name (*dicee.models.flock.Flock attribute*), 125
name (*dicee.models.flock.FlockRelation attribute*), 126
name (*dicee.models.FlockRelation attribute*), 245
name (*dicee.models.FMult attribute*), 239
name (*dicee.models.FMult2 attribute*), 241
name (*dicee.models.function_space.FMult attribute*), 129
name (*dicee.models.function_space.FMult2 attribute*), 131
name (*dicee.models.function_space.GFMult attribute*), 130
name (*dicee.models.function_space.LFMult attribute*), 132
name (*dicee.models.function_space.LFMult1 attribute*), 131
name (*dicee.models.GFMult attribute*), 240
name (*dicee.models.graph_model.GraphKGE attribute*), 134
name (*dicee.models.Keci attribute*), 222
name (*dicee.models.KeciTransformer attribute*), 229
name (*dicee.models.LFMult attribute*), 242
name (*dicee.models.LFMult1 attribute*), 241
name (*dicee.models.MuRE attribute*), 195
name (*dicee.models.octonion.AConvO attribute*), 140
name (*dicee.models.octonion.ConvO attribute*), 140
name (*dicee.models.octonion.OMult attribute*), 139
name (*dicee.models.OMult attribute*), 220
name (*dicee.models.Pyke attribute*), 197
name (*dicee.models.pykeen_models.PykeenKGE attribute*), 142
name (*dicee.models.PykeenKGE attribute*), 234
name (*dicee.models.QMult attribute*), 213
name (*dicee.models.quaternion.AConvQ attribute*), 145
name (*dicee.models.quaternion.ConvQ attribute*), 144
name (*dicee.models.quaternion.QMult attribute*), 143
name (*dicee.models.real.CoKE attribute*), 153
name (*dicee.models.real.DistMult attribute*), 146
name (*dicee.models.real.MuRE attribute*), 150
name (*dicee.models.real.Pyke attribute*), 152
name (*dicee.models.real.Shallom attribute*), 151
name (*dicee.models.real.TransE attribute*), 148
name (*dicee.models.real.TransH attribute*), 149
name (*dicee.models.RotatE attribute*), 207
name (*dicee.models.Shallom attribute*), 196
name (*dicee.models.TransE attribute*), 193
name (*dicee.models.transformers.BytE attribute*), 155
name (*dicee.models.TransH attribute*), 194
name (*dicee.models.TRIX attribute*), 245
name (*dicee.models.TRIXRelation attribute*), 245
name (*dicee.models.trix.TRIX attribute*), 163
name (*dicee.models.trix.TRIXRelation attribute*), 164
name (*dicee.models.ULTRA attribute*), 244
name (*dicee.models.ultra.ULTRA attribute*), 167
named_children() (*dicee.models.ensemble.EnsembleKGE method*), 120
names (*dicee.losses.custom_losses.ACLS property*), 87
names (*dicee.models.ACLS property*), 170
Namespace (*class in dicee.config*), 28
neg_lambda (*dicee.losses.custom_losses.ACLS attribute*), 87
neg_lambda (*dicee.models.ACLS attribute*), 170
neg_ratio (*dicee.config.Namespace attribute*), 30
neg_ratio (*dicee.dataset_classes.BPE_NegativeSamplingDataset attribute*), 35
neg_ratio (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
neg_ratio (*dicee.dataset_classes.KvsSampleDataset attribute*), 39

neg_sample_ratio (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
neg_sample_ratio (*dicee.dataset_classes.OnevsSample attribute*), 42
neg_sample_ratio (*dicee.dataset_classes.TriplePredictionDataset attribute*), 42
negnorm() (*dicee.abstracts.InteractiveQueryDecomposition method*), 18
net (*dicee.models.flock.FlockBase attribute*), 125
neural_searcher (*in module dicee.scripts.index_serve*), 256
NeuralSearcher (*class in dicee.scripts.index_serve*), 256
node_init (*dicee.models.flock.FlockBase attribute*), 125
node_logit (*dicee.models.flock.FlockBase attribute*), 125
NodeTrainer (*class in dicee.trainer.torch_trainer_ddp*), 270
norm_fc1 (*dicee.models.AConEx attribute*), 205
norm_fc1 (*dicee.models.AConvO attribute*), 221
norm_fc1 (*dicee.models.complex.AConEx attribute*), 115
norm_fc1 (*dicee.models.complex.ConEx attribute*), 114
norm_fc1 (*dicee.models.ConEx attribute*), 204
norm_fc1 (*dicee.models.ConvO attribute*), 221
norm_fc1 (*dicee.models.octonion.AConvO attribute*), 141
norm_fc1 (*dicee.models.octonion.ConvO attribute*), 140
normalization (*dicee.analyse_experiments.Experiment attribute*), 22
normalization (*dicee.config.Namespace attribute*), 31
normalization_params (*dicee.dataset_classes.LiteralDataset attribute*), 40
normalization_type (*dicee.dataset_classes.LiteralDataset attribute*), 40
normalize_head_entity_embeddings (*dicee.models.base_model.BaseKGE attribute*), 101
normalize_head_entity_embeddings (*dicee.models.BaseKGE attribute*), 182, 188, 200, 208, 216, 230, 236
normalize_relation_embeddings (*dicee.models.base_model.BaseKGE attribute*), 101
normalize_relation_embeddings (*dicee.models.BaseKGE attribute*), 182, 188, 200, 208, 216, 230, 236
normalize_tail_entity_embeddings (*dicee.models.base_model.BaseKGE attribute*), 101
normalize_tail_entity_embeddings (*dicee.models.BaseKGE attribute*), 182, 188, 200, 208, 216, 230, 236
normalizer_class (*dicee.models.base_model.BaseKGE attribute*), 101
normalizer_class (*dicee.models.BaseKGE attribute*), 182, 188, 200, 208, 216, 230, 236
num_bpe_entities (*dicee.dataset_classes.BPE_NegativeSamplingDataset attribute*), 35
num_bpe_entities (*dicee.knowledge_graph.KG attribute*), 75
num_classes (*dicee.losses.custom_losses.ACLS attribute*), 87
num_classes (*dicee.models.ACLS attribute*), 170
num_core (*dicee.config.Namespace attribute*), 31
num_datapoints (*dicee.dataset_classes.BPE_NegativeSamplingDataset attribute*), 35
num_datapoints (*dicee.dataset_classes.MultiLabelDataset attribute*), 36
num_direct_relations (*dicee.models.graph_model.GraphKGE attribute*), 134
num_embeddings (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 127
num_ent (*dicee.models.DualE attribute*), 243
num_ent (*dicee.models.dualE.DualE attribute*), 119
num_entities (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
num_entities (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
num_entities (*dicee.dataset_classes.KvsSampleDataset attribute*), 39
num_entities (*dicee.dataset_classes.LiteralDataset attribute*), 40
num_entities (*dicee.dataset_classes.OnevsSample attribute*), 42
num_entities (*dicee.dataset_classes.TriplePredictionDataset attribute*), 43
num_entities (*dicee.evaluation.Evaluator attribute*), 60, 61
num_entities (*dicee.evaluation.evaluator.Evaluator attribute*), 49, 50
num_entities (*dicee.Evaluator attribute*), 292, 293
num_entities (*dicee.evaluator.Evaluator attribute*), 67, 68
num_entities (*dicee.knowledge_graph.KG attribute*), 74
num_entities (*dicee.models.base_model.BaseKGE attribute*), 101
num_entities (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 215, 230, 235
num_epochs (*dicee.abstracts.AbstractPPECallback attribute*), 19
num_epochs (*dicee.analyse_experiments.Experiment attribute*), 21
num_epochs (*dicee.config.Namespace attribute*), 30
num_epochs (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
num_epochs (*dicee.weight_averaging.ASWA attribute*), 277
num_folds_for_cv (*dicee.config.Namespace attribute*), 31
num_layers (*dicee.models.ULTRA attribute*), 244
num_layers (*dicee.models.ultra.ULTRA attribute*), 167
num_negatives (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
num_nodes (*dicee.models.flock_walks.WalkGraph attribute*), 126
num_of_data_points (*dicee.dataset_classes.MultiClassClassificationDataset attribute*), 36
num_of_data_properties (*dicee.models.literal.LiteralEmbeddings attribute*), 137, 138
num_of_epochs (*dicee.callbacks.PseudoLabellingCallback attribute*), 25
num_of_output_channels (*dicee.config.Namespace attribute*), 32

num_of_output_channels (*dicee.models.base_model.BaseKGE attribute*), 101
num_of_output_channels (*dicee.models.BaseKGE attribute*), 182, 188, 200, 208, 216, 230, 236
num_params (*dicee.analyse_experiments.Experiment attribute*), 21
num_relations (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
num_relations (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
num_relations (*dicee.dataset_classes.OnevsSample attribute*), 42
num_relations (*dicee.dataset_classes.TriplePredictionDataset attribute*), 43
num_relations (*dicee.evaluation.Evaluator attribute*), 60, 61
num_relations (*dicee.evaluation.evaluator.Evaluator attribute*), 49, 50
num_relations (*dicee.Evaluator attribute*), 292, 293
num_relations (*dicee.evaluator.Evaluator attribute*), 67, 68
num_relations (*dicee.knowledge_graph.KG attribute*), 74
num_relations (*dicee.models.base_model.BaseKGE attribute*), 101
num_relations (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 215, 230, 235
num_sample (*dicee.models.FMult attribute*), 240
num_sample (*dicee.models.function_space.FMult attribute*), 129
num_sample (*dicee.models.function_space.GFMult attribute*), 130
num_sample (*dicee.models.GFMult attribute*), 240
num_samples (*dicee.weight_averaging.TWA attribute*), 280
num_tokens (*dicee.knowledge_graph.KG attribute*), 75
num_tokens (*dicee.models.base_model.BaseKGE attribute*), 101
num_tokens (*dicee.models.BaseKGE attribute*), 182, 187, 199, 208, 215, 230, 235
num_types (*dicee.models.flock_walks.WalkGraph attribute*), 126
num_workers (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
numpy_data_type_changer() (*in module dicee.static_funcs*), 259

O

octonion_mul() (*in module dicee.models*), 220
octonion_mul() (*in module dicee.models.octonion*), 139
octonion_mul_norm() (*in module dicee.models*), 220
octonion_mul_norm() (*in module dicee.models.octonion*), 139
octonion_normalizer() (*dicee.models.AConvO static method*), 222
octonion_normalizer() (*dicee.models.ConvO static method*), 221
octonion_normalizer() (*dicee.models.octonion.AConvO static method*), 141
octonion_normalizer() (*dicee.models.octonion.ConvO static method*), 140
octonion_normalizer() (*dicee.models.octonion.OMult static method*), 139
octonion_normalizer() (*dicee.models.OMult static method*), 220
OMult (*class in dicee.models*), 220
OMult (*class in dicee.models.octonion*), 139
on_epoch_end() (*dicee.callbacks.KGESaveCallback method*), 24
on_epoch_end() (*dicee.callbacks.PseudoLabellingCallback method*), 25
on_fit_end() (*dicee.abstracts.AbstractCallback method*), 19
on_fit_end() (*dicee.abstracts.AbstractPPECallback method*), 19
on_fit_end() (*dicee.abstracts.AbstractTrainer method*), 13
on_fit_end() (*dicee.callbacks.AccumulateEpochLossCallback method*), 23
on_fit_end() (*dicee.callbacks.Eval method*), 25
on_fit_end() (*dicee.callbacks.KGESaveCallback method*), 24
on_fit_end() (*dicee.callbacks.LRScheduler method*), 28
on_fit_end() (*dicee.callbacks.PeriodicEvalCallback method*), 27
on_fit_end() (*dicee.callbacks.PrintCallback method*), 23
on_fit_end() (*dicee.weight_averaging.AMWA method*), 276
on_fit_end() (*dicee.weight_averaging.ASWA method*), 277
on_fit_end() (*dicee.weight_averaging.EMA method*), 280
on_fit_end() (*dicee.weight_averaging.SWA method*), 278
on_fit_end() (*dicee.weight_averaging.SWAG method*), 279
on_fit_end() (*dicee.weight_averaging.TWA method*), 281
on_fit_start() (*dicee.abstracts.AbstractCallback method*), 18
on_fit_start() (*dicee.abstracts.AbstractPPECallback method*), 19
on_fit_start() (*dicee.abstracts.AbstractTrainer method*), 13
on_fit_start() (*dicee.callbacks.Eval method*), 25
on_fit_start() (*dicee.callbacks.KGESaveCallback method*), 24
on_fit_start() (*dicee.callbacks.KronE method*), 26
on_fit_start() (*dicee.callbacks.PrintCallback method*), 23
on_init_end() (*dicee.abstracts.AbstractCallback method*), 18
on_init_start() (*dicee.abstracts.AbstractCallback method*), 18
on_train_batch_end() (*dicee.abstracts.AbstractCallback method*), 19
on_train_batch_end() (*dicee.abstracts.AbstractTrainer method*), 13

on_train_batch_end() (*dicee.callbacks.Eval method*), 26
on_train_batch_end() (*dicee.callbacks.KGESaveCallback method*), 24
on_train_batch_end() (*dicee.callbacks.LRScheduler method*), 28
on_train_batch_end() (*dicee.callbacks.PrintCallback method*), 23
on_train_batch_start() (*dicee.callbacks.Perturb method*), 27
on_train_epoch_end() (*dicee.abstracts.AbstractCallback method*), 19
on_train_epoch_end() (*dicee.abstracts.AbstractTrainer method*), 13
on_train_epoch_end() (*dicee.callbacks.Eval method*), 25
on_train_epoch_end() (*dicee.callbacks.KGESaveCallback method*), 24
on_train_epoch_end() (*dicee.callbacks.PeriodicEvalCallback method*), 27
on_train_epoch_end() (*dicee.callbacks.PrintCallback method*), 23
on_train_epoch_end() (*dicee.models.base_model.BaseKGELightning method*), 97
on_train_epoch_end() (*dicee.models.BaseKGELightning method*), 179
on_train_epoch_end() (*dicee.weight_averaging.AMWA method*), 276
on_train_epoch_end() (*dicee.weight_averaging.ASWA method*), 277
on_train_epoch_end() (*dicee.weight_averaging.EMA method*), 280
on_train_epoch_end() (*dicee.weight_averaging.SWA method*), 278
on_train_epoch_end() (*dicee.weight_averaging.SWAG method*), 279
on_train_epoch_end() (*dicee.weight_averaging.TWA method*), 281
on_train_epoch_start() (*dicee.abstracts.AbstractTrainer method*), 13
on_train_epoch_start() (*dicee.weight_averaging.AMWA method*), 276
on_train_epoch_start() (*dicee.weight_averaging.EMA method*), 280
on_train_epoch_start() (*dicee.weight_averaging.SWA method*), 278
on_train_epoch_start() (*dicee.weight_averaging.SWAG method*), 279
on_train_epoch_start() (*dicee.weight_averaging.TWA method*), 281
on_train_start() (*dicee.callbacks.LRScheduler method*), 28
OnevsAllDataset (*class in dicee.dataset_classes*), 39
OnevsSample (*class in dicee.dataset_classes*), 41
optim (*dicee.config.Namespace attribute*), 30
OptimizedModule (*in module dicee.trainer.torch_trainer_fsdp*), 271
optimizer (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 270
optimizer (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
optimizer (*dicee.trainer.torch_trainer.TorchTrainer attribute*), 269
optimizer_name (*dicee.models.base_model.BaseKGE attribute*), 101
optimizer_name (*dicee.models.BaseKGE attribute*), 182, 187, 200, 208, 216, 230, 235
ordered_bpe_entities (*dicee.dataset_classes.BPE_NegativeSamplingDataset attribute*), 35
ordered_bpe_entities (*dicee.knowledge_graph.KG attribute*), 75
ordered_shaped_bpe_tokens (*dicee.knowledge_graph.KG attribute*), 74

P

p (*dicee.config.Namespace attribute*), 32
p (*dicee.models.clifford.DeCaL attribute*), 111
p (*dicee.models.clifford.Keci attribute*), 106
p (*dicee.models.DeCaL attribute*), 226
p (*dicee.models.Keci attribute*), 222
P (*dicee.weight_averaging.TWA attribute*), 281
pack_walks (*dicee.models.flock.FlockBase attribute*), 125
padding (*dicee.knowledge_graph.KG attribute*), 75
pandas_dataframe_indexer() (*in module dicee.read_preprocess_save_load_kg.util*), 251
param_init (*dicee.models.base_model.BaseKGE attribute*), 101
param_init (*dicee.models.BaseKGE attribute*), 182, 188, 200, 209, 216, 230, 236
parameters() (*dicee.abstracts.BaseInteractiveKGE method*), 17
parameters() (*dicee.models.ensemble.EnsembleKGE method*), 120
parse_types() (*dicee.models.flock_walks.WalkGraph method*), 126
path (*dicee.abstracts.AbstractPPECallback attribute*), 19
path (*dicee.callbacks.AccumulateEpochLossCallback attribute*), 23
path (*dicee.callbacks.Eval attribute*), 25
path (*dicee.callbacks.KGESaveCallback attribute*), 24
path (*dicee.weight_averaging.AMWA attribute*), 276
path (*dicee.weight_averaging.ASWA attribute*), 277
path_dataset_folder (*dicee.analyse_experiments.Experiment attribute*), 21
path_for_deserialization (*dicee.knowledge_graph.KG attribute*), 74
path_for_serialization (*dicee.knowledge_graph.KG attribute*), 74
path_single_kg (*dicee.config.Namespace attribute*), 28
path_single_kg (*dicee.knowledge_graph.KG attribute*), 74
path_to_store_single_run (*dicee.config.Namespace attribute*), 28
PeriodicEvalCallback (*class in dicee.callbacks*), 27

Perturb (*class in dicee.callbacks*), 26
pl_trainer_kwargs (*dicee.config.Namespace attribute*), 32
polars_dataframe_indexer() (*in module dicee.read_preprocess_save_load_kg.util*), 250
poly_NN() (*dicee.models.function_space.LFMult method*), 132
poly_NN() (*dicee.models.LFMult method*), 242
polynomial() (*dicee.models.function_space.LFMult method*), 133
polynomial() (*dicee.models.LFMult method*), 243
pop() (*dicee.models.function_space.LFMult method*), 133
pop() (*dicee.models.LFMult method*), 243
pos_emb (*dicee.models.CoKE attribute*), 198
pos_emb (*dicee.models.real.CoKE attribute*), 153
pos_lambda (*dicee.losses.custom_losses.ACLS attribute*), 87
pos_lambda (*dicee.models.ACLS attribute*), 170
pq (*dicee.analyse_experiments.Experiment attribute*), 21
predict() (*dicee.KGE method*), 285
predict() (*dicee.knowledge_graph_embeddings.KGE method*), 78
predict_dataloader() (*dicee.models.base_model.BaseKGELightning method*), 99
predict_dataloader() (*dicee.models.BaseKGELightning method*), 180
predict_literals() (*dicee.KGE method*), 289
predict_literals() (*dicee.knowledge_graph_embeddings.KGE method*), 81
predict_missing_head_entity() (*dicee.KGE method*), 284
predict_missing_head_entity() (*dicee.knowledge_graph_embeddings.KGE method*), 76
predict_missing_relations() (*dicee.KGE method*), 285
predict_missing_relations() (*dicee.knowledge_graph_embeddings.KGE method*), 77
predict_missing_tail_entity() (*dicee.KGE method*), 285
predict_missing_tail_entity() (*dicee.knowledge_graph_embeddings.KGE method*), 77
predict_topk() (*dicee.KGE method*), 286
predict_topk() (*dicee.knowledge_graph_embeddings.KGE method*), 78
prefetch_factor (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
prefetch_walks (*dicee.models.flock.FlockBase attribute*), 125
prepare_graph_model() (*dicee.DICE_Trainer method*), 291
prepare_graph_model() (*dicee.trainer.DICE_Trainer method*), 274
prepare_graph_model() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 267
prepare_ultra (*dicee.DICE_Trainer attribute*), 291
prepare_ultra (*dicee.trainer.DICE_Trainer attribute*), 274
prepare_ultra (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 267
preprocess_with_byte_pair_encoding() (*dicee.read_preprocess_save_load_kg.PreprocessKG method*), 253
preprocess_with_byte_pair_encoding() (*dicee.read_preprocess_save_load_kg.preprocess.PreprocessKG method*), 248
preprocess_with_byte_pair_encoding_with_padding() (*dicee.read_preprocess_save_load_kg.PreprocessKG method*), 253
preprocess_with_byte_pair_encoding_with_padding() (*dicee.read_preprocess_save_load_kg.preprocess.PreprocessKG method*), 248
preprocess_with_pandas() (*dicee.read_preprocess_save_load_kg.PreprocessKG method*), 253
preprocess_with_pandas() (*dicee.read_preprocess_save_load_kg.preprocess.PreprocessKG method*), 248
preprocess_with_polars() (*dicee.read_preprocess_save_load_kg.PreprocessKG method*), 254
preprocess_with_polars() (*dicee.read_preprocess_save_load_kg.preprocess.PreprocessKG method*), 248
preprocesses_input_args() (*in module dicee.static_preprocess_funcs*), 264
PreprocessKG (*class in dicee.read_preprocess_save_load_kg*), 253
PreprocessKG (*class in dicee.read_preprocess_save_load_kg.preprocess*), 247
prev_loss (*dicee.losses.custom_losses.AdaptiveLabelRelaxationLoss attribute*), 85
prev_loss (*dicee.losses.custom_losses.AdaptiveLabelSmoothingLoss attribute*), 84
prev_loss (*dicee.models.AdaptiveLabelRelaxationLoss attribute*), 171
prev_loss (*dicee.models.AdaptiveLabelSmoothingLoss attribute*), 171
PrintCallback (*class in dicee.callbacks*), 23
process (*dicee.trainer.torch_trainer.TorchTrainer attribute*), 269
projection_cache_mb (*dicee.models.ULTRA attribute*), 245
projection_cache_mb (*dicee.models.ultra.ULTRA attribute*), 167
PseudoLabellingCallback (*class in dicee.callbacks*), 24
ptdtype (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
Pyke (*class in dicee.models*), 197
Pyke (*class in dicee.models.real*), 152
pykeen_model_kwargs (*dicee.config.Namespace attribute*), 32
PykeenKGE (*class in dicee.models*), 234
PykeenKGE (*class in dicee.models.pykeen_models*), 141

Q

q (*dicee.config.Namespace attribute*), 32
q (*dicee.models.clifford.DeCaL attribute*), 111
q (*dicee.models.clifford.Keci attribute*), 106

q (*dicee.models.DeCaL attribute*), 226
q (*dicee.models.Keci attribute*), 222
qdrant_client (*dicee.scripts.index_serve.NeuralSearcher attribute*), 256
QMult (*class in dicee.models*), 212
QMult (*class in dicee.models.quaternion*), 143
quaternion_mul() (*in module dicee.models*), 212
quaternion_mul() (*in module dicee.models.static_funcs*), 154
quaternion_mul_with_unit_norm() (*in module dicee.models*), 212
quaternion_mul_with_unit_norm() (*in module dicee.models.quaternion*), 143
quaternion_multiplication_followed_by_inner_product() (*dicee.models.QMult method*), 213
quaternion_multiplication_followed_by_inner_product() (*dicee.models.quaternion.QMult method*), 143
quaternion_normalizer() (*dicee.models.QMult static method*), 213
quaternion_normalizer() (*dicee.models.quaternion.QMult static method*), 143
queries (*dicee.scripts.index_serve.StringListRequest attribute*), 256
query_batch_size (*dicee.models.graph_model.GraphKGE attribute*), 134
query_name_to_struct (*dicee.query_generator.QueryGenerator attribute*), 246
query_name_to_struct (*dicee.QueryGenerator attribute*), 290
QueryGenerator (*class in dicee*), 289
QueryGenerator (*class in dicee.query_generator*), 246

R

r (*dicee.models.clifford.DeCaL attribute*), 111
r (*dicee.models.clifford.Keci attribute*), 106
r (*dicee.models.DeCaL attribute*), 226
r (*dicee.models.Keci attribute*), 222
random_seed (*dicee.config.Namespace attribute*), 31
range_constraints_per_rel (*dicee.evaluation.Evaluator attribute*), 61
range_constraints_per_rel (*dicee.evaluation.evaluator.Evaluator attribute*), 50
range_constraints_per_rel (*dicee.Evaluator attribute*), 293
range_constraints_per_rel (*dicee.evaluator.Evaluator attribute*), 68
rank (*dicee.Execute attribute*), 282
rank (*dicee.executer.Execute attribute*), 71
rank (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 127
ratio (*dicee.callbacks.Perturb attribute*), 26
raw_model (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
raw_test_set (*dicee.knowledge_graph.KG attribute*), 75
raw_train_set (*dicee.knowledge_graph.KG attribute*), 75
raw_valid_set (*dicee.knowledge_graph.KG attribute*), 75
re (*dicee.models.clifford.DeCaL attribute*), 111
re (*dicee.models.DeCaL attribute*), 226
re_vocab (*dicee.evaluation.Evaluator attribute*), 60
re_vocab (*dicee.evaluation.evaluator.Evaluator attribute*), 49
re_vocab (*dicee.Evaluator attribute*), 292
re_vocab (*dicee.evaluator.Evaluator attribute*), 67, 68
read_from_disk() (*in module dicee.read_preprocess_save_load_kg.util*), 252
read_from_triple_store_with_pandas() (*in module dicee.read_preprocess_save_load_kg.util*), 252
read_from_triple_store_with_polars() (*in module dicee.read_preprocess_save_load_kg.util*), 252
read_only_few (*dicee.config.Namespace attribute*), 31
read_only_few (*dicee.knowledge_graph.KG attribute*), 74
read_or_load_kg() (*in module dicee.static_funcs*), 260
read_with_pandas() (*in module dicee.read_preprocess_save_load_kg.util*), 252
read_with_polars() (*in module dicee.read_preprocess_save_load_kg.util*), 252
ReadFromDisk (*class in dicee.read_preprocess_save_load_kg*), 254
ReadFromDisk (*class in dicee.read_preprocess_save_load_kg.read_from_disk*), 248
reducer (*dicee.scripts.index_serve.StringListRequest attribute*), 256
refinements (*dicee.models.flock.FlockBase attribute*), 125
reg_lambda (*dicee.weight_averaging.TWA attribute*), 281
rel2id (*dicee.query_generator.QueryGenerator attribute*), 246
rel2id (*dicee.QueryGenerator attribute*), 289
relation_cache_mb (*dicee.models.TRIX attribute*), 245
relation_cache_mb (*dicee.models.trix.TRIX attribute*), 164
relation_cache_mb (*dicee.models.ULTRA attribute*), 244
relation_cache_mb (*dicee.models.ultra.ULTRA attribute*), 167
relation_embeddings (*dicee.models.AConvQ attribute*), 214
relation_embeddings (*dicee.models.complex.RotatE attribute*), 118
relation_embeddings (*dicee.models.ConvQ attribute*), 214
relation_embeddings (*dicee.models.DualE attribute*), 243

relation_embeddings (*dicee.models.dualE.DualE attribute*), 119
relation_embeddings (*dicee.models.FMult attribute*), 240
relation_embeddings (*dicee.models.FMult2 attribute*), 241
relation_embeddings (*dicee.models.function_space.FMult attribute*), 129
relation_embeddings (*dicee.models.function_space.FMult2 attribute*), 131
relation_embeddings (*dicee.models.function_space.GFMult attribute*), 130
relation_embeddings (*dicee.models.function_space.LFMult attribute*), 132
relation_embeddings (*dicee.models.function_space.LFMult1 attribute*), 131
relation_embeddings (*dicee.models.GFMult attribute*), 240
relation_embeddings (*dicee.models.LFMult attribute*), 242
relation_embeddings (*dicee.models.LFMult1 attribute*), 241
relation_embeddings (*dicee.models.pykeen_models.PykeenKGE attribute*), 142
relation_embeddings (*dicee.models.PykeenKGE attribute*), 234
relation_embeddings (*dicee.models.quaternion.AConvQ attribute*), 145
relation_embeddings (*dicee.models.quaternion.ConvQ attribute*), 144
relation_embeddings (*dicee.models.RotatE attribute*), 207
relation_graph (*dicee.models.trix.TRIXBase property*), 163
relation_model (*dicee.models.TRIX attribute*), 245
relation_model (*dicee.models.TRIXRelation attribute*), 245
relation_model (*dicee.models.trix.TRIX attribute*), 163
relation_model (*dicee.models.trix.TRIXRelation attribute*), 164
relation_model (*dicee.models.ULTRA attribute*), 244
relation_model (*dicee.models.ultra.ULTRA attribute*), 167
relation_normal_translations (*dicee.models.real.TransH attribute*), 149
relation_normal_translations (*dicee.models.TransH attribute*), 194
relation_prediction (*dicee.models.flock.FlockBase attribute*), 124
relation_prediction (*dicee.models.flock.FlockRelation attribute*), 126
relation_prediction (*dicee.models.FlockRelation attribute*), 245
relation_to_idx (*dicee.knowledge_graph.KG attribute*), 74
relation_translations (*dicee.models.MuRE attribute*), 195
relation_translations (*dicee.models.real.MuRE attribute*), 150
RelationalConv (*class in dicee.models.ultra*), 165
RelationGraphKGE (*class in dicee.models.graph_model*), 136
RelationReasoner (*class in dicee.models.trix*), 162
relations_str (*dicee.knowledge_graph.KG property*), 75
RelNBFNet (*class in dicee.models.ultra*), 165
reload_dataset() (*in module dicee.dataset_classes*), 37
report (*dicee.DICE_Trainer attribute*), 291
report (*dicee.evaluation.Evaluator attribute*), 60, 61
report (*dicee.evaluation.evaluator.Evaluator attribute*), 49, 50
report (*dicee.Evaluator attribute*), 292, 293
report (*dicee.evaluator.Evaluator attribute*), 67, 68
report (*dicee.Execute attribute*), 282, 283
report (*dicee.executer.Execute attribute*), 71, 72
report (*dicee.trainer.DICE_Trainer attribute*), 273
report (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 266
reports (*dicee.callbacks.Eval attribute*), 25
reports (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
requires_grad_for_interactions (*dicee.models.CKeci attribute*), 226
requires_grad_for_interactions (*dicee.models.clifford.CKeci attribute*), 111
requires_grad_for_interactions (*dicee.models.clifford.Keci attribute*), 106
requires_grad_for_interactions (*dicee.models.Keci attribute*), 222
resid_dropout (*dicee.models.clifford.TransformerSelfAttention attribute*), 110
resid_dropout (*dicee.models.transformers.SelfAttention attribute*), 157
residual_convolution() (*dicee.models.AConEx method*), 205
residual_convolution() (*dicee.models.AConvO method*), 222
residual_convolution() (*dicee.models.AConvQ method*), 215
residual_convolution() (*dicee.models.complex.AConEx method*), 115
residual_convolution() (*dicee.models.complex.ConEx method*), 114
residual_convolution() (*dicee.models.ConEx method*), 204
residual_convolution() (*dicee.models.ConvO method*), 221
residual_convolution() (*dicee.models.ConvQ method*), 214
residual_convolution() (*dicee.models.octonion.AConvO method*), 141
residual_convolution() (*dicee.models.octonion.ConvO method*), 140
residual_convolution() (*dicee.models.quaternion.AConvQ method*), 145
residual_convolution() (*dicee.models.quaternion.ConvQ method*), 145
retrieve_embedding() (*dicee.scripts.index_serve.NeuralSearcher method*), 256
retrieve_embeddings() (*in module dicee.scripts.index_serve*), 256

return_multi_hop_query_results() (*dicee.KGE method*), 287
return_multi_hop_query_results() (*dicee.knowledge_graph_embeddings.KGE method*), 79
RMSNorm (*class in dicee.models.flock*), 121
root() (*in module dicee.scripts.index_serve*), 256
roots (*dicee.models.FMult attribute*), 240
roots (*dicee.models.function_space.FMult attribute*), 130
roots (*dicee.models.function_space.GFMult attribute*), 130
roots (*dicee.models.GFMult attribute*), 240
RotatE (*class in dicee.models*), 206
RotatE (*class in dicee.models.complex*), 117
RowWiseShardedEmbedding (*class in dicee.models.fsdp_models*), 127
runtime (*dicee.analyse_experiments.Experiment attribute*), 22

S

sample() (*dicee.models.flock_walks.WalkGraph method*), 126
sample() (*dicee.weight_averaging.SWAG method*), 279
sample_counter (*dicee.abstracts.AbstractPPECallback attribute*), 19
sample_entity() (*dicee.abstracts.BaseInteractiveKGE method*), 15
sample_relation() (*dicee.abstracts.BaseInteractiveKGE method*), 15
sample_triples_ratio (*dicee.config.Namespace attribute*), 31
sample_triples_ratio (*dicee.knowledge_graph.KG attribute*), 74
sample_walks() (*dicee.models.flock.FlockBase method*), 125
sample_weights() (*dicee.weight_averaging.TWA method*), 281
sampling_ratio (*dicee.dataset_classes.LiteralDataset attribute*), 40
sanity_check_callback_args() (*in module dicee.sanity_checkers*), 255
sanity_checking_with_arguments() (*in module dicee.sanity_checkers*), 255
save() (*dicee.abstracts.BaseInteractiveKGE method*), 15
save() (*dicee.read_preprocess_save_load_kg.LoadSaveToDisk method*), 254
save() (*dicee.read_preprocess_save_load_kg.save_load_disk.LoadSaveToDisk method*), 249
save_checkpoint() (*dicee.abstracts.AbstractTrainer static method*), 13
save_checkpoint_model() (*in module dicee.static_funcs*), 259
save_embeddings() (*in module dicee.static_funcs*), 260
save_embeddings_as_csv (*dicee.config.Namespace attribute*), 28
save_every_n_epochs (*dicee.config.Namespace attribute*), 33
save_experiment() (*dicee.analyse_experiments.Experiment method*), 22
save_graph() (*dicee.models.graph_model.GraphKGE method*), 135
save_local_shard_checkpoint() (*dicee.models.fsdp_models.FSDPShardedEntityModel method*), 128
save_model_at_every_epoch (*dicee.config.Namespace attribute*), 31
save_model_every_n_epoch (*dicee.callbacks.PeriodicEvalCallback attribute*), 27
save_numpy_ndarray() (*in module dicee.read_preprocess_save_load_kg.util*), 253
save_numpy_ndarray() (*in module dicee.static_funcs*), 259
save_pickle() (*in module dicee.read_preprocess_save_load_kg.util*), 253
save_pickle() (*in module dicee.static_funcs*), 259
save_queries() (*dicee.query_generator.QueryGenerator method*), 247
save_queries() (*dicee.QueryGenerator method*), 290
save_queries_and_answers() (*dicee.query_generator.QueryGenerator static method*), 247
save_queries_and_answers() (*dicee.QueryGenerator static method*), 290
save_trained_model() (*dicee.Execute method*), 283
save_trained_model() (*dicee.executer.Execute method*), 72
scalar_batch_NN() (*dicee.models.function_space.LFMult method*), 132
scalar_batch_NN() (*dicee.models.LFMult method*), 242
scaler (*dicee.callbacks.Perturb attribute*), 27
scaler (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
scaler (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
scheduler (*dicee.callbacks.LRScheduler attribute*), 28
score() (*dicee.models.clifford.Keci method*), 109
score() (*dicee.models.CoKE method*), 199
score() (*dicee.models.ComplEx static method*), 206
score() (*dicee.models.complex.ComplEx static method*), 117
score() (*dicee.models.complex.RotatE method*), 118
score() (*dicee.models.DistMult method*), 192
score() (*dicee.models.Keci method*), 225
score() (*dicee.models.octonion.OMult method*), 139
score() (*dicee.models.OMult method*), 220
score() (*dicee.models.QMult method*), 213
score() (*dicee.models.quaternion.QMult method*), 144
score() (*dicee.models.real.CoKE method*), 153

score() (*dicee.models.real.DistMult method*), 147
score() (*dicee.models.real.TransE method*), 148
score() (*dicee.models.RotatE method*), 207
score() (*dicee.models.TransE method*), 193
score_func (*dicee.models.FMult2 attribute*), 241
score_func (*dicee.models.function_space.FMult2 attribute*), 131
score_walks() (*dicee.models.flock.FlockBase method*), 125
scoring_technique (*dicee.analyse_experiments.Experiment attribute*), 22
scoring_technique (*dicee.config.Namespace attribute*), 30
search() (*dicee.scripts.index_serve.NeuralSearcher method*), 256
search_embeddings() (*in module dicee.scripts.index_serve*), 256
search_embeddings_batch() (*in module dicee.scripts.index_serve*), 256
seed (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
seed (*dicee.dataset_classes.TriplePredictionDataset attribute*), 43
seed (*dicee.models.flock.FlockBase attribute*), 125
seed (*dicee.query_generator.QueryGenerator attribute*), 246
seed (*dicee.QueryGenerator attribute*), 289
select_model() (*in module dicee.static_funcs*), 259
selected_optimizer (*dicee.models.base_model.BaseKGE attribute*), 101
selected_optimizer (*dicee.models.BaseKGE attribute*), 182, 188, 200, 208, 216, 230, 236
SelfAttention (*class in dicee.models.transformers*), 157
separator (*dicee.config.Namespace attribute*), 30
separator (*dicee.knowledge_graph.KG attribute*), 75
seq_len (*dicee.models.clifford.KeciTransformer attribute*), 109
seq_len (*dicee.models.KeciTransformer attribute*), 229
sequential_vocabulary_construction() (*dicee.read_preprocess_save_load_kg.PreprocessKG method*), 254
sequential_vocabulary_construction() (*dicee.read_preprocess_save_load_kg.preprocess.PreprocessKG method*), 248
serve() (*in module dicee.scripts.index_serve*), 256
set_global_seed() (*dicee.query_generator.QueryGenerator method*), 246
set_global_seed() (*dicee.QueryGenerator method*), 290
set_graph() (*dicee.models.graph_model.GraphKGE method*), 135
set_inference_backend() (*dicee.models.graph_model.GraphKGE method*), 134
set_model_eval_mode() (*dicee.abstracts.BaseInteractiveKGE method*), 14
set_model_train_mode() (*dicee.abstracts.BaseInteractiveKGE method*), 14
setup_distributed_training() (*in module dicee.static_funcs*), 258
setup_executor() (*dicee.Execute method*), 283
setup_executor() (*dicee.executer.Execute method*), 72
setup_fsdp_training() (*dicee.models.fsdp_models.FSDPShardedEntityModel method*), 128
Shallom (*class in dicee.models*), 196
Shallom (*class in dicee.models.real*), 151
shallom (*dicee.models.real.Shallom attribute*), 151
shallom (*dicee.models.Shallom attribute*), 196
shard_size (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 127
sharding_strategy (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
single_hop_query_answering() (*dicee.KGE method*), 287
single_hop_query_answering() (*dicee.knowledge_graph_embeddings.KGE method*), 79
smoothing_factor (*dicee.losses.custom_losses.AdaptiveLabelSmoothingLoss attribute*), 84
smoothing_factor (*dicee.models.AdaptiveLabelSmoothingLoss attribute*), 171
smoothing_factor_step (*dicee.losses.custom_losses.AdaptiveLabelSmoothingLoss attribute*), 84
smoothing_factor_step (*dicee.models.AdaptiveLabelSmoothingLoss attribute*), 171
smoothness_ratio (*dicee.losses.custom_losses.AggregatedLSandLR attribute*), 86
smoothness_ratio (*dicee.losses.custom_losses.CombinedLSandLR attribute*), 85
smoothness_ratio (*dicee.losses.custom_losses.LabelSmoothingLoss attribute*), 83
smoothness_ratio (*dicee.models.AggregatedLSandLR attribute*), 172
smoothness_ratio (*dicee.models.CombinedLSandLR attribute*), 172
smoothness_ratio (*dicee.models.LabelSmoothingLoss attribute*), 174
snapshot_dir (*dicee.callbacks.LRScheduler attribute*), 27
snapshot_loss (*dicee.callbacks.LRScheduler attribute*), 28
sparql_endpoint (*dicee.config.Namespace attribute*), 29
sparql_endpoint (*dicee.knowledge_graph.KG attribute*), 74
sq_mean (*dicee.weight_averaging.SWAG attribute*), 279
stable_history (*dicee.weight_averaging.AMWA attribute*), 276
stable_state_dict (*dicee.weight_averaging.AMWA attribute*), 276
start() (*dicee.DICE_Trainer method*), 291
start() (*dicee.Execute method*), 284
start() (*dicee.executer.Execute method*), 72
start() (*dicee.read_preprocess_save_load_kg.PreprocessKG method*), 253
start() (*dicee.read_preprocess_save_load_kg.preprocess.PreprocessKG method*), 248

start() (*dicee.read_preprocess_save_load_kg.read_from_disk.ReadFromDisk method*), 248
start() (*dicee.read_preprocess_save_load_kg.ReadFromDisk method*), 254
start() (*dicee.trainer.DICE_Trainer method*), 274
start() (*dicee.trainer.dice_trainer.DICE_Trainer method*), 267
start_row (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 127
start_time (*dicee.callbacks.PrintCallback attribute*), 23
start_time (*dicee.Execute attribute*), 283
start_time (*dicee.executer.Execute attribute*), 72
state_dict() (*dicee.models.ensemble.EnsembleKGE method*), 120
step() (*dicee.models.ADOPT method*), 176
step() (*dicee.models.adopt.ADOPT method*), 91
step() (*dicee.models.ensemble.EnsembleKGE method*), 120
step() (*dicee.models.trix.RelationReasoner method*), 163
step_count (*dicee.callbacks.LRScheduler attribute*), 28
storage_path (*dicee.config.Namespace attribute*), 28
storage_path (*dicee.DICE_Trainer attribute*), 291
storage_path (*dicee.trainer.DICE_Trainer attribute*), 273
storage_path (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 266
store() (*in module dicee.static_funcs*), 259
store_ensemble() (*dicee.abstracts.AbstractPPECallback method*), 20
strategy (*dicee.abstracts.AbstractTrainer attribute*), 13
strict_negative_sampling (*dicee.config.Namespace attribute*), 29
StringListRequest (*class in dicee.scripts.index_serve*), 256
SWA (*class in dicee.weight_averaging*), 277
swa (*dicee.config.Namespace attribute*), 32
swa_c_epochs (*dicee.config.Namespace attribute*), 33
swa_c_epochs (*dicee.weight_averaging.SWA attribute*), 278
swa_c_epochs (*dicee.weight_averaging.SWAG attribute*), 279
swa_lr (*dicee.weight_averaging.SWA attribute*), 278
swa_lr (*dicee.weight_averaging.SWAG attribute*), 279
swa_model (*dicee.weight_averaging.SWA attribute*), 278
swa_n (*dicee.weight_averaging.SWA attribute*), 278
swa_start_epoch (*dicee.config.Namespace attribute*), 33
swa_start_epoch (*dicee.weight_averaging.SWA attribute*), 278
swa_start_epoch (*dicee.weight_averaging.SWAG attribute*), 279
SWAG (*class in dicee.weight_averaging*), 278
swag (*dicee.config.Namespace attribute*), 32

T

T() (*dicee.models.DualE method*), 244
T() (*dicee.models.dualE.DualE method*), 120
t_conorm() (*dicee.abstracts.InteractiveQueryDecomposition method*), 17
t_norm() (*dicee.abstracts.InteractiveQueryDecomposition method*), 17
target_dim (*dicee.dataset_classes.AllvsAll attribute*), 37
target_dim (*dicee.dataset_classes.MultiLabelDataset attribute*), 36
target_dim (*dicee.dataset_classes.OnevsAllDataset attribute*), 39
target_dim (*dicee.knowledge_graph.KG attribute*), 75
temperature (*dicee.models.transformers.BytE attribute*), 155
tensor_t_norm() (*dicee.abstracts.InteractiveQueryDecomposition method*), 17
TensorParallel (*class in dicee.trainer.model_parallelism*), 268
test_dataloader() (*dicee.models.base_model.BaseKGELightning method*), 98
test_dataloader() (*dicee.models.BaseKGELightning method*), 179
test_epoch_end() (*dicee.models.base_model.BaseKGELightning method*), 98
test_epoch_end() (*dicee.models.BaseKGELightning method*), 179
test_h1 (*dicee.analyse_experiments.Experiment attribute*), 22
test_h3 (*dicee.analyse_experiments.Experiment attribute*), 22
test_h10 (*dicee.analyse_experiments.Experiment attribute*), 22
test_mrr (*dicee.analyse_experiments.Experiment attribute*), 22
test_path (*dicee.query_generator.QueryGenerator attribute*), 246
test_path (*dicee.QueryGenerator attribute*), 289
test_samples (*dicee.models.flock.FlockBase attribute*), 125
test_set (*dicee.knowledge_graph.KG attribute*), 74, 75
timeit() (*in module dicee.read_preprocess_save_load_kg.util*), 252
timeit() (*in module dicee.static_funcs*), 258
timeit() (*in module dicee.static_preprocess_funcs*), 264
to() (*dicee.KGE method*), 284
to() (*dicee.knowledge_graph_embeddings.KGE method*), 76

to() (*dicee.models.ensemble.EnsembleKGE method*), 120
to_df() (*dicee.analyse_experiments.Experiment method*), 22
to_node (*dicee.models.flock.FlockBase attribute*), 125
to_type (*dicee.models.flock.FlockBase attribute*), 125
topk (*dicee.models.transformers.BytE attribute*), 155
topk (*dicee.scripts.index_serve.NeuralSearcher attribute*), 256
torch_ordered_shaped_bpe_entities (*dicee.dataset_classes.MultiLabelDataset attribute*), 36
TorchDDPTrainer (*class in dicee.trainer.torch_trainer_ddp*), 270
TorchFSDPTrainer (*class in dicee.trainer.torch_trainer_fsdp*), 271
TorchTrainer (*class in dicee.trainer.torch_trainer*), 269
total_epochs (*dicee.callbacks.LRScheduler attribute*), 27
total_steps (*dicee.callbacks.LRScheduler attribute*), 27
train() (*dicee.abstracts.BaseInteractiveTrainKGE method*), 20
train() (*dicee.models.graph_model.GraphKGE method*), 134
train() (*dicee.trainer.torch_trainer_ddp.NodeTrainer method*), 271
train_data (*dicee.dataset_classes.AllvsAll attribute*), 37
train_data (*dicee.dataset_classes.FSDP1vsSampleDataset attribute*), 38
train_data (*dicee.dataset_classes.KvsAll attribute*), 38
train_data (*dicee.dataset_classes.KvsSampleDataset attribute*), 39
train_data (*dicee.dataset_classes.MultiClassClassificationDataset attribute*), 35
train_data (*dicee.dataset_classes.OnevsAllDataset attribute*), 39
train_data (*dicee.dataset_classes.OnevsSample attribute*), 42
train_dataloader() (*dicee.models.base_model.BaseKGELightning method*), 99
train_dataloader() (*dicee.models.BaseKGELightning method*), 180
train_dataloaders (*dicee.trainer.torch_trainer.TorchTrainer attribute*), 269
train_dataset_loader (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 271
train_dataset_loader (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272
train_file_path (*dicee.dataset_classes.LiteralDataset attribute*), 40
train_h1 (*dicee.analyse_experiments.Experiment attribute*), 21
train_h3 (*dicee.analyse_experiments.Experiment attribute*), 21
train_h10 (*dicee.analyse_experiments.Experiment attribute*), 21
train_indices_target (*dicee.dataset_classes.MultiLabelDataset attribute*), 36
train_k_vs_all() (*dicee.abstracts.BaseInteractiveTrainKGE method*), 20
train_literals() (*dicee.abstracts.BaseInteractiveTrainKGE method*), 20
train_mode (*dicee.models.ensemble.EnsembleKGE attribute*), 120
train_mrr (*dicee.analyse_experiments.Experiment attribute*), 21
train_path (*dicee.query_generator.QueryGenerator attribute*), 246
train_path (*dicee.QueryGenerator attribute*), 289
train_set (*dicee.dataset_classes.BPE_NegativeSamplingDataset attribute*), 35
train_set (*dicee.dataset_classes.MultiLabelDataset attribute*), 36
train_set (*dicee.knowledge_graph.KG attribute*), 74, 75
train_set_target (*dicee.knowledge_graph.KG attribute*), 75
train_target (*dicee.dataset_classes.AllvsAll attribute*), 37
train_target (*dicee.dataset_classes.KvsAll attribute*), 38
train_target (*dicee.dataset_classes.KvsSampleDataset attribute*), 39
train_target_indices (*dicee.knowledge_graph.KG attribute*), 75
train_triples (*dicee.dataset_classes.FixedNegSampleDataset attribute*), 41
train_triples() (*dicee.abstracts.BaseInteractiveTrainKGE method*), 20
trained_model (*dicee.Execute attribute*), 282, 283
trained_model (*dicee.executer.Execute attribute*), 71
trainer (*dicee.config.Namespace attribute*), 30
trainer (*dicee.DICE_Trainer attribute*), 291
trainer (*dicee.Execute attribute*), 282, 283
trainer (*dicee.executer.Execute attribute*), 71
trainer (*dicee.trainer.DICE_Trainer attribute*), 273
trainer (*dicee.trainer.dice_trainer.DICE_Trainer attribute*), 266
trainer (*dicee.trainer.torch_trainer_ddp.NodeTrainer attribute*), 270
training_step (*dicee.trainer.torch_trainer.TorchTrainer attribute*), 269
training_step() (*dicee.models.base_model.BaseKGELightning method*), 96
training_step() (*dicee.models.BaseKGELightning method*), 178
training_step() (*dicee.models.transformers.BytE method*), 156
training_step_outputs (*dicee.models.base_model.BaseKGELightning attribute*), 96
training_step_outputs (*dicee.models.BaseKGELightning attribute*), 178
training_technique (*dicee.knowledge_graph.KG attribute*), 75
TransE (*class in dicee.models*), 193
TransE (*class in dicee.models.real*), 147
transformer (*dicee.models.clifford.KeciTransformer attribute*), 109
transformer (*dicee.models.KeciTransformer attribute*), 229

transformer (*dicee.models.transformers.BytE attribute*), 155
transformer (*dicee.models.transformers.GPT attribute*), 160
TransformerBlock (*class in dicee.models.clifford*), 110
TransformerMLP (*class in dicee.models.clifford*), 110
TransformerSelfAttention (*class in dicee.models.clifford*), 110
TransH (*class in dicee.models*), 194
TransH (*class in dicee.models.real*), 148
trapezoid() (*dicee.models.FMult2 method*), 241
trapezoid() (*dicee.models.function_space.FMult2 method*), 131
tri_score() (*dicee.models.function_space.LFMult method*), 132
tri_score() (*dicee.models.function_space.LFMult1 method*), 132
tri_score() (*dicee.models.LFMult method*), 243
tri_score() (*dicee.models.LFMult1 method*), 242
triple_score() (*dicee.KGE method*), 287
triple_score() (*dicee.knowledge_graph_embeddings.KGE method*), 79
TriplePredictionDataset (*class in dicee.dataset_classes*), 42
TRIX (*class in dicee.models*), 245
TRIX (*class in dicee.models.trix*), 163
trix_checkpoint (*dicee.config.Namespace attribute*), 29
trix_dim (*dicee.config.Namespace attribute*), 29
trix_query_batch_size (*dicee.config.Namespace attribute*), 29
TRIXBase (*class in dicee.models.trix*), 163
TRIXRelation (*class in dicee.models*), 245
TRIXRelation (*class in dicee.models.trix*), 164
tuple2list() (*dicee.query_generator.QueryGenerator method*), 246
tuple2list() (*dicee.QueryGenerator method*), 290
TWA (*class in dicee.weight_averaging*), 280
twa (*dicee.config.Namespace attribute*), 32
twa_c_epochs (*dicee.weight_averaging.TWA attribute*), 281
twa_model (*dicee.weight_averaging.TWA attribute*), 281
twa_start_epoch (*dicee.weight_averaging.TWA attribute*), 280
type_init (*dicee.models.flock.FlockBase attribute*), 125
type_logit (*dicee.models.flock.FlockBase attribute*), 125

U

ULTRA (*class in dicee.models*), 244
ULTRA (*class in dicee.models.ultra*), 167
ultra_checkpoint (*dicee.config.Namespace attribute*), 29
ultra_dim (*dicee.config.Namespace attribute*), 29
ultra_num_layers (*dicee.config.Namespace attribute*), 29
ultra_query_batch_size (*dicee.config.Namespace attribute*), 29
unlabelled_size (*dicee.callbacks.PseudoLabellingCallback attribute*), 25
unmap() (*dicee.query_generator.QueryGenerator method*), 247
unmap() (*dicee.QueryGenerator method*), 290
unmap_query() (*dicee.query_generator.QueryGenerator method*), 247
unmap_query() (*dicee.QueryGenerator method*), 290
update_hits() (*in module dicee.evaluation.utils*), 58
UPSTREAM_COMMIT (*in module dicee.models.flock*), 121
UPSTREAM_COMMIT (*in module dicee.models.trix*), 161
UPSTREAM_COMMIT (*in module dicee.models.ultra*), 165
use_clifford_mul (*dicee.models.clifford.KeciTransformer attribute*), 109
use_clifford_mul (*dicee.models.KeciTransformer attribute*), 229
use_compile (*dicee.trainer.torch_trainer_fsdp.TorchFSDPTrainer attribute*), 272

V

val_aswa (*dicee.weight_averaging.ASWA attribute*), 277
val_dataloader() (*dicee.models.base_model.BaseKGELightning method*), 98
val_dataloader() (*dicee.models.BaseKGELightning method*), 180
val_h1 (*dicee.analyse_experiments.Experiment attribute*), 22
val_h3 (*dicee.analyse_experiments.Experiment attribute*), 22
val_h10 (*dicee.analyse_experiments.Experiment attribute*), 22
val_mrr (*dicee.analyse_experiments.Experiment attribute*), 21
val_path (*dicee.query_generator.QueryGenerator attribute*), 246
val_path (*dicee.QueryGenerator attribute*), 289
VALID_SCORING_TECHNIQUES (*in module dicee.evaluation.evaluator*), 49
valid_set (*dicee.knowledge_graph.KG attribute*), 74, 75
validate_knowledge_graph() (*in module dicee.sanity_checkers*), 255

var_clamp (*dicee.weight_averaging.SWAG attribute*), 279
vocab_preparation() (*dicee.evaluation.Evaluator method*), 61
vocab_preparation() (*dicee.evaluation.evaluator.Evaluator method*), 50
vocab_preparation() (*dicee.Evaluator method*), 293
vocab_preparation() (*dicee.evaluator.Evaluator method*), 68
vocab_size (*dicee.models.CoKEConfig attribute*), 197, 198
vocab_size (*dicee.models.real.CoKEConfig attribute*), 152, 153
vocab_size (*dicee.models.transformers.GPTConfig attribute*), 159
vocab_to_parquet() (*in module dicee.static_funcs*), 260
vtp_score() (*dicee.models.function_space.LFMult method*), 133
vtp_score() (*dicee.models.function_space.LFMult1 method*), 132
vtp_score() (*dicee.models.LFMult method*), 243
vtp_score() (*dicee.models.LFMult1 method*), 242

W

w1 (*dicee.models.flock.FeedForward attribute*), 123
w2 (*dicee.models.flock.FeedForward attribute*), 123
w3 (*dicee.models.flock.FeedForward attribute*), 123
walk() (*dicee.models.flock_walks.WalkGraph method*), 126
walk_len (*dicee.models.flock.FlockBase attribute*), 125
walk_num (*dicee.models.flock.FlockBase attribute*), 125
WalkGraph (*class in dicee.models.flock_walks*), 126
warmup_steps (*dicee.callbacks.LRScheduler attribute*), 28
weight (*dicee.models.flock.RMSNorm attribute*), 122
weight (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 128
weight (*dicee.models.transformers.LayerNorm attribute*), 157
weight_decay (*dicee.config.Namespace attribute*), 31
weight_decay (*dicee.models.base_model.BaseKGE attribute*), 101
weight_decay (*dicee.models.BaseKGE attribute*), 182, 188, 200, 208, 216, 230, 236
weight_samples (*dicee.weight_averaging.TWA attribute*), 281
WeightedBCELoss (*class in dicee.losses.custom_losses*), 82
WeightedBCELoss (*class in dicee.models*), 174
weights (*dicee.models.FMult attribute*), 240
weights (*dicee.models.function_space.FMult attribute*), 130
weights (*dicee.models.function_space.GFMult attribute*), 130
weights (*dicee.models.GFMult attribute*), 240
world_size (*dicee.Execute attribute*), 282
world_size (*dicee.executer.Execute attribute*), 71
world_size (*dicee.models.fsdp_models.RowWiseShardedEmbedding attribute*), 127
write_csv_from_model_parallel() (*in module dicee.static_funcs*), 261
write_links() (*dicee.query_generator.QueryGenerator method*), 247
write_links() (*dicee.QueryGenerator method*), 290
write_report() (*dicee.Execute method*), 284
write_report() (*dicee.executer.Execute method*), 72

X

x_values (*dicee.models.function_space.LFMult attribute*), 132
x_values (*dicee.models.LFMult attribute*), 242